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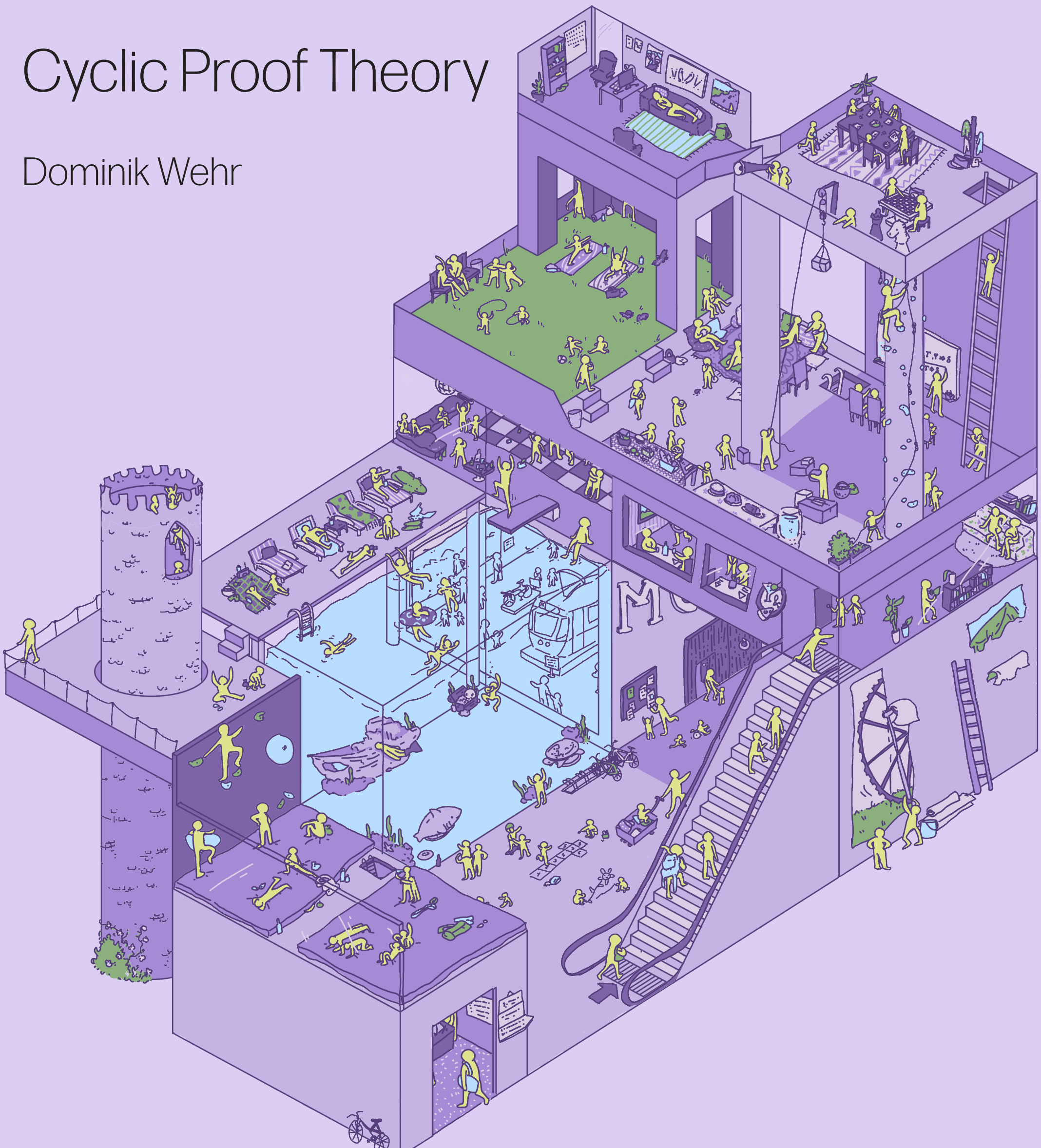
Cyclic Proof Theory

Dominik Wehr



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Cover by Amelie Wehr, dortselbst.
The cover depicts a building with paths leading up and back down, akin to a cyclic tree.

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Abstract

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Cyclic proofs are graphs, rather than trees, requiring additional soundness conditions to distinguish proofs from mere derivations. Using an abstract notion of cyclic proof system (CPS), this dissertation presents uniform results about different methods of representing the same CPS.

We prove that every traditional CPS can be represented as two other types of CPS: Reset proofs, which 'localise' soundness using a mechanism of sequent annotations, and stack-controlled proofs, introduced in this thesis, which align the cyclic and inductive structure of proofs.

Using stack-controlled proofs, we obtain multiple further results: a new translation from cyclic to 'plain' Heyting arithmetic; a correspondence between fragments of cyclic Peano arithmetic and the inductive fragments $I\Pi_n$; a proof-theoretic consistency argument for and ordinal annotation of cyclic Peano arithmetic; a general soundness argument for CPS which avoids the use of classical principles.

Reset proofs are employed to give a constructive analysis of the most common soundness condition for CPS. The condition is equivalent to the additive Ramsey theorem, the combination of two novel principles and, under unique choice, the LPO. The soundness condition can be constructivised by restricting it to periodic paths.

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First and foremost, I want to thank the two people without whom I would not even be here, writing this dissertation: Graham E. Leigh and Bahareh Afshari! Thank you so much for giving me this opportunity! I am well aware that I am not easy to supervise, being rather set in my own ways and resistant to advice. Thank you, Graham and Bahareh, for putting up with this and nonetheless giving me the kind of supervision I need, letting me go off on my own while also keeping me from getting lost. This cannot have been easy! Beyond the formal supervisor-supervisee-relationship, I also consider you two my friends! In that capacity, thank you, especially for all the bike rides, board gaming and climbing sessions!

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[1] Names are roughly grouped in friend groups and then sorted alphabetically; the order of groups is randomised.

dir meine Verbindung zur deutschen Kultur nicht komplette abreißt, und so vieles mehr! Du verdienst alles Glück der Welt und mehr!

Lastly, I want to thank everyone who had a part in creating and defending this dissertation. Thank you to everyone who collaborated with me on research, regardless of whether it bore fruit in this dissertation: Bahareh Afshari, Gianluca Curzi, Anupam Das, Dominik Kirst, Graham E. Leigh and Reuben Rowe. Further, thank you to everyone who helped with the creation of this document: Amelie for the *incredible* cover, Orvar for the "Sammanfattning" and Rasmus for the template. I want to thank Rajeev Gore for green reading the thesis, his feedback greatly affecting the final shape and orientation of this text. I also want to thank everyone who proof read this thesis: Bahareh, Dominik, Graham and Paaras. An especially great thanks is owed by me to Reuben Rowe, who has agreed to be the opponent at my defence.[2] I could not ask for a better person to do this! I want to thank everyone who agreed to be on the committee of my defence: Eleni Gregoromichelaki, Anders Mörtberg, Mernoosh Sadrzadeh and Thomas Studer. Thank you to Susanna Radovic for moderating the defence.

2 In Sweden, it is the opponent, not the defendant, who presents the dissertation to the committee and the public. Thus, this position entails a surprising amount of work.

Contents

Preface

It is often expected that a dissertation should be 'telling a coherent story'. I will not even attempt to do this. To me, that would feel like hiding the reality of my PhD: there is no grand plan, only problems which piqued my interest, some of which I managed to solve, which now populate my dissertation. The structure of my thesis witnesses this: the final two chapters are a rather random-seeming collection of individual results. Instead of trying to invent some common themes for these results post-hoc, I would much rather discuss how I have come across the various topics and questions covered in this thesis. This also makes for a somewhat more coherent story. Readers are welcome to skip ahead to chapter 1 if they are not interested.

I was introduced to cyclic proofs by Bahareh Afshari, while writing my Master's thesis with her at the ILLC in Amsterdam. We originally set out to transform a cyclic proof system for $\mathrm{HFL}_{\mathbb{N}}$ into one with a path condition.[3] Quickly, we realised that building such a proof system for $\mathrm{HFL}_{\mathbb{N}}$ would be about as hard as building one for *almost all* cyclic proof systems. Thus, we began by building an intricate, general notion of cyclic proof system. Then we ran out of time and I had to submit, never having worked on path conditions. When I started my PhD on cyclic proof theory, finally figuring out this construction was on top of my list. Thankfully Graham Leigh and I did manage this eventually, resulting in chapter 5.

While looking for PhD positions, I also talked to Anupam Das. He asked me to pick out an article to informally discuss research ideas around. I selected Sprenger and Dam's article on translating cyclic to inductive proofs for μFOL. My idea was to adapt the result to the setting of to cyclic arithmetics.[4] After starting my PhD in Gothenburg, I remained interested in the idea and continued working on it with Graham, leading to the results in sections 3.4 and 7.1. The

[3] Really, this meant a reset proof system, but we didn't know better back then.

[4] Back then, I was under the incorrect impression that the translation results obtained via Simpson's or Berardi–Tatsuta's methods were unconstructive. Of course, constructing proofs in classical proof systems is still constructive. This misunderstanding is especially embarrassing because results in my Bachelor's thesis relied on the exact same idea of 'pushing classical reasoning into the proof system'.

notion of stack-controlled proofs in chapter 6 arose directly from the desire to find a nicer representation for the normal forms relied upon by Sprenger and Dam. The generic presentation of stack-controlled proofs is just an extension of our work on reset proofs and my Master's thesis.

Graham originally proposed the idea of an ordinal analysis of cyclic arithmetic to me. While reading up on the topic, I noticed that following Gentzen's method, the closed end-part of a proof in cyclic arithmetic is the same as in regular Peano arithmetic, meaning most of the proof already works in the cyclic setting. However, the remaining task, closing the end-part by eliminating companions from it, turned out to be shockingly intricate and took us many months and multiple false starts to puzzle out. In the end, the result making it into the thesis at all (in section 7.3) was a true buzzer-beater.

As is common among youths interested in functional programming, learning Rocq[5] early during my Bachelor's quickly radicalized me into a Constructivist.[6] During that time, I was made to believe that all proof theory is naturally constructive.[7] To my shock and discomfort, I discovered that *cyclic* proof theory is typically quite unconstructive. I had already worked on constructive reverse mathematics in my Bachelor's thesis in Saarbrücken, under the supervision of Yannick Forster and Dominik Kirst. Thus, pinning down the unconstructivity of cyclic proof theory became a natural research question, which Dominik and I worked out in chapter 8. I had also always been discomforted by the fact that the usual soundness argument, even for intuitionistic theories like Heyting arithmetic, is highly classical. This, too, is thankfully assuaged by this thesis in section 7.4.

5 Formerly known as Coq.
6 I have since recovered; thanks for asking!
7 As someone raised on Rocq, I did not care about issues of (im)predicativity.

1 Introduction

Traditionally, proofs in logic are finite trees constructed according to certain derivation rules (see fig. 1a). That such proof trees embody sound arguments is easily verified: their leaves are axioms, which are always true, and the derivation rules are locally sound, meaning they derive true conclusions from true premises. If infinite proof trees are also considered, they are traditionally required to be well-founded i.e., possibly featuring infinitely branching derivation rules but no infinite branches (see fig. 1b). As each branch of a well-founded tree still ends in an axiom, the aforementioned soundness argument remains valid.

Ill-founded proofs break with this tradition, allowing for proof trees with infinite branches but typically no infinitely branching derivation rules (see fig. 1c). *Cyclic proofs* are rooted finite graphs, constructed according to derivation rules akin to the usual proof trees (see fig. 1d, where the proof nodes labelled by $A^\star$ are identical, the $\star$ thus indicating a cycle). Such a cyclic proof serves as a finitary representation of the ill-founded proof tree obtained by infinitely *unfolding* its cycles or, equivalently, generated by collecting all paths through it in a tree. The aforementioned soundness argument does not extend to such ill-founded derivation trees, finitely represented or otherwise, because these trees may now contain infinite branches which never end in axioms. Usually, this issue cannot be overcome: not all derivations of an ill-founded or cyclic proof system embody sound arguments. Instead, such proof systems employ a *soundness condition* which delineates between proofs and mere derivations.

$$\dfrac{\mathrm{B} \quad \dfrac{\mathrm{D}}{\mathrm{C}}}{\mathrm{A}}$$

(a) finite

$$\dfrac{\dots\ \mathrm{B} \quad \dfrac{\mathrm{D}}{\mathrm{C}}\ \dots}{\mathrm{A}}$$

(b) well-founded

$$\dfrac{\mathrm{B} \quad \dfrac{\dfrac{\vdots}{\mathrm{D}}}{\mathrm{C}}}{\mathrm{A}}$$

(c) ill-founded

$$\dfrac{\mathrm{B} \quad \dfrac{\dfrac{\mathrm{A}^\star}{\mathrm{D}}}{\mathrm{C}}}{\mathrm{A}^\star}$$

(d) cyclic

Figure 1: A schematic depiction of a ... proof.

From now on we only discuss cyclic proofs and cyclic proof systems as they are the focus of the thesis. However, everything mentioned in this section extends to ill-founded proof systems just as well.

Cyclic proof systems are most naturally compared to proof systems which derive finite proof trees because both only feature rules with finitely many premises. Such a comparison between proof systems for a given logic typically reveals a *trade-off*: The rules of the cyclic and traditional proof system are often the same, except for the cyclic proof system 'replacing' induction rules by *simpler proof rules*. Such simpler rules might be case-distinction rules for inductively defined objects (see e.g. Brotherston 2006) or unfolding rules for logical fixed points (see e.g. Niwiński and Walukiewicz 1996). Hence, cyclic proof systems are usually considered for logics or theories which contain some (co-)inductive features. This simplification of rules is offset by imposing a *more complex notion of "proof"*: Instead of simple well-formedness according to the derivation rules, an additional soundness condition must also be considered. For this reason, deriving cyclic proof transformations is more involved than usual because of the 'added obligation' of preserving soundness.

This dissertation was submitted during a 'high period' of the young field of *cyclic proof theory*. Generally, cyclic proof systems are attractive to proof theorists because the aforementioned simplification of derivation rules often makes them amenable to simpler proof techniques. Indeed, many cyclic proof systems are cut-free complete (see e.g. Niwiński and Walukiewicz (1996), Shamkanov (2014), Das and Girlando (2023), Afshari, Leigh, and Menéndez Turata (2023)), yielding *fully analytic proof systems*. At the moment, the literature mostly centres around a few types of results:

- *interpolation results* for various logics, see e.g. Shamkanov (2014), Afshari, Leigh, and Menéndez Turata (2021), Kloibhofer and Venema (2025), Kloibhofer, Trucco Dalmas, et al. (2026).
- *equivalences between cyclic and traditional proof systems*, see e.g. Simpson (2017), Berardi and Tatsuta (2017), Curzi and Das (2021), Afshari, Leigh, and Menéndez Turata (2024), Ikebuchi (2025), Horvat et al. (2026).
- *decidability results* via proof search, see e.g. Jungteerapanich (2010), Das, De, et al. (2022), Rooduijn, Kozen, et al. (2024), Afshari and Grotenhuis (2026).
- *cut-elimination results*, often using variants of continuous cut-elimination laid out by Mints (1978), see e.g. Fortier and Santocanale (2013), Savateev and Shamkanov (2021), Saurin (2023), Afshari, Leigh, and Menéndez Turata (2024), Miranda and Studer (2024), and Afshari and Kloibhofer (2025).

Note that the above is just a selection of results of the mentioned types, rather than an exhaustive enumeration, although we try to select references in such a way as to represent the diversity of proof strategies in the literature. The last item of the preceding list also illustrates the current rapid development of the field: A mere two years ago, Wehr (2023) still mentions general techniques for cut-elimination in cyclic and ill-founded proofs as an open question. Lastly, two possible landmark results bear mentioning.[8] Towsner (2024) gives an ordinal analysis of full second-order arithmetic using ill-founded proof trees. Borzechowski et al. (2025) prove Craig-interpolation for propositional dynamic logic. Both of these results solve long-standing open problems in their respective fields.

Outside of proof theory, cyclic proof systems have also been of interest to the community working on *automatic theorem proving*, aiming to develop programs which can prove theorems without human interaction. There exist multiple experimental theorem provers relying on some form of cyclic proof search, see e.g. Brotherston, Distefano, et al. (2011), Brotherston, Gorogiannis, et al. (2012), Tellez and Brotherston (2017), Stratulat (2020), Itzhaky et al. (2021), Jones et al. (2022), and Cohen, Rowe, et al. (2025).

Key concepts & ideas

This section introduces the concepts and ideas which are at the core to understanding this dissertation's approach to cyclic proof theory.

Every cyclic proof system features a soundness condition which distinguishes between cyclic proofs and mere cyclic derivations (usually called *preproofs*). By far the most common soundness condition in the literature is the *global trace condition* (GTC). While the specific details differ between proof systems, this condition always takes the same 'shape': along *every infinite branch* of a proof, some syntactic feature must be *tracked* which *progresses* infinitely often. This general shape is not: the infinite branches of cyclic proofs are precisely what 'broke' the usual soundness argument, so it stands to reason that they would be the target of any soundness. Typically, the syntactic feature to be tracked is some inductively generated object or logical fixed point and 'progress' is a case-distinction or a fixed point unfolding. For example, in simple cyclic Heyting arithmetic (covered in section 2.1) one follows variables denoting natural numbers and 'progress' occurs when the case-distinction rule, between 0 and successor, is applied.

The global trace condition serves three different purposes in this dissertation.

[8] At time of writing, neither result has been peer-reviewed.

For one, it is often *the source of cyclic soundness*: essentially every other soundness condition in the literature can be viewed as a mechanism to ensure the global trace condition holds. When constructing cyclic proof systems with more intricate soundness conditions, we justify the correctness of said constructions by proving that they ensure the GTC. Second, the global trace condition serves as a sort of *antagonist* in this dissertation. While widespread, the GTC has many shortcomings which we discuss in great detail in section 3.1. Many of the results of this dissertation concern the construction of cyclic proof systems with 'better' soundness conditions. Lastly, the global trace condition is a key witness of an important observation which underlies much of the technical work in this thesis: any cyclic proof system may be divided into its *logical* and *trace components*. As mentioned above, the GTC in some way features in the vast majority of cyclic proof systems. The trace component of a cyclic proof system essentially only contains the 'combinatorial' aspects of the GTC while the logical component contains 'everything else' (e.g. the choice of sequents, derivation rules, models and the semantic reason why the GTC ensures soundness of the proof system). The mechanism underlying soundness conditions different from the GTC can usually be understood solely through the 'combinatorial lens' of the trace perspective.

This trace perspective motivates both the methods and the results of this thesis: Because the trace components are structurally very similar, even between cyclic proof systems for wildly different logics, one can uniformly study cyclic proof systems through this 'lens'. Indeed, the main goal of this dissertation is to further develop *the general theory of cyclic proofs* from this abstract perspective. It should be noted that we are not the only ones working from this perspective, see e.g. Brotherston (2006) or Cohen, Jabarin, et al. (2024).

Another key concept of this thesis are *representations* of cyclic proof systems and cyclic proofs. Consider a cyclic proof system $\mathcal{R}$ whose soundness condition is a GTC. There usually exist other cyclic proof systems $\mathcal{R}'$, which can be regarded as variants of $\mathcal{R}$, with different soundness conditions, serving as a mechanism to ensure the GTC of $\mathcal{R}$. We call such $\mathcal{R}'$ representations of $\mathcal{R}$ because they are just different ways of presenting the 'underlying' cyclic proof system $\mathcal{R}$. Typically, the proofs of $\mathcal{R}'$ will also be representations of certain $\mathcal{R}$-proofs, often in some kind of normal form. A key idea of this thesis is that, just as between cyclic and non-cyclic proofs, there are trade-offs between different representations, which we discuss more thoroughly in chapter 3. The new results presented in this thesis focus on cyclic proof representations which are especially well-suited to proof theoretic investigations: reset proofs and stack-controlled proofs.

Reset proofs were originally introduced for the modal μ-calculus by Jungteerapanich (2010) and Stirling (2013). They ensure soundness through a mechanism of sequent annotations, inspired by the automata determinisation construction of Safra (1988). To verify the GTC on a preproof, one might have to check uncountably many branches. Contrasting this, the soundness condition of reset proofs is a *path condition*: to ensure soundness, a simple condition must only be checked on finitely many *local cycles* of a preproof. Such path conditions present soundness as a local condition as opposed to the *global* trace condition. This makes cyclic proof systems with path conditions, such as reset proof systems, much better suited to both reasoning about proof transformations and 'data extraction', such as finding interpolants.

Stack-controlled proofs are a new representation proposed in this thesis. Their proofs embody a normal form which was first studied by Sprenger and Dam (2003b) for μFOL, first-order logic with least and greatest fixed points. To understand this normal form, note that the local cycles of a cyclic proof can be ordered along a *physical* and a *logical order*.[9] The physical ordering records along which local cycle a 'detour' through which other local cycle can occur. The logical ordering, interpreting each local cycle as embodying an instance of induction, records which local cycle's induction is 'inside' which other local cycle's. Usually, these two orderings need not be compatible and indeed such 'misalignments' are a key source of complexity in cyclic proof theory. Stack-controlled proofs embody the cyclic proofs in which these two orderings align. The name "stack-controlled" derives from their mechanism for ensuring soundness: each sequent is annotated by a stack which tracks both the available 'inductive hypotheses' and the nodes which the proof may 'cycle back to' at any given point. Because of their alignment of the physical and inductive proof structure, stack-controlled proofs are closely related to finite proof trees with induction rules. We exploit this connection in various ways in chapter 7 to derive multiple results about cyclic proof systems for arithmetic.

Outline & contributions

We close this chapter with an outline of the remaining chapters of the dissertation. We also take this opportunity to highlight our main contributions.

Chapters 2 and 3 provide an extended introduction to cyclic proof theory and the perspective of cyclic proof representation which is central to this dissertation.

[9] Note that both of these orders can be characterised solely in terms of the trace component.

Two cyclic proof systems for Heyting arithmetic, CHA and $\mathrm{CHA}_<$, are introduced in chapter 2 and serve as running examples throughout the rest of the thesis. These chapters provide an accessible introduction to the wider field of cyclic proof theory, requiring only a light proof theory background as provided by an introductory proof theory course or book, such as Mancosu et al. (2021).

From chapter 4 onwards, the dissertation's focus shifts to presenting our new results. As such, we write for a narrower audience of experts from the field of cyclic proof theory.

Chapter 4 introduces an abstract rendition of cyclic proof systems and their soundness conditions, which we rely on throughout the rest of this dissertation. Special attention is paid to representing the global trace condition via so-called activation algebras. It should be noted that the material of this chapter cannot be counted towards the examination of this thesis because most notions are taken from Wehr's Master's thesis (Wehr 2021).

Chapter 5 is concerned with reset proof systems. We give a general definition of reset proof systems and show that every cyclic proof system with the GTC as its soundness condition can be transformed into an equivalent reset proof system. We prove a strong equivalence between the two cyclic proof systems which preserves all proof theoretic content of proofs in both directions. While this result is stated abstractly, it can be used to obtain concrete reset proof systems corresponding to existing cyclic proof, as we demonstrate at the examples of CHA and $\mathrm{CHA}_<$.

In chapter 6 we introduce stack-controlled proofs. As for reset proof systems, we prove that every cyclic proof system with the GTC as its soundness condition can be transformed into an equivalent stack-controlled proof system. This equivalence again preserves all proof theoretic content in both directions. These constructions of this chapter rely on the results of chapter 5 because stack-controlled proofs can be seen as representing a normal form of reset proofs. We again demonstrate the use of the abstract result by deriving stack-controlled proof systems corresponding to CHA and $\mathrm{CHA}_<$.

In chapter 7, we obtain various results of cyclic proof theory for CHA, $\mathrm{CHA}_<$ and variants thereof. These results all utilise the completeness of their stack-controlled proof systems and the aforementioned connection between stack-controlled proofs and finite proof trees with induction rules:

- In section 7.1, we translate proofs of $\mathrm{CHA}_<$ into finitary proofs of Heyting arithmetic. While the equivalence of the two systems is known (Berardi and Tatsuta 2017), this result demonstrates a new technique for obtaining such equivalences which we expect to adapt well to other settings.

- Section 7.2 refines the aforementioned result into a characterisation of fragments $\mathrm{CI}\Pi_n$ of the cyclic arithmetic of Simpson (2017) which correspond the fragments $\mathrm{I}\Pi_n$ of restricted induction of Peano arithmetic. This extends prior work of Das (2020) and Beklemishev et al. (2025).
- In section 7.3, we adapt the proof of the consistency of PA given by Gentzen (1938) to cyclic Peano arithmetic directly, giving an ordinal annotation scheme of stack-controlled CPA. To the best of our knowledge, this is the first result of its kind in cyclic proof theory.
- Section 7.4 presents an abstract rendition of a direct[10] soundness argument for cyclic proof systems. That is, an abstract argument showing that a locally sound cyclic proof system are sound. This is an extension of a similar undertaking by Brotherston (2006), which gives a highly classical abstract soundness argument.

We close in chapter 8 with a constructive analysis of the usual global trace condition. We identify its non-constructivity with the additive Ramsey principle and put forward a suitable constructive alternative: a version of the GTC which only considers periodic branches. These results can be seen as an extension of the constructive analysis of infinite word automata by Lichter and Smolka (2018). We also contribute two new characterisations of the additive Ramsey principle: It is equivalent to the combination of two novel but natural principles which we call ILPO and TWKL. Under unique choice, it is also equivalent to the limited principle of omniscience (LPO).

[10] One could be tempted to call it constructive. However, there are some details involving ordinals which make this classification a little more subtle.

2 Cyclic arithmetics

This chapter introduces two cyclic proof systems for Heyting arithmetic. In this chapter, we use them to introduce the basic notions of cyclic proofs. The coming chapters rely on them as running examples.

2.1 Simple cyclic Heyting arithmetic CHA

The theory of Heyting arithmetic employs the language of first-order arithmetic, which is given below. We fix a countable set of variables $\textsc{Var}$ and write $\mathrm{FV}(\varphi)$ for the set of *free variables* of φ.

$$
\begin{aligned}
s, t \in \textsc{Term} &::= x \mid 0 \mid Ss \mid s + t \mid s \cdot t && x \in \textsc{Var} \\
\varphi, \psi \in \textsc{Form} &::= s = t \mid \bot \mid \varphi \wedge \psi \mid \varphi \vee \psi \mid \varphi \to \psi \mid \forall x.\varphi \mid \exists x.\varphi
\end{aligned}
$$

We define negation as $\neg\varphi \coloneqq \varphi \to \bot$. Denote by $[t/x]$ the usual *substitution operation*, replacing all free occurrences of the variable x by the term t. On formulas, this is a partial operation, $\varphi[t/x]$ being undefined if the free variables in t are not distinct from the bound variables in φ. Henceforth, writing $\varphi[t/x]$ will double as an assertion of the resulting formula being defined. When x is unimportant or clear from context, we sometimes write $\varphi(t)$ for $\varphi[t/x]$.

Before presenting a cyclic proof system, we begin by recalling an inductive proof system for Heyting arithmetic i.e., one presented in the common manner of finite derivation trees.

Definition 2.1.1 (Heyting arithmetic)**.** The *sequents* of *Heyting arithmetic* are expressions $\Gamma \Rightarrow \delta$ where Γ is a finite set of formulas and δ a single formula of first-order arithmetic. Write Γ, φ for $\Gamma \cup \{\varphi\}$ and Γ, Γ' for $\Gamma \cup \Gamma'$. The *derivation rules* of HA comprise of the following choice of standard rules for classical first-order logic with equality,

$$
\mathrm{Ax}\,\frac{}{\Gamma, \delta \Rightarrow \delta}
\qquad
{\to}\mathrm{L}\,\frac{\Gamma \Rightarrow \varphi \quad \Gamma, \psi \Rightarrow \delta}{\Gamma, \varphi \to \psi \Rightarrow \delta}
\qquad
{\to}\mathrm{R}\,\frac{\Gamma, \varphi \Rightarrow \psi}{\Gamma \Rightarrow \varphi \to \psi}
$$

$$
{\wedge}\mathrm{L}\,\frac{\Gamma, \varphi, \psi \Rightarrow \delta}{\Gamma, \varphi \wedge \psi \Rightarrow \delta}
\qquad
{\wedge}\mathrm{R}\,\frac{\Gamma \Rightarrow \varphi \quad \Gamma \Rightarrow \psi}{\Gamma \Rightarrow \varphi \wedge \psi}
$$

$$\vee\mathrm{L}\frac{\Gamma, \varphi \Rightarrow \delta \quad \Gamma, \psi \Rightarrow \delta}{\Gamma, \varphi \vee \psi \Rightarrow \delta} \qquad \vee\mathrm{R}\frac{\Gamma \Rightarrow \varphi_i}{\Gamma \Rightarrow \varphi_1 \vee \varphi_2}$$

$$\forall\mathrm{L}\frac{\Gamma, \varphi[t/x] \Rightarrow \delta}{\Gamma, \forall x.\varphi \Rightarrow \delta} \qquad \forall\mathrm{R}\frac{\Gamma \Rightarrow \varphi \quad x \notin \mathrm{FV}(\Gamma)}{\Gamma \Rightarrow \forall x.\varphi}$$

$$\exists\mathrm{L}\frac{\Gamma, \varphi \Rightarrow \delta \quad x \notin \mathrm{FV}(\Gamma)}{\Gamma, \exists x.\varphi \Rightarrow \delta} \qquad \exists\mathrm{R}\frac{\Gamma \Rightarrow \varphi[t/x]}{\Gamma \Rightarrow \exists x.\varphi} \qquad \bot\mathrm{L}\frac{}{\Gamma, \bot \Rightarrow \delta}$$

$$=\!\mathrm{L}\frac{\Gamma[t/x, s/y] \Rightarrow \delta[t/x, s/y] \quad x, y \notin \mathrm{FV}(s, t)}{\Gamma[s/x, t/y], s = t \Rightarrow \delta[s/x, t/y]} \qquad =\!\mathrm{R}\frac{}{\Gamma \Rightarrow t = t}$$

with the following two structural rules,

$$\textsc{Wk}\frac{\Gamma \Rightarrow \delta}{\Gamma, \Gamma' \Rightarrow \delta} \qquad \textsc{Cut}\frac{\Gamma \Rightarrow \varphi \quad \Gamma, \varphi \Rightarrow \delta}{\Gamma \Rightarrow \delta}$$

the following axiomatic sequents, which characterise the function symbols of first-order arithmetic,

$$0 = St \Rightarrow \bot \qquad Ss = St \Rightarrow s = t \qquad \Rightarrow s + 0 = 0$$
$$\Rightarrow s + St = S(s + t) \qquad \Rightarrow s \cdot 0 = 0 \qquad \Rightarrow s \cdot St = (s \cdot t) + s$$

and the axiomatic sequents of *induction*, for any formula φ,

$$\varphi(0), \forall x.\varphi(x) \to \varphi(Sx) \Rightarrow \varphi(s).$$

If a proof with endsequent $\Gamma \Rightarrow \delta$ exists then $\Gamma \Rightarrow \delta$ is *provable in Heyting arithmetic*, denoted by $\mathrm{HA} \vdash \Gamma \Rightarrow \delta$.

Remark 2.1.1. Observe that, as the left-hand side of sequent in HA is a set, no explicit contraction rule is needed. Instead, such set sequents allow for *implicit contraction*. For example, the following is an instance of $\forall$L

$$\forall\mathrm{L}\frac{\forall x.\varphi, \varphi[t/x] \Rightarrow \delta}{\forall x.\varphi \Rightarrow \delta}$$

in which we choose $\Gamma \coloneqq \{\forall x.\varphi\}$, observing that $\Gamma \cup \{\forall x.\varphi\} = \Gamma$.

To illustrate the contrast with cyclic proof systems, it is helpful to consider the representation of proofs in Heyting arithmetic. Formally, a proof in HA is a pair $\Pi = (T, \lambda)$ consisting of a finite tree T and a function $\lambda : \mathrm{Node}(T) \to \textsc{Seq}$ labelling the nodes by sequents, in a manner consistent with the derivation rules given above. Considering the conclusions of rules without premises as axiomatic, if $t \in \mathrm{Leaf}(T)$ then $\lambda(t)$ is an axiomatic sequent.

The underlying structure of cyclic proofs is somewhat more complicated. Often, this structure is described simply as 'finite graphs'. However, in practice, a stricter notion consisting of a finite tree with 'added cycles' often proves more useful, retaining common notions, such as 'root', 'leaf' and 'subtree'. A *cyclic tree* is a pair $C = (T, \beta)$ consisting of a finite tree T and a partial function $\beta\colon \mathrm{Leaf}(T) \to \mathrm{Inner}(T)$ mapping some leaves of T onto inner nodes of T. A leaf $b \in \mathrm{dom}(\beta)$ is called a *bud* and $\beta(b)$ its *companion*.

Commonly, a further restriction is imposed on the cyclic trees underlying cyclic proofs. A cyclic tree is said to be in *cycle normal form* if for every bud $b \in \mathrm{dom}(\beta)$ its companion $\beta(b)$ occurs along the path from the tree's root to b. This restriction has various useful consequences. For example, all buds of an inner node must be located in the subtree above it, easing recursive proof translations. Furthermore, trees in cycle normal form possess a well-defined notion of the 'inside' of a cycle: the path from $\beta(t)$ to t for any bud $t \in \mathrm{dom}(\beta)$. Every cyclic tree can be unfolded into a cyclic tree in cycle normal form, although this may cause a super-exponential increase in the number of nodes (see Brotherston 2006, theorem 6.3.6).

We proceed by defining a cyclic proof system CHA for Heyting arithmetic. It eschews the induction axioms in favour of *cycles* which allow leaves to be considered closed if they are labelled by a sequent that has previously appeared on the branch between the root and said node.

Definition 2.1.2 (CHA-preproofs)**.** The sequents of *cyclic Heyting arithmetic* (CHA) are the same as in regular Heyting arithmetic. A *preproof* of CHA is a pair $\Pi = (C, \lambda)$ consisting of a cyclic tree $C = (T, \beta)$ and a labelling function $\lambda : \mathrm{Node}(T) \to \textsc{Seq}$. The labelling function λ labels the nodes of T with sequents such that

- inner nodes are labelled according to the derivation rules of HA or the CHA-specific derivation rules:

$$\textsc{Case}_x\,\frac{\Gamma[0/x] \Rightarrow \delta[0/x] \quad \Gamma[Sx/x] \Rightarrow \delta[Sx/x]}{\Gamma \Rightarrow \delta}$$

$$\textsc{Sub}\,\frac{\Gamma \Rightarrow \delta}{\Gamma[t/x] \Rightarrow \delta[t/x]}$$

  in which $x \in \mathrm{FV}(\Gamma, \delta)$ for $\textsc{Case}_x$ and $x \notin \mathrm{FV}(t)$ for $\textsc{Sub}$.
- leaves, with the exception of buds, must be labelled with HA axioms, excluding the induction axioms.

- each bud and companion share labels i.e., $\lambda(s) = \lambda(\beta(s))$ for $s \in \mathrm{dom}(\beta)$.

For an example of a CHA-preproof, consider fig. 2. It derives the induction scheme of HA. We employ the shorthand $\overline{\varphi} := \varphi(0), \forall x.\varphi(x) \to \varphi(Sx)$ for the left-hand side of the induction scheme. The two sequents marked with a ★ form a cycle.

$$
\begin{prooftree}
\AxiomC{}
\LeftLabel{\textsc{Ax}}
\UnaryInfC{$\overline{\varphi}, \varphi(0) \Rightarrow \varphi(0)$}
\AxiomC{$\overline{\varphi} \Rightarrow \varphi(x)\ \star$}
\LeftLabel{\textsc{Wk}}
\UnaryInfC{$\overline{\varphi} \Rightarrow \varphi(x), \varphi(Sx)$}
\AxiomC{}
\RightLabel{\textsc{Ax}}
\UnaryInfC{$\overline{\varphi}, \varphi(Sx) \Rightarrow \varphi(Sx)$}
\RightLabel{$\forall$L,$\to$L}
\BinaryInfC{$\overline{\varphi}, \forall x.\varphi(x) \to \varphi(Sx) \Rightarrow \varphi(Sx)$}
\LeftLabel{\textsc{Case}$_x$}
\BinaryInfC{$\overline{\varphi} \Rightarrow \varphi(x)\ \star$}
\LeftLabel{\textsc{Sub}}
\UnaryInfC{$\overline{\varphi} \Rightarrow \varphi(s)$}
\end{prooftree}
$$

Figure 2: A CHA-preproof of the induction axiom, ★ marking a cycle

However, not every CHA-preproof is a sound argument. Figure 3 presents a counterexample, a cyclic 'argument' deriving the inconsistency $\Rightarrow \bot$.

$$
\begin{prooftree}
\AxiomC{}
\LeftLabel{$\bot$L}
\UnaryInfC{$\bot \Rightarrow \bot$}
\AxiomC{$\Rightarrow \bot\ \star$}
\LeftLabel{\textsc{Cut}}
\BinaryInfC{$\Rightarrow \bot\ \star$}
\end{prooftree}
$$

Figure 3: An unsound CHA-preproof, ★ marking a cycle

To ensure the soundness of CHA, an additional condition beyond well-formedness, called a *soundness condition*, must be imposed. The soundness condition given in definition 2.1.3 is called a *global trace condition*. We consider alternative soundness conditions for cyclic Heyting arithmetic in chapter 3.

Definition 2.1.3 (CHA-proofs)**.** Let $\Pi = (C, \lambda)$ be a preproof. An *infinite branch* through Π is an infinite sequence $t \in \mathrm{Node}(T)^\omega$ with t_0 being the root node of T and such that for each $i \in \omega$ either $t_{i+1} \in \mathrm{Chld}(t_i)$ or $t_i \in \mathrm{dom}(\beta)$ and $t_{i+1} = \beta(t_i)$. This induces an infinite sequence $(\Gamma_i \Rightarrow \delta_i)_{i\in\omega}$ of sequents with $\lambda(t_i) = \Gamma_i \Rightarrow \delta_i$, which we use interchangeably with $(t_i)_{i\in\omega}$ to denote an infinite branch. A variable x is said to have *a trace along* $(\Gamma_i \Rightarrow \delta_i)_{i\in\omega}$ if there exists an $n \in \omega$ such that $x \in \mathrm{FV}(\Gamma_i, \delta_i)$ for all $i \geq n$. Such a trace on x is said to be *progressing* if its branch passes through the right-hand side of the $\textsc{Case}_x$-rule (for the same variable x) infinitely often.

A CHA-preproof Π is a *proof* in cyclic Heyting arithmetic if for every infinite branch through Π there exists a variable which has a progressing trace along it. CHA $\vdash \Gamma \Rightarrow \delta$ denotes provability of $\Gamma \Rightarrow \delta$ in CHA.

Observe that the preproof of $\Rightarrow \bot$ in fig. 3 is no CHA-proof because it does not contain any free variables. The preproof of the induction scheme in fig. 2 is a proper proof: It only has one infinite branch, obtained by following the $\star$-cycle indefinitely. This branch has an x-trace, starting above the Sub-application, which passes through the $\textsc{Case}_x$-rule infinitely often.

From the perspective of proofs-as-programs, cyclic proofs correspond to recursively defined functions, the soundness conditions corresponding to termination conditions. This connection has been used to bring notions and results from the field of program termination (Lee et al. 2001) to the setting of cyclic proof theory (e.g. Nollet et al. 2019; Ikebuchi 2025).

The global trace condition is the most common soundness condition in cyclic proof theory. It usually follows a common template: Along every infinite branch, a syntactic feature (e.g. a term, fixed-point quantifier, formula) can be traced which *progresses* (e.g. decreases, is unfolded, is principal in certain derivation rules) infinitely often. Based on this observation, abstract notions of trace, progress and cyclic proof can be formulated which allow the study of cyclic proofs from a general perspective (Brotherston 2006; Afshari and Wehr 2024). We discuss the latter framework in chapter 4 and rely on it in chapters 5, 6 and 8 to present arguments and constructions for many cyclic proof systems at once.

The system CHA is merely one possible cyclic proof system for Heyting arithmetic. Even when restricting one's attention to cyclic proof systems with global trace conditions, there are other renditions of Heyting arithmetic as a cyclic proof system in the literature, one of which we present in section 2.3.

The cyclic proof system CHA can be transformed into a cyclic proof system CPA of Peano arithmetic via the usual broadening of the right-hand side of sequents to finite sets of formulas Δ. The global trace condition, including the employed notions of branch, trace and progress, remain unchanged.

The purpose of the global trace condition in definition 2.1.3 is most easily illustrated by proving the soundness of CHA over the standard model of the natural numbers i.e., the structure ω of first-order arithmetic whose domain are the natural numbers and which interprets the function symbols in the usual manner.

Theorem 2.1.4 (Soundness)**.** If CHA $\vdash \Gamma \Rightarrow \delta$ then $\omega \vDash \Gamma \Rightarrow \delta$.

Proof. Let Π be a CHA-proof of $\Gamma \Rightarrow \delta$ and suppose, towards contradiction, that there is an assignment $\rho : \textsc{Var} \to \omega$ such that $\rho \nvDash \Gamma \Rightarrow \delta$. Consider the last CHA-rule applied to derive $\Gamma \Rightarrow \delta$ in Π. It must have a premise $\Gamma_1 \Rightarrow \delta_1$ and there must be an assignment ρ_1 such that $\rho_1 \nvDash \Gamma_1 \Rightarrow \delta_1$. We treat a few illustrative cases:

$\wedge$R: Then $\delta = \varphi_0 \wedge \varphi_1$ and the premises of the rule are $\Gamma \Rightarrow \varphi_i$ for $i \in \{0, 1\}$. Fix $\rho_1 := \rho$. If $\rho \nvDash \Gamma \Rightarrow \delta$ then $\rho \vDash \Gamma$ and $\rho \nvDash \delta$. Thus there must be $i \in \{0, 1\}$ such that $\rho \nvDash \varphi_i$ and thus $\rho \nvDash \Gamma \Rightarrow \varphi_i$.

Sub: Then $\Gamma = \Gamma'[t/x]$ and $\delta = \delta'[t/x]$ for $x \notin \mathrm{FV}(t)$. Furthermore, the premise of the rule is $\Gamma' \Rightarrow \delta'$. Fix $\rho_1 := \rho[x \mapsto t^\rho]$ i.e., the environment which assigning the same values to $y \neq x$ as ρ and the evaluation of t under ρ to x. Observe that for every term s, $(s[t/x])^\rho = s^{\rho_1}$. Thus, $\rho \nvDash \Gamma[t/x] \Rightarrow \delta[t/x]$ entails $\rho_1 \nvDash \Gamma \Rightarrow \delta$.

$\forall$R: Then $\delta = \forall x.\varphi$ with $x \notin \mathrm{FV}(\Gamma)$ and the premise is $\Gamma \Rightarrow \varphi$. By $\rho \nvDash \Gamma \Rightarrow \forall x.\varphi$ there must exist $n \in \omega$ such that $\rho[x \mapsto n] \nvDash \varphi$. Hence, fixing $\rho_1 := \rho[x \mapsto n]$, obtains $\rho_1 \nvDash \Gamma \Rightarrow \varphi$ because $x \notin \mathrm{FV}(\Gamma, \delta)$.

$\forall$L: Then $\Gamma = \Gamma', \forall x.\varphi$ and the premise is $\Gamma, \varphi[t/x] \Rightarrow \delta$. From $\rho \nvDash \Gamma, \forall x.\varphi$ it follows that $\rho \vDash \forall x.\varphi$ and thus $\rho[x \mapsto t^\rho] \vDash \varphi$, or equivalently, $\rho \vDash \varphi[t/x]$. Hence, fixing $\rho_1 := \rho$, it follows that $\rho_1 \nvDash \Gamma', \varphi[t/x] \Rightarrow \delta$.

=L: Then $\Gamma = \Gamma'[s/x, t/y], s = t$ and $\delta = \delta'[s/x, t/y]$ with $x, y \notin \mathrm{FV}(s, t)$ and the premise is $\Gamma'[t/x, s/y] \Rightarrow \delta'[t/x, s/y]$. From $\rho \nvDash \Gamma \Rightarrow \delta$ it follows that $\rho \vDash s = t$ and hence $s^\rho = t^\rho$. Hence, for any term u and $v \in \{x, y\}$, $(u[s/v])^\rho = (u[t/v])^\rho$ and thus $\rho \nvDash \Gamma'[s/x, t/y], s = t \Rightarrow \delta'[s/x, t/y]$ entails $\rho \nvDash \Gamma'[t/x, s/y] \Rightarrow \delta'[t/x, s/y]$.

Case$_x$: Then $x \in \mathrm{FV}(\Gamma, \delta)$. Observe that either $\rho(x) = 0$ or $\rho(x) = n + 1$. In the former case, $\rho \nvDash \Gamma[0/x] \Rightarrow \delta[0/x]$, in the latter case $\rho[x \mapsto n] \nvDash \Gamma[Sx/x] \Rightarrow \delta[Sx/x]$.

The remaining cases are analogous to those considered above. This argument can be applied to any non-axiomatic node of Π. The contradicted premises arrived at cannot be axioms as they cannot be contradicted in the standard model. Iterating this argument, 'following' buds to their companions, yields an infinite branch $(\Gamma_i \Rightarrow \delta_i)_{i\in\omega}$ through Π and a sequence $(\rho_i)_{i\in\omega}$ of contradicting assignments such that $\rho_i \nvDash \Gamma_i \Rightarrow \delta_i$. As Π is a proof, there exists a progressing x-trace along $(\Gamma_i \Rightarrow \delta_i)_{i\in\omega}$ (w.l.o.g. we assume it starts at $\Gamma_0 \Rightarrow \delta_0$). Now consider the sequence $(\rho_i(x))_{i\in\omega}$ in ω. From the cases above it follows that an increase i.e.,

$\rho_i(x) < \rho_{i+1}(x)$, can only occur in the cases of ∀R and ∃L with x being the bound variable. However, as x is free in all $\Gamma_i \Rightarrow \delta_i$, this cannot occur along the branch. On the other hand, decreases i.e., $\rho_i(x) > \rho_{i+1}(x)$, take place whenever the right-hand side of a $\textsc{Case}_x$-rule is passed. As this takes place infinitely often, $(\rho_i(x))_{i\in\omega}$ is a never increasing, infinitely decreasing sequence of natural numbers, which cannot exist. Hence, the conclusion of Π could not have been invalid to begin with. □

One way of understanding the global trace condition is as 'reintroducing' the well-foundedness, usually required of the proofs themselves, at the level of the logic. For example, the global trace condition of CHA ensures that all infinite branches of a proof produce an infinitely decreasing sequence and can thus be disregarded when considering soundness. A similar strategy, for example adopted by cyclic proof systems for μ-calculi, is to allow for coinductive arguments, by requiring greatest fixed points to be unfolded infinitely along infinite branches, thereby 'making' all infinite branches sound, even though they do not end in axioms.

The proof strategy of theorem 2.1.4 is employed throughout the literature to prove soundness of cyclic proof systems with a global trace condition. Often, the demonstrated decreasing sequence is a sequence of ordinals, rather than just natural numbers. Brotherston (2006, section 4.2.1) gives a general account of this style of soundness proof. It should be noted the soundness proof is highly classical: deducing the existence of a contradicted premise requires variants of the law of the excluded middle and 'constructing' the contradicted branch by iterating the argument requires a variant of the axiom of choice. In section 7.4, we give a similarly general, direct soundness proof, which avoids the sources of classicality pointed out above.[11]

2.2 Relating proof systems

This section discusses how cyclic proof systems may be related to other kinds of proof systems. We continue to work with CHA as the running example. Because HA is incomplete with regards to the standard model, the soundness result in theorem 2.1.4 does not contribute towards proving the equivalence between the 'inductive' proof system HA and the cyclic proof system CHA. Instead, this

[11] The argument is essentially constructive, except for some operations performed on ordinals, whose constructivity hinges on the precise representation/class of ordinals required for the soundness of the specific logics.

equivalence is witnessed by proof theoretic translations back and forth. One of the two directions can be proven using the notions introduced so far.

Theorem 2.2.1. If HA $\vdash \Gamma \Rightarrow \delta$ then CHA $\vdash \Gamma \Rightarrow \delta$.

Proof. Recall that fig. 2 gives CHA-proofs for every instance of the induction axiom of HA. Now, suppose HA $\vdash \Gamma \Rightarrow \delta$. A HA-proof Π differs from a CHA-preproof only by the fact that some of its leaves may be labelled with instances of the induction axioms. Thus, a CHA-preproof Π' of $\Gamma \Rightarrow \delta$ may be obtained by 'grafting' copies of the CHA-proof in fig. 2 onto each such leaf of Π. As there are no cycles in Π, every infinite branch of Π' must follow the $\star$-cycle of one of the 'grafted on' induction axiom proofs and thus has a successful trace. Hence Π' is a CHA-proof witnessing CHA $\vdash \Gamma \Rightarrow \delta$. □

Whereas the translation given in theorem 2.2.1 is rather simple, the converse direction of the equivalence is much more complicated. The result was first obtained by Berardi and Tatsuta (2017). We give a proof of it, relying on an alternative representation of CHA-proofs, in theorem 3.4.9. We thus contend ourselves with simply stating the result for now.

Fact 2.2.2. If CHA $\vdash \Gamma \Rightarrow \delta$ then HA $\vdash \Gamma \Rightarrow \delta$.

In cyclic proof systems for logics, rather than first-order theories like Heyting or Peano arithmetic, the equivalence to inductive proof systems is usually established by proving the soundness and completeness of the cyclic proof system directly in terms of a canonical semantics of the logic. Comparing such results for cyclic and inductive proof systems, cyclic proof systems often require more intricate proofs of soundness (as witnessed by theorem 2.1.4) but allow for simpler completeness proofs. One intuition behind this phenomenon is that ill-founded proofs (and hence cyclic proofs) are *coinductive* objects whereas inductive proofs are *inductive* objects. Framed in the language of category theory, algebras ease the construction of morphisms 'out of them' via recursion (corresponding to the 'direction' of soundness) whereas coalgebras ease the construction of morphisms 'into them' via corecursion (corresponding to the 'direction' of completeness).

Cyclic proofs can be viewed as finite representations of ill-founded proofs. In the setting of first-order arithmetic, such ill-founded proofs are interesting in their own right.

Definition 2.2.3 (CHA-∞-proofs)**.** The *sequents* of CHA-∞-preproofs are the same as those of Heyting arithmetic. An ∞*-preproof* is a pair $\Pi = (T, \lambda)$ consisting of a (possibly) ill-founded tree T and a labelling function $\lambda : \mathrm{Node}(T) \to \textsc{Seq}$. The labelling function λ labels the nodes of T with sequents such that

- inner nodes are labelled according to the derivation rules of HA or the CHA-specific derivation rule

$$\textsc{Case}_x \frac{\Gamma[0/x] \Rightarrow \delta[0/x] \quad \Gamma[Sx/x] \Rightarrow \delta[Sx/x]}{\Gamma \Rightarrow \delta}$$

$$\textsc{Sub} \frac{\Gamma \Rightarrow \delta}{\Gamma[t/x] \Rightarrow \delta[t/x]}$$

  in which $x \notin \mathrm{FV}(\Gamma, \delta)$ for $\textsc{Case}_x$ and $x \notin \mathrm{FV}(t)$ for $\textsc{Sub}$.
- leaves labelled with HA axioms, excluding the induction axioms.

A CHA-∞-preproof Π is a CHA-∞*-proof* if for every infinite branch through Π there exists a variable which has a progressing trace along it. Provability via CHA-∞-proofs is denoted $\mathrm{CHA} \vdash^{\infty} \Gamma \Rightarrow \delta$.

Every CHA-preproof can be 'unfolded' into an CHA-∞-preproof by continuously replacing each bud with the subtree above, and including, its companion. Because the soundness condition imposed on CHA-proofs and CHA-∞-proofs is 'the same', this method transforms CHA-proofs into ∞-proofs. CHA-proof correspond to a class of ∞-proofs: the *regular* CHA-∞-proofs i.e., those with finitely many distinct subproofs. The notion of ∞-proof is rather general and can be considered in the context of any cyclic proof system. Conversely, the regular proofs of an ∞-proof system always induce a cyclic proof systems.

From Gödel's incompleteness theorems, it follows that HA (and thus CHA) is incomplete i.e., it cannot prove all theorems of *true arithmetic*, the first-order theory of the standard model ω of arithmetic. Completeness can be obtained by extending HA with an infinitary ω-rule which allows universal statements to be deduced from derivations of the statement for every numeral $S^n 0$ for $n \in \omega$.

$$\omega \frac{\Gamma \Rightarrow \varphi[0/x] \quad \Gamma \Rightarrow \varphi[S0/x] \quad \dots \quad \Gamma \Rightarrow \varphi[S^n 0/x] \quad \dots}{\Gamma \Rightarrow \forall x.\varphi}$$

Simpson (2017, theorem 4) observes that the ω-rule is admissible in ∞-proofs.

Proposition 2.2.4. The ω-rule is admissible in CHA-∞-proofs.

Proof. Suppose there were ∞-proofs Π_n of $\Gamma \Rightarrow \varphi[S^n 0/x]$ for every $n \in \omega$. Consider the ∞-preproof Π below, in which w.l.o.g. $x \notin \mathrm{FV}(\Gamma, \forall x.\varphi)$.

$$
\textsc{Case}_x\,\dfrac{\dfrac{\Pi_0}{\Gamma \Rightarrow \varphi[0/x]} \qquad \dfrac{\dfrac{\Pi_1}{\Gamma \Rightarrow \varphi[S0/x]} \qquad \dfrac{\begin{array}{c}\dfrac{\dfrac{\Pi_n}{\Gamma \Rightarrow \varphi[S^n0/x]} \qquad \dfrac{\vdots}{\Gamma \Rightarrow \varphi[S^{n+1}x/x]}}{\Gamma \Rightarrow \varphi[S^nx/x]}\,\textsc{Case}_x \\ \vdots\end{array}}{\Gamma \Rightarrow \varphi[S^2x/x]}}{\Gamma \Rightarrow \varphi[Sx/x]}\,\textsc{Case}_x}{\forall\mathrm{R}\,\dfrac{\Gamma \Rightarrow \varphi}{\Gamma \Rightarrow \forall x.\varphi}}
$$

It remains to argue that Π is a proof. Thus, consider an infinite path through it. Such a path must either infinitely pass through the right-hand sides of the $\textsc{Case}_x$-applications or enter one of the Π_n. Per assumption, every branch through the Π_n has a progressing trace. As the notion of progressing trace is closed under taking prefixes, all the branches through Π entering one of the Π_n exhibit one. The branch passing infinitely many right-hand sides of $\textsc{Case}_x$ clearly has a progressing x-trace. Hence Π is a proof. □

Corollary 2.2.5 (Completeness)**.** If $\omega \vDash \Gamma \Rightarrow \delta$ then $\mathrm{CHA} \vdash^\infty \Gamma \Rightarrow \delta$.

For the cyclic proofs for first-order logic with inductive definitions considered by Brotherston (2006), a similar mismatch is exhibited, the ∞-proofs being complete for standard second-order models. However, this disagreement between cyclic and ∞-proofs is not universal. For example, in modal logics like the modal μ-calculus (Niwiński and Walukiewicz 1996), Gödel-Löb logic (Shamkanov 2014) and Grzegorczyk logic (Savateev and Shamkanov 2021), cyclic proofs and ∞-proofs prove exactly the same theorems. More generally, cyclic and ∞-proof systems in which only finitely many distinct sequents can occur in a proof are always equivalent (Wehr 2021, theorem 4.6). This result applies to many proof systems which exhibit a subformula property, such as the modal logics listed above.

Returning to the setting of first-order arithmetic, Heyting and true arithmetic corresponding to regular and unrestricted ∞-proofs, respectively, naturally raises the question whether correspondences between other classes of ∞-proofs and fragments of first-order arithmetic can be found. An interesting extension of regular trees are the classes of hyper-regular trees generated by higher-order recursion schemes as considered in Ong (2006). The article proves that acceptance of ω-regular conditions, such as the global trace condition, on such hyper-regular trees is decidable, extending the property of decidable proof checking to hyper-regular ∞-proofs. However, the trees considered may only employ a finite supply of labels, corresponding to the case of proof systems with finitely many sequents discussed in the previous paragraph. Hence, higher-order recursion system generated ∞-proofs (for virtually every cyclic proof system in the literature) prove the same theorems as cyclic proofs. This is the only other such result we are aware of.

2.3 Simpson's cyclic Heyting arithmetic $\mathrm{CHA}_<$

The running example of this chapter so far has been the simple cyclic Heyting arithmetic CHA. Its 'simplicity' stems from its trace condition: following an individual variable which may never change. In this section, we introduce another cyclic proof system for Heyting arithmetic, in the style of Simpson 2017. We do this, both to demonstrate another way in which inductive systems for Heyting arithmetic can be 'turned into' a cyclic proof systems and also to provide one more concrete cyclic proof systems for use as running examples in the rest of this thesis.

Simpson (2017) presents a proof system over the language of first-order arithmetic extended by a relation $s < t$. Traces in his system are sequences of terms appearing along the branch which are, at each 'step', either equal or ordered by a formula $s < t$ to the left of $\Rightarrow$. Each $<$-step is considered a progression. Strictly speaking, Simpson presents a system for Peano arithmetic, not Heyting arithmetic. We shall instead consider the easily derived 'intuitionistic sibling' in this section and the rest of the thesis.

The term and formula languages of $\mathrm{CHA}_<$ are given below. The formula language is non-standard, treating inequality $s < t$ as a primitive, rather than a defined notion. As will become clear below, this eases the definition of the global trace condition of $\mathrm{CHA}_<$.

$$s, t \in \textsc{Term} ::= x \mid 0 \mid Ss \mid s + t \mid s \cdot t$$
$$\varphi, \psi \in \textsc{Form} ::= s = t \mid s < t \mid \bot \mid \varphi \wedge \psi \mid \varphi \vee \psi \mid \varphi \to \psi \mid \forall x.\varphi \mid \exists x.\varphi$$

We still denote by $[t/x]$ the *substitution operation* described in section 2.1.

Definition 2.3.1. $\mathrm{CHA}_<$-preproofs The *sequents* of $\mathrm{CHA}_<$ are expressions $\Gamma \Rightarrow \delta$ where Γ is a finite sets of formulas and δ a single formula. Write Γ, φ for $\Gamma \cup \{\varphi\}$ and Γ, Γ' for $\Gamma \cup \Gamma'$. The *derivation rules* of $\mathrm{CHA}_<$ comprise of the following choice of standard rules for intuitionistic first-order logic,

$$\text{Ax}\frac{}{\Gamma, \delta \Rightarrow \delta} \qquad \to\text{L}\frac{\Gamma \Rightarrow \varphi \quad \Gamma, \psi \Rightarrow \delta}{\Gamma, \varphi \to \psi \Rightarrow \delta} \qquad \to\text{R}\frac{\Gamma, \varphi \Rightarrow \psi}{\Gamma \Rightarrow \varphi \to \psi}$$

$$\wedge\mathrm{L}\frac{\Gamma, \varphi, \psi \Rightarrow \delta}{\Gamma, \varphi \wedge \psi \Rightarrow \delta} \qquad \wedge\mathrm{R}\frac{\Gamma \Rightarrow \varphi \quad \Gamma \Rightarrow \psi}{\Gamma \Rightarrow \varphi \wedge \psi}$$

$$\vee\mathrm{L}\frac{\Gamma, \varphi \Rightarrow \delta \quad \Gamma, \psi \Rightarrow \delta}{\Gamma, \varphi \vee \psi \Rightarrow \delta} \qquad \vee\mathrm{R}\frac{\Gamma \Rightarrow \varphi_i}{\Gamma \Rightarrow \varphi_1 \vee \varphi_2} \qquad \forall\mathrm{L}\frac{\Gamma, \varphi[t/x] \Rightarrow \delta}{\Gamma, \forall x.\varphi \Rightarrow \delta}$$

$$\forall\mathrm{R}\frac{\Gamma \Rightarrow \varphi \quad x \notin \mathrm{FV}(\Gamma, \forall x.\varphi)}{\Gamma \Rightarrow \forall x.\varphi} \qquad \exists\mathrm{L}\frac{\Gamma, \varphi \Rightarrow \delta \quad x \notin \mathrm{FV}(\Gamma, \exists x.\varphi, \delta)}{\Gamma, \exists x.\varphi \Rightarrow \delta}$$

$$\exists\mathrm{R}\frac{\Gamma \Rightarrow \varphi[t/x]}{\Gamma \Rightarrow \exists x.\varphi} \qquad \bot\mathrm{L}\frac{}{\Gamma, \bot \Rightarrow \delta}$$

$$=\mathrm{L}\frac{\Gamma[t/x, s/y] \Rightarrow \delta[t/x, s/y] \quad x, y \notin \mathrm{FV}(s, t)}{\Gamma[s/x, t/y], s = t \Rightarrow \delta[s/x, t/y]} \qquad =\mathrm{R}\frac{}{\Gamma \Rightarrow t = t}$$

with the following structural rules,

$$\textsc{Wk}\frac{\Gamma \Rightarrow \delta}{\Gamma, \Gamma' \Rightarrow \delta} \qquad \textsc{Cut}\frac{\Gamma \Rightarrow \varphi \quad \Gamma, \varphi \Rightarrow \delta}{\Gamma \Rightarrow \delta} \qquad \textsc{Sub}\frac{\Gamma \Rightarrow \delta}{\Gamma[s/x] \Rightarrow \delta[s/x]}$$

with $x \notin \mathrm{FV}(t)$ for Sub, the following arithmetic-specific axioms

$$\begin{array}{rr}
s < t, t < u \Rightarrow s < u & s < t \Rightarrow Ss < St \\
\Rightarrow s + St = S(s + t) & s < t, t < s \Rightarrow \bot \\
\Rightarrow s < t \vee s = t \vee t < s & \Rightarrow t \cdot 0 = 0 \\
s < t, t < Ss \Rightarrow \bot & \Rightarrow t < St \\
\Rightarrow s \cdot St = (s \cdot t) + s & t < 0 \Rightarrow \bot \\
\Rightarrow t + 0 = t &
\end{array}$$

and the arithmetic-specific derivation rule

$$\mathrm{S}\frac{\Gamma, t = Sx \Rightarrow \delta \quad x \notin \mathrm{FV}(\Gamma, 0 < t, \delta)}{\Gamma, 0 < t \Rightarrow \delta}.$$

A *preproof* of $\mathrm{CHA}_<$ is a pair $\Pi = (C, \lambda)$ consisting of a cyclic tree $C = (T, \beta)$ and a labelling function $\lambda : \mathrm{Node}(T) \to \textsc{Seq}$. The labelling function λ labels the nodes of T with sequents such that

- inner nodes are labelled according to the derivation rules given above.
- leaves, with the exception of buds, must be labelled by axioms.
- each bud and companion share labels i.e., $\lambda(s) = \lambda(\beta(s))$ for $s \in \mathrm{dom}(\beta)$.

The notion of trace in $\mathrm{CHA}_<$ is much more complex than that in CHA: traces may follow arbitrary terms, not just variables, and the term being followed may change along the trace. At the core of the definition of trace is the rule-dependent notion of predecessor $t' \leftarrow^i_R t$. Intuitively, a term t' in the ith premise of a rule R is a predecessor of a term t in its conclusion if their value, in the standard model ω, is the same. Thus, the value of the terms along a trace remains constant, except when 'passing through $<$', as will become clear below.

Definition 2.3.2 (Trace)**.** A term t *occurs* in a sequent $\Gamma \Rightarrow \delta$ if it appears, possibly as a subterm of another term, in a formula in Γ or δ. Write $\textsc{Term}(\Gamma \Rightarrow \delta)$ for the set of terms occurring in $\Gamma \Rightarrow \delta$.

Let R be a rule of $\mathrm{CHA}_<$ of the form

$$\mathrm{R}\frac{\Gamma_1 \Rightarrow \delta_1 \qquad \dots \qquad \Gamma_n \Rightarrow \delta_n}{\Gamma \Rightarrow \delta}$$

Fix $t \in \textsc{Term}(\Gamma \Rightarrow \delta)$ and $t' \in \textsc{Term}(\Gamma_i \Rightarrow \delta_i)$. The term t' is called a *precursor* of t, denoted $t' \leftarrow^i_R t$ if one of the following three conditions holds:

- R is an instance of $(\textsc{Sub})$ and $\Gamma = \Gamma'[s/x], \delta = \delta'[s/x]$ and $t = t'[s/x]$;
- or R is an instance of a rule *other than* $(\textsc{Sub})$ and $t = t'$;
- or R is an instance of $(=\mathrm{L})$ and $\Gamma = \Gamma'[s/x, t/y], \Gamma_1 = \Gamma'[t/x, s/y]$ (analogously for δ) and there exists a term t'' such that $t = t''[s/x, t/y]$ and $t' = t''[t/x, s/y]$.

Now let Π be $\mathrm{CHA}_<$-preproof and let $(\Gamma_i \Rightarrow \delta_i)_{i\in\omega}$ be an infinite branch of Π. A trace along said branch is a sequence of terms $(t_i)_{i\in\omega}$ and an offset n such that for every $i \in \omega$ we have $t_i \in \textsc{Term}(\Gamma_{n+i} \Rightarrow \delta_{n+i})$ and furthermore, if $\Gamma_{n+i+1} \Rightarrow \delta_{n+i+1}$ is the jth premise of $\mathrm{CHA}_<$-rule R then

(i) either $t_{i+1} \leftarrow^j_R t_i$
(ii) or $s < t_i \in \Gamma_{N+i}$ and $t_{i+1} \leftarrow^j_R s$.

A trace is said to be progressing if case (ii) applies infinitely often along it.

The soundness condition of $\mathrm{CHA}_<$ is the global trace condition induced by the notion of trace from definition 2.3.2.

Definition 2.3.3 ($\mathrm{CHA}_<$-proof)**.** A $\mathrm{CHA}_<$-preproof Π is a proof if every infinite branch of Π has a progressing trace. We write $\mathrm{CHA}_< \vdash \Gamma \Rightarrow \delta$ to mean that there exists a $\mathrm{CHA}_<$-proof Π with endsequent $\Gamma \Rightarrow \delta$.

Example 2.3.4. For an example of a $\mathrm{CHA}_<$-proof, consider the proof of the induction axiom for a formula $\varphi(x)$ with $\overline{\varphi} := \forall x.\ \varphi(x) \rightarrow \varphi(Sx)$ given below. As before, the $\star$ marks the bud and companion of a cycle.

$$
\begin{prooftree}
\AxiomC{$\varphi(0), \overline{\varphi} \Rightarrow \varphi(x) \ \ (\star)$}
\LeftLabel{\textsc{Sub}}
\UnaryInfC{$\varphi(0), \overline{\varphi} \Rightarrow \varphi(y)$}
\AxiomC{$\varphi(Sy) \Rightarrow \varphi(Sy)$}
\LeftLabel{$\to$L}
\BinaryInfC{$\varphi(0), \overline{\varphi}, \varphi(y) \to \varphi(Sy) \Rightarrow \varphi(Sy)$}
\LeftLabel{$\forall$L}
\UnaryInfC{$\varphi(0), \overline{\varphi} \Rightarrow \varphi(Sy)$}
\LeftLabel{\textsc{Wk}}
\UnaryInfC{$y < Sy, \varphi(0), \overline{\varphi} \Rightarrow \varphi(Sy)$}
\LeftLabel{\textsc{Cut} + axiom}
\UnaryInfC{$\varphi(0), \overline{\varphi} \Rightarrow \varphi(Sy)$}
\LeftLabel{=L}
\UnaryInfC{$x = Sy, \varphi(0), \overline{\varphi} \Rightarrow \varphi(x)$}
\LeftLabel{S}
\UnaryInfC{$0 < x, \varphi(0), \overline{\varphi} \Rightarrow \varphi(x)$}
\AxiomC{$\varphi(0) \Rightarrow \varphi(0)$}
\LeftLabel{=L}
\UnaryInfC{$0 = x, \varphi(0) \Rightarrow \varphi(x)$}
\AxiomC{}
\RightLabel{axiom}
\UnaryInfC{$x < 0 \Rightarrow$}
\LeftLabel{$\vee$L}
\TrinaryInfC{$0 < x \vee 0 = x \vee x < 0, \varphi(0), \overline{\varphi} \Rightarrow \varphi(x)$}
\LeftLabel{\textsc{Cut} + axiom}
\UnaryInfC{$\varphi(0), \overline{\varphi} \Rightarrow \varphi(x) \ \ (\star)$}
\LeftLabel{$\forall$R}
\UnaryInfC{$\varphi(0), \overline{\varphi} \Rightarrow \forall x.\, \varphi(x)$}
\LeftLabel{$\to$R}
\UnaryInfC{$\Rightarrow \varphi(0) \to \overline{\varphi} \to \forall x.\, \varphi(x)$}
\end{prooftree}
$$

To verify it is a proof, we must find a progressing trace along every infinite branch. There is only one infinite branch, taking the $\star$-cycle infinitely. The infinite trace, starting at the premise of the $\forall$R-rule, is given below. Unless specified otherwise, each step is taken by case (i) of definition 2.3.2.

$$
x \xrightarrow{\textsc{Cut} + \text{ax}} x \xrightarrow{\vee\text{L}} x \xrightarrow{S} x \xrightarrow{=\text{L}} Sy \xrightarrow{\textsc{Cut} + \text{ax}} Sy \xrightarrow[(ii)]{\textsc{Wk}} y \xrightarrow{\forall\text{L}} y \xrightarrow{\to\text{L}} y \xrightarrow{\textsc{Sub}} x
$$

This trace progresses infinitely because, when crossing the Wk-rule, case (ii) of definition 2.3.2 applies, decreasing the value of the term being followed.

Observe that the derivation steps of Cut, the axiom and Wk are only present in the proof above to 'present' the decrease, via the introduced assumption $y < Sy$, to the trace condition. While removing these steps would leave the argument embodied by the preproof intuitively sound, the trace condition would not be able to recognise it at such.

Similar to CHA, $\text{CHA}_<$ proves the same theorems as HA when accounting for the fact that the $<$-relation is not part of the language of HA. The soundness argument from theorem 2.1.4 could also be adapted to $\text{CHA}_<$ in a straightforward manner, but we choose to omit this for brevity.

Proposition 2.3.5. For Γ, δ not mentioning $<$, $\text{HA} \vdash \Gamma \Rightarrow \delta$ iff $\text{CHA}_< \vdash \Gamma \Rightarrow \delta$.

Proof. The argument for transforming a HA-proof into a $\text{CHA}_<$-proof is analogous to theorem 2.2.1: Simply replace every instance of the induction axiom with the cyclic proof from example 2.3.4. As for CHA, the opposite direction is much more involved. We give a proof of it in section 7.1. □

As we have observed before, the trace condition of CHA is very simple: only a single, unchanging variable can be followed by a trace. The traces of $\text{CHA}_<$

are more complex: arbitrary terms can may be followed. A further observation adding to the complexity of $CHA_<$-traces is their ancestry. For example, in the =L rule, it is both possible for one term in the conclusion to have multiple different precursors in the premise and, conversely, for a term in the premise to be the precursor to multiple different terms in the conclusion. Thus, much more complicated 'trace patterns' can occur in $CHA_<$-proofs than in CHA-proofs.

In fact, the traces of $CHA_<$ are 'maximally' complex in a certain sense: any possible trace pattern is 'produced' by some $CHA_<$-derivation. We leave this statement vague for now but make it more concrete in chapter 4 as lemma 4.3.11. Crucially, this means that $CHA_<$ serves as a good example for the generic cyclic proof constructions we give in chapters 5 and 6, as it serves as a 'maximally complex' example.

3 Representing cyclic proofs

This chapter illustrates what we mean when speaking of ‘representations of cyclic proofs’ and some of the considerations around this notion which we deem important.

There are usually many different cyclic proof systems for a logic or logical theory. For example, we have presented two different cyclic proof systems for Heyting arithmetic in chapter 2. These can each be thought of as ‘cyclic representations’ of Heyting arithmetic. Both of these cyclic proof systems employ a global trace condition as their soundness condition. It is possible to adapt these proof systems to other soundness conditions. Often, this leaves the derivation rules unchanged, only modifying which preproofs are considered proofs. Sometimes, however, it also requires the derivation rules to be changed, for example by introducing new non-logical rules. In either case, adapting a cyclic proof system to a different soundness condition can yield a different ‘representation’ of the ‘same’ proof system. This notion of ‘representations’ can also be extended to the level of individual cyclic proofs. For example, a cyclic proof satisfying a certain soundness condition might need to be unfolded or otherwise slightly modified to satisfy a different soundness condition. This usually does not change the ‘proof theoretic content’ of the cyclic proof.

Considerations relating to representation of cyclic proofs can be crucial when working in cyclic proof theory i.e., when one is looking to prove proof theoretical results about cyclic proof systems. Carefully choosing which cyclic representation of a logic to consider can often considerably ease the effort required to derive certain results.

In the coming sections, we discuss some of the ideas around the representations of cyclic proof systems for the examples of CHA, specifically those relating to different soundness conditions. Section 3.1 focuses on global trace conditions, highlighting their shortcomings and thus motivating the consideration of other soundness conditions. In section 3.2 we present induction orders, a soundness condition especially well-suited to explaining the soundness of cyclic proofs to human readers. Section 3.3 introduces reset proof systems, which are particularly well-suited for computational or proof theoretic uses of cyclic proof systems. In section 3.4, we cover stack-controlled proofs, a novel cyclic proof representation introduced in this dissertation, which are well-suited to obtaining translations 'out of' cyclic proof systems due to their structural similarities to 'inductive' proofs. In section 3.5, we cover simplistic cyclic proofs, which only accept cyclic proofs with 'simple cycle patters'.

3.1 Global trace conditions

The global trace condition (GTC) is the prevalent soundness condition in the literature of cyclic proof theory. The definitions of GTCs in the literature follow a simple pattern: Every infinite branch of a cyclic proof must have a suffix along which a progressing trace exists. While the notions of 'trace' and 'progress' vary between logics, the shape of the condition remains the same.

Most of this section focuses on the shortcomings of the GTC. Nonetheless, the positive aspects of GTC deserve mentioning. The GTC's central property is its *simplicity*. When constructing a cyclic proof system for a new logic, finding a GTC can be quite easy: Simply identify suitable notions of 'trace' and 'progress'. These notions often arise naturally from the semantics of the logic. Furthermore, some properties of cyclic proofs are most easily proven for proof systems with a GTC. Among such properties are those which are expected to be covered in any article proposing forward a new cyclic proof system, such as soundness, completeness and decidability of proof checking. These strengths of the GTC make it the 'default' soundness condition in many settings.

One disadvantage of the global trace condition is that is *difficult to verify for humans*. For an example, consider the CHA-proof in fig. 4. It proves the totality of a recursive function, coded as a three-place relation $A(x, y, z)$. The preproof has three buds, all sharing the same companion, marked with $\star$. We encourage the reader to try to convince themselves that the preproof indeed is a CHA-proof. The difficulty of this endeavour stems from the fact that there are many (in fact, uncountably many) different infinite branches through the preproof. To verify the GTC, humans would usually first group these branches into finitely many distinct groups and then verify that each such group satisfies the trace condition.

The verification is somewhat eased by revealing which function is coded by $A(x, y, z)$: the Ackermann function. By scrutinising the definition given below, one finds that the function terminates according to the lexicographic ordering of the parameters x and y. With this knowledge, GTC satisfaction in fig. 4 is easily

$$
\begin{prooftree}
\AxiomC{Π}
\AxiomC{$\Rightarrow \exists z.\, A(x,y,z) \star$}
\AxiomC{}
\UnaryInfC{$\Rightarrow A(0,Sy,SSy)$}
\UnaryInfC{$\Rightarrow \exists z.\, A(0,Sy,z)$}
\AxiomC{$\Rightarrow \exists z.\, A(x,y,z) \star$}
\RightLabel{\textsc{Sub}}
\UnaryInfC{$\Rightarrow \exists z.\, A(x,z',z)$}
\UnaryInfC{$A(Sx,y,z') \Rightarrow \exists z.\, A(Sx,Sy,z)$}
\RightLabel{$\exists$L}
\UnaryInfC{$\exists z'.\, A(Sx,y,z') \Rightarrow \exists z.\, A(Sx,Sy,z)$}
\RightLabel{\textsc{Case}$_x$}
\BinaryInfC{$\exists z'.\, A(x,y,z') \Rightarrow \exists z.\, A(x,Sy,z)$}
\RightLabel{\textsc{Cut}}
\BinaryInfC{$\Rightarrow \exists z.\, A(x,Sy,z)$}
\RightLabel{\textsc{Case}$_y$}
\BinaryInfC{$\Rightarrow \exists z.\, A(x,y,z) \star$}
\end{prooftree}
$$

$$
\Pi : \quad
\begin{prooftree}
\AxiomC{}
\UnaryInfC{$\Rightarrow A(0,0,1)$}
\UnaryInfC{$\Rightarrow \exists z.\, A(0,0,z)$}
\AxiomC{$\Rightarrow \exists z.\, A(x,y,z) \star$}
\RightLabel{\textsc{Sub}}
\UnaryInfC{$\Rightarrow \exists z.\, A(x,1,z)$}
\UnaryInfC{$\Rightarrow \exists z.\, A(Sx,0,z)$}
\LeftLabel{\textsc{Case}$_x$}
\BinaryInfC{$\Rightarrow \exists z.\, A(x,0,z)$}
\end{prooftree}
$$

Figure 4: A CHA-proof, $\star$ marking cycles

verified.

$$
A(x,y) \coloneqq \begin{cases} y+1 & x = 0 \\ A(x-1,1) & x > 0, y = 0 \\ A(x-1, A(x,y-1)) & x > 0, y > 0 \end{cases}
$$

Note that fig. 4 is still a rather simple example of a cyclic proof. Cyclic proofs in proof systems with more complex traces, such as the system $\mathrm{CHA}_{<}$, can make verification of the global trace condition even more difficult.

Generally, a key property of proof systems is that proofs should be easy to verify, usually by simply checking whether each rule has been applied correctly. Sometimes this comes at the cost of proofs being difficult to construct. This notion has, for example, been made formal by Cook and Reckhow (1979) who require poly-time verifiability as part of their abstract notion of 'proof system'. In cyclic proof systems, especially those with a GTC, this principle is turned on its head: as proof verification also requires checking of the GTC, it becomes more difficult. On the flip side, the simplification of the induction rules in many cyclic proof systems, as discussed in chapter 1, tends to ease the construction of cyclic proofs.

The difficulty of verifying the global trace condition experienced by humans extends to the realm of machines: the global trace condition is *computationally expensive* to verify. Indeed, for some proof systems, such as multiplicative-additive linear logic with fixed points (Nollet et al. 2019) or the system $\mathrm{CHA}_{<}$ for Heyting arithmetic, the problem of verifying the GTC is known to be PSPACE-complete (an abstract rendition of the argument can be found in (Afshari and Wehr 2024, section 7)). We conjecture that most cyclic proof systems in the literature exhibit full PSPACE-completeness. While the reduction for μMALL and Simpson's system rely on rather complex 'trace patterns', we believe they can be replicated in most cyclic proof systems.

One might argue that the difficulty of verification by humans is just a consequence of the sheer computational complexity of the task. However, in section 3.2 we present a soundness condition which, in practice, is often easy for humans to verify while still being in coNP.

Lastly, and most crucially to this thesis, cyclic proof systems with a global trace condition are often *ill-suited to proof theoretic investigations*. For a simple example, consider the task of proving the admissibility of the following induction rule in CHA, in which $x \notin \mathrm{FV}(\Gamma)$:

$$\textsc{Ind}\frac{\Gamma, \varphi \Rightarrow \varphi[Sx/x]}{\Gamma, \varphi[0/x] \Rightarrow \varphi[t/x]}$$

It should be noted that 'admissibility' in cyclic proof systems is more subtle than in inductive proof systems. Here, we understand admissibility of Ind in CHA to mean that the extension of CHA by said rule, with traces through it being defined in terms of free variable occurrences analogously to the 'proper' rules of CHA, proves no more sequents than CHA. To prove this, one may supply a preproof with conclusion $\Gamma, \varphi[0/x] \Rightarrow \varphi[t/x]$ and an open leaf $\Gamma, \varphi \Rightarrow \varphi[Sx/x]$ and then prove that every Ind-application in a CHA-proof may be replaced by said preproof without invalidating the soundness condition. A suitable preproof can quickly be found, such as the one depicted in fig. 5. When the derivation is 'inserted' into a preproof, the subderivation above the Ind-premise is inserted above the right-most leaf. It remains to verify that a CHA-proof employing the Ind-rule remains a proof i.e., satisfies the GTC, once the Ind-instance is replaced by the derivation above. The simplest method of showing this is proving that every branch through the modified proof 'corresponds' to a branch through the original proof. Even for CHA, which employs a comparably simple notion of trace, this requires a subtle argument relying on a technical notion of branch bisimulation.

Most articles in the cyclic proof theory literature which carry out more sophisticated proof theoretic investigations rely on soundness conditions different from the global trace condition. This can be taken as another piece of evidence of the unsuitability of the global trace condition to carrying out proof theoretic investigations. For example, most interpolation results employing cyclic proof systems rely on cyclic proof systems with *path conditions* (see e.g. Afshari, Leigh,

$$
\begin{prooftree}
\AxiomC{}
\LeftLabel{\textsc{Ax}}
\UnaryInfC{$\Gamma, \varphi[0/x] \Rightarrow \varphi[0/x]$}
\AxiomC{$\Gamma, \varphi[0/x] \Rightarrow \varphi \;\; \star$}
\LeftLabel{\textsc{Wk}}
\UnaryInfC{$\Gamma, \varphi[0/x] \Rightarrow \varphi, \varphi[Sx/x]$}
\AxiomC{$\Gamma, \varphi \Rightarrow \varphi[Sx/x]$}
\RightLabel{\textsc{Wk}}
\UnaryInfC{$\Gamma, \varphi[0/x], \varphi \Rightarrow \varphi[Sx/x]$}
\LeftLabel{\textsc{Cut}}
\BinaryInfC{$\Gamma, \varphi[0/x] \Rightarrow \varphi[Sx/x]$}
\LeftLabel{\textsc{Case}$_x$}
\BinaryInfC{$\Gamma, \varphi[0/x] \Rightarrow \varphi \;\; \star$}
\LeftLabel{\textsc{Sub}}
\UnaryInfC{$\Gamma, \varphi[0/x] \Rightarrow \varphi[t/x]$}
\end{prooftree}
$$

Figure 5: A derivation of Ind in CHA, ⋆ marking a cycle

and Menéndez Turata 2021; Marti and Venema 2021b; Shamkanov 2014; Kloibhofer, Trucco Dalmas, et al. 2026). Path conditions are soundness conditions on cyclic preproofs in cycle normal form which are phrased in terms of the local cycles, i.e. the paths between each companion $\beta(b)$ and bud b. In a sense, these conditions are much more 'local' than the global trace condition.

These disadvantages of the global trace condition can be circumvented by choosing to represent cyclic proofs in terms of different soundness conditions. We present three such soundness conditions in sections 3.2 to 3.4.

3.2 Induction orders⋆

This section describes *induction orders*, a soundness condition first put forward by Sprenger and Dam (2003a). The main advantage of induction orders over global trace conditions is that they tend to be much easier to verify for humans.

To define induction orders, we must first introduce some auxiliary, combinatorial notions. Here, we aim only to provide an intuitive presentation of these concepts, deferring a fully formal treatment to section 4.1. A *strongly connected component* of a cyclic tree (T, β) is a set of nodes $S \subseteq \mathrm{Node}(T)$ such that between every $s, t \in S$ there is a path from s to t through (T, β), which only passes nodes in S. We shall from now on only consider cyclic trees in cycle normal form. Let us write $\gamma(t)$, for any bud $t \in \mathrm{dom}(\beta)$, for the set of nodes which occur on the shortest path from $\beta(t)$ to t. We call the path whose nodes are in $\gamma(t)$ the *local cycle* of t. If the cyclic tree is in cycle normal form, its strongly connected components correspond to certain sets of buds $\eta \subseteq \mathrm{dom}(\beta)$ via

$$S = C[\eta] \coloneqq \{s \in \mathrm{Node}(T) \mid \exists t \in \eta.\, s \in \gamma(t)\}.$$

The crucial property of strongly connected components, which we prove in lemma 4.1.4, is that the set of nodes passed infinitely by any infinite branch

⋆ Some of the explanations/observations in this section are based on joint work with Graham E. Leigh; see Leigh and Wehr (2025).

$s \in \mathrm{Node}(T)^\omega$ through (T, β) always form a strongly connected component i.e., $\mathrm{Inf}(s) = C[\eta]$ for some $\eta \subseteq \mathrm{dom}(\beta)$.

We define induction orders for CHA-preproofs. Because the notions of the previous paragraph only applies to cyclic trees in cycle normal form, we restrict our attention to CHA-preproofs in cycle normal form.

Definition 3.2.1 (Induction order)**.** Let $\Pi = (C, \lambda)$ be a preproof in cycle normal form. An *induction order* is a pre-order $\preceq \subseteq \mathrm{dom}(\beta) \times \mathrm{dom}(\beta)$ together with an associated mapping $x : \mathrm{dom}(\beta) \to \textsc{Var}$ such that $x_s \in \mathrm{FV}(\lambda(s))$ for every $s \in \mathrm{dom}(\beta)$ and further

- every strongly connected component $C[\eta]$ has a $\preceq$-maximal element,
- if $s \preceq t$ then $\gamma(s)$ *preserves* x_t i.e., $x_t \in \mathrm{FV}(\lambda(u))$ for every $u \in \gamma(s)$,
- the cycle $\gamma(s)$ *progresses* x_s i.e., $\textsc{Case}_{x_s}$ is applied along the path $\gamma(s)$

The idea behind induction orders is to fix, for each local cycle $\gamma(s)$ with $s \in \mathrm{dom}(\beta)$, a variable x_s on which induction is being 'performed' along said cycle. The induction order $\preceq$ ensures that along any infinite branch through the proof some induction variable is preserved and progressed infinitely.

Theorem 3.2.2. If a CHA-preproof in cycle normal form has an induction order, it satisfies the global trace condition.

Proof. Consider such a preproof (C, λ) with induction order $(\preceq, x_{-})$. An infinite branch $s \in \mathrm{Node}(T)^\omega$ of C passes precisely the nodes of some strongly connected component $C[\eta]$, for $\eta \subseteq \mathrm{dom}(\beta)$, infinitely. Thus $s_i \in C[\eta]$ for $i > N$. There must be some $\preceq$-maximal element $t \in \eta$. As every local cycle of $C[\eta]$ preserves x_t, we know that $x_t \in \mathrm{FV}(\lambda(u))$ for every $u \in C[\eta]$ and thus that $x_t \in \mathrm{FV}(\lambda(s_i))$ for $i > N$. As the local cycle $\gamma(t)$ contains a $\textsc{Case}_{x_t}$-instance, this instance is passed infinitely by s. As x_t is preserved, $\gamma(t)$ must pass through the right-hand side of the $\textsc{Case}_{x_t}$ rule. Hence, s has a progressing x_t-trace. □

Induction orders are most easily defined for cyclic proof systems with simple notions of 'trace', such as CHA, in which a trace always follows a fixed variable. Other cyclic proof systems require a more intricate notion of induction order, for example the system $\mathrm{CHA}_<$, in which the term being followed may change along a trace.

The next theorem is a consequence of the simplicity of the notion of 'trace' employed by CHA. Its proof is based on (Sprenger and Dam 2003a, theorem 5.14) which proves the same property for the μFOL-system Sprenger and Dam consider. Note that the property, as stated in theorem 3.2.4 is rather rare among cyclic proof systems. Sometimes, preproofs satisfying the global trace condition must first be unfolded to exhibit an induction order. Even more commonly, this property completely fails: many cyclic proof systems, including $\mathrm{CHA}_<$, contain cyclic proofs which cannot even be unfolded to exhibit an induction order (see remark 6.1.2 for details).

The result requires the following rather technical notion and lemma, which we prove in section 4.1. A strongly connected η is said to be *B-maximal* for $B \subseteq \mathrm{dom}(\beta)$ if $\eta \subseteq B$ and there is no $\eta \subsetneq \eta' \subseteq B$ which is strongly connected. A key property of B-maximal strongly connected subgraphs is that they are disjoint iff they are distinct.

Fact 3.2.3. For a cyclic tree $C = (T, \beta)$ let $B \subseteq \mathrm{dom}(\beta)$ and let $\eta, \eta' \subseteq B$ be B-maximal strongly connected components. Then either $\eta = \eta'$ or $\eta \cap \eta' = \emptyset$.

Theorem 3.2.4. Every CHA-proof in cycle normal form has an induction order.

Proof. Suppose Π is a proof CHA. We obtain an induction order on Π through an iterative process. This process produces a sequence $S_0, S_1, \ldots$ of sets of pairwise disjoint strongly connected components of Π such that at each step, some bud $s \in \mathrm{dom}(\beta)$ is removed from $\bigcup S_i$. To begin, define

$$S_0 := \{\eta \subseteq \mathrm{dom}(\beta) \mid \eta \text{ is a } \mathrm{dom}(\beta)\text{-maximal strongly connected component}\}.$$

To obtain S_{i+1}, select some $\eta_i \in S_i$ and consider a branch $t \in T^\omega$ through Π such that $\mathrm{Inf}(t) = \eta_i$. As Π is a proof, there must be an $n \in \omega$ such that for some $x \in \bigcap_{n<j} \textsc{Var}(\Gamma_i, \delta_i)$ there exists a progressing x-trace along p. Hence, x must be preserved along the cycle $\gamma(t)$ for every $t \in \eta$ and there must be an $s \in \eta$ for which the cycle $\gamma(s)$ progresses x. Fix $x_s := x$ and set

$$S_{i+1} := (S_i \setminus \eta_i) \cup \{\eta' \subseteq (\eta_i \setminus \{s\}) \mid \eta' \text{ is a } (\eta_i \setminus \{s\})\text{-maximal scc}\}.$$

That is, S_{i+1} is obtained by replacing η_i in S_i by the maximal strongly connected components partitioning η_i after removing s. Because each S_i consists of pairwise disjoint strongly connected components, each S_i partitions $\bigcup S_i$. This process terminates at $S_{|\mathrm{dom}(\beta)|} = \emptyset$ as every step removes a bud from $\bigcup S_i$. It remains to define the pre-order $\preceq$ on $\mathrm{dom}(\beta)$. For $s, t \in \mathrm{dom}(\beta)$, let i be the least index such that $t \notin \bigcup S_{i+1}$ i.e., t was removed at step i of the procedure. We set $s \preceq t$ iff $s \in \eta_i$. Now consider a strongly connected $\eta \subseteq \mathrm{dom}(\beta)$. We claim that there is a $\preceq$-maximum of η. As the elements of S_0 are $\mathrm{dom}(\beta)$-maximal, there must be $\eta' \in S_0$ such that $\eta \subseteq \eta'$. Now at each step of the process, if $\eta' \in S_i$ then either

$\eta' = \eta_i$ or $\eta' \in S_{i+1}$. Because $S_{|\mathrm{dom}(\beta)|} = \emptyset$ there thus must be $\eta_i = \eta'$. Consider the bud s which is removed from η_i to construct S_{i+1}; if $s \notin \eta$ then there is a $(\eta_i \setminus \{s\})$-maximal $\eta'' \in S_{i+1}$ such that $\eta \subseteq \eta''$. Again, because $S_{|\mathrm{dom}(\beta)|} = \emptyset$ there eventually must be a step i such that $\eta \subseteq \eta_i$ and the bud s removed from η_i is in η. Then s is a $\preceq$-maximum in η_i, and thus in η, per definition of $\preceq$. Reflexivity of $\preceq$ is trivial. Transitivity of $\preceq$ follows from an argument, similar to that for $\preceq$-maxima, along the iterative process, observing that for $s \preceq t \preceq u$ with corresponding indexes $t \notin \bigcup S_{i+1}, u \notin \bigcup S_{j+1}$ and $s \in \eta_i, t \in \eta_j$, $\eta_i \subsetneq \eta_j$ because each step 'separates' some η_k into strongly connected sub-components $\eta' \subsetneq \eta_k$. $\square$

Even though the global trace condition and the existence of induction orders guarantee the soundness of the same CHA-preproofs, the 'information' contained in an induction order often makes it easier to verify to humans. For example, reconsider the CHA-proof of the totality of the Ackermann function, recalled in fig. 6. To distinguish between the local cycles of the proof, each bud is annotated with a number $\star^i$. The buds still share the same companion marked with $\star$. Both $\star^2 \prec \star^1 \prec \star^3$ and $\star^2 \prec \star^3 \prec \star^1$ are Possible induction orders for the proof, associating to $\star^2$ the variable y and to $\star^1$ and $\star^3$ the variable x. As the order is linear, every strongly connected component has a $\preceq$-maximal bud. Verifying that this is indeed an induction order thus reduces to checking the progress and preservation conditions, which is easily done.

$$
\begin{prooftree}
\AxiomC{Π}
\AxiomC{$\Rightarrow \exists z.\, A(x,y,z) \star^2$}
\AxiomC{}
\UnaryInfC{$\Rightarrow A(0,Sy,SSy)$}
\UnaryInfC{$\Rightarrow \exists z.\, A(0,Sy,z)$}
\AxiomC{$\Rightarrow \exists z.\, A(x,y,z) \star^3$}
\RightLabel{\textsc{Sub}}
\UnaryInfC{$\Rightarrow \exists z.\, A(x,z',z)$}
\UnaryInfC{$A(Sx,y,z') \Rightarrow \exists z.\, A(Sx,Sy,z)$}
\RightLabel{$\exists$L}
\UnaryInfC{$\exists z'.\, A(Sx,y,z') \Rightarrow \exists z.\, A(Sx,Sy,z)$}
\RightLabel{\textsc{Case}$_x$}
\BinaryInfC{$\exists z'.\, A(x,y,z') \Rightarrow \exists z.\, A(x,Sy,z)$}
\RightLabel{\textsc{Cut}}
\BinaryInfC{$\Rightarrow \exists z.\, A(x,Sy,z)$}
\RightLabel{\textsc{Case}$_y$}
\BinaryInfC{$\Rightarrow \exists z.\, A(x,y,z) \star$}
\end{prooftree}
$$

$$
\Pi : \quad
\begin{prooftree}
\AxiomC{}
\UnaryInfC{$\Rightarrow A(0,0,1)$}
\UnaryInfC{$\Rightarrow \exists z.\, A(0,0,z)$}
\AxiomC{$\Rightarrow \exists z.\, A(x,y,z) \star^1$}
\RightLabel{\textsc{Sub}}
\UnaryInfC{$\Rightarrow \exists z.\, A(x,1,z)$}
\UnaryInfC{$\Rightarrow \exists z.\, A(Sx,0,z)$}
\LeftLabel{\textsc{Case}$_x$}
\BinaryInfC{$\Rightarrow \exists z.\, A(x,0,z)$}
\end{prooftree}
$$

Figure 6: A CHA-proof, $\star$ marking cycles

Induction orders are somewhat more computationally tractable than the worst case of global trace conditions, being at least in coNP. We do not know whether there are more efficient algorithms than the one we give below.

Proposition 3.2.5. The problem of determining whether $(\preceq, x_{-})$ is not an induction order for CHA-preproof $\Pi = (C, \lambda)$ can be decided by a non-deterministic Turing machine in polynomial time.

Proof. Well-derivedness of Π according to the derivation rules of CHA as well as the progress and preservation conditions for $(\preceq, x_{-})$ can be verified deterministically in polynomial time. If one of these properties is not satisfied, accept. Otherwise, 'guess' a subset $\eta \subseteq \mathrm{dom}(\beta)$. If $C[\eta]$ is no strongly connected component of, which can be verified in polynomial time (Tarjan 1972), reject. Otherwise, accept iff η has no $\preceq$-maximal element.

The algorithm above clearly runs in polynomial time on a non-deterministic Turing machine. It accepts iff Π is not well-formed or there are issues surrounding progress and preservation or there is a strongly connected component $C[\eta]$ of Π with no $\preceq$-minimal element. In other words, it accepts iff Π is no proof justified by $(\preceq, x_{-})$. $\square$

Induction orders are better suited than global trace conditions for verifying proof transformations. For example, reconsider the problem of proving the admissibility of Ind-rule. This can now be achieved with a rather simple argument.

Lemma 3.2.6. The following rule is admissible in CHA for $x \notin \mathrm{FV}(\Gamma)$:

$$\textsc{Ind}\ \dfrac{\Gamma, \varphi \Rightarrow \varphi[Sx/x]}{\Gamma, \varphi[0/x] \Rightarrow \varphi[t/x]}$$

Proof. Let Π be a CHA-preproof which w.l.o.g. makes use of the Ind-rule precisely once. Replace the Ind-application with the following derivation:

$$\textsc{Sub}\ \dfrac{\textsc{Case}_x\ \dfrac{\textsc{Ax}\ \dfrac{}{\Gamma, \varphi[0/x] \Rightarrow \varphi[0/x]} \qquad \textsc{Cut}\ \dfrac{\textsc{Wk}\ \dfrac{\Gamma, \varphi[0/x] \Rightarrow \varphi \ \star}{\Gamma, \varphi[0/x] \Rightarrow \varphi, \varphi[Sx/x]} \qquad \dfrac{\Gamma, \varphi \Rightarrow \varphi[Sx/x]}{\Gamma, \varphi[0/x], \varphi \Rightarrow \varphi[Sx/x]}\ \textsc{Wk}}{\Gamma, \varphi[0/x] \Rightarrow \varphi[Sx/x]}}{\Gamma, \varphi[0/x] \Rightarrow \varphi \ \star}}{\Gamma, \varphi[0/x] \Rightarrow \varphi[t/x]}$$

As Π is a CHA-proof, it has an induction order $(\preceq, x_{-})$. We extend this induction order to justify the modified proof. Fix the induction variable $x_\star$ of the $\star$-cycle to be x, which is clearly preserved and progressed along $\gamma(\star)$. Set $\star \prec t$ iff $t \in \mathrm{dom}(\beta)$ is a bud whose local cycle $\gamma(t)$ passes through the Ind-application i.e., any bud which is above the Ind-application and whose companion is below the Ind-application. By the preservation properties of the induction order of the original Π, $x_t \in \mathrm{FV}(\Gamma, \varphi \Rightarrow \varphi[Sx/x])$ if $\star \prec t$ and hence x_t is free along every node of $\gamma(\star)$.

It remains to prove that every strongly connected component $C[\eta]$ of the modified proof has a $\preceq$-maximal element. It suffices to consider the case of $\{\star\} \subsetneq \eta$. There must be a bud $t \in \eta$ which is above the IND-application, as this corresponds to the only kind of cycle $\star$ can 'access'. Furthermore, one of the buds $t' \in \eta$ above the IND-application must have a companion below the $\star$-companion, as there could not be a path from such buds to the $\star$-companion otherwise, and hence $\star \prec t'$. $\eta \setminus \{\star\}$ induces a strongly connected component of the original CHA-proof and thus has a $\preceq$-maximal element s. Then $\star \prec t' \preceq s$, meaning s is also $\preceq$-maximal for η in the modified proof. □

Induction orders have seen little use in deriving proof theoretic results. The only instances we are aware of is the method for proving the equivalence of cyclic and inductive proof systems introduced by Sprenger and Dam (2003b), which has been transferred to the setting of Heyting and Peano arithmetic in (Leigh and Wehr 2025).

3.3 Reset proofs

This section describes *reset proofs*, another representation of cyclic proofs. They were first introduced by Jungteerapanich (2010) and Stirling (2013) for the modal μ-calculus. To convert a cyclic proof system into a reset proof, additional derivation rules must be introduced and the existing ones must be modified. In chapter 5, we give a generic description of this process. Reset proof systems employ *annotations* which are used to track preservation and progress in a syntactic manner. They are best suited to computational and proof theoretic purposes.

Below, we give a reset proof system for Heyting arithmetic. It was obtained from CHA by the method described in chapter 5.

Definition 3.3.1 (Reset Heyting arithmetic)**.** The *sequents* of *reset Heyting arithmetic* (RHA) are expressions $\Theta; \sigma \mid \Gamma \Rightarrow \delta$ consisting of a sequent $\Gamma \Rightarrow \delta$ of first-order arithmetic, a finite, repetition-free sequence Θ of distinct letters called the *control* and an *assignment* $\sigma : \mathrm{FV}(\Gamma, \delta) \to \textsc{Sub}(\Theta)$ mapping the free variables of $\Gamma \Rightarrow \delta$ to subsequences of Θ. The sequence $\sigma(x)$ is called the *annotation* of x.

The derivation rules of RHA are given below.

$$\text{Ax}\,\frac{}{\Theta; \sigma \mid \Gamma, \delta \Rightarrow \delta} \qquad \bot\text{L}\,\frac{}{\Theta; \sigma \mid \Gamma, \bot \Rightarrow \delta}$$

$$\to\text{L}\,\frac{\Theta'; \sigma' \mid \Gamma \Rightarrow \varphi \quad \Theta'; \sigma' \mid \Gamma, \psi \Rightarrow \delta}{\Theta; \sigma \mid \Gamma, \varphi \to \psi \Rightarrow \delta} \qquad \to\text{R}\,\frac{\Theta; \sigma \mid \Gamma, \varphi \Rightarrow \psi}{\Theta; \sigma \mid \Gamma \Rightarrow \varphi \to \psi}$$

$$\wedge\mathrm{L}\frac{\Theta; \sigma \mid \Gamma, \varphi, \psi \Rightarrow \delta}{\Theta; \sigma \mid \Gamma, \varphi \wedge \psi \Rightarrow \delta} \qquad \wedge\mathrm{R}\frac{\Theta'; \sigma' \mid \Gamma \Rightarrow \varphi \quad \Theta'; \sigma' \mid \Gamma \Rightarrow \psi}{\Theta; \sigma \mid \Gamma \Rightarrow \varphi \wedge \psi}$$

$$\vee\mathrm{L}\frac{\Theta'; \sigma' \mid \Gamma, \varphi \Rightarrow \delta \quad \Theta'; \sigma' \mid \Gamma, \psi \Rightarrow \delta}{\Theta; \sigma \mid \Gamma, \varphi \vee \psi \Rightarrow \delta} \qquad \vee\mathrm{R}\frac{\Theta; \sigma \mid \Gamma \Rightarrow \varphi_i}{\Theta; \sigma \mid \Gamma \Rightarrow \varphi_1 \vee \varphi_2}$$

$$\forall\mathrm{L}\frac{\Theta; \sigma^+ \mid \Gamma, \varphi[t/x] \Rightarrow \delta}{\Theta; \sigma \mid \Gamma, \forall x.\varphi \Rightarrow \delta} \qquad \forall\mathrm{R}\frac{\begin{array}{c} x \notin \mathrm{FV}(\Gamma, \forall x.\varphi) \\ \Theta; \sigma^+ \mid \Gamma \Rightarrow \varphi \end{array}}{\Theta; \sigma \mid \Gamma \Rightarrow \forall x.\varphi}$$

$$\exists\mathrm{L}\frac{\begin{array}{c} x \notin \mathrm{FV}(\Gamma, \exists x.\varphi, \delta) \\ \Theta; \sigma^+ \mid \Gamma, \varphi \Rightarrow \delta \end{array}}{\Theta; \sigma \mid \Gamma, \exists x.\varphi \Rightarrow \delta} \qquad \exists\mathrm{R}\frac{\Theta; \sigma^+ \mid \Gamma \Rightarrow \varphi[t/x]}{\Theta; \sigma \mid \Gamma \Rightarrow \exists x.\varphi}$$

$$=\mathrm{L}\frac{\Theta; \sigma[t/x, s/y] \mid \Gamma[t/x, s/y] \Rightarrow \delta[t/x, s/y] \quad x, y \notin \mathrm{FV}(s, t)}{\Theta; \sigma[s/x, t/y] \mid \Gamma[s/x, t/y], s = t \Rightarrow \delta[s/x, t/y]}$$

$$=\mathrm{R}\frac{}{\Theta; \sigma \mid \Gamma \Rightarrow t = t} \qquad \textsc{Wk}\frac{\Theta'; \sigma' \mid \Gamma \Rightarrow \delta}{\Theta; \sigma, \sigma^* \mid \Gamma, \Gamma' \Rightarrow \delta}$$

$$\textsc{Cut}\frac{\Theta'; \sigma'^+ \mid \Gamma \Rightarrow \varphi \quad \Theta; \sigma^+ \mid \Gamma, \varphi \Rightarrow \delta}{\Theta; \sigma \mid \Gamma \Rightarrow \delta}$$

$$\textsc{Sub}\frac{x \notin \mathrm{FV}(t) \quad \Theta'; \sigma'^+ \mid \Gamma \Rightarrow \delta}{\Theta; \sigma \mid \Gamma[t/x] \Rightarrow \delta[t/x]}$$

$$\textsc{Case}_x\frac{\begin{array}{c} a \notin \Theta \\ \Theta'; \sigma' \mid \Gamma(0) \Rightarrow \delta(0) \quad \Theta a; \sigma[x \mapsto ua] \mid \Gamma(Sx) \Rightarrow \delta(Sx) \end{array}}{\Theta; \sigma[x \mapsto u] \mid \Gamma(x) \Rightarrow \delta(x)}$$

$$\textsc{Clear}\frac{\Theta'; \sigma[x \mapsto \varepsilon] \mid \Gamma \Rightarrow \delta}{\Theta; \sigma[x \mapsto u] \mid \Gamma \Rightarrow \delta} \qquad \textsc{Reset}_a\frac{\begin{array}{c} u' \text{non-empty} \\ \Theta'; \sigma[x \mapsto ua] \mid \Gamma \Rightarrow \delta \end{array}}{\Theta; \sigma[x \mapsto uau'] \mid \Gamma \Rightarrow \delta}$$

In some rules of the system, the free variables in the premises and conclusion might differ. In such situations, σ' denotes the restriction of σ to the remaining free variables and Θ' the subsequence of Θ which only contains letters which occur in at least one annotation $\sigma'(x)$. Conversely, in rules in which new free variables might be introduced in the premise, σ^+ denotes the extension of σ which annotates each new variable with the empty sequence ε. Furthermore, the sequents $\Theta; \sigma \mid \Gamma \Rightarrow \delta$, in which $\Gamma \Rightarrow \delta$ are as below, are axioms.

$$0 = St \Rightarrow \bot \qquad Ss = St \Rightarrow s = t \qquad \Rightarrow s + 0 = 0$$
$$\Rightarrow s + St = S(s + t) \qquad \Rightarrow s \cdot 0 = 0 \qquad \Rightarrow s \cdot St = (s \cdot t) + s$$

The preproofs of RHA are defined analogously to those of CHA, except we only consider proofs in cycle normal form. That is, they are a pair $\Pi = (C, \lambda)$ consisting a cyclic tree C in cycle normal form and a labelling function λ. The labelling λ must be according to the derivation rules given above, labelling non-bud leaves with axioms and satisfying $\lambda(\beta(s)) = \lambda(s)$ for $s \in \mathrm{dom}(\beta)$.

A RHA-preproof is a RHA-proof if every local cycle $\gamma(t)$ has an *invariant:* a sequence θa which is a prefix to all controls along the local cycle $\gamma(t)$ of t and $\beta(t)$, $\textsc{Reset}_a$ rule being applied at some point along $\gamma(t)$. We often refer to the longest invariant of a local cycle as *the* invariant of said cycle.

Write RHA $\vdash \Gamma \Rightarrow \delta$ if there is an RHA-proof of $\varepsilon; \{x \mapsto \varepsilon\} \mid \Gamma \Rightarrow \delta$ i.e., a proof whose endsequent has an empty control and in which all variables are assigned the empty sequence.

Proofs in RHA employ the control Θ and assignment σ to record the progress and ensure the preservation of the free variables in the sequent. Most of the derivation rules of RHA are those of first-order logic, modified to ensure the invariants governing the control Θ and the assignment σ. The three most interesting derivation rules of RHA are $\textsc{Case}_x$, $\textsc{Clear}$ and $\textsc{Reset}_a$. $\textsc{Case}_x$ corresponds the eponymous rule of CHA, modified to extend annotation of the variable x in the right-hand premise, thereby recording that progress has occurred for x. The rules $\textsc{Clear}$ and $\textsc{Reset}_a$ allow for the 'management' of assignments, each erasing some of the progress recorded by an annotation.

For any given RHA-preproof Π, there exists a corresponding CHA preproof $erase(\Pi)$ which is obtained by erasing $\Theta; \sigma$ from each sequent and removing the applications of the $\textsc{Clear}$- and $\textsc{Reset}_a$-rules from Π. The soundness condition of RHA is easiest explained by connecting the annotations of RHA-proofs Π to induction orders of CHA-proofs $erase(\Pi)$. While presented in terms of Heyting arithmetic, the following argument can be carried out in any setting in which a reset proof system is given for a cyclic proof system with induction orders.

Proposition 3.3.2. If Π is an RHA-proof then $erase(\Pi)$ has an induction order.

Proof. Consider a local cycle $\gamma(t)$ from $\beta(t)$ to t of Π with invariant θa. Because the $\textsc{Case}_x$-rule is the only method of extending annotations, any letter b in θa must annotate precisely one variable x along $\gamma(t)$. This variable must be preserved along $\gamma(t)$ as the annotation of x, and thus the letter b, would otherwise disappear at some point along $\gamma(t)$, meaning the invariant θa would not be prefix to all annotations along $\gamma(t)$. Now fix x_t for each such local cycle with invariant θa to be the variable annotated by a. Observe that in $\beta(t)$ and t, x_t must be annotated with the same sequence. Because the $\textsc{Reset}_a$-application along $\gamma(t)$ removes some letters from the annotation of x_t, said annotation is must also be extended along $\gamma(t)$. Because $\textsc{Case}_{x_t}$ is the only method of extending annotations, $\gamma(t)$ must thus pass through the right-hand side of a $\textsc{Case}_{x_t}$-application, progressing x_t.

Define the order $\preceq$ on $\mathrm{dom}(\beta)$ according to which $s \preceq t$ holds iff the invariant θa of t is a prefix of the invariant of s and a annotates the variable x_t in s. From the previous arguments, we may already conclude that $\preceq$ satisfies the conditions regarding progress and preservation for induction orders. That $(\preceq, x_-)$ is an induction order on $erase(\Pi)$ follows from the proposition 5.3.3: every strongly connected component of Π, and thus of $erase(\Pi)$, has a $\preceq$-maximal element. □

The proofs of RHA can be viewed as a normal form of CHA-proofs. One can annotate a CHA-proof with controls and assignments and insert suitable applications of $\textsc{Reset}_a$ to obtain an RHA-proof. However, the original CHA-proof must usually first be unfolded. The result below is a corollary of the more general theorem 5.3.8.

Fact 3.3.3. If Π is a CHA-proof of $\Gamma \Rightarrow \Delta$ then an unfolding of Π can be annotated into an RHA-proof of $\Gamma \Rightarrow \delta$.

Note that no RHA-proof produced by the result above will contain any $\textsc{Clear}$-applications. The inclusion of the $\textsc{Clear}$-rule does not hamper the soundness of RHA but often allows for smaller RHA proofs to be constructed (see lemma 3.3.5 for example).

The key property of reset proofs is that their soundness condition is a *path condition*: it is formulated purely in terms of the local cycles of preproofs. A path condition can hence be verified by simply considering each individual local cycle. It should be noted that, while induction orders are formulated using the notion of local cycles, the preservation condition still is somewhat global, which is why we do not consider it a path condition.

Path conditions posses many desirable properties. For example, they can usually be verified in polynomial time. In the case of reset proofs, verification can be carried out in linear time, assuming a reasonable representation of preproofs. This is significantly faster than the usual verification of global trace conditions or induction orders.

Proposition 3.3.4. The problem of determining whether an RHA-preproof is a proof can be decided in linear time.

Proof. For each local cycle $\gamma(t)$, compute the largest prefix θ_t of all controls along $\gamma(t)$, which requires one pass per local cycle. Then verify that every local cycle $\gamma(t)$ contains a $\textsc{Reset}_a$-application for some letter a in θ_t, which requires one pass per local cycle. The preproof is a proof iff such an application can be found for every cycle. ☐

Converting a CHA-proof to an RHA-proof thus reduces the coNP-problem of checking CHA-proofs to a poly-time problem. If this reduction could be carried out in polynomial time and with a polynomial size increase, one may be able to conclude P = coNP and thus P = NP. Unsurprisingly, all known techniques of translating CHA-proofs to RHA-proofs lead to at least an exponential size increase, yielding no breakthroughs in complexity theory. For example, we present one such technique with a $O(n!)$ size increase in section 5.3.2.

Cyclic proof systems with path conditions have played key roles in many proof theoretic investigations of cyclic proof systems. For example, many interpolation result in cyclic proof theory rely on a path condition (see e.g. Afshari, Leigh, and Menéndez Turata 2021; Marti and Venema 2021b; Shamkanov 2014; Kloibhofer, Trucco Dalmas, et al. 2026). Path conditions allow the interpolant to be 'closed' at each companion in a manner accounting for the cycle, yielding fixed point equations characterising the interpolants per recursion on the tree underlying the derivation, the solution of which yielding the interpolants. Another example is the cut-free completeness result for Kozen's axiomatisation of the modal μ-calculus given by (Afshari and Leigh 2016; Afshari, Leigh, and Menéndez Turata 2024) employs various reset proof systems.

Reset proof systems are more generally well-suited to carrying out proof theoretic investigations of cyclic proof systems, as the 'global' nature of global trace conditions is 'localised' into the control and annotation. For example, admissibility arguments and other proof transformations can often be verified easily, as demonstrated by the proof below.

Lemma 3.3.5. The following rule is admissible in RHA for variables $x \notin \mathrm{FV}(\Gamma)$:

$$\textsc{Ind}\,\frac{\Theta'; \sigma'^{+} \mid \Gamma, \varphi \Rightarrow \varphi[Sx/x]}{\Theta; \sigma \mid \Gamma, \varphi[0/x] \Rightarrow \varphi[t/x]}$$

Proof. Observe that $\sigma'^{+} = \sigma'[x \mapsto \varepsilon]$ in $\Theta'; \sigma'^{+} \mid \Gamma, \varphi \Rightarrow \varphi[Sx/x]$. A suitable preproof Π_1 is given below:

$$
\begin{prooftree}
\AxiomC{}
\RightLabel{\textsc{Ax}}
\UnaryInfC{$\Theta'; \sigma' \mid \Gamma, \varphi[0/x] \Rightarrow \varphi[0/x]$}
\AxiomC{Π_2}
\AxiomC{$\Theta'; \sigma'[x \mapsto \varepsilon] \mid \Gamma, \varphi \Rightarrow \varphi[Sx/x]$}
\RightLabel{\textsc{Clear}}
\UnaryInfC{$\Theta' a; \sigma'[x \mapsto a] \mid \Gamma, \varphi \Rightarrow \varphi[Sx/x]$}
\RightLabel{\textsc{Cut, Wk}}
\BinaryInfC{$\Theta' a; \sigma'[x \mapsto a] \mid \Gamma, \varphi[0/x] \Rightarrow \varphi[Sx/x]$}
\LeftLabel{\textsc{Case}$_x$}
\BinaryInfC{$\Theta'; \sigma'[x \mapsto \varepsilon] \mid \Gamma, \varphi[0/x] \Rightarrow \varphi$}
\LeftLabel{\textsc{Sub}}
\UnaryInfC{$\Theta; \sigma \mid \Gamma, \varphi[0/x] \Rightarrow \varphi[t/x]$}
\end{prooftree}
$$

The preproof Π_1 relies on another preproof Π_2 given below, $\star$ marking a cycle:

$$
\begin{prooftree}
\AxiomC{}
\LeftLabel{\textsc{Ax}}
\UnaryInfC{$\vdots$}
\AxiomC{$\Theta' a; \sigma[x \mapsto a] \mid \Gamma, \varphi[0/x] \Rightarrow \varphi \ \star$}
\LeftLabel{\textsc{Reset}$_a$}
\UnaryInfC{$\Theta' ab; \sigma[x \mapsto ab] \mid \Gamma, \varphi[0/x] \Rightarrow \varphi$}
\AxiomC{$\Theta'; \sigma'[x \mapsto \varepsilon] \mid \Gamma, \varphi \Rightarrow \varphi[Sx/x]$}
\RightLabel{\textsc{Clear}}
\UnaryInfC{$\Theta' ab; \sigma'[x \mapsto ab] \mid \Gamma, \varphi \Rightarrow \varphi[Sx/x]$}
\LeftLabel{\textsc{Cut, Wk}}
\BinaryInfC{$\Theta' ab; \sigma'[x \mapsto ab] \mid \Gamma, \varphi[0/x] \Rightarrow \varphi[Sx/x]$}
\LeftLabel{\textsc{Case}$_x$}
\BinaryInfC{$\Theta' a; \sigma'[x \mapsto a] \mid \Gamma, \varphi[0/x] \Rightarrow \varphi \ \star$}
\end{prooftree}
$$

We claim that any IND-application in an RHA-proof Π may be replaced by the preproof Π_1, inserting the subtree above the premise of the IND-application above the right-most leaf of both Π_1 and Π_2. The local $\star$-cycle satisfies the reset condition as it has invariant $\Theta' a$ and a RESET$_a$ takes place along $\gamma(\star)$. Furthermore, any prefix θ of both Θ and Θ' is also a prefix of every control occurring in Π_1 and Π_2. Thus, no invariants in the original RHA-proof Π are 'disturbed' by replacing the IND-application by Π_1, preserving the reset condition of Π. $\square$

The preproofs Π_i used in the proof of lemma 3.3.5 illustrate one source of the exponential size-increase associated with RHA-proofs: Because an annotation must contain at least two letters to be subject to the RESET-rule, the CASE$_x$-applications introducing the letters must be 'duplicated'. In the course of this, the subproof above the premise of the IND-rule is duplicated as well. Thus, applying the above admissibility result to all instances of IND in an RHA-proof can result in another exponential size increase.

Another core property of reset proof systems is that they tend to induce a *terminating proof search procedure*. That is, one can bound the size of the smallest existing reset proof of a given endsequent in terms of the size of the endsequent. By simply enumerating all reset proofs of this size, one can thus decide provability of a given sequent. Because RHA contains a cut-rule, it does not exhibit this property. Examples which do exhibit this property are reset proof systems for the modal μ-calculus. Notably, the first reset proof system in the literature was given by Jungteerapanich (2010) to derive a tableaux-style decision procedure for the modal μ-calculus. Terminating proof search is also closely connected to the uniform interpolation property (see e.g. Iemhoff (2019)). A reset proof system has been used by Afshari, Leigh, and Menéndez Turata (2021) to establish this property for the modal μ-calculus proof theoretically.

While highly suitable for computational and proof theoretic uses, reset proof systems are arguably not as well-suited to human use as, say, induction orders. Because proofs in RHA can be exponentially larger than CHA-proofs, deriving these proofs by hand quickly becomes infeasible. This is also the reason why the totality proof of $A(x, y, z)$ is not recalled in this section: Its RHA-proof is simply too large. Furthermore, the syntactic clutter of the annotation mechanism often makes RHA-proofs difficult to read and subsequently verify.

3.4 Stack-controlled proofs

A core component of an induction order, as discussed in section 3.2, is the ordering $t \prec s$ of the local cycles of a CHA-proof, capturing which interleaving cycles preserve the progress of others. There exists another natural ordering on the cycles of a CHA-proof: the structural dependency ordering. For two buds $s, t \in \mathrm{dom}(\beta)$ we set $t \sqsubseteq s$ if $\beta(s) \leq \beta(t) < s$, in other words, if the companion of t occurs along $\gamma(s)$ meaning the t-cycle 'directly forks off' the s-cycle. To illustrate, consider fig. 7: There we have $u \sqsubset t \sqsubset s$. Observe that the ordering is not necessarily transitive: $u \not\sqsubset s$.

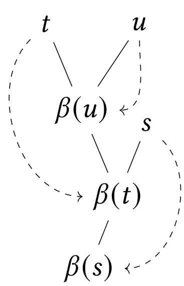


Figure 7: Some cycles in a cyclic tree. Based on the structural ordering, $u \sqsubset t \sqsubset s$ but not $u \sqsubset s$.

Another way of thinking of $t \sqsubset s$ is that 'while travelling from $\beta(s)$ to s, arbitrarily many detours along the t-cycles may be taken'. In fact, this intuition is even preserved under the transitive closure: Because $u \sqsubseteq^* s$ in fig. 7, arbitrarily many 'loops' along $\gamma(u)$ may occur along an s-cycle.

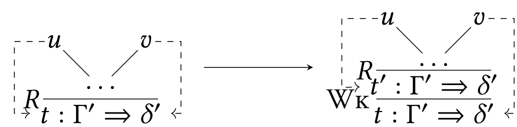
Thinking of each cycle as embodying an instance of induction in the proof, the two aforementioned orderings give rise to intuitive readings in terms of said inductions:

- $t \prec s$: the t-induction is 'inside' the s-induction, *logically*, and
- $t \sqsubset s$: the t-induction is 'inside' the s-induction, *physically*.

Crucially, the two orderings $\prec$ and $\sqsubset$ need not be aligned in a CHA-proof: It might be that a cycle t is 'logically inside' s ($t \prec s$) while s is 'physically inside' of t ($s \sqsubset t$). This leads to an important result, initially due to Sprenger and Dam (2003b, theorem 5), which allows one to always bring the two orderings into alignment:[13]

Fact 3.4.1. Let Π be a CHA-proof such that β is an injection. Then it may be unfolded into a CHA-proof Π' which is justified by an induction order with $\sqsubseteq^*$ as its underlying ordering.

Remark 3.4.1. The injectivity of β required above is a rather weak precondition: Any CHA-proof can be transformed in an 'equivalent' one with injective β by inserting 'spurious' instances of Wk, which do not remove any formulas, and 'redirecting' β as illustrated below.



This section is concerned with stack-controlled Heyting arithmetic (SHA), which is a syntactic representation of the CHA-normal form where $\prec\, = \,\sqsubseteq^*$. The key insight is that in such a setting, preservation behaves 'stack-like': Cycles further from the root preserve the progress of cycles closer to the root.

The sequents of SHA are expressions $\Lambda \mid \Gamma \Rightarrow \delta$, consisting of an HA sequent $\Gamma \Rightarrow \delta$ and an ordered list Λ of companion labels, which is treated as a stack. We denote the empty list by ε. Intuitively, the companion labels in Λ records the sequents of companions to which one may 'cycle back' at the current point of the proof. Formally, a *companion label* is an expression of the shape $x^\bullet \mapsto \Gamma \Rightarrow \delta$ for a HA-sequent $\Gamma \Rightarrow \delta$, a variable $x \in \mathrm{FV}(\Gamma, \delta)$ and a mark $\bullet \in \{+, -\}$. We write $\textsc{Var}(\Lambda)$ for the set $\{x \mid (x^\bullet \mapsto \Gamma \Rightarrow \delta) \in \Lambda\}$ of variables associated to companion labels in Λ.

A SHA-sequent $\Lambda \mid \Gamma \Rightarrow \delta$ is *well-formed* if for every $(x^\bullet \mapsto \Delta \Rightarrow \gamma) \in \Lambda$ we have $x \in \mathrm{FV}(\Gamma, \delta)$. Going forward, we assume that every SHA-sequent $\Lambda \mid \Gamma \Rightarrow \delta$ under consideration is well-formed, unless stated otherwise.

13 Formally, the result is only proven for the μFOL-system they consider. However, the result transfers easily to other cyclic proof systems with induction orders, like CHA.

Before presenting the rules of the calculus SHA, we fix further operations on stacks of companion labels. Given a variable x, we define Λ^{+x} to be the result of replacing in Λ all occurrences of x^- by x^+:

$$\varepsilon^{+x} := \varepsilon \qquad (y^\bullet \mapsto \Gamma \Rightarrow \delta; \Lambda)^{+x} := \begin{cases} x^- \mapsto \Gamma \Rightarrow \delta; \Lambda^{+x}, & \text{if } y = x, \\ y^\bullet \mapsto \Gamma \Rightarrow \delta; \Lambda^{+x}, & \text{otherwise.} \end{cases}$$

Definition 3.4.2. The *derivation rules* of SHA are of split into three categories:

LOGICAL RULES, which do not interact with companion labels:

$$\text{Ax}\,\frac{}{\varepsilon \mid \Gamma, \delta \Rightarrow \delta} \qquad \to\text{L}\,\frac{\Lambda^* \mid \Gamma \Rightarrow \varphi \quad \Lambda^* \mid \Gamma, \psi \Rightarrow \delta}{\Lambda \mid \Gamma, \varphi \to \psi \Rightarrow \delta}$$

$$\to\text{R}\,\frac{\Lambda \mid \Gamma, \varphi \Rightarrow \psi}{\Lambda \mid \Gamma \Rightarrow \varphi \to \psi} \qquad \wedge\text{L}\,\frac{\Lambda \mid \Gamma, \varphi, \psi \Rightarrow \delta}{\Lambda \mid \Gamma, \varphi \wedge \psi \Rightarrow \delta}$$

$$\wedge\text{R}\,\frac{\Lambda^* \mid \Gamma \Rightarrow \varphi \quad \Lambda^* \mid \Gamma \Rightarrow \psi}{\Lambda \mid \Gamma \Rightarrow \varphi \wedge \psi}$$

$$\vee\text{L}\,\frac{\Lambda^* \mid \Gamma, \varphi \Rightarrow \delta \quad \Lambda^* \mid \Gamma, \psi \Rightarrow \delta}{\Lambda \mid \Gamma, \varphi \vee \psi \Rightarrow \delta} \qquad \vee\text{R}\,\frac{\Lambda \mid \Gamma \Rightarrow \varphi_i}{\Lambda \mid \Gamma \Rightarrow \varphi_1 \vee \varphi_2}$$

$$\forall\text{L}\,\frac{\Lambda \mid \Gamma, \varphi[t/x] \Rightarrow \delta}{\Lambda \mid \Gamma, \forall x.\varphi \Rightarrow \delta} \qquad \forall\text{R}\,\frac{\Lambda \mid \Gamma \Rightarrow \varphi \quad x \notin \mathrm{FV}(\Gamma, \forall x.\varphi)}{\Lambda \mid \Gamma \Rightarrow \forall x.\varphi}$$

$$\exists\text{L}\,\frac{\Lambda \mid \Gamma, \varphi \Rightarrow \delta \quad x \notin \mathrm{FV}(\Gamma, \exists x.\varphi, \delta)}{\Lambda \mid \Gamma, \exists x.\varphi \Rightarrow \delta} \qquad \exists\text{R}\,\frac{\Lambda \mid \Gamma \Rightarrow \varphi[t/x]}{\Lambda \mid \Gamma \Rightarrow \exists x.\varphi}$$

$$=\text{L}\,\frac{\Lambda^* \mid \Gamma[x/y] \Rightarrow \delta[x/y] \quad x, y \notin \mathrm{FV}(s,t)}{\Lambda \mid \Gamma[s/x, t/y], s = t \Rightarrow \delta[s/x, t/y]}$$

$$=\text{R}\,\frac{}{\Lambda \mid \Gamma \Rightarrow t = t} \qquad \bot\text{L}\,\frac{}{\Lambda \mid \Gamma, \bot \Rightarrow \delta} \qquad \text{Wk}\,\frac{\Lambda \mid \Gamma \Rightarrow \delta}{\Lambda \mid \Gamma, \Gamma' \Rightarrow \delta}$$

$$\text{Cut}\,\frac{\Lambda^* \mid \Gamma \Rightarrow \varphi \quad \Lambda \mid \Gamma, \varphi \Rightarrow \delta}{\Lambda \mid \Gamma \Rightarrow \delta} \qquad \text{Sub}\,\frac{\Lambda \mid \Gamma \Rightarrow \delta}{\Lambda \mid \Gamma[t/x] \Rightarrow \delta[t/x]}$$

where Λ^* marks the adjustment of Λ which ensures well-formedness of the sequent. That is, for a HA-sequent $\Gamma \Rightarrow \delta$ and some companion label stack Λ we define the well-formed sequent $\Lambda^* \mid \Gamma \Rightarrow \delta$ recursively with

$\varepsilon^* := \varepsilon$ and

$$(x^\bullet \mapsto \Gamma' \Rightarrow \delta'; \Lambda)^* := \begin{cases} x^\bullet \mapsto \Gamma' \Rightarrow \delta'; \Lambda^* & \text{if } x \in \mathrm{FV}(\Gamma, \delta) \\ \varepsilon & \text{otherwise} \end{cases}.$$

Arithmetical axioms of the form:

$$\begin{array}{ll} \varepsilon \mid \Gamma, 0 = St \Rightarrow \bot & \varepsilon \mid \Gamma, Ss = St \Rightarrow s = t \\ \varepsilon \mid \Gamma \Rightarrow s + 0 = 0 & \varepsilon \mid \Gamma \Rightarrow s + St = S(s + t) \\ \varepsilon \mid \Gamma \Rightarrow s \cdot 0 = 0 & \varepsilon \mid \Gamma \Rightarrow s \cdot St = (s \cdot t) + s \end{array}$$

Label rules which manipulate companion labels:

$$\textsc{Comp}\frac{\Lambda; (x^- \mapsto \Gamma \Rightarrow \delta) \mid \Gamma \Rightarrow \delta}{\Lambda \mid \Gamma \Rightarrow \delta}$$

$$\textsc{Bud}\frac{}{\Lambda; (x^+ \mapsto \Gamma \Rightarrow \delta) \mid \Gamma \Rightarrow \delta} \qquad \textsc{Pop}\frac{\Lambda_0 \mid \Gamma \Rightarrow \delta}{\Lambda_0; \Lambda_1 \mid \Gamma \Rightarrow \delta}$$

$$\textsc{Case}_x\frac{\Lambda^* \mid \Gamma(0) \Rightarrow \delta(0) \qquad \Lambda^{+x} \mid \Gamma(S\,x) \Rightarrow \delta(S\,x)}{\Lambda \mid \Gamma(x) \Rightarrow \delta(x)}$$

We only consider finite, non-cyclic derivations in SHA to be proofs. That is, a proof in SHA is a pair $\pi = (T, \rho)$ consisting of a finite tree T and a function $\rho : T \mapsto \mathcal{R}$, assigning instances of SHA-derivation rules to nodes of T. If $t \in \mathrm{Nodes}(T)$ and $\rho(t)$ is a rule with n premises $\Lambda_1 \mid \Gamma_1 \Rightarrow \delta_1, \ldots, \Lambda_n \mid \Gamma_n \Rightarrow \delta_n$ then $\mathrm{Chld}(t)$ must be $t_1, \ldots, t_n$ and $\Lambda_i \mid \Gamma_i \Rightarrow \delta_i$ must be the conclusion of each $\rho(t_i)$. We consider axiomatic sequents as premiseless derivation rules. We call the conclusion of the rule labelling the root of T the *endsequent*. If such a proof with endsequent $\varepsilon \mid \Gamma \Rightarrow \delta$ exists then $\Gamma \Rightarrow \delta$ is *provable in stack-controlled simple Heyting arithmetic*, denoted $\mathrm{SHA} \vdash \Gamma \Rightarrow \delta$.

While SHA technically is a 'standard' proof system, one should still consider it a cyclic proof system. Previously presented cyclic proof systems, such as CHA and SHA, impose a bud-companion structure on a proof tree T by means of a partial map $\beta : \mathrm{Leaf}(T) \rightharpoonup \mathrm{Inner}(T)$. In SHA, this is instead achieved using the Comp- and Bud-rules. The Comp-rule 'stores' the current sequent on the companion label stack, 'remembering' it as an available companion further up in the proof. The Bud-rule, exhibits that the current sequent matches that of a previously 'stored' companion (often first 'digging it out of the stack' using an instance of Pop), closing the branch of the proof. Such an application of the Bud-rule corresponds, in the presentation of cyclic trees via $\beta : \mathrm{Leaf}(T) \rightharpoonup \mathrm{Inner}(T)$, to mapping the conclusion of the Bud-rule to the conclusion of the conclusion corresponding Comp-rule.

In fact, every SHA-derivation is a SHA-proof, as the stack-control ensures that the 'global trace property' is satisfied. To make this a bit more precise, observe that from an SHA-proof Π one can obtain a CHA-preproof $erase(\Pi)$ by 'stripping off' the stack-control and reintroducing cyclic edges as described above (for a more formal specification of this operation, consult definition 6.1.1). We can prove that any SHA-*derivation* embodies a CHA-proof.

Proposition 3.4.3. If Π is a SHA-derivation then $erase(\Pi)$ is a CHA-proof.

Proof. Consider an infinite branch π through $erase(\Pi)$. We know $\mathrm{Inf}(\pi)$ is a strongly connected component of $erase(\Pi)$ which corresponds to a strongly connected component of Π. The scc in Π must have a lowest node b which must be the premise of a Comp-rule, adding a companion label $x^- \mapsto \Gamma \Rightarrow \delta$ to the stack. We claim that the branch π has a progressing x-trace. It can be shown that this label $x^\bullet \mapsto \Gamma \Rightarrow \delta$ (with $\bullet \in \{-, +\}$) must be present in the stacks of every sequent in the scc. Thus, as the sequents of Π must be well-formed, x must occur free in every HA-sequent of the scc, both in Π but thus also in $erase(\Pi)$, and thus such an x-trace exists along π. Because the node b is in the scc, π must pass through Bud-instances 'jumping to it' infinitely often. As the aforementioned label at such Buds is $x^+ \mapsto \Gamma \Rightarrow \delta$, π must have passed through the right-hand side of a $\textsc{Case}_x$-rule between b and any such Bud-instance. Thus, the x-trace must be progressing. ☐

Differing from the various cyclic proof representations discussed so far in this chapter, 'SHA-proof checking' thus is a trivial affair.

Corollary 3.4.4. It can be determined in constant time whether a SHA-derivation is an SHA-proof.

Proof. Every SHA-derivation is a SHA-proof. ☐

Of course, this also makes 'proof checking' also very easy for humans. For example, fig. 8 is the CHA-proof of the totality of the Ackermann function, this time presented as an SHA-proof. Here, we employ the companion label notation $v^\sigma := v^\sigma \mapsto \Rightarrow \exists z.A(x, y, z)$. To verify that it is a SHA-proof, one only needs to check that every rule is applied correctly, which is easily done.

$$
\dfrac{\Pi_1 \qquad \dfrac{\dfrac{}{x^-; y^+ \mid \;\Rightarrow \exists z.\, A(x,y,z)}\,\textsc{Bud} \qquad \dfrac{\dfrac{\dfrac{}{\varepsilon \mid \;\Rightarrow A(0,Sy,SSy)}}{\varepsilon \mid \;\Rightarrow \exists z.\, A(0,Sy,z)} \qquad \Pi_2}{x^-; y^+ \mid \exists z'.\, A(x,y,z') \Rightarrow \exists z.\, A(x,Sy,z)}}{x^-; y^+ \mid \;\Rightarrow \exists z.\, A(x,Sy,z)}\,\textsc{Cut}}{\dfrac{x^-; y^- \mid \;\Rightarrow \exists z.\, A(x,y,z)}{\dfrac{x^- \mid \;\Rightarrow \exists z.\, A(x,y,z)}{\varepsilon \mid \;\Rightarrow \exists z.\, A(x,y,z)}\,\textsc{Comp}}\,\textsc{Comp}}\,\textsc{Case}_y
$$

$$
\Pi_1 : \quad \dfrac{\dfrac{\dfrac{}{\varepsilon \mid \;\Rightarrow A(0,0,1)}}{\varepsilon \mid \;\Rightarrow \exists z.\, A(0,0,z)} \qquad \dfrac{\dfrac{\dfrac{}{x^+ \mid \;\Rightarrow \exists z.\, A(x,y,z)}\,\textsc{Bud}}{x^+ \mid \;\Rightarrow \exists z.\, A(x,1,z)}\,\textsc{Sub}}{x^+ \mid \;\Rightarrow \exists z.\, A(Sx,0,z)}}{x^- \mid \;\Rightarrow \exists z.\, A(x,0,z)}\,\textsc{Case}_x
$$

$$
\Pi_2 : \quad \dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{x^+ \mid \;\Rightarrow \exists z.\, A(x,y,z)}\,\textsc{Bud}}{x^+ \mid \;\Rightarrow \exists z.\, A(x,z',z)}\,\textsc{Sub}}{x^+ \mid A(Sx,y,z') \Rightarrow \exists z.\, A(Sx,Sy,z)}}{x^+; y^+ \mid A(Sx,y,z') \Rightarrow \exists z.\, A(Sx,Sy,z)}\,\textsc{Pop}}{x^+; y^+ \mid \exists z'.\, A(Sx,y,z') \Rightarrow \exists z.\, A(Sx,Sy,z)}\,\exists\mathrm{L}
$$

Figure 8: A SHA-proof the totality of $A(x, y, z)$.

It should be noted that the stack-controlled proof system presented in this section is exceptionally simple. This is in the same vein as the fact that every CHA-proof admits an induction order which, as discussed in section 3.2, is a property few cyclic proof systems in the literature exhibit. In chapter 6 we present a general construction to obtain a stack-controlled cyclic proof system which can be applied to most cyclic proof systems in the literature. The notions of control stack and companion label involved in these systems is far more complex.

Similarly to reset proofs, SHA-proofs can be viewed as a normal form of CHA-proofs (namely the one where $\preceq \; = \; \sqsubseteq^*$). One can always unfold a CHA-proof and add suitable stack-annotations and instances of the rules Comp, Bud and Pop to obtain a SHA-proof. The result below is a corollary of the more general theorem 6.3.8.

Fact 3.4.5. If Π is a CHA-proof of $\Gamma \Rightarrow \delta$ then an unfolding of Π can be annotated into a SHA-derivation of $\Gamma \Rightarrow \delta$.

Remark 3.4.2. While the proof of fact 3.4.5 can be given in terms of induction orders, as outlined above, the more general version instead relies on reset proofs, as not every cyclic proof system admits an induction order. The result relies on lemmas proven in Afshari and Leigh (2016) which can be viewed as the results of Sprenger and Dam (2003b) 'transported along' the observation from proposition 3.3.2 that every reset proof 'contains' an induction ordering.

It should be noted that, similarly to reset proofs, the SHA-derivation corresponding to a CHA-proof can be exponentially larger.

Stack-controlled proof systems seem to be quite well-suited to proof theoretic investigations of cyclic proof systems. As they are a quite young notion,[14] there are no examples of them being used for proof theory in the literature. However, we obtain a handful of results via them in chapter 7.

To demonstrate the ease of carrying out research in cyclic proof theory using stack-controlled proofs, let us return to our running example of the admissibility of Ind. The proof of lemma 3.4.7 requires the following SHA-specific weakening principle:

Proposition 3.4.6. The following rule is admissible in SHA:

$$\textsc{Drop}\,\frac{\Lambda \mid \Gamma \Rightarrow \delta}{\Lambda^{+x} \mid \Gamma \Rightarrow \delta}.$$

Proof. Induction on the derivation of $\Lambda \mid \Gamma \Rightarrow \delta$. □

Lemma 3.4.7. The following rule is admissible in SHA for variables $x \notin \mathrm{FV}(\Gamma)$:

$$\textsc{Ind}\,\frac{\Lambda \mid \Gamma, \varphi \Rightarrow \varphi[Sx/x]}{\Lambda \mid \Gamma, \varphi[0/x] \Rightarrow \varphi[t/x]}$$

Proof. Let Π be a SHA-derivation of $\Lambda \mid \Gamma, \varphi \Rightarrow \varphi[Sx/x]$. Then below is a SHA-derivation of the desired conclusion:

$$\begin{prooftree}
\AxiomC{}
\RightLabel{}\LeftLabel{\textsc{Ax}}
\UnaryInfC{$\Lambda_* \upharpoonright x \mid \Gamma, \varphi[0/x] \Rightarrow \varphi[0/x]$}
\AxiomC{}
\LeftLabel{\textsc{Bud}}
\UnaryInfC{$\Lambda_*^{+x} \mid \Gamma, \varphi[0/x] \Rightarrow \varphi$}
\AxiomC{$\Lambda \mid \Gamma, \varphi \Rightarrow \varphi[Sx/x]$}
\RightLabel{\textsc{Pop}}
\UnaryInfC{$\Lambda_* \mid \Gamma, \varphi \Rightarrow \varphi[Sx/x]$}
\RightLabel{\textsc{Drop}}
\UnaryInfC{$\Lambda_*^{+x} \mid \Gamma, \varphi \Rightarrow \varphi[Sx/x]$}
\LeftLabel{\textsc{Cut, Wk}}
\BinaryInfC{$\Lambda_*^{+x} \mid \Gamma, \varphi[0/x] \Rightarrow \varphi[Sx/x]$}
\LeftLabel{$\textsc{Case}_x$}
\BinaryInfC{$\Lambda; (x \mapsto \Gamma, \varphi[0/x] \Rightarrow \varphi) \mid \Gamma, \varphi[0/x] \Rightarrow \varphi$}
\LeftLabel{\textsc{Comp}}
\UnaryInfC{$\Lambda \mid \Gamma, \varphi[0/x] \Rightarrow \varphi$}
\LeftLabel{\textsc{Sub}}
\UnaryInfC{$\Lambda \mid \Gamma, \varphi[0/x] \Rightarrow \varphi[t/x]$}
\end{prooftree}$$

where $\Lambda_* := \Lambda; (x \mapsto \Gamma, \varphi[0/x] \Rightarrow \varphi)$. □

Because any SHA-derivation is a proof, no further argument beyond the derivation above is needed to establish admissibility. Compare this to every other

14 At the time of printing, the article (Leigh and Wehr 2025) has not yet been published in a peer-reviewed venue. There thus are no peer-reviewed articles on stack-controlled proofs yet.

cyclic proof representation discussed in this chapter, where further arguments about the soundness condition were required.

Beyond the aforementioned absence of a soundness condition separate from well-derivedness, the core property of stack-controlled proofs which makes them so well-suited to proof theoretic investigations is the fact that the ordering of their cycles and the logical hierarchy of the inductions they embody align (again, that $\preceq\, = \sqsubseteq^*$). In the setting of CHA, this means that SHA-proofs are 'nearly' HA-proofs. Indeed, the translation back into HA, which we have previously mentioned is very difficult, becomes quite simple if one starts from SHA.[15] We give a sketch of the argument below. Again, for the proof construction in HA, some admissible rules are required.

Fact 3.4.8. The following rules are admissible in HA:

$$\textsc{Sub}\,\frac{\Gamma \Rightarrow \delta}{\Gamma[s/x] \Rightarrow \delta[s/x]} \qquad \textsc{Ind}\,\frac{\Gamma, \forall x' < x.\varphi[x'/x] \Rightarrow \varphi}{\Gamma \Rightarrow \forall x.\varphi}$$

$$\textsc{Case}_x\,\frac{\Gamma[0/x] \Rightarrow \delta[0/x] \quad \Gamma[Sx/x] \Rightarrow \delta[Sx/x]}{\Gamma \Rightarrow \delta}$$

where $x \notin \mathrm{FV}(\Gamma, \forall x.\varphi)$ in $\textsc{Ind}$.

Theorem 3.4.9. If SHA $\vdash \Gamma \Rightarrow \delta$ then HA $\vdash \Gamma \Rightarrow \delta$.

Proof. For a companion label stack Λ, let $x_i^{\sigma_i} \mapsto \Gamma_i \Rightarrow \delta_i$ be the ith companion label in Λ, counting from the left and starting at 1, and let Λ_i be all the labels *before* the ith label. We define the *inductive hypotheses* of Λ as $\mathcal{H}_\Lambda := \{\varphi_i \mid 1 \leq i \leq |\Lambda|\}$ where

$$\varphi_i := \forall x_i' <^{\sigma_i} x_i.\forall \vec{u} \leq \vec{v}.\forall \vec{y}.(\Gamma_i \to \delta_i)[x_i'/x_i, \vec{u}/\vec{v}]$$

with $\vec{v} = \textsc{Var}(\Lambda_i) \setminus \{x\}$, $\vec{y} = \mathrm{FV}(\Gamma_i, \delta_i) \setminus \{\vec{v}, x\}$ and $x_i', \vec{u}$ fresh. We write $\Phi \to \psi$ for $\bigwedge \Phi \to \psi$ and $<^- := <$ and $<^+ := \leq$. We prove that HA $\vdash \mathcal{H}_\Lambda, \Gamma \Rightarrow \delta$ per induction on a SHA-derivation of $\Lambda \mid \Gamma \Rightarrow \delta$. The claim directly follows as SHA $\vdash \Gamma \Rightarrow \delta$ means SHA derives $\varepsilon \mid \Gamma \Rightarrow \delta$.

$\textsc{Comp}$: Let $\Lambda' := \Lambda; x^- \mapsto \Gamma \Rightarrow \delta$, then per IH we know that HA $\vdash \mathcal{H}_{\Lambda'}, \Gamma \Rightarrow \delta$ where $\mathcal{H}_{\Lambda'} = \mathcal{H}_\Lambda, \varphi^*$ with

$$\varphi^* = \forall x' < x.\psi \qquad \varphi^i = z \leq x \to \psi[z/x]$$

$$\psi := \forall \vec{u} \leq \vec{v}.\forall \vec{y}.(\Gamma \to \delta)[x'/x, \vec{u}/\vec{v}]$$

[15] The difficulty of the translation from CHA to HA is now 'hidden' in the translation from CHA to SHA.

where z does not clash with any variable used so far. Then derive

$$
\textsc{Cut}\ \dfrac{
  \textsc{Ind}\ \dfrac{
    {\to}\mathrm{R}, \dagger\ \dfrac{
      \forall\mathrm{R}, {\to}\mathrm{R}\ \dfrac{
        \star\ \dfrac{
          \textsc{Sub}\ \dfrac{\dfrac{\mathrm{IH}}{\mathcal{H}_\Lambda, \varphi^*, \Gamma \Rightarrow \delta}}
          {\mathcal{H}_\Lambda[\vec{u}/\vec{v}, z/x], \forall z' < z.\psi[\vec{u}/\vec{v}, z'/x], \Gamma[\vec{u}/\vec{v}, z/x] \Rightarrow \delta[\vec{u}/\vec{v}, z/x]}
        }{\mathcal{H}_\Lambda, \forall z' < z.\psi[z'/x], z \leq x, \vec{u} \leq \vec{v}, \Gamma[\vec{u}/\vec{v}, z/x] \Rightarrow \delta[\vec{u}/\vec{v}, z/x]}
      }{\mathcal{H}_\Lambda, \forall z' < z.\psi[z'/x], z \leq x \Rightarrow \psi[z/x]}
    }{\mathcal{H}_\Lambda, \forall z' < z.\varphi^i[z'/z] \Rightarrow \varphi^i}
  }{\mathcal{H}_\Lambda \Rightarrow \forall z.\varphi^i}
  \qquad
  \dfrac{\dfrac{}{\Gamma \to \delta, \Gamma \Rightarrow \delta}}{\forall z.\varphi^i, \Gamma \Rightarrow \delta}\ \forall\mathrm{L}
}{\mathcal{H}_\Lambda, \Gamma \Rightarrow \delta}
$$

Here, the $\dagger$-step makes use of

$$z \leq x, \forall z' < z.\varphi^i[z'/z] \Rightarrow \forall z' < z.\varphi[z'/x].$$

The $\forall\mathrm{R}$ above $\dagger$ may be applied for the $\vec{y}$ as $\vec{y} \notin \mathrm{FV}(\mathcal{H}_\Lambda, \varphi^*) \cup \{z\}$. The $\star$-step lowers the 'upper bounds' in $\mathcal{H}_\Lambda$ (and analogously in $\psi[z'/x]$) as follows: Let $\varphi_i = \forall x_i' <^{\sigma_i} x_i.\forall \vec{a} \leq \vec{b}.(\Gamma_i \to \delta_i)[x_i'/x_i, \vec{a}/\vec{b}] \in \mathcal{H}_\Lambda$. We know that the $x_i, \vec{b}$ are among $x, \vec{v}$. By $z \leq x, \vec{u} \leq \vec{v}$, we know that there are $x_i'' \leq x_i$ and $\vec{c} \leq \vec{b}$. Transitivity of $\leq$ thus yields

$$\varphi_i, z \leq x, \vec{u} \leq \vec{v} \Rightarrow \varphi_i[x_i''/x_i, \vec{c}/\vec{b}]$$

for each $i \leq |\Lambda|$, where $\varphi_i[x_i''/x_i, \vec{c}/\vec{b}] = \varphi_i[\vec{u}/\vec{v}, z/x]$ as desired.

$\textsc{Bud}$: We must prove $\mathrm{HA} \vdash \mathcal{H}_\Lambda, \varphi^*, \Gamma \Rightarrow \delta$ where

$$\varphi^* = \varphi_{|\Lambda|+1} = \forall x' \leq x.\forall \vec{u} \leq \vec{v}.\forall \vec{y}.(\Gamma \to \delta)[x'/x, \vec{u}/\vec{v}].$$

This can be achieved as in the right-hand part of the $\textsc{Comp}$-case:

$$\dfrac{\dfrac{}{\mathcal{H}_\Lambda, \Gamma \to \delta, \Gamma \Rightarrow \delta}}{\mathcal{H}_\Lambda, \varphi^*, \Gamma \Rightarrow \delta}$$

$\textsc{Case}_x$: We may assume, per IH, that

$$\mathrm{HA} \vdash \mathcal{H}_{\Lambda\restriction x}, \Gamma[0/x] \Rightarrow \delta[0/x] \text{ and } \mathrm{HA} \vdash \mathcal{H}_{\Lambda^{+x}}, \Gamma[Sx/x] \Rightarrow \delta[Sx/x].$$

Then derive

$$
\textsc{Case}_x\ \dfrac{
  \star, \textsc{Wk}\ \dfrac{\dfrac{\mathrm{IH}}{\mathcal{H}_{\Lambda\restriction x}, \Gamma[0/x] \Rightarrow \delta[0/x]}}{\mathcal{H}_\Lambda[0/x], \Gamma[0/x] \Rightarrow \delta[0/x]}
  \quad
  \dfrac{\dfrac{\mathrm{IH}}{\mathcal{H}_{\Lambda^{+x}}, \Gamma[Sx/x] \Rightarrow \delta[Sx/x]}}{\mathcal{H}_\Lambda[Sx/x], \Gamma[Sx/x] \Rightarrow \delta[Sx/x]}\ \dagger
}{\mathcal{H}_\Lambda, \Gamma \Rightarrow \delta}
$$

where $\star$ relies on the fact that $\mathcal{H}_{\Lambda\restriction x}[0/x] = \mathcal{H}_{\Lambda\restriction x}$ as $x \notin \textsc{Var}(\Lambda)$. The step $\dagger$ arrives at $\mathcal{H}_{\Lambda^{+x}}$ from $\mathcal{H}_\Lambda[Sx/x]$ via a sequence of $\textsc{Cut}$s, 'adjusting' some bounds of the form $x' < Sx$ to $x' \leq x$ and some bounds of the form $u \leq Sx$ to $u \leq x$ (analogously to the $\star$-step of $\textsc{Comp}$).

Other rules: The remaining cases are all straightforward. □

The proof above illustrates the aforementioned similarity between stack-controlled and inductive proof systems, generally, and SHA and HA, specifically. Observe that the translation from SHA to HA is carried out in one 'recursive pass' over the SHA-proof. Furthermore, each Comp-rule is translated into one Ind-instance and this is (except for reasoning about $<$ in HA) the only source of inductions in the resulting HA-proof. In this manner, the cyclic structure of SHA and the inductive structure of HA correspond to each other.

In chapter 7, we derive further proof theoretic results about CHA, $\mathrm{CHA}_<$ and variants thereof. All results rely on this close connection between stack-controlled proofs and those with explicit induction rules. This connection sometimes allows proof theoretic strategies and notions from proof systems with induction rules to be 'transported', along the opposite direction of theorem 3.4.9, back into the realm cyclic proof systems (see for example section 7.3).

Lastly, we would claim that stack-controlled proofs are quite well-suited for 'human' use. While the stacks introduce some syntactic clutter to the sequents, these can usually be accounted for with a little bit of notation as we do in fig. 8. Thus, usability for writing proofs mostly comes down to the size of the proofs. Like reset proof systems, translating from CHA to SHA can introduce an exponential size-increase. However, experience has shown that for many concrete examples (like fig. 8) one can still find 'small' SHA-proofs. This is different from reset-proofs, where the 'inefficiency' of the annotation mechanism always introduces some unfolding when compared with a 'plain' CHA-proof.

3.5 Simplistic cyclic proofs

So far, we have only considered representations of cyclic proofs which, possibly after some unfolding and additional annotation, fundamentally agree with the global trace condition on which CHA-preproofs are proofs. By instead giving up on this ambition and accepting only *some* GTC-verified CHA-preproofs, for example those with a simpler cyclic structure, one can also arrive at a notion of cyclic proof which is easier to work with.

We are aware of two instances in the literature where the attention is restricted to structurally simpler cyclic proofs: the systems S_n of Beklemishev et al. (2025) and the system CNB of Curzi and Das (2021). Simplifying a little, the mentioned systems only accept cyclic proofs with the following property: For any two cycles s and t in a strongly connected component, each must preserve the other's progressing thread.

To illustrate the idea of restricting the notion of cyclic proof, we present what can only be described as a 'cartoonish exaggeration' of the restriction taken in the aforementioned articles: we only consider CHA-proofs without interleaving cycles.

Definition 3.5.1 (Simplistic CHA)**.** A cyclic tree (T, β) is *simplistic* if it is in cycle normal form and it has no interleaving cycles. That is, for every $s \in \mathrm{dom}(\beta)$ no node $t \in \gamma(s)$ is a companion to a node different from s i.e., if $t \in \gamma(s)$ then $\beta^{-1}(t) \subseteq \{s\}$.

A CHA-proof is a *simplistic* CHA-*proof* if its underlying cyclic tree is simplistic. We write sCHA $\vdash \Gamma \Rightarrow \delta$ if there is a simplistic CHA-proof (a sCHA-proof) of $\Gamma \Rightarrow \delta$.

Like the reset proof system RHA, the soundness condition of sCHA is a path condition: Soundness can be verified by only considering the individual cycles of a proof. Unlike RHA, this is not achieved by a clever annotation scheme but instead as interaction between different cycles simply cannot occur in a 'late enough' suffix of any infinite branch.

Proposition 3.5.2. A simplistic CHA-preproof is a proof iff the following condition holds: Along every $\gamma(t)$ for $t \in \mathrm{dom}(\beta)$, there is a variable x which is free in every sequent and $\gamma(t)$ contains both the conclusion and the right-hand premise of the same $\textsc{Case}_x$-instance.

Proof. Let Π be a simplistic CHA-preproof and π be a path through it. Observe that π must eventually repeat only one cycle of Π infinitely: Suppose, w.l.o.g., that π eventually interleaved two cycles $s, t \in \mathrm{dom}(\beta)$. For this to be possible, either $\beta(s) \in \gamma(t)$ or $\beta(t) \in \gamma(s)$. But either case contradicts C being simplistic unless $s = t$.

Thus, if π eventually only repeats one cycle, which per assumption has a progressing x-trace for some variable x along it, then π has a progressing trace. □

Because sCHA has a path condition, proof checking is computationally easy. Indeed, verifying that a sCHA-preproof is a proof is also very easy for humans: The procedure described below is not at all cognitively difficult to carry out.

Corollary 3.5.3. It can be determined in linear time whether a simplistic CHA-preproof is a proof.

Proof. For each cycle $\gamma(t)$, compute the set V_t of variables x for which the right-hand side of $\textsc{Case}_x$ is passed by $\gamma(t)$. Then verify that for each cycle $\gamma(t)$, some variable $x \in V_t$ occurs free in each of its sequents. The preproof is a proof iff this can be established for every cycle. □

A great risk of excluding some CHA-proofs is that it may come at the cost of completeness: There may be theorems of HA which cannot be proven by CHA without interleaving cycles. The core observation 'justifying' the restriction of sCHA is that this does not happen. In fact, the proofs one usually produces in translating from HA to CHA already are sCHA-proofs. Indeed, this is a pattern common to the cyclic proof theory literature: Translating from an inductive to a cyclic proof system generally produces cyclic proofs with a very simple cyclic structure.

Proposition 3.5.4. If $\mathrm{HA} \vdash \Gamma \Rightarrow \delta$ then $\mathrm{sCHA} \vdash \Gamma \Rightarrow \delta$.

Proof. Simply observe that theorem 2.2.1 already produces sCHA-proofs: Every leaf with an induction axiom is replaced by a subproof containing a single cycle and no other cycles are introduced anywhere else. □

While sCHA can prove every HA-theorem, a lot of 'nice' CHA-proofs are still excluded. For example, recall the cyclic proof of totality of the Ackermann function, depicted in fig. 9, which has been a running example of this chapter. It is not a sCHA-proof and there is no way of transforming it into one which would not fundamentally change the 'argument embodied by the proof'.

$$
\begin{prooftree}
\AxiomC{Π}
\AxiomC{$\Rightarrow \exists z.\, A(x,y,z) \star$}
\AxiomC{}
\UnaryInfC{$\Rightarrow A(0,Sy,SSy)$}
\UnaryInfC{$\Rightarrow \exists z.\, A(0,Sy,z)$}
\AxiomC{$\Rightarrow \exists z.\, A(x,y,z) \star$}
\RightLabel{$\textsc{Sub}$}
\UnaryInfC{$\Rightarrow \exists z.\, A(x,z',z)$}
\UnaryInfC{$A(Sx,y,z') \Rightarrow \exists z.\, A(Sx,Sy,z)$}
\RightLabel{$\exists\mathrm{L}$}
\UnaryInfC{$\exists z'.\, A(Sx,y,z') \Rightarrow \exists z.\, A(Sx,Sy,z)$}
\RightLabel{$\textsc{Case}_x$}
\BinaryInfC{$\exists z'.\, A(x,y,z') \Rightarrow \exists z.\, A(x,Sy,z)$}
\RightLabel{$\textsc{Cut}$}
\BinaryInfC{$\Rightarrow \exists z.\, A(x,Sy,z)$}
\RightLabel{$\textsc{Case}_y$}
\BinaryInfC{$\Rightarrow \exists z.\, A(x,y,z) \star$}
\end{prooftree}
$$

$$
\Pi : \quad
\begin{prooftree}
\AxiomC{}
\UnaryInfC{$\Rightarrow A(0,0,1)$}
\UnaryInfC{$\Rightarrow \exists z.\, A(0,0,z)$}
\AxiomC{$\Rightarrow \exists z.\, A(x,y,z) \star$}
\RightLabel{$\textsc{Sub}$}
\UnaryInfC{$\Rightarrow \exists z.\, A(x,1,z)$}
\UnaryInfC{$\Rightarrow \exists z.\, A(Sx,0,z)$}
\LeftLabel{$\textsc{Case}_x$}
\BinaryInfC{$\Rightarrow \exists z.\, A(x,0,z)$}
\end{prooftree}
$$

Figure 9: A non-simplistic CHA-proof, ★ marking cycles

In a similar vein, deriving proof theoretic results *within* sCHA will be rather cumbersome or even impossible. For example, let us return to the running example of the admissibility of the following Ind-rule in sCHA. While the theorem remains true, a somewhat more awkward construction must be made to ensure that the transformed proof remains simplistic.

Lemma 3.5.5. The following rule is admissible in sCHA for variables $x \notin \mathrm{FV}(\Gamma)$:

$$\textsc{Ind}\ \frac{\Gamma, \varphi \Rightarrow \varphi[Sx/x]}{\Gamma, \varphi[0/x] \Rightarrow \varphi[t/x]}$$

Proof. Let Π be a sCHA-proof which w.l.o.g. makes use of the Ind-rule precisely once. Replace the Ind-application with the following derivation:

$$\textsc{Cut}\ \dfrac{\begin{array}{c}\vdots\\ \varphi[0/x] \Rightarrow \iota\end{array} \qquad \dfrac{\forall\mathrm{R}, \to\mathrm{R}\ \dfrac{\Gamma, \varphi \Rightarrow \varphi[Sx/x]}{\Gamma \Rightarrow \forall x.\varphi \to \varphi[Sx/x]} \qquad \forall x.\varphi \Rightarrow \varphi[t/x]}{\iota, \Gamma \Rightarrow \varphi[t/x]}\ {\to}\mathrm{L}}{\Gamma, \varphi[0/x] \Rightarrow \varphi[t/x]}$$

where $\iota \coloneqq (\forall x.\varphi \to \varphi[Sx/x]) \to \forall x.\varphi$ and the left premise of the Cut is the usual, simplistic proof of induction.

First, the resulting CHA-preproof remains a proof: Because $x \notin \mathrm{FV}(\Gamma)$, any variable $y \in \mathrm{FV}(\Gamma, \varphi[0/x], \varphi[t/x])$ remains free along the path from the conclusion to the premise $\Gamma, \varphi \Rightarrow \varphi[Sx/x]$. Furthermore, the resulting proof remains simplistic: The subproof of $\varphi[0/x] \Rightarrow \iota$ is simplistic and a cycle running through the Ind-instance remains separated from the $\varphi[0/x] \Rightarrow \iota$ subproof, meaning no new interleaving cycles are added. ◻

Remark 3.5.1. Observe that the proof construction used in the proofs of this property for other representations (depicted below) cannot be used as it could transform a simplistic proof to a non-simplistic one: The new ⋆-cycle can be interleaved with a cycle running from the conclusion of Ind to its premise.

$$\textsc{Sub}\ \dfrac{\textsc{Case}_x\ \dfrac{\textsc{Ax}\ \dfrac{}{\Gamma, \varphi[0/x] \Rightarrow \varphi[0/x]} \qquad \textsc{Cut}\ \dfrac{\textsc{Wk}\ \dfrac{\Gamma, \varphi[0/x] \Rightarrow \varphi \ \star}{\Gamma, \varphi[0/x] \Rightarrow \varphi, \varphi[Sx/x]} \qquad \dfrac{\Gamma, \varphi \Rightarrow \varphi[Sx/x]}{\Gamma, \varphi[0/x], \varphi \Rightarrow \varphi[Sx/x]}\ \textsc{Wk}}{\Gamma, \varphi[0/x] \Rightarrow \varphi[Sx/x]}}{\Gamma, \varphi[0/x] \Rightarrow \varphi \ \star}}{\Gamma, \varphi[0/x] \Rightarrow \varphi[t/x]}$$

While proof-theoretic transformations within sCHA might be cumbersome, one proof theoretic argument becomes very easy with it: translating to HA. Indeed, this might be one of the main proof theoretic advantages of simplified systems like sCHA. Both of the aforementioned works in the literature with simplified trace conditions (Beklemishev et al. 2025; Curzi and Das 2021) have translations from cyclic to inductive proof systems among their main results. To illustrate, we present the argument for sCHA below.

Proposition 3.5.6. If sCHA $\vdash \Gamma \Rightarrow \delta$ then HA $\vdash \Gamma \Rightarrow \delta$.

Proof. Let Π be a simplistic CHA-proof of $\Gamma \Rightarrow \delta$. By proposition 3.5.2, every cycle $s \in \mathrm{dom}(\beta)$ has a variable x_s which is free and progressed along $\gamma(s)$. For each node $t \in T$ write $\lambda(t) = \Gamma_t \Rightarrow \delta_t$ and define the set of $\mathcal{H}_t$ of inductive hypotheses at t by

$$\mathcal{H}_t := \begin{cases} \{\forall x' < x_s.\forall\vec{u}.\ (\Gamma_s \to \delta_s)[x'/x_s]\} & \text{if } t \in \gamma(s) \text{ and } [\beta(s), t] \text{ passes} \\ & \quad \textit{no} \text{ right-hand side of } \textsc{Case}_{x_s} \\ \{\forall x' \leq x_s.\forall\vec{u}.\ (\Gamma_s \to \delta_s)[x'/x_s]\} & \text{if } t \in \gamma(s) \text{ and } [\beta(s), t] \text{ passes} \\ & \quad \textit{at least one } \textsc{Case}_{x_s}\text{-rhs} \\ \emptyset & \text{otherwise} \end{cases}$$

where $\vec{u} = \mathrm{FV}(\Gamma_s, \delta_s) \setminus \{x_s\}$. Observe that $\mathcal{H}_t$ is well-defined as $t \in \gamma(s)$ for at most one $s \in \mathrm{dom}(\beta)$ because Π is simplistic.

Now we prove that HA $\vdash \mathcal{H}_t, \Gamma_t \Rightarrow \delta_t$ for every node $t \in T$ by induction along the tree structure of T.

$t \in \mathrm{Leaf}(T) \setminus \mathrm{dom}(\beta)$: Then $\mathcal{H}_t = \emptyset$ and $\Gamma_t \Rightarrow \delta_t$ is an axiom of HA, meaning HA $\vdash \mathcal{H}_t, \Gamma_t \Rightarrow \delta_t$.

$t \in \mathrm{dom}(\beta)$: By proposition 3.5.2, we know that a $\textsc{Case}_{x_t}$-rhs is passed along $[\beta(t), t] = \gamma(t)$ and thus that

$$\mathcal{H}_t = \{\forall x' \leq x_t.\forall\vec{u}.\ (\Gamma_t \to \delta_t)[x'/x_t]\} \text{ where } \vec{u} = \mathrm{FV}(\Gamma_t, \delta_t) \setminus \{x_t\}.$$

Then HA $\vdash \mathcal{H}_t, \Gamma_t \Rightarrow \delta_t$ by instantiating the quantifiers as $x_t := x_t; \vec{u} := \vec{u}$ and simple propositional reasoning.

$t \in \mathrm{Inner}(t)$: Let $\mathrm{Chld}(t) = \{t_1, \ldots, t_n\}$, meaning the sequent at t was derived from those at t_i via some rule $R \in$ HA. Let us, for now, assume that $R \neq \textsc{Case}$. Per IH, we know that HA $\vdash \mathcal{H}_{t_i}, \Gamma_{t_i} \Rightarrow \delta_{t_i}$ for all $i \leq n$. First, we prove HA $\vdash \mathcal{H}_t, \Gamma_{t_i} \Rightarrow \delta_{t_i}$ for each premise. There are two cases to consider:

$\mathcal{H}_{t_i} = \mathcal{H}_t$: Trivial.

$\mathcal{H}_{t_i} \neq \mathcal{H}_t$: Let w.l.o.g. $\mathcal{H}_{t_i} \neq \emptyset$: Because $R \neq \textsc{Case}$, $t_i \in \gamma(s)$ for some $s \in \mathrm{dom}(\beta)$ with $t \notin \gamma(s)$, which is only possible if $t_i = \beta(s)$.

Then

$$\mathcal{H}_{t_i} = \{\varphi\} \qquad \varphi := \forall x' < x_s.\psi[x'/x_s] \qquad \psi := \forall \vec{u}.\, \Gamma_s \to \delta_s$$

where $\vec{u} = \mathrm{FV}(\Gamma_s, \delta_s) \setminus \{x_s\}$. Then derive $\mathrm{HA} \vdash \Gamma_{t_i} \Rightarrow \delta_{t_i}$:

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\mathrm{IH}}{\mathcal{H}_{t_i}, \Gamma_{t_i} \Rightarrow \delta_{t_i}}}{\varphi, \Gamma_s \Rightarrow \delta_s}}{\varphi \Rightarrow \psi}\,\forall\mathrm{R}, \to\mathrm{R}}{\Rightarrow \forall x_s.\psi}\,\textsc{Ind} \qquad \forall x_s.\psi, \Gamma_s \Rightarrow \delta_s}{\dfrac{\Gamma_s \Rightarrow \delta_s}{\Gamma_{t_i} \Rightarrow \delta_{t_i}}}\,\textsc{Cut}$$

where $\forall x_s.\psi, \Gamma_s \Rightarrow \delta_s$ is a tautology of first-order logic. If $\mathcal{H}_t \neq \emptyset$ conclude $\mathrm{HA} \vdash \mathcal{H}_t, \Gamma_{t_i} \Rightarrow \delta_{t_i}$ via weakening.

Now derive

$$\dfrac{\mathcal{H}_t, \Gamma_{t_1} \Rightarrow \delta_{t_1} \qquad \ldots \qquad \mathcal{H}_t, \Gamma_{t_n} \Rightarrow \delta_{t_n}}{\mathcal{H}_t, \Gamma_t \Rightarrow \delta_t}\,R$$

noting that every rule of CHA is admissible in HA in a context-sharing manner.

It remains to treat the case of $R = \textsc{Case}$. We shall treat only the most interesting case of $t \in \gamma(s)$ and $R = \textsc{Case}_{x_s}$, meaning the situation looks as follows:

$$\dfrac{t_1 : \Gamma_t[0/x_s] \Rightarrow \delta_t[0/x_s] \qquad t_2 : \Gamma_t[Sx_s/x_s] \Rightarrow \delta_t[Sx_s/x_s]}{\Gamma_t \Rightarrow \delta_t}\,\textsc{Case}_{x_s}$$

Suppose w.l.o.g. that no $\textsc{Case}_{x_s}$-rhs is passed along $[\beta(s), t]$ and thus

$$\mathcal{H}_t = \{\forall x' < x_s.\psi\} \quad \mathcal{H}_{t_2} = \{\forall x' \leq x_s.\psi\} \quad \psi = \forall \vec{u}.\, (\Gamma_s \to \delta_s)[x'/x_s].$$

Then derive

$$\dfrac{\dfrac{\Gamma_t[0/x_s] \Rightarrow \delta_t[0/x_s]}{\mathcal{H}_t[0/x_s], \Gamma_t[0/x_s] \Rightarrow \delta_t[0/x_s]}\,\textsc{Wk} \qquad \dfrac{\dfrac{\mathrm{IH}}{\mathcal{H}_{t_2}, \Gamma_t[Sx_s/x] \Rightarrow \delta_t[Sx_s/x]}}{\mathcal{H}_t[Sx_s/x], \Gamma_t[Sx_s/x] \Rightarrow \delta_t[Sx_s/x]}\,\star}{\mathcal{H}_t, \Gamma_t \Rightarrow \delta_t}\,\textsc{Case}_{x_s}$$

where $\mathrm{HA} \vdash \Gamma_t[0/x_s] \Rightarrow \delta_t[0/x_s]$ follows from the IH for t_1 as in the case of $R \neq \textsc{Case}$. The $\star$-step uses

$$\mathcal{H}_t[Sx_s/x] = \{\forall x' < Sx_s.\psi\} \Rightarrow \{\forall x' \leq x_s.\psi\} = \mathcal{H}_{t_2}. \qquad \square$$

Remark 3.5.2. Observe that a simplistic CHA-proof can be transformed into a SHA-proof in a straightforward manner: the control stack is at most of height 1 and no unfolding is needed. The proof of proposition 3.5.6 can be understood as the 'composition' of the aforementioned transformation and the translation from SHA to HA as described in theorem 3.4.9.

In a sense, proposition 3.5.6 can be considered a 'toy version' of theorem 3.4.9: Because there are no interleaving cycles, the inductive invariants become much simpler to define and the arguments about the 'adjustments of bounds' become much less intricate.

All of this leaves one obvious question: given that cyclic proof systems which restrict the cyclic complexity of their proofs, such as sCHA, have so many desirable properties, why would anyone choose to work with 'unrestricted' cyclic proofs? There are at least three reasons (vaguely sorted from weakest to strongest):

1. The complexity of unrestricted cycles is an aspect that makes cyclic proof systems so fascinating. Certainly, the appeal of 'taming' said complexity was a driving motivation for the results in chapters 5 and 6 of this dissertation.
2. Allowing for more complex cycles might allow for smaller proofs. The example of the totality of the Ackermann function already illustrates this, but it can also be made more formal: Every translation from CHA (or CPA) to their inductive counterpart in the literature induces at least an exponential increase in proof size, while translating from the inductive to the cyclic systems induces no such increase and results in simplistic proofs. Thus, complex cycles can be removed 'at the cost of exponential size increase'. Das (2020) conjectures that this might indicate a 'feature' of complex cycles: Could it be that complex cycles allow for exponentially smaller proofs? While this question remains open, the application of cyclic proofs in automatic theorem proving somewhat relies on the belief that this is true: the 'cost' of performing expensive GTC checks on candidate proofs should be offset by the fact that these smaller proofs are found much faster during proof search (recall that every 'level' of a proof that has to be searched induces an exponential cost).
3. Lastly, while it is true that sCHA is complete for HA, this observation does not extend to every cyclic proof system. For many logics, such as the modal μ-calculus, cyclic proof systems are valued because they are cut-free complete. However, without cut, simplistic proofs, as considered in this section, do not suffice for completeness. For example, Kloibhofer (2023, section 3) gives a sequent of the alternation-free modal μ-calculus which requires multiple interleaving cycles to be proven in the absence of cut. It should be noted that it still is 'simple' akin to the proofs considered by Curzi and Das (2021) and Beklemishev et al. (2025): the whole proof

can be justified by the progress of a single trace value. We conjecture that one should also be able to find a μ-calculus sequent which requires a proof which does not even satisfy this more general notion of simplicity. However, we have not been able to construct such an example yet.

4 Abstract cyclic proofs

Warning. This chapter marks a shift in writing style, which will remain in effect for the rest of the thesis. While chapters 2 and 3 were written with the aim of introducing a broad audience to cyclic proof theory, the rest of this dissertation is instead focused on communicating new results, mainly targeting the community of cyclic proof theorists. This means the writing will be less pedagogical and instead focus on pinning down all important details, even those which are technical or intricate. While anyone who has read and understood chapters 2 and 3 should also be capable to read and understand the rest of this thesis, 'the going will get tougher'.

All subsequent chapters contain results which apply to most cyclic proof systems in the literature.[16] To be able to state and prove such results, we rely on an abstract notion of cyclic proof system. This chapter is concerned with defining and illustrating this notion, connecting it to the cyclic proof systems CHA and $\mathrm{CHA}_<$ presented in chapter 2. We begin by making the notions of tree and cyclic tree, which have been left somewhat vague so far, concrete in section 4.1. We also prove some 'combinatorial' results, mostly related to the strongly connected components of cyclic trees. In section 4.2, we define an abstract notion of derivation system which induces generic definitions of preproofs and cyclic proof systems. The aforementioned notion of cyclic proof system leaves the soundness condition rather unspecified, we give a general method of specifying the global trace condition in section 4.3.

4.1 Cyclic tree combinatorics

In this thesis, we work with trees represented by subsets of ω^*. We denote both concatenation of sequences $s, t \in \omega^*$ and concatenation of 'single letters' $i \in \omega$ to sequences $s \in \omega^*$ by st and si. We rely on the 'types' of the concatenated objects to disambiguate. Let us write $s < t$ for finite sequences $s, t \in \omega^*$ if s is a strict prefix of t. For $t \in \omega^*$ and $i, j \in \omega$ we call $tj \in \omega^*$ a *left neighbour* of ti if $j < i$. A *tree* is a non-empty set $T \subseteq \omega^*$ which is closed under taking prefixes and left

[16] At least most of those with a global trace condition, which is still the vast majority.

neighbours i.e., a $T \subseteq \omega^*$ with

$$T = \{s < t \mid t \in T\} \cup \{tj \mid ti \in T, i \in \omega, j < i\}.$$

Every $t \in T$ is a *node* of the tree T, represented by the path to it from the root. This means $\varepsilon \in T$ is the root of T. This is much easier illustrated than explained, thus see fig. 10 below.

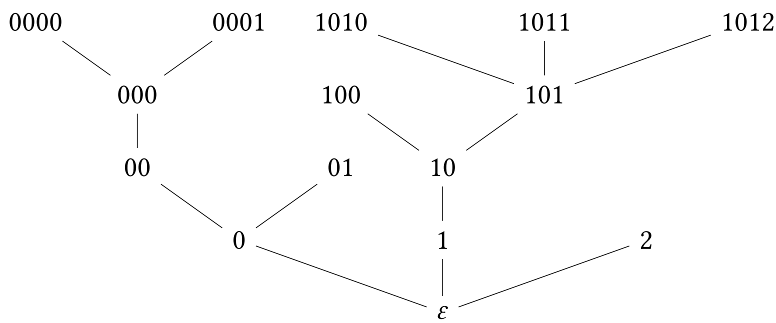


Figure 10: A tree $T \subseteq \omega^*$, each node labeled by the sequence in ω^* representing it.

The children of $t \in T$ are those nodes which can be reached by taking one more 'step' i.e., $\mathrm{Chld}(t) := \{ti \in T \mid i \in \omega\}$. A node $t \in T$ is a *leaf* of T if $\mathrm{Chld}(t) = \emptyset$ and an *inner node* otherwise. For nodes $s \leq t$ we write $[s, t] := \{u \in \omega^* \mid s \leq u \leq t\}$ for the set of nodes along the path from s to t.

As before, a *cyclic tree* is a pair (T, β) of a finite tree T and a partial function $\beta\colon \mathrm{Leaf}(T) \rightharpoonup \mathrm{Inner}(T)$ mapping some leaves of T onto inner nodes of T. Unlike before, we require every cyclic tree be in *cycle normal form* i.e., $\beta(t) < t$ for every $t \in \mathrm{dom}(\beta)$. A $t \in \mathrm{dom}(\beta)$ is called a *bud* and $\beta(t)$ its *companion*. For $t \in \mathrm{dom}(\beta)$ we call $\gamma(t) := [\beta(t), t]$ the *local cycle* of t.

Fix a cyclic tree $C = (T, \beta)$. A finite sequence $\pi \in T^+$ is a *path* through C if at every index $i < |\pi| - 2$:

(a) if $\pi_i \notin \mathrm{Leaf}(T)$ then $\pi_{i+1} \in \mathrm{Chld}(\pi_i)$ and

(b) if $\pi_i \in \mathrm{Leaf}(T)$ then $\pi_i \in \mathrm{dom}(\beta)$ and $\pi_{i+1} = \beta(\pi_i)$.

A sequence $\pi \in T^\omega$ is an *infinite branch* through C if $\pi_0 = \varepsilon$ and $\pi_0\pi_1 \ldots \pi_n$ is a path through C for every $n \in \omega$. For both paths and infinite branches through C, we write $\mathrm{Occ}(\pi) = \{\pi_i \mid i < |\pi|\}$ for the set of nodes occurring along them. For an infinite branch $\pi \in T^\omega$ we write $\mathrm{Inf}(\pi) \subseteq T$ for the occurring infinitely along π i.e., $t \in \mathrm{Inf}(\pi)$ iff there are infinitely many $i \in \omega$ with $\pi_i = t$.

A key notion of cyclic trees are their strongly connected components (scc). There are many different ways in which one can define the sccs of a cyclic tree. One option is to define them as sets of nodes such that there is a path between any two nodes in the set which does not leave the set. Thus, intuitively, an scc is a 'self contained' part of a cyclic tree. Because we only consider cyclic trees in cycle normal form, the sccs of a cyclic tree can be identified with subsets $\eta \subseteq \mathrm{dom}(\beta)$.

Definition 4.1.1 (Strongly connected component)**.** Given a cyclic tree C, a *strongly connected component* is a set $\eta \subseteq \mathrm{dom}(\beta)$ of buds of C such that

(i) there exists some *base element* $b(\eta) \in \eta$ such that $\beta(b(\eta)) \leq \beta(t)$ for every $t \in \eta$

(ii) for every $t_0 \in \eta$ there exist $t_1, \ldots, t_n \in \eta$ (where possibly $n = 0$) such that for each $i < n$, $\beta(t_i) \leq t_{i+1}$ and $t_n = b(\eta)$

Let us first connect our notion of strongly connected component with the notion of "self contained part" discussed before. For a cyclic tree $C = (T, \beta)$, a *cyclic subtree* is a set $T' \subseteq T$ with a minimal element such that if $s, t \in T'$ with $s < t$ then $[s, t] \subseteq T$ and furthermore if $s \in T' \cap \mathrm{dom}(\beta)$ then $\beta(s) \in T'$. For our notion of strongly connected component η, the aforementioned "self contained part" is the cyclic subtree $C[\eta] = \{s \in T \mid \exists t \in \eta.\ \beta(t) \leq s \leq t\}$.

To illustrate the definition of strongly connected component further, we prove the following proposition, recovering the previous description of scc:

Proposition 4.1.2. Let C be a cyclic tree and $\eta \subseteq \mathrm{dom}(\beta)$ a strongly connected component. Let $s, s' \in C[\eta]$ then there exists a path $s\pi s'$ through C such that $\mathrm{Occ}(\pi) \subseteq C[\eta]$.

Proof. First we prove that $C[\eta] = \{[\beta(b(\eta)), t] \mid t \in \eta\}$. For the $\subseteq$-direction, let $s \in \gamma(t)$ for $t \in \eta$ then $\beta(b(\eta)) \leq \beta(t) \leq s \leq t$ and thus $s \in [\beta(b(\eta)), t]$. For the $\supseteq$-direction, let $s \in [\beta(b(\eta)), t_0]$ for some $t_0 \in \eta$ but $s \notin \gamma(t)$ for any $t \in \eta$. By (ii) we know there is a sequence $t_0 t_1 \ldots t_n \in \eta^+$ with $t_n = b(\eta)$ such that $\beta(t_i) \leq t_{i+1}$ for all $i < n$. We prove that $s < \beta(t_i)$ for all $i \leq n$ per induction on i. For $i = 0$, as $s, \beta(t_0) < t_0$, they must both be prefixes of t_0 and thus comparable by $<$. As $\beta(t_0) \leq s$ would imply $s \in \gamma(t_0)$, this must mean $s < \beta(t_0)$. Now suppose $s < \beta(t_i)$. It follows that $s < \beta(t_i) \leq t_{i+1}$ and $\beta(t_{i+1}) < t_{i+1}$. Analogously to the base case, this must mean $s < \beta(t_i + 1)$. But this means that $s < \beta(t_n) = \beta(b(\eta))$, contradicting the assumed $\beta(b(\eta)) \leq s$. Thus, $s \in C[\eta]$ in the first place.

Returning to constructing the path from s to s' through $C[\eta]$, observe that $s' \in \gamma(t')$ for some $t' \in \eta$. Then by (i), $\beta(b(\eta)) \leq \beta(t') \leq s' \leq t'$, meaning the path $b(\eta)\beta(b(\eta))\ldots s'$ lies within $C[\eta]$. It thus suffices to find a path from s to $b(\eta)$. Again, $s \in \gamma(t_0)$ for some $t_0 \in \eta$. By (ii) there are is a sequence $t_0 \ldots t_n \in \eta^+$ such that $\beta(t_i) \leq t_{i+1}$. Then $s \ldots t_0\beta(t_0)\ldots t_1\beta(t_1)\ldots t_n$ is the desired path from s to $t_n = b(\eta)$. It is easily observed that this path remains within $C[\eta]$. □

Another way of understanding the strongly connected components η of C is as the cyclic subtrees of C which can be traversed exactly by cyclic paths i.e., by paths which start and finish at the same node.

Lemma 4.1.3. Let $C = (T, \beta)$ and $s \in T$ with $\pi = s\pi's$ a path through C. Then $\eta := \mathrm{Occ}(\pi) \cap \mathrm{dom}(\beta)$ is a strongly connected component with $C[\eta] = \mathrm{Occ}(\pi)$.

Proof. First, observe that if $s' \in \mathrm{Occ}(\pi) \setminus \{s\}$ then $\pi = s\pi_0 s'\pi_1 s$ for some sequences π_0, π_1. Then $s'\pi 1 s\pi_0 s'$ is also a path through C. We may thus arbitrarily choose the starting point of π from among $\mathrm{Occ}(\pi)$.

We begin by checking that the two conditions required of η to be a strongly connected component:

(i) *there is a base element:* We know that π is of the form $\alpha_1 t_1 \alpha_2 t_2 \ldots \alpha_n t_n \alpha_{n+1}$ where all $t_i \in \eta$ and $\mathrm{Occ}(\alpha_i) \cap \mathrm{dom}(\beta) = \emptyset$. We prove that $\{\beta(t_i) \mid i \leq n\}$ has a minimal element (even if π is no cycle). For $n = 1$ this is trivial. For the inductive step, let there be a path $t_n\alpha_{n+1}t_{n+1}$. Because $t_n \in \eta$ we know that $\alpha_{n+1} = \beta(t_n)\alpha'_{n+1}$ and as $\mathrm{Occ}(\alpha'_{n+1}) \cap \mathrm{dom}(\beta) = \emptyset$ it follows that $\beta(t_n) < t_{n+1}$. Furthermore, $\beta(t_{n+1}) < t_{n+1}$ per definition of cyclic trees, and thus $\beta(t_n) \leq \beta(t_{n+1})$ or $\beta(t_{n+1}) \leq \beta(t_n)$ as both are prefixes of t_{n+1}. We distinguish the two cases:

$\beta(t_n) \leq \beta(t_{n+1})$: Per IH, there is t_i with $i \leq n$ and $\beta(t_i) = \min\{\beta(t_j) \mid j \leq n\}$. Then t_i remains the base element as $\beta(t_i) \leq \beta(t_n) \leq \beta(t_{n+1})$.

$\beta(t_{n+1}) \leq \beta(t_n)$: Per IH there is a t_i with $i \leq n$ and $\beta(t_i) = \min\{\beta(t_j) \mid j \leq n\}$. Because $\beta(t_i) \leq \beta(t_n)$ we have $\beta(t_{n+1}) \leq \beta(t_i)$ or $\beta(t_i) \leq \beta(t_{n+1})$ as both are prefix of $\beta(t_n)$. In the former case, $\beta(t_i)$ remains the minimum, in the latter $\beta(t_{n+1})$ becomes the new joint minimum.

(ii) *t-sequence to base:* Let w.l.o.g. $\pi = b(\eta)\pi'b(\eta)$. For any $t \in \eta$ with $t \neq b(\eta)$ we know that $\pi' = \pi_0 t\alpha_1 t_1 \ldots t_n\alpha_{n+1}b(\eta)$ with $t_i \in \eta$ and

$\mathrm{Occ}(\alpha_i) \cap \mathrm{dom}(\beta) = \emptyset$. It is easily observed that $\beta(t) < t_1$ and $\beta(t_i) < t_{i+1}$ as well as $\beta(t_n) < b(\eta)$. Thus, $tt_1 \ldots t_n b(\eta)$ is the desired sequence.

We now prove $\mathrm{Occ}(\pi) = C[\eta]$:

$\subseteq$: Let $s \in \mathrm{Occ}(\pi)$ and suppose there was no $t \in \eta$ with $s \in \gamma(t)$. Let w.l.o.g. $\pi = s\pi' s$ then we can show per induction on $|\pi^*|$ with $\pi^* := \pi' s$ that $\pi^*_i > s$ for ever $i < |\pi^*|$: For the base, observe that $\pi^*_0 > s$ as, if $\pi^*_0 = \beta(s)$ then $s \in \gamma(s)$. For the inductive step, let $\pi^*_i > s$ then either $\pi^*_{i+1} > \pi^*_i > s$ or $\pi^*_{i+1} = \beta(\pi^*_i)$. In the latter case, we must have $\pi^*_{i+1} \leq s$ or $s < \pi^*_{i+1}$ as both are prefixes of π^*_i. As in the former case, $s \in \gamma(\pi^*_i)$, the latter must be the case.

But, as $s \in \mathrm{Occ}(\pi^*)$, this means $s < s$, a contradiction.

$\supseteq$: We begin by proving a lemma: *Let $s\alpha$ be a finite path, $s < t$ and $t \notin \mathrm{Occ}(\alpha)$ then $\mathrm{Occ}(\alpha) \cap \mathrm{Up}(t) = \emptyset$:* Proof per induction on $|\alpha|$. If $|\alpha| = 1$ then $s <_+ \alpha_0$ as $s \notin \mathrm{dom}(\beta)$ because $s < t$. In such a situation, $\alpha_0 \in \mathrm{Up}(t)$ is only possible if $\alpha_0 = t$, which contradicts the assumption. If $|\alpha| = n + 1$ suppose $\alpha_n \in \mathrm{Up}(t)$. By the same argument as for $|\alpha| = 1$, this means $\alpha_{n-1} \not<_+ \alpha_n$. Thus $\alpha_n = \beta(\alpha_{n-1})$ and $t \leq \alpha_n \leq \alpha_{n-1}$, contradicting the induction hypothesis.

Returning to $C[\eta] \subseteq \mathrm{Occ}(\pi)$, let $s \in \gamma(t)$ for some $t \in \eta$. Let w.l.o.g. $\pi = t\pi' t$. Observing that $\pi' = \beta(t)\pi''$ necessarily, there must be $s \in \mathrm{Occ}(\pi'' t)$ as by the previous result, t cannot be reached from $\beta(t)$ otherwise. □

Strongly connected components are important in cyclic proof theory because they characterise the infinite branches of a cyclic tree: the nodes visited infinitely by an infinite branch always form a strongly connected component.

Lemma 4.1.4. Let $\pi \in T^\omega$ be an infinite path through a cyclic tree $C = (T, \beta)$ in cyclic normal form. Then there exists a strongly connected component η of C such that $\mathrm{Inf}(\pi) = C[\eta]$.

Proof. Pick $s \in \mathrm{Inf}(\pi)$, there must be a segment $s\alpha s$ of π such that $\mathrm{Occ}(s\alpha s) = \mathrm{Inf}(\pi)$. By lemma 4.1.3 that means there is a strongly connected $\eta \subseteq \mathrm{dom}(\beta)$ with $C[\eta] = \mathrm{Occ}(s\alpha s) = \mathrm{Inf}(\pi)$. □

Lastly, let us prove a result we have used in section 3.2. A strongly connected η is said to be *B-maximal* for $B \subseteq \mathrm{dom}(\beta)$ if $\eta \subseteq B$ and there is no $\eta \subsetneq \eta' \subseteq B$ which is strongly connected. A key property of B-maximal strongly connected component is that they are disjoint iff they are distinct.

Lemma 4.1.5. For a cyclic tree $C = (T, \beta)$ let $B \subseteq \mathrm{dom}(\beta)$ and let $\eta, \eta' \subseteq B$ be B-maximal strongly connected components. Then either $\eta = \eta'$ or $\eta \cap \eta' = \emptyset$.

Proof. Suppose there was $s \in \eta \cap \eta'$. Then, w.l.o.g. there exists paths π and π' starting and ending at s such that π and π' visit precisely the nodes of $C[\eta]$ and $C[\eta']$, respectively. 'Pasting together' π and π' yields a path from s to s visiting precisely the elements of $C[\eta] \cup C[\eta'] = C[\eta \cup \eta']$, meaning $\eta \cup \eta'$ is strongly connected. By the B-maximality of η and η' it then follows that $\eta = \eta \cup \eta' = \eta'$. □

4.2 Abstract cyclic proof systems

This section defines general notions of derivation system, preproof and cyclic proof system. A derivation system simply is a collection of rules deriving sequents from sequents.

Definition 4.2.1 (Derivation system)**.** A *derivation system* is a triple $(\textsc{Seq}, \mathcal{R}, \rho)$ consisting of a set of sequents $\textsc{Seq}$ and a set $\mathcal{R}$ of derivation rules and a rule-interpretation $\rho\colon \mathcal{R} \longrightarrow \textsc{Seq}^+$ such that for each $r \in \mathcal{R}$, $\rho(r) = (\Gamma, \Delta_1, \ldots, \Delta_n) \in \textsc{Seq}^n$ for $n \in \omega$. The sequent Γ is the *conclusion* of r and the Δ_i its *premises*. Henceforth, we refer to a derivation system $(\textsc{Seq}, \mathcal{R}, \rho)$ simply by $\mathcal{R}$.

The derivation systems of HA, CHA and $\mathrm{CHA}_<$ presented in chapter 2 easily fit these definitions: For each, $\textsc{Seq}$ is the set of sequents of first-order arithmetic (in the case of $\mathrm{CHA}_<$ including $s < t$ atoms). Their respective rule sets $\mathcal{R}$ simply are all instances of the rules specified in their definitions. In our formalism, axioms are represented by rules without premises.

As we ultimately want to use derivation systems to define a general notion of cyclic proof, allowing different derivation rules $r, r' \in \mathcal{R}$ with the same interpretation i.e., $\rho(r) = \rho(r')$, is useful when considering cyclic proof systems in which there can sometimes be two rule-instances with identical premises and conclusion but which have different 'effects' on traces through them.

As one might expect, a derivation in a derivation system is a tree, labelled by derivation rules in such a way that their sequents 'match up'.

Definition 4.2.2 ($\mathcal{R}$-derivation)**.** Let $\mathcal{R}$ be a derivation system. An *$\mathcal{R}$-derivation* is a pair $\Pi = (T, \delta)$ consisting of a finite tree T together with a function $\delta\colon T \longrightarrow \mathcal{R}$ labelling each node with the rule applied to derive it. Write $\lambda(t) := \pi_1(\rho(\delta(t)))$ i.e., the sequent labelling the conclusion of the rule applied at t. For Π to be a derivation, we require that for each $t \in T$ if $\rho(\delta(t)) = (\Gamma, \Delta_1, \ldots, \Delta_n)$ then $\mathrm{Chld}(t) = \{t0, \ldots, t(n-1)\}$ and $\lambda(ti) = \Delta_{i+1}$. We call $\lambda(\varepsilon)$ the *endsequent* of Π. Denote by $\mathrm{D}(\mathcal{R})$ the set of $\mathcal{R}$-derivations.

Similarly, a preproof is a cyclic tree labelled by derivation rules and sequents. We also consider preproofs with assumptions i.e., with open leaves. These are only needed for the notion of cyclic proof morphism we define later. Generally, the preproofs we consider in this dissertation do not feature assumptions.

Definition 4.2.3 ($\mathcal{R}$-preproof)**.** An *$\mathcal{R}$-preproof with assumptions* is a triple $\Pi = (C, \lambda, \delta)$ consisting of a cyclic tree $C = (T, \beta)$ together with a labelling $\lambda \colon T \longrightarrow \textsc{Seq}$ such that for every $t \in \mathrm{dom}(\beta)$ one has $\lambda(t) = \lambda(\beta(t))$ and a partial function $\delta \colon (T \setminus \mathrm{dom}(\beta)) \rightharpoonup \mathcal{R}$ such that for each $t \in T \setminus \mathrm{dom}(\beta)$

- either $t \in \mathrm{dom}(\delta)$ with $\rho(\delta(t)) = (\Gamma, \Delta_1, \ldots, \Delta_n)$ and $\lambda(t) = \Gamma$ and furthermore $\mathrm{Chld}(t) = \{t0, \ldots, t(n-1)\}$ and $\lambda(ti) = \Delta_{i+1}$
- or $t \in \mathrm{Leaf}(T)$.

The sequent $\lambda(\varepsilon)$ is called the *endsequent* of Π. Each leaf $o \in \mathrm{Leaf}(T) \setminus \mathrm{dom}(\delta)$ is called *open* and its associated sequent $\lambda(o)$ is a *assumption* of Π. We denote by $\textsc{Pp}_a(\mathcal{R})$ the set of $\mathcal{R}$-preproofs with assumptions and by $\textsc{Pp}(\mathcal{R}) \subseteq \textsc{Pp}_a(\mathcal{R})$ the set of $\mathcal{R}$-preproofs without open leaves. If we speak of an $\mathcal{R}$-preproof Π without mentioning assumptions, we mean $\Pi \in \textsc{Pp}(\mathcal{R})$.

We can now define a general notion of cyclic proof system. These simply are derivation systems together with a soundness condition on preproofs. The nature of the soundness condition is left intentionally vague: as demonstrated in chapter 3, there are many different soundness conditions for cyclic proof systems and our definition is intended to 'leave room' them.

Definition 4.2.4 (Cyclic proof system)**.** A *cyclic proof system* is given by a tuple $(\textsc{Seq}, \mathcal{R}, \rho, \textsc{Pfs})$ consisting of a derivation system $(\textsc{Seq}, \mathcal{R}, \rho)$ and a set $\textsc{Pfs} \subseteq \textsc{Pp}(\mathcal{R})$ of $\mathcal{R}$-preproofs without assumptions called *$\mathcal{R}$-proofs*. Any $\Pi \in \textsc{Pfs}$ with endsequent Γ is called *an $\mathcal{R}$-proof of* Γ. Such a preproof is said to satisfy the *soundness condition* of $\mathcal{R}$. We extend the naming convention for derivation systems to cyclic derivation systems, referring to $(\textsc{Seq}, \mathcal{R}, \rho, \textsc{Pfs})$ by $\mathcal{R}$.

Observe that the specification of CHA we give in section 2.1 fits the definitions above. In that case, $\textsc{Pfs}$ contains precisely those CHA preproofs which satisfy the global trace condition.

With a general notion of preproof, we can make precise the notion of unfolding which has so far been left informal. The definition below is still rather intricate but it captures the following intuition: A $\mathcal{R}$-preproof Π' is an unfolding of an $\mathcal{R}$-preproof Π if Π' can be obtained from Π by unfolding cycles a finite number of times. That is, picking a companion and replacing it by the subtree above its companion.

Definition 4.2.5 (Unfolding preproofs)**.** Let $\mathcal{R}$ be a derivation system and $\Pi = (C, \lambda, \delta)$ and $\Pi' = (C', \lambda', \delta')$ be $\mathcal{R}$-preproofs. Then Π' is an *unfolding* of Π if there exists a function $u : T' \to T$ with the following properties:

(i) $u(\varepsilon) = \varepsilon$

(ii) If $s \in T'$ and $u(s)i \in T \setminus \mathrm{dom}(\beta)$ then $si \in T'$ and $u(si) = u(s)i$.

(iii) If $s \in T'$ and $u(s)i \in \mathrm{dom}(\beta)$ then $si \in T'$ and $u(si) = \beta(u(s)i)$.

(iv) If $s \in \mathrm{dom}(\beta')$ then $u(s) \in \mathrm{dom}(\beta)$ and $u(\beta'(s)) = \beta(u(s))$.

(v) For every $s \in T'$ we have $\lambda(u(s)) = \lambda'(s)$.

(vi) If $s \in T' \setminus \mathrm{dom}(\beta')$ then $u(s) \in T \setminus \mathrm{dom}(\beta)$ and $\delta'(s) = \delta(u(s))$.

As noted before, every $\mathcal{R}$-preproof can be considered a finitary representation of the ill-founded, infinitary derivation obtained by repeatedly unfolding it (its 'unfolding limit', so to speak). For two $\mathcal{R}$-preproofs Π, Π', if Π' is an unfolding of Π then they represent the same infinitary derivation. In that sense, their proof theoretic content is the same. This observation is important in the setting of reset proof and stack controlled proofs, which we present in chapters 5 and 6, respectively. Both are proof annotation constructions in which, at the level of pure $\mathcal{R}$-preproofs (without the annotations), are simply finite unfoldings. The preceding argument means that the resulting annotated proofs do not change the proof theoretic content of the original proofs, thereby 'still' representing the same proofs. This is important when applying these methods to cyclic proof systems in which various forms of proof theoretic content are highly valued, such as μMALL (Nollet et al. 2018) or Gödel's T (Das 2021).

We close this section by defining preproof and proof morphisms. They are a formalism to describe translations between cyclic proof systems and play a key role in the arguments in chapter 5. A preproof morphism $f : \mathcal{R} \to \mathcal{R}'$ witnesses that the derivation rules of $\mathcal{R}$ are admissible in $\mathcal{R}'$.

Definition 4.2.6. Let $(\textsc{Seq}, \mathcal{R}, \rho, \textsc{Pfs})$ and $(\textsc{Seq}', \mathcal{R}', \rho', \textsc{Pfs}')$ be cyclic proof systems. A *preproof morphism* $f : \mathcal{R} \to \mathcal{R}'$ consists of a function $f_0 : \textsc{Seq} \to \textsc{Seq}'$ mapping $\mathcal{R}$-sequents to $\mathcal{R}'$-sequents and a function $f_1 : \mathcal{R} \to \textsc{Pp}_a(\mathcal{R}')$ assigning to each $\mathcal{R}$-rule an $\mathcal{R}'$-preproof with assumptions. Furthermore, these two functions must agree: For $R \in \mathcal{R}$ with $\rho(R) = (\Gamma, \Delta_1, \ldots, \Delta_n)$, the preproof $f_1(R)$

must have $f_0(\Gamma)$ as its endsequent and $f_0(\Delta_1), \dots, f_0(\Delta_n)$ as its only assumptions. Henceforth, we denote both $f_0\colon \textsc{Seq} \to \textsc{Seq}'$ and $f_1\colon \mathcal{R} \to \textsc{Pp}(\mathcal{R}')$ by f.

This gives rise to a method for translating $\mathcal{R}$-preproofs into $\mathcal{R}'$-preproofs: Simply replace each application of a derivation rule in the $\mathcal{R}$-proof by the $\mathcal{R}'$-preproof witnessing its admissibility to obtain a $\mathcal{R}'$-preproof. It is easiest to formally describe this in terms of preproof composition. Thus, suppose $\mathcal{R}$ is a cyclic derivation system and $\Pi = ((C, \beta), \lambda, \delta)$ is an $\mathcal{R}$-preproof with open leaves $o_1, \dots, o_n$. Furthermore, suppose there are $\mathcal{R}$-preproofs $\Pi_1 = ((T_1, \beta_1), \lambda_1, \delta_1), \dots, \Pi_n = ((T_n, \beta_n), \lambda_n, \delta_n)$ such that the endsequent of Π_i is $\lambda(o_i)$ and its assumptions are $\Xi^i_1, \dots, \Xi^i_{m_i}$. Then one may compose this material into a preproof $\Pi[\Pi_1, \dots, \Pi_n] = ((T_c, \beta_c), \lambda_c, \delta_c)$ with endsequent Γ and assumptions $\Xi^1_1, \dots \Xi^1_{m_1}, \Xi^2_1, \dots, \Xi^2_{m_2}, \dots, \Xi^n_1, \dots, \Xi^n_{m_n}$ as follows:

$$T_c := T \cup \bigcup_{i=1}^{n} \{l_i t \mid t \in T_i\}$$

$$\lambda_c(t) := \begin{cases} \lambda(t) & t \in T \\ \lambda_i(s) & t = o_i s \text{ for } s \in T_i \end{cases}$$

$$\delta_c(t) := \begin{cases} \delta(t) & t \in \mathrm{dom}(\delta) \\ \delta_i(s) & t = o_i s \text{ for } s \in T_i \text{ and } s \in \mathrm{dom}(\delta_i) \end{cases}$$

$$\beta_c(t) := \begin{cases} \beta(t) & t \in \mathrm{dom}(\beta) \\ o_i\beta_i(s) & t = o_i s \text{ for } s \in T_i \text{ and } s \in \mathrm{dom}(\beta_i) \end{cases}$$

Suppose there was a preproof morphism $f\colon \mathcal{R} \to \mathcal{R}'$ and a $\mathcal{R}$-preproof $\Pi = ((T, \beta), \lambda, \delta)$ with endsequent Γ and assumptions $\Delta_1, \dots, \Delta_n$. This induces an $\mathcal{R}'$-preproof $f(\Pi)$ with assumptions $f(\Delta_1), \dots, f(\Delta_n)$ and endsequent $f(\Gamma)$. It is defined recursively on T as by associating to each node $t \in T$ a preproof Π_t of $\lambda(t)$. Start setting for each $t \in \mathrm{Leaf}(T) \setminus \mathrm{dom}(\delta)$ the preproof $\Pi_t := (\{\varepsilon\}, \varepsilon \mapsto \lambda(t), \emptyset)$ i.e. the preproof deriving $\lambda(t)$ as an open leaf. Now for each $t \in \mathrm{dom}(\delta)$ such that all $\{t_1, \dots, t_n\} = \mathrm{Chld}(t)$ have associated preproofs Π_{t_i}, define $\Pi'_t := \Pi_{f(\delta(t))}[\Pi_{t_1}, \dots, \Pi_{t_n}]$ (where $\Pi_{\delta(t)}$ is given by the morphism $f\colon \mathcal{R} \to \mathcal{R}'$). If $t \notin \mathrm{im}(\beta)$ then $\Pi_t = \Pi'_t$. Otherwise, Π_t is obtained from Π'_t by adding β-cycles from each open leaf of Π'_t corresponding to a leaf in $\beta^{-1}(t)$ to ε. Then $f(\Pi) := \Pi_\varepsilon$.

If the aforementioned method translates all $\mathcal{R}$-proofs into $\mathcal{R}'$-proofs, $f\colon \mathcal{R} \to \mathcal{R}'$ is considered a proof morphism. Note for the definition below that the method translates $\mathcal{R}$-preproofs without assumptions into $\mathcal{R}'$-preproofs without assumptions.

Definition 4.2.7. Let $(\textsc{Seq}, \mathcal{R}, \rho, \textsc{Pfs})$ and $(\textsc{Seq}', \mathcal{R}', \rho', \textsc{Pfs}')$ be cyclic proof systems. A preproof morphism $f : \mathcal{R} \to \mathcal{R}'$ is a *proof morphism* if it preserves the soundness condition of $\mathcal{R}$. That is, if for every $\Pi \in \textsc{Pfs}$ one has $f(\Pi) \in \textsc{Pfs}'$.

4.3 Trace maps and activation algebras★

The previous section's notion of cyclic proof system left the soundness condition vague. In this section we describe a general way of specifying the global trace condition, the most common kind of soundness condition in the literature. Our presentation here is based on Afshari and Wehr (2024) but similar notions have been presented elsewhere in the literature, for example by Brotherston (2006) and Cohen, Jabarin, et al. (2024). The goal of such formalisms is to give an account of 'traces along branches' and 'progress of traces' which abstracts away all logic-specific details. This allows one to study most cyclic proof systems from the literature at once.

The global trace condition is usually formulated in terms of a notion of 'trace', or rather some syntactic feature which is 'traced' along a branch, and a notion of 'progress'. *Trace maps* are relations which describe the individual 'steps' traces may take going from the conclusion to the premise of a rule. The notion of 'progress' is modelled by *activation algebras*. An activation algebra is a tuple $\mathcal{A} = (A, \leq, \vee, 0, \alpha)$ consisting of a finite semilattice $(A, \leq, \vee, 0)$ and an *activation element* $\alpha \in A$ where $0 \neq \alpha$. We often write $\mathcal{A}$ to refer to the carrier set A.

Definition 4.3.1 (Trace map)**.** Let $\mathcal{A}$ be an activation algebra. A trace map over $\mathcal{A}$ between finite sets X and Y is a relation $\tau \subseteq X \times \mathcal{A} \times Y$. We also write $\tau : X \to Y$.

Given two trace maps $\tau\colon X \to Y, \tau'\colon Y \to Z$ their composition is given by

$$\tau' \circ \tau := \{(x, a \vee b, z) \mid (x, a, y) \in \tau, (y, b, z) \in \tau'\}.$$

As the astute reader may already suspect, the trace maps over a fixed $\mathcal{A}$ indeed form a category! We do not go into further detail about the category theoretical aspects of trace maps, as they are not really relevant to the rest of the thesis. Instead, we direct the interested reader to (Afshari and Wehr 2024).

★ The contents of this section are based on Wehr's Master's thesis (Wehr 2021) under Bahareh Afshari, whose results have been published in (Afshari and Wehr 2024).

Example 4.3.2. To illustrate the use of trace maps, recall the $\textsc{Case}_x$-rule from CHA:

$$\textsc{Case}_x \frac{\Gamma[0/x] \Rightarrow \delta[0/x] \quad \Gamma[Sx/x] \Rightarrow \delta[Sx/x]}{\Gamma \Rightarrow \delta}$$

The possible trace steps from its conclusion to its premises can be modelled with two trace maps, one per premise. More concretely, we use trace maps over the Booleans $\mathbb{B}$. The Booleans $\mathbb{B} = \{0, 1\}$ form an activation algebra with $\vee$ and 0 defined as usual and $\alpha := 1$ (observe that this choice is forced by $\alpha \neq 0$).

To begin, we must associate a finite set of 'trace values' to each sequent of $\textsc{Case}_x$-rule, which contains the syntactic features that CHA-traces follow. In CHA, these are the free variables in each sequent i.e., we will define the two trace maps

$$\tau_L : \mathrm{FV}(\Gamma, \delta) \longrightarrow \mathrm{FV}(\Gamma[0/x], \delta[0/x]), \; \tau_R : \mathrm{FV}(\Gamma, \delta) \longrightarrow \mathrm{FV}(\Gamma[Sx/x], \delta[Sx/x])$$

from the conclusion to the left and right premise, respectively.

A trace map $\tau : X \longrightarrow Y$ captures two kinds of information:

(i) which trace values $x \in X$ of the conclusion 'may be followed' to which trace values $y \in Y$ of the premise, represented by the existence of triples (x, a, y) regardless of the value of $a \in \mathcal{A}$ and

(ii) how the trace 'progresses' from such connected x and y, represented by the value of $a \in \mathcal{A}$.

In $\mathbb{B}$ (and CHA), moving from the conclusion to the premise of a rule, a trace value can either progress (represented by $a = 1$) or not (represented by $a = 0$).

In CHA there is exactly one way for a trace to progress: passing through the right-hand side of $\textsc{Case}_x$ while following the variable x. Because traces must always follow the same variable, τ_R is easily defined as

$$\tau_R := \{(y, a_y, y) \mid y \in \mathrm{FV}(\Gamma, \delta)\} \quad a_y := \begin{cases} 1 & \text{if } y = x \\ 0 & \text{otherwise} \end{cases}$$

noting that $\mathrm{FV}(\Gamma, \delta) = \mathrm{FV}(\Gamma[Sx/x], \delta[Sx/x])$. The definition of τ_L is even simpler, noting that traces can never progress along the left-hand side of $\textsc{Case}_x$ (or any other rule, for that matter):

$$\tau_L := \{(y, 0, y) \mid y \in \mathrm{FV}(\Gamma, \delta) \cap \mathrm{FV}(\Gamma[0/x], \delta[0/x])\}.$$

As the previous example illustrates, a key use of trace maps is assigning them to the conclusion-premise pairs of each derivation rule in a derivation system. We continue by making this intuition formal by means of trace interpretations.

Definition 4.3.3 (Trace interpretation)**.** Let $(\textsc{Seq}, \mathcal{R}, \rho)$ be a derivation system and $\mathcal{A}$ an activation algebra. A *trace interpretation of* $\mathcal{R}$ *over* $\mathcal{A}$ consists of a map ι with $\mathrm{dom}(\iota) = \textsc{Seq}$ such that $\iota(\Gamma)$ is a finite set for every $\Gamma \in \textsc{Seq}$. Further, the trace interpretation picks, for each rule $r \in \mathcal{R}$ with $\rho(r) = (\Gamma, \Delta_1, \ldots, \Delta_n)$, trace maps $r_i \colon \iota(\Gamma) \longrightarrow \iota(\Delta_i)$ over $\mathcal{A}$ for each $1 \leq i \leq n$. We write $\iota : \mathcal{R} \longrightarrow \mathcal{A}$ for a trace interpretation of $\mathcal{R}$ over $\mathcal{A}$.

Example 4.3.2 can now be extended to a full trace interpretation of CHA.

Example 4.3.4. The trace interpretation $\iota\colon \mathrm{CHA} \longrightarrow \mathbb{B}$ for CHA is given by $\iota(\Gamma \Rightarrow \delta) := \mathrm{FV}(\Gamma, \delta)$. If $r \in \mathrm{CHA}$ with $\rho(r) = (\Gamma \Rightarrow \delta, \Gamma_1 \Rightarrow \delta_1, \ldots, \Gamma_n \Rightarrow \delta_n)$ the trace maps $r_i \colon \iota(\Gamma \Rightarrow \delta) \longrightarrow \iota(\Gamma_i \Rightarrow \delta_i)$ are defined by

$$r_i := \{(x, a(r, i, x), x) \mid x \in \mathrm{FV}(\Gamma, \delta) \cap \mathrm{FV}(\Gamma_i, \delta_i)\}$$

where

$$a(r, i, x) := \begin{cases} 1, & \text{if } r \text{ instance of } \textsc{Case}_x \text{ and } i = 2 \\ & \text{(i.e., } \Gamma_i \Rightarrow \delta_i \text{ is the right-hand premise)} \\ 0, & \text{otherwise.} \end{cases}$$

Crucially, a trace interpretation $\iota : \mathcal{R} \longrightarrow \mathcal{A}$ allows one to extend $\mathcal{R}$ into a cyclic proof system $\iota(\mathcal{R})$ is a natural manner.

Definition 4.3.5 ($\iota(\mathcal{R})$)**.** Let $\mathcal{A}$ be an activation algebra. A *path over* $\mathcal{A}$ is an infinite sequence $(X_i, \tau_i)_{i \in \omega}$ where all X_i are finite and all τ_i are trace maps $\tau_i : X_i \longrightarrow X_{i+1}$ over $\mathcal{A}$.

We say a path satisfies the *trace condition* if there exists a strictly increasing sequence $n : \omega \longrightarrow \omega$ and a sequence $(x_i \in X_{n_i})_{i \in \omega}$ such that $(x_i, \alpha, x_{i+1}) \in \tau_{n_{i+1}-1} \circ \ldots \circ \tau_{n_i}$ for every $i \in \omega$.

Let $(\textsc{Seq}, \mathcal{R}, \rho)$ be a derivation system with a trace interpretation $\iota\colon \mathcal{R} \longrightarrow \mathcal{A}$. Let $\Pi = (C, \lambda, \delta)$ be a preproof and π be an infinite branch through C. This induces a path $\hat{\pi} := (X_i, \tau_i)_{i \in \omega}$ with $X_i := \iota(\lambda(\pi_i))$ and

$$\tau_i \colon \iota(\lambda(\pi_i)) \longrightarrow \iota(\lambda(\pi_{i+1})) := \begin{cases} r_i : X_i \rightarrow X_{i+1} & \pi_{i+1} = \pi_i j, \delta(\pi_i) = r \\ \{(x, 0, x) \mid x \in X_i\} & \pi_i \in \mathrm{dom}(\beta) \end{cases}$$

This induces a cyclic proof system $\iota(\mathcal{R}) := (\textsc{Seq}, \mathcal{R}, \rho, \textsc{Pfs})$ with $\Pi \in \textsc{Pfs}$ iff for every infinite branch π through Π the path $\hat{\pi}$ satisfies the trace condition.

In other words: Every infinite branch through an $\mathcal{R}$-preproof induces a path over $\mathcal{A}$ via the trace interpretation $\iota : \mathcal{R} \to \mathcal{A}$. An $\mathcal{R}$-preproof is a $\iota(\mathcal{R})$-proof if every such path satisfies the trace condition.

While the definition of the trace condition in terms of compositions is rather complex, it can be restated in slightly more 'elementary' terms. The proof of this is essentially lemma 5.4 of Wehr (2021).

Proposition 4.3.6. Let $\hat{\pi} = (X_i, \tau_i)_{i\in\omega}$ be a path over $\mathcal{A}$. Then $\hat{\pi}$ satisfies the trace condition iff if there exists a strictly increasing sequence $n : \omega \to \omega$ and sequences $(x_i \in X_i)_{i\geq n(0)}$ and $(a_i \in \mathcal{A})_{i\geq n(0)}$ such that

1. $(x_i, a_i, x_{i+1}) \in \tau_i$ for all $i \geq n(0)$ and
2. $\alpha = \bigvee_{j=n(i)}^{j<n(i+1)} a_j$ for all $i \in \omega$.

For $\mathcal{A} = \mathbb{B}$, which covers all examples we consider in this dissertation, the previous result simplifies to a simpler, hopefully more familiar condition.

Proposition 4.3.7. Let $\hat{\pi} = (X_i, \tau_i)_{i\in\omega}$ be a path over $\mathbb{B}$. Then $\hat{\pi}$ satisfies the trace condition iff there exists an offset k and a sequence $(x_i \in X_{k+i})_{i\in\omega}$ such that

(i) for every $i \in \omega$ there is $a \in \mathbb{B}$ with $(x_i, a, x_{i+1}) \in \tau_{k+i}$ and
(ii) $(x_i, 1, x_{i+1}) \in \tau_{k+i}$ for infinitely many $i \in \omega$.

It is now easy to observe that the cyclic proof system $\iota(\mathrm{CHA})$ induced by the trace interpretation from example 4.3.4 is exactly the same as the cyclic proof system CHA defined in chapter 2. Of course, their underlying derivation systems are the same. By proposition 4.3.7 and the definition of $\iota : \mathrm{CHA} \to \mathbb{B}$, an infinite branch π of a $\iota(\mathrm{CHA})$-preproof satisfies the trace condition of $\iota(\mathrm{CHA})$ iff there exists a suffix of π in which some variable x occurs free in all sequents (noting that $(x, a, y) \in \tau$ for some trace map τ from $\iota : \mathrm{CHA} \to \mathbb{B}$ means $x = y$) and infinitely many right-hand sides of $\textsc{Case}_x$ are passed by said suffix. This is exactly the trace condition given for CHA in chapter 2.

The trace condition of $\mathrm{CHA}_<$ can be captured by a trace interpretation $\iota : \mathrm{CHA}_< \to \mathbb{B}$ in a manner similar to that of CHA.

Definition 4.3.8. The *trace interpretation* ι: $\mathrm{CHA}_< \to \mathbb{B}$ is given by $\iota(\Gamma \Rightarrow \delta) := \textsc{Term}(\Gamma \Rightarrow \delta)$ and for any $R \in \mathrm{CHA}_<$ with $\rho(R) = (\Gamma \Rightarrow \delta, \Gamma_1 \Rightarrow \delta_1, \ldots, \Gamma_n \Rightarrow \delta_n)$ the trace map r_i: $\textsc{Term}(\Gamma \Rightarrow \delta) \to \textsc{Term}(\Gamma_i \Rightarrow \delta_i)$ is given by the following, where $\leftarrow_r^i$ is as in definition 2.3.2:

$$r_i := \{(t, 0, t') \mid t \in \textsc{Term}(\Gamma \Rightarrow \delta), t' \in \textsc{Term}(\Gamma_i \Rightarrow \delta_i) \text{ and } t' \leftarrow_r^i t\} \cup \left\{ (t, 1, s) \;\middle|\; \begin{array}{l} t \in \textsc{Term}(\Gamma \Rightarrow \delta) \text{ and } t', s \in \textsc{Term}(\Gamma_i \Rightarrow \delta_i) \\ \text{such that } t \leftarrow_r^i t' \text{ and } s < t' \in \Gamma_i \end{array} \right\}$$

For much of the rest of this thesis, cyclic proof systems $\iota(\mathcal{R})$ will play an important role as they occupy a 'sweet spot', abstract enough to capture most cyclic proof systems with a global trace condition from the literature while also containing enough information about the system's notion of 'trace' to derive interesting results about them. To illustrate, consider the decidability result below.

Proposition 4.3.9. Fix $\mathcal{R}$ and a computable $\iota\colon \mathcal{R} \to \mathcal{A}$ where the operations of $\mathcal{A}$ are computable and its equality decidable. There exists a computable procedure which, given any $\mathcal{R}$-preproof Π, decides whether Π is a proof of $\iota(\mathcal{R})$.

Proof. There are various ways of proving this. For example by appealing to infinite word automata (see e.g. proposition 5.1.3) or to Ramsey's theorem (Afshari and Wehr 2024, theorem 5.3). □

For the sake of completeness, we mention one more activation algebra which lends itself most naturally to modelling certain trace conditions from the literature.

Example 4.3.10. Some logics, especially μ-calculi, require more complex activation algebras to model their trace conditions. An example is the *failure algebra* $\mathbb{F} := (\{0, 1, 2\}, \leq, \vee, 0, \alpha)$ where $\alpha = 1$. The value $2 > \alpha$ can be used to represent 'failure' of trace condition satisfaction. That is, traces through $\mathcal{T}_{\mathbb{F}}$ satisfy the trace condition iff they have infinitely many progress points (triples $(x, 1, y)$ in trace maps over $\mathbb{F}$) and *no* failure points (triples $(x, 2, y)$ in maps over $\mathbb{F}$). The failure algebra can be used to represent one of the common trace conditions for various μ-calculi (see e.g. definition 3.4 of Afshari and Wehr (2024)) where the failure element is used to rule out the unfolding of certain higher fixed point quantifiers.

To close, we make formal a claim about Simpson's $\mathrm{CHA}_<$ that has so far been informal. In section 2.3 we claim that the traces of $\mathrm{CHA}_<$ are 'as complicated as possible'. That means it serves as a 'maximally complicated' example for the cyclic proof system representations later in this thesis. We now make this formal: For every trace map $\tau : X \to Y$ over $\mathbb{B}$, there exists a $\mathrm{CHA}_<$-preproof Π without cycles and with precisely one open premise such that the traces along Π are 'exactly' those of τ.

For the following construction, we assume for every finite set X the existence of two injections $v^X : X \to \textsc{Var}$ and $u^X : X \to \textsc{Var}$ such that for any finite X, Y we have $\mathrm{im}(v^X) \cap \mathrm{im}(u^Y) = \emptyset$. We remark at the end of this section how to go about obtaining them. Using these injections, we define two kinds of atomic formulas which we call *name gadgets* for $x \in X$ where X finite:

$$n^X_x := (v^X_x = v^X_x) \qquad m^X_x := (u^X_x = u^X_x)$$

and write $n(X) := \{n^X_x \mid x \in X\}$ and $m(X) := \{m^X_x \mid x \in X\}$.

Lemma 4.3.11. Let $\tau : X \to Y$ be a trace map over $\mathbb{B}$. Then there exists a cycle-free $\mathrm{CHA}_<$-preproof Π_τ of $\bot, n(X) \Rightarrow \bot$ with a unique open premise $\bot, n(Y) \Rightarrow \bot$. Furthermore, the trace map induced between the root of Π_τ and its premise is exactly $\{(v_x, b, v_y) \mid (x, b, y) \in \tau\}$.

Proof. The derivation Π_τ looks as follows:

$$\dfrac{\dfrac{\dfrac{\dfrac{\bot, n(Y) \Rightarrow \bot}{\bot, m(Y) \Rightarrow \bot}\ \text{(iii)}}{\bot, n(X), m(Y) \Rightarrow \bot}\ \textsc{Wk}}{\bot, n(X), m(Y) \Rightarrow \bot}\ \text{(ii)}}{\bot, n(X) \Rightarrow \bot}\ \text{(i)}$$

with the following steps:

(i) Cut in all the $m(Y)$ i.e., if $Y = \{y_1, \dots, y_n\}$ we derive

$$\dfrac{\dfrac{}{\Rightarrow u^Y_{y_1} = u^Y_{y_1}} \qquad \dfrac{\dfrac{\dfrac{}{\Rightarrow u^Y_{y_n} = u^Y_{y_n}} \qquad \bot, n(X), m(Y) \Rightarrow \bot}{\bot, n(X), m^Y_{y_1}, \dots, m^Y_{y_{n-1}} \Rightarrow \bot}\ \textsc{Cut}}{\bot, n(X), m^Y_{y_1} \Rightarrow \bot}\ \vdots}{\bot, n(X) \Rightarrow \bot}\ \textsc{Cut}$$

(ii) Mirror all the trace-steps in τ. That is, if $\tau = \{t_1, \dots, t_n\}$ where $t_i = (x_i, b_i, y_i)$, we derive

$$\begin{array}{c} \bot, n(X), m(Y) \Rightarrow \bot \\ \vdots\ \Pi_n \\ \bot, n(X), m(Y) \Rightarrow \bot \\ \vdots \\ \hline \bot, n(X), m(Y) \Rightarrow \bot \\ \vdots\ \Pi_1 \\ \bot, n(X), m(Y) \Rightarrow \bot \end{array}$$

where there the nature of Π_i depends on b_i:

$b_i = 0$: Then we want to establish a non-progressing trace-step between $v^X_{x_i}$ and $u^Y_{y_i}$. We do this with the following construction:

$$\dfrac{\dfrac{}{\bot \Rightarrow v^X_{x_i} = u^Y_{y_i}}\,\bot\mathrm{L} \qquad \dfrac{\bot, n(X), m(Y) \Rightarrow \bot}{v^X_{x_i} = u^Y_{y_i}, \bot, n(X), m(Y) \Rightarrow \bot}\,{=}\mathrm{L}}{\bot, n(X), m(Y) \Rightarrow \bot}$$

where =L-rule does not actually make any changes to the sequent. Observe that by definition 4.3.8, the induced trace map from the root to the open premise of the derivation above nonetheless contains $(v^X_{x_i}, 0, u^Y_{y_i})$.

$b_i = 1$: Then we want to establish a non-progressing trace-step between $v^X_{x_i}$ and $u^Y_{y_i}$. We do this with the following construction:

$$\dfrac{\dfrac{}{\bot \Rightarrow u^Y_{y_i} < v^X_{x_i}}\,\bot\mathrm{L} \qquad \dfrac{\bot, n(X), m(Y) \Rightarrow \bot}{u^Y_{y_i} < v^X_{x_i}, \bot, n(X), m(Y) \Rightarrow \bot}\,\textsc{Wk}}{\bot, n(X), m(Y) \Rightarrow \bot}$$

(iii) Substitute every u^Y_y with v^Y_y i.e., if $Y = \{y_1, \dots, y_n\}$ we derive

$$\textsc{Sub}\,\dfrac{\bot, n(X), n(Y) \Rightarrow \bot}{\dfrac{\bot, n^Y_{y_1}, \dots, n^Y_{y_{n-1}}, m^Y_{y_n} \Rightarrow \bot}{\dfrac{\vdots}{\textsc{Sub}\,\dfrac{\bot, n^Y_{y_1}, m^Y_{y_2}, \dots, m^Y_{y_n} \Rightarrow \bot}{\bot, m(Y) \Rightarrow \bot}}}}$$

Towards the claim of the induced trace map, observe that the trace map τ' induced by step (ii) is such that $(v^X_x, b, u^Y_y) \in \tau'$ iff $(x, b, y) \in \tau$ because $\mathrm{im}(v^X)$ and $\mathrm{im}(u^Y)$ are disjoint. These must be the only trace steps in the trace map induced along (i – ii). The substitutions in (iii) then ensure the claim. □

Remark 4.3.1. To obtain the injection $v^X : X \longrightarrow \textsc{Var}$ and $u^X : X \longrightarrow \textsc{Var}$ as described above i.e., with $\mathrm{im}(v^X) \cap \mathrm{im}(v^Y) = \emptyset$ for all finite X, Y, first observe that there must be injections $v : \omega \longrightarrow \textsc{Var}$, $u : \omega \longrightarrow \textsc{Var}$ with $\mathrm{im}(v) \cap \mathrm{im}(u) = \emptyset$ because $\textsc{Var}$ is countably infinite. Now, for every finite X we know there must be a bijection $f_X : X \longrightarrow n$ for some $n \in \omega$. Then simply define $v^X := v \circ f_X$ and $u^X := u \circ f_X$.

Note that combining all of the, say, v^X for every finite X into $v^\bullet : \textsc{FinSet} \longrightarrow \textsc{Var}$ clearly leads to size-issues. However, as every instance of lemma 4.3.11 only makes use of finitely many finite sets $T := \{\iota(s) \mid s \in T\}$, the required $v^\bullet \restriction T : T \longrightarrow \textsc{Var}$ can always be obtained by finite choice.

5 Reset proof systems*

Reset proofs are cyclic proofs that, in place of a global trace condition, employ a mechanism of annotating sequents and a specific 'Reset' rule of inference marking 'progress'. Reset proofs were first introduced for the modal μ-calculus by Jungteerapanich (2010) and Stirling (2013). The soundness condition only considers each local cycle of the proof. This eases both proof checking, making it a polynomial time problem, as well as simplifying reasoning about proof transformations. For a more approachable introduction to reset proofs and their benefits, we refer readers to section 3.3.

At the time of writing (Leigh and Wehr 2024), which this chapter is based on, it was already understood that reset proof systems can be obtained from global-trace-based cyclic proof systems by annotating sequents with Safra automata (see e.g. Jungteerapanich (2010) and Marti and Venema (2021a)). However, all reset proof systems in the literature were designed as variations of Jungteerapanich's original reset system for the modal μ-calculus, rather than 'directly' via the Safra construction. This is likely because the modal μ-calculus has an 'unusual' trace condition.[19]

This chapter describes how to construct reset proof systems in a general manner more generally, thus illustrating reset proofs independently of the modal μ-calculus. Working in terms of the abstract notions of trace and cyclic proof system from chapter 4, we show that any cyclic proof system induced by a trace interpretation naturally induces an equivalent reset proof system.

This is not solely an abstract result but also provides a 'recipe' for deriving a corresponding reset proof system from any suitable cyclic proof system, including those unrelated to the modal μ-calculus. We demonstrate the method on the two running examples of simple CHA and Simpson's $\mathrm{CHA}_<$, obtaining in each case an equivalent reset-style proof system. Further examples can be found in (Leigh and Wehr 2024), namely cyclic Gödel's T (Das 2021) and two variants of the

* The contents of this chapter are joint work with Graham E. Leigh and have been published in Leigh and Wehr (2024).

19 Using the notions from section 4.3, it is most naturally described in terms of the activation algebra $\mathbb{F}$ rather than the more common $\mathbb{B}$. Equivalently, it is most naturally expressed as a parity condition, rather than a Büchi condition.

modal μ-calculus. The latter systems also illustrate the difference between our construction and that of Jungteerapanich and Stirling.

Outline of the arguments We show that every cyclic proof system whose soundness condition can be specified via a trace interpretation over some activation algebra $\mathcal{A}$ can be associated a reset proof system i.e., a cyclic proof system in which the soundness condition is wholly determined by local cycles.

Given a cyclic proof system $\mathcal{R}$ whose soundness condition is specified by an activation algebra $\mathcal{A}$, the reset system $\mathrm{R}(\mathcal{R})$ is obtained by annotating the sequents of $\mathcal{R}$ with *Safra boards*. Introduced in section 5.2, Safra boards are inspired by the Safra construction for determinising infinite word automata (Safra 1988), following the presentation of Kozen (2006).

Roughly, a Safra board for a given sequent Γ with trace objects $\iota(\Gamma)$ consists of 'squares' $(x, a) \in \iota(\Gamma) \times \mathcal{A}$ on which stacks of 'playing chips' are placed, tracking the progress of the trace values. Given a derivation rule with conclusion Γ and Γ' as one of its premises, there are rules describing how to move and extend the stacks on a board for Γ to obtain a board for Γ', which take into account the 'progress' made in the trace step from Γ to Γ'. Additionally, there are certain bookkeeping operations that may be performed on such Safra boards, including a *reset* operation which 'resets' some of the progress tallied on a Safra board. This machinery allows for a simpler trace condition: A preproof is a proof if along every infinite branch, infinitely many reset-steps take place, indicating that a trace value has progressed infinitely along said branch. Differing from the global trace condition, this property can be established by simply verifying that appropriate resets are part of each individual local cycle of a preproof, yielding a 'local' soundness condition.

Section 5.3 defines the reset system $\mathrm{R}(\mathcal{R})$ induced by a cyclic proof system $\mathcal{R}$ with a global trace condition specified by an activation algebra $\mathcal{A}$. There are two key properties we prove relating $\mathcal{R}$ to $\mathrm{R}(\mathcal{R})$: *soundness* and *completeness*. Soundness states that any annotated sequent provable in the reset system $\mathrm{R}(\mathcal{R})$ is provable, without annotations, in the original system $\mathcal{R}$. Conversely, completeness is the property that any sequent provable in $\mathcal{R}$ is provable in $\mathrm{R}(\mathcal{R})$. The names of these two properties are apt because they allow soundness and completeness of $\mathcal{R}$, relative to some semantics, to be 'lifted' to $\mathrm{R}(\mathcal{R})$.

Both results are established by providing suitable cyclic proof system homomorphisms, a formalism for specifying and verifying translations between cyclic proof systems. In section 5.3.1, we consider the translation $\mathit{erase}\colon \mathrm{R}(\mathcal{R}) \to \mathcal{R}$

which strips an $\mathrm{R}(\mathcal{R})$-proof of its Safra board annotations and removes the derivation steps corresponding to the various bookkeeping operations on Safra boards. By showing this to be a homomorphism, we can conclude that every $\mathrm{R}(\mathcal{R})$-proof induces a naturally corresponding $\mathcal{R}$-proof of the 'same' sequent, yielding soundness.

The proof of completeness in section 5.3.2 is less direct: For every finite subsystem $\mathcal{F}$ of $\mathcal{R}$ we define a finite subsystem $\mathrm{S}(\mathcal{F})$ of $\mathrm{R}(\mathcal{R})$ which enjoys the proof search property: For every sequent Γ provable in $\mathcal{F}$ via cyclic proof Π, there exists an annotated finite unfolding of Π in $\mathrm{S}(\mathcal{F})$ which is a proof of Γ. As $\mathrm{S}(\mathcal{F})$ can be embedded into $\mathrm{R}(\mathcal{R})$ via a homomorphism $\mathit{embed}\colon \mathrm{S}(\mathcal{F}) \to \mathrm{R}(\mathcal{R})$ this yields a proof of Γ in the reset system.

Section 5.4 applies the above results to obtain reset systems for the two running examples of this thesis: Simpson's $\mathrm{CHA}_{<}$ (section 5.4.1) and simple CHA (section 5.4.2).

While the system $\mathrm{R}(\mathcal{R})$ is sound and complete for any suitable cyclic system, it tends to not be pleasant to 'use'. This state of affairs can usually be assuaged with a few ergonomic adjustments. This is precisely the task undertaken in section 5.4: For both $\mathrm{CHA}_{<}$ and CHA, we design a bespoke reset system $\mathcal{S}$ inspired by $\mathrm{R}(\mathcal{R})$. Soundness and completeness of $\mathcal{S}$ is obtained via a pair of homomorphisms $\mathit{embed}\colon \mathcal{S} \to \mathrm{R}(\mathcal{R})$ and $\mathit{expand}\colon \mathrm{S}(\mathcal{F}) \to \mathcal{S}$. Importantly, the construction of these bespoke systems $\mathcal{S}$ and the homomorphisms *embed* and *expand* require very little additional work beyond the results of the previous sections. We hope these examples are sufficiently illuminating that they can be used as templates for designing bespoke reset systems from compatible cyclic proof systems.

5.1 Automata theory

The theory of infinite word and tree automata has always been an important tool in cyclic proof theory. In this regard, this dissertation is no exception: The notion of Safra boards (section 5.2) central to our construction of reset systems is based on the Safra tree construction (Safra 1988) developed for the efficient determinisation of certain infinite word automata. The completeness proofs we give in this chapter also crucially rely on a theorem about the inhabitation of languages described by infinite tree automata (proposition 5.1.5). We thus begin by recalling some key notions from automata theory relied upon in this chapter.

Definition 5.1.1 (Büchi automata)**.** A *Büchi automaton* is a quintuple $\mathfrak{B} = (Q, \Sigma, \Delta, S, F)$ where Q is a finite set of *states*, Σ is a finite *alphabet*, $S \subseteq Q$ is the set of *starting states*, the relation $\Delta \subseteq Q \times \Sigma \times Q$ is the *transition relation* and $F \subseteq Q$ is the *acceptance condition*.

Given a word $\sigma \in \Sigma^\omega$, the sequence $\rho \in Q^\omega$ is called a *run of* $\mathfrak{B}$ *on* σ if $\rho_0 \in S$ and for each $i \in \omega$ one has $(\rho_i, \sigma_i, \rho_{i+1}) \in \Delta$. A run ρ is *accepting* if there is some $q \in F$ such that $\rho_i = q$ infinitely often. A word σ *is accepted by* $\mathfrak{B}$ if there exists an accepting run of $\mathfrak{B}$ on σ. The set $L(\mathfrak{B}) \coloneqq \{\sigma \in \Sigma^\omega \mid \sigma \text{ is accepted by } \mathfrak{B}\}$ is the *language of* $\mathfrak{B}$.

An important result connecting the theories of cyclic proof theory and infinite word automata is that the branches satisfying the trace conditions of many cyclic proof systems from the literature can be recognised by certain infinite word automata. A generic construction for such recognising automata can be given using activation algebras and used to prove proposition 4.3.9. The construction below is a variant of that given in Wehr (2021, proposition 5.11) which also gives a proof of proposition 5.1.3. Note that $\mathfrak{B}(\mathcal{A}, M)$ plays a key role in our construction of Safra boards in section 5.2.

Definition 5.1.2. Let $\mathcal{A}$ be an activation algebra, M be a finite set of trace maps over $\mathcal{A}$ and fix $O \coloneqq \bigcup_{\tau\in M}\{\mathrm{dom}(\tau), \mathrm{cod}(\tau)\}$. The Büchi automaton $\mathfrak{B}(\mathcal{A}, M) = (M, Q, \Delta, O, F)$ is defined below, fixing some arbitrary $0^* \notin \mathcal{A}$.

$$\begin{aligned}
Q &\coloneqq O \cup \{(X, x, a) \mid X \in O, x \in X, a \in \mathcal{A} \cup \{0^*\}\} \\
\Delta &\coloneqq \{(X, R\colon X \to Y, Y) \mid R\colon X \to Y \in M\} \\
&\quad \cup \{(X, R\colon X \to Y, (Y, y, 0)) \mid Y \in O, x \in Y\} \\
&\quad \cup \{((X, x, a), R\colon X \to Y, (Y, y, a \vee b)) \mid xR^b y, a \vee b \neq \alpha, R \in M\} \\
&\quad \cup \{((X, x, a), R\colon X \to Y, (Y, y, 0^*)) \mid xR^b y, a \vee b = \alpha, R \in M\} \\
&\quad \cup \{((X, x, 0^*), R\colon X \to Y, (Y, y, a)) \mid xR^a y, R \in M, a \neq \alpha\} \\
&\quad \cup \{((X, x, 0^*), R\colon X \to Y, (Y, y, 0^*)) \mid xR^a y, R \in M, a = \alpha\} \\
F &\coloneqq \{(X, x, 0^*) \mid X \in O, x \in X\}
\end{aligned}$$

Proposition 5.1.3. Let $\mathcal{A}$ be an activation algebra and M be a finite set of trace maps over $\mathcal{A}$. Then $\mathfrak{B}(\mathcal{A}, M)$ recognises the trace condition of $\mathcal{A}$ i.e.,

$$L(\mathfrak{B}(\mathcal{A}, M)) = \left\{ \tau \in M^\omega \;\middle|\; \begin{array}{l} (\mathrm{dom}(\tau_i), \tau_i)_{i\in\omega} \text{ is a valid path and} \\ \text{satisfies the trace condition of } \mathcal{A} \end{array} \right\}.$$

For an alphabet Σ, a Σ-*labelled tree* is a pair $(T, \lambda\colon T \to \Sigma)$ for a, possibly infinite, tree T. A Σ-labelled tree (T, λ) is a *subtree* of Σ-labelled (T', λ') if is a 'suffix' of T' i.e., there exists some $t \in T'$ such that $T = \{ts \in T' \mid s \in T'\}$ and $\lambda(s) = \lambda'(ts)$.

Definition 5.1.4 (Rabin tree automata)**.** A *Rabin tree automaton* is a tuple $\mathfrak{A} = (\Sigma, Q, \Delta, s, R)$ consisting of a finite alphabet Σ, a set of *states* Q, a set of *transitions* $\Delta \subseteq Q \times \Sigma \times Q^*$, a *starting state* $s \in Q$ and an *acceptance condition* $R = \{(G_0, B_0), \dots (G_n, B_n)\}$ where $G_i \cap B_i = \emptyset$ and $G_i \cup B_i \subseteq Q$.

Let (T, λ) be a Σ-labelled tree. A *run* of $\mathfrak{A}$ on (T, λ) is a Q-labelling $\rho \colon T \to Q$ of T such that $\rho(\varepsilon) = s$ and for each $t \in T$ with $\mathrm{Chld}(t) = \{t0, \dots, tn\}$ the transition $(\rho(t), \lambda(t), \rho(t0), \dots, \rho(tn)) \in \Delta$. A run is *accepting* if for every infinite branch $b \in T^\omega$ of T there exists $(G, B) \in R$ such that $\rho(b_i) \in G$ for infinitely many $i \in \omega$ and $\rho(b_i) \in B$ for only finitely many $i \in \omega$. A Σ-labelled tree (T, λ) is *accepted* by $\mathfrak{A}$ if there is an accepting run of $\mathfrak{A}$ on it. The set $L(\mathfrak{A}) := \{(T, \lambda \colon T \to \Sigma) \mid (T, \lambda) \text{ is accepted by } \mathfrak{A}\}$ is the *language of* $\mathfrak{A}$.

The following a corollary of the memoryless determinacy of Rabin games; see, e.g. Piterman and Pnueli (2006).

Proposition 5.1.5. If $\mathfrak{A}$ is a Rabin tree automaton with non-empty language then $\mathfrak{A}$ accepts a regular tree via a regular run.

5.2 Safra boards

This section introduces Safra boards, a variant of the tree construction introduced by Safra (Safra 1988) to determinise Büchi automata. Our presentation of Safra boards has been adapted specifically to the automata $\mathfrak{B}(\mathcal{A}, M)$ or, equivalently, the trace condition of $\mathcal{A}$. Inspired by Kozen's account of Safra automata in Kozen (2006), we present the construction in terms of boards with stacks of chips on them rather than trees. Safra boards can recognise whether a sequence τ of trace maps is a path satisfying the trace condition, similar to the automata $\mathfrak{B}(\mathcal{A}, M)$. They serve as building blocks of the abstract cyclic reset proofs presented in section 5.3.

For the following definitions fix some countable set C with $\omega \subseteq C$, which we call the set of *chips*.

Definition 5.2.1. A *Safra board* on an activation algebra $\mathcal{A}$ and a finite set X is a tuple (Θ, σ) consisting of a *control* Θ, a finite linear order $(\Theta, \leq)$ on a set $\Theta \subset C$, and a map $\sigma \colon X \times (\mathcal{A} \setminus \{\alpha\}) \to \mathcal{P}(\mathcal{P}(\Theta))$. Furthermore, it is required that for every $\gamma \in \Theta$ there are $a \in \mathcal{A}$ and $x \in X$ such that $\gamma \in S \in \sigma(x, a)$. Elements of Θ are called *chips*. The sets $S \in \sigma(x, a)$ represent *stacks* of chips with their $\leq$-least element the bottom and their $\leq$-greatest on top.

A chip $\gamma \in \Theta$ is *covered* if for all $x \in X$ and $a \in A$, γ is not on top of any $S \in \sigma(x, a)$.

The stacks of chips in any given control Θ are *linearly ordered* by the relation $S <_\Theta S'$ which holds iff S contains the $\leq$-least element of the symmetric difference $S \Delta S' = (S \setminus S') \cup (S' \setminus S)$. That is, $<_\Theta$ is the lexicographical ordering, except $S <_\Theta S'$ if S' is a proper prefix on S.

We write $\textsc{Sb}(\mathcal{A}, X)$ for the set of Safra boards on $\mathcal{A}$ and X.

Similar to the automaton $\mathfrak{B}(\mathcal{A}, M)$, Safra boards give rise to a state transition system with a notion of 'accepting run' which recognises sequences $\tau \in M^\omega$ describing paths over $\mathcal{A}$ which satisfy the trace condition. The transitions the shape $(\Theta, \sigma) \overset{X}{\rightsquigarrow} (\Theta', \sigma')$: from Safra board to Safra board. Here, the letter X denotes the type of transition, of which there are five: τ-successors, weakenings, thinnings, γ-resets and populations. We proceed by defining each kind of transition.

Definition 5.2.2. Let (Θ, σ) be a Safra board on $\mathcal{A}$ and X and let $\tau\colon X \to Y$ be a trace map over $\mathcal{A}$. The *τ-successor* of (Θ, σ) is a Safra board (Θ', σ') on Y, which is obtained in two steps:

Move Move all of the stacks around the board according to τ to obtain the intermediate "board" (with $\mathrm{dom}(\sigma^*) = Y \times \mathcal{A}$) (Θ^*, σ^*) on Y as follows:

$$\sigma^*(y, a) := \{S \in \sigma(x, b) \mid \exists c \in \mathcal{A}.\ (x, c, y) \in \tau \text{ and } a = b \vee c\}$$

where $\Theta^* := \{\gamma \in \Theta \mid \exists y \in Y, a \in \mathcal{A}, S \in \sigma^*(y, a) \mid \gamma \in S\}$ is a suborder of Θ.

Cover Cover all stacks that have landed on α. First, fix some linearly ordered set $\Theta^\circ \subset C \setminus \Theta$ and bijection $\iota\colon \{y \in Y \mid \sigma^*(y, \alpha) \neq \emptyset\} \simeq \Theta^\circ$. Then set

$$\sigma'(y, a) := \begin{cases} \sigma^*(y, a) \cup \{S \cup \{\iota(y)\} \mid S \in \sigma^*(y, \alpha)\} & a = 0 \\ \sigma^*(y, a) & \text{otherwise} \end{cases}$$

observing that $\mathrm{dom}(\sigma') = Y \times (\mathcal{A} \setminus \{\alpha\})$ again. Now fix $\Theta' := \Theta^* \oplus \Theta^\circ$ where $\oplus$ denotes the concatenation of linear orders.

We write $(\Theta, \sigma) \overset{\tau}{\rightsquigarrow} (\Theta', \sigma')$ to signal that (Θ', σ') is a τ-successor of (Θ, σ).

Example 5.2.3. For an example, denote by $\mathbb{F}$ the three-value *failure algebra* $(\{0, 1, 2\}, \leq, \vee, 0, 1)$ and the set $\{w, x, y, z\}$. A Safra board in $\textsc{Sb}(\mathbb{F}, X)$ may be thought of as a square game board, akin to a chess board, as pictured in fig. 11. Indeed, fig. 11 gives an example of a τ-successor transition for $\tau := \{(x, 1, x), (x, 1, y), (y, 0, y), (y, 1, y)(z, 2, z)\}$ and $\Theta := \{a, b, c, d, e\}$ (ordered alphabetically) and

$$\sigma(w, 0) := \{\{a\}\} \quad \sigma(x, 0) := \{\emptyset, \{b\}\} \quad \sigma(y, 0) := \{\{c, d\}\} \quad \sigma(z, 0) := \{\{e\}\}$$

where $\sigma(u, v) := \emptyset$ for $u \in X$ and $v \in \{1, 2\}$, as pictured in fig. 11a. To obtain one of its τ-successors, one first needs to carry out the MOVE-step, moving the stacks on the board according to τ. The board (Θ^*, σ^*) resulting from this step is pictured in fig. 11b. Note that the stacks from $(y, 0)$ were moved both to $(y, 1)$ and stayed on $(y, 0)$. Note furthermore that the stack on $(w, 0)$ was removed, as there is no trace triplet for w in τ. Lastly, observe that the MOVE-step is fully deterministic for a fixed board (Θ, σ) and trace map τ. To obtain (Θ', σ'), one needs to carry out the COVER-step: The stack on each (u, α) in (Θ^*, σ^*) need to be moved back to $(u, 0)$. To mark that α has been attained, a new chip (which was not present in Θ) is placed on each such stack that was moved back. If multiple stacks are moved from (v, α) to $(v, 0)$, the same chip is placed on top of each. In this case, as $\alpha = 1$, one moves the stacks on $(x, 1)$ and $(y, 1)$, introducing the new chips g and h.

$X \setminus \mathcal{A}$	0	2
w	a	
x	ε, b	
y	cd	
z	e	

(a) Board (Θ, σ)

$X \setminus \mathcal{A}$	0	1	2
w			
x		ε, b	
y	c	ε, b, cd	
z			e

(b) “Board” (Θ^*, σ^*)

$X \setminus \mathcal{A}$	0	2
w		
x	g, bg	
y	c, h, bh, cdh	
z		e

(c) Board (Θ', σ')

Figure 11: Example of $(\Theta, \sigma) \overset{\tau}{\rightsquigarrow} (\Theta', \sigma')$

Definition 5.2.4. Let (Θ, σ) be a Safra board on $\mathcal{A}$ and X. Another Safra board (Θ', σ') on X is a *weakening* of (Θ, σ) if $\sigma'(x, a) \subseteq \sigma(x, a)$ for every $x \in X$ and $a \in \mathcal{A}$. Furthermore, it is required that $\Theta' \subseteq \Theta$ is such that every $\gamma \in \Theta'$ occurs in some $S \in \sigma'(x, a)$ for some $x \in X, a \in \mathcal{A}$. We write $(\Theta, \sigma) \overset{W}{\rightsquigarrow} (\Theta', \sigma')$ to express that the latter board is a weakening of the former.

The *thinning* of (Θ, σ) is the special weakening (Θ', σ') induced by $\sigma'(x, a) := \{\min_{<_\Theta} \sigma(x, a)\}$ if $\sigma(x, a) \neq \emptyset$ and $\sigma'(x, a) = \emptyset$ otherwise. We write $(\Theta, \sigma) \overset{T}{\rightsquigarrow} (\Theta', \sigma')$ to express that the latter board is the thinning of the former.

Example 5.2.5. Figure 12 pictures the result of a thinning transition starting from the result of example 5.2.3. When multiple stacks are present on a space on the board, a thinning removes all but the $<_\Theta$-least. On the board in fig. 12a, the thinning thus modifies only $\sigma(y, 0)$. Observe that $c <_\Theta cdh$ as $\{c\}\Delta\{c, d, h\} = \{d, h\}$, meaning cdh contains the Θ-least element of the symmetric difference.

Indeed, whenever $S \subset S'$ for two stacks on Θ, one has $S' <_\Theta S$. Secondly, $bh <_\Theta cdh$ as bh contains b and cdh does not.

$X \setminus \mathcal{A}$	0	2
w		
x	bg	
y	c, bh, cdh	
z		e

(a) Board (Θ, σ)

$X \setminus \mathcal{A}$	0	2
w		
x	bg	
y	bh	
z		e

(b) Board (Θ', σ')

Figure 12: Example of $(\Theta, \sigma) \overset{T}{\rightsquigarrow} (\Theta', \sigma')$

Definition 5.2.6. Let (Θ, σ) be a Safra board on $\mathcal{A}$ and X and let $\gamma \in \Theta$ be covered. The *γ-reset* $S \upharpoonright \gamma$ of a stack S is defined as

$$S \upharpoonright \gamma := \begin{cases} \{z \in S \mid z \leq \gamma\} & \text{if } \gamma \in S \\ S & \text{otherwise} \end{cases}$$

Then the *γ-reset* of (Θ, σ) is (Θ', σ') where $\sigma'(x, a) := \{S \upharpoonright \gamma \mid S \in \sigma(x, a)\}$ and $\Theta' := \{\gamma \in \Theta \mid \exists x \in X, a \in \mathcal{A}, S \in \sigma'(x, a).\ \gamma \in S\}$. We write $(\Theta, \sigma) \overset{R_\gamma}{\rightsquigarrow} (\Theta', \sigma')$ to express this.

Example 5.2.7. Pictured in fig. 13 is the result of a b-reset applied to the resulting board from example 5.2.5. Note that a reset on e, g or h would not be possible on that board as none of them are covered.

$X \setminus \mathcal{A}$	0	2
w		
x	bg	
y	bh	
z		e

(a) Board (Θ, σ)

$X \setminus \mathcal{A}$	0	2
w		
x	b	
y	b	
z		e

(b) Board (Θ', σ')

Figure 13: Example of $(\Theta, \sigma) \overset{R_b}{\rightsquigarrow} (\Theta', \sigma')$

Definition 5.2.8. Let (Θ, σ) be a Safra board on $\mathcal{A}$ and X. The board (Θ, σ') is a *population* of (Θ, σ), denoted by $(\Theta, \sigma) \overset{P}{\rightsquigarrow} (\Theta, \sigma')$, if for each $x \in X$ one has $\sigma(x, 0) \subseteq \sigma'(x, 0) \subseteq \sigma(x, 0) \cup \{\emptyset\}$ and $\sigma'(x, a) = \sigma(x, a)$ for all $a \in \mathcal{A} \setminus \{0\}$.

Example 5.2.9. Pictured in fig. 14 is the result of a population transition on the board resulting from example 5.2.7. Here, the empty stack (denoted ε here) has been added to $(w, 0)$ and $(x, 0)$. It would also have been legal to additionally add it to $(y, 0)$ and $(z, 0)$.

$X \setminus \mathcal{A}$	0	2
w		
x	b	
y	b	
z		e

(a) Board (Θ, σ)

$X \setminus \mathcal{A}$	0	2
w	ε	
x	b, ε	
y	b	
z		e

(b) Board (Θ', σ')

Figure 14: Example of $(\Theta, \sigma) \overset{P}{\rightsquigarrow} (\Theta', \sigma')$

Safra board runs are sequences of the different kinds of transitions we have defined. Importantly, for such a sequence to be considered a run on some $\tau \in M^\omega$ it is crucial that it 'consumes' all 'letters' of τ.

Definition 5.2.10. Fix a set M of trace maps of over and some $\tau \in M^\omega$. A sequence $(\Theta_i, \sigma_i)_{i\in\omega}$ of Safra boards is called a *run of* τ if there exists a strictly monotone function $\iota : \omega \longrightarrow \omega$ and for every $i \in \omega$ either

- $i = \iota(n)$ for some $n \in \omega$ and $(\Theta_i, \sigma_i) \overset{\tau_n}{\rightsquigarrow} (\Theta_{i+1}, \sigma_{i+1})$ or
- $i \neq \iota(n)$ and $(\Theta_i, \sigma_i) \overset{W}{\rightsquigarrow} (\Theta_{i+1}, \sigma_{i+1})$ or $(\Theta_i, \sigma_i) \overset{P}{\rightsquigarrow} (\Theta_{i+1}, \sigma_{i+1})$ or $(\Theta_i, \sigma_i) \overset{R_\gamma}{\rightsquigarrow} (\Theta_{i+1}, \sigma_{i+1})$ for some $\gamma \in \Theta_i$

A run $(\Theta_i, \sigma_i)_{i\in\omega}$ is *accepting* if there exists some N and some $\gamma \in \bigcap_{N\leq n} \Theta_n$ such that infinitely many γ-resets take place along $(\Theta_i, \sigma_i)_{i\in\omega}$.

There is a lot of leeway when constructing a Safra board run because of the many different kinds of transitions that may be taken at any point it time (for example, it is always possible to take a weakening transition which leaves σ unchanged). For some proofs in this chapter, it will prove useful to be stricter about the ordering of transitions a long a run. This is accomplished by the concept of greedy runs, runs whose ordering of transitions is deterministic for any given τ. Such runs are called *greedy* because it can be shown that whenever there exists an accepting Safra board run of τ, the greedy run of τ is accepting as well. In many cases, it thus suffices to restrict ones attention to greedy runs. Dually, when constructing runs, one may always follow the greedy construction strategy.

The concept of greedy runs is also closely linked to the runs on determinised $\mathfrak{B}(\mathcal{A}, M)$ (see definition 5.2.19).

Definition 5.2.11. Fix X, Y finite and $\mathcal{A}$, some trace map $\tau \colon X \longrightarrow Y$ over $\mathcal{A}$ and a Safra board (Θ, σ) on X. Then (Θ', σ') is the result of a *greedy τ-transition* from (Θ, σ), denoted by $(\Theta, \sigma) \overset{\tau}{\rightsquigarrow}_g (\Theta', \sigma')$, if

$$(\Theta, \sigma) = (\Theta_0, \sigma_0) \overset{R_{\gamma_k}}{\rightsquigarrow} \dots \overset{R_{\gamma_1}}{\rightsquigarrow} (\Theta_k, \sigma_k) \overset{P}{\rightsquigarrow}$$
$$(\Theta_{k+1}, \sigma_{k+1}) \overset{\tau}{\rightsquigarrow} (\Theta_{k+2}, \sigma_{k+2}) \overset{T}{\rightsquigarrow} (\Theta_{k+3}, \sigma_{k+3}) = (\Theta', \sigma')$$

is the transition sequence produced according to the following instructions, starting at step 1.

1. If there exist covered chips $\gamma_1 < \dots < \gamma_k$ in (Θ_0, σ_0) then perform γ_i-resets in descending order, that is:
$$(\Theta_0, \sigma_0) \overset{R_{\gamma_k}}{\rightsquigarrow} (\Theta_1, \sigma_1) \overset{R_{\gamma_{k-1}}}{\rightsquigarrow} \dots \overset{R_{\gamma_1}}{\rightsquigarrow} (\Theta_k, \sigma_k)$$
then continue with step 2.
2. Continue with a population $(\Theta_k, \sigma_k) \overset{P}{\rightsquigarrow} (\Theta_{k+1}, \sigma_{k+1})$ in such a way that every $\sigma_i(x, 0) = \emptyset$ is populated to $|\sigma_{i+1}(x, 0)| = 1$ and all other $\sigma_{i+1}(x, a)$ remain unchanged. Continue with step 3.
3. Carry out the τ-transition $(\Theta_{k+1}, \sigma_{k+1}) \overset{\tau}{\rightsquigarrow} (\Theta_{k+2}, \sigma_{k+2})$ then continue with step 4.
4. Carry out a thinning $(\Theta_{k+2}, \sigma_{k+2}) \overset{T}{\rightsquigarrow} (\Theta_{k+3}, \sigma_{k+3})$.

We write $(\Theta, \sigma) \overset{\tau}{\rightsquigarrow}_g (\Theta', \sigma')$ to denote the full transition sequence described above.

Definition 5.2.12. A run $(\Theta_i, \sigma_i)_{i \in \omega}$ of some τ is *greedy* if $\Theta_0 = \emptyset$ and $\sigma_0(x, a) = \emptyset$ and furthermore the run is a sequence of greedy τ_i-transitions i.e.,

$$(\Theta_0, \sigma_0) \overset{\tau_0}{\rightsquigarrow}_g (\Theta_{\iota(0)+2}, \sigma_{\iota(0)+2}) \overset{\tau_1}{\rightsquigarrow}_g (\Theta_{\iota(1)+2}, \sigma_{\iota(1)+2}) \overset{\tau_2}{\rightsquigarrow}_g \dots$$

Fact 5.2.13. If $\tau \in M^\omega$ describes a path over $\mathcal{A}$, there exists a greedy run of τ which is unique up-to the choice of chips for the Θ_i. If τ does not describe such a path, no greedy run of τ exists.

Proof. For the existence of the greedy run, observe that the transitions as prescribed by clauses 1., 2. and 4. of definition 5.2.11 can always be taken. The only reason why constructing such a run might thus fail is if some prescribed τ_i-successor transition could not be taken. The only reason for this would be that the current Safra board is on a set different from $\mathrm{dom}(\tau_i)$. But if $(\Theta_0, \sigma_0) \in \mathrm{SB}(\mathcal{A}, \mathrm{dom}(\tau_0))$, it is easily observed that this problem will not arise as τ is assumed to describe a path over $\mathcal{A}$. Hence a greedy run can be constructed and it indeed is a run because all 'letters' of τ are read eventually. As the clauses of definition 5.2.11 always prescribe a unique transition to be taken next, the order of transitions along the greedy runs of τ is always fixed, meaning they can only differ by the choice of chips as claimed.

For the second claim, observe that if τ does not describe a path then there must exist τ_i and τ_{i+1} such that $\mathrm{cod}(\tau_i) \neq \mathrm{dom}(\tau_{i+1})$. In such a cases, the τ_{i+1}-successor step cannot be taken as elaborated above, meaning no run (and thus no greedy run) on τ can exist. □

We continue by proving that the definitions we have given above are correct in the following sense: Any $\tau \in M^\omega$ describes a path satisfying the trace condition if and only if there exists an accepting Safra board run on τ. Our arguments rely on the correspondence between Safra board runs and runs on $\mathfrak{B}(\mathcal{A}, M)$.

Lemma 5.2.14. Fix a finite set M of trace maps over $\mathcal{A}$. If $\tau \in M^\omega$ describes a path which satisfies the trace condition then the greedy run on τ exists and is accepting.

Proof. The greedy run $(\Theta_i, \sigma_i)_{i\in\omega}$ exists by fact 5.2.13. Recall that there exists a function $\iota\colon \omega \to \omega$ indicating the index at which τ_i is read i.e., $(\Theta_{\iota(i)}, \sigma_{\iota(i)}) \overset{\tau_i}{\rightsquigarrow} (\Theta_{\iota(i)+1}, \sigma_{\iota(i)+1})$. Furthermore, observe that, as τ satisfies the trace condition, there exists an accepting run $\rho \in Q^\omega$ of τ on $\mathfrak{B}(\mathcal{A}, M)$. As the run ρ is accepting, it must, from some point R onwards, 'track' a trace along τ through the states $\Sigma X \in O.\, X \times (\mathcal{A} \cup 0^*) \subseteq Q$. Such a state (X, x, a) corresponds to the spot (x, a) on a Safra board on X and this connection is vital to this proof. For any $R \leq i$ we thus write $\sigma_{\iota(i)}(\rho_i)$ to mean $\sigma_{\iota(i)}(x, a)$ where $\rho_i = (X, x, a)$, treating 0^* as $0 \in \mathcal{A}$ (it is easily observed that the set $X_{\iota(i)}$ on which $\Theta_{\iota(i)}$ is defined must always be identical with X).

We make a few observations about the Safra boards along greedy runs just before the next letter of τ is read i.e., the boards $(\Theta_{\iota(i)}, \sigma_{\iota(i)})$. For this, fix $|X| := \sup_{i\in\omega} |X_i|$ which is finite as M is a finite set oftrace maps and there thus exist only finitely many distinct X_i.

1. $|\sigma_{\iota(i)}(x, a)| \leq 1$ at every $\iota(i)$: This is ensured by the thinning after the τ_{i-1}-successor (or the fact that the greedy run starts on the empty board). We thus treat the $\sigma_{\iota(i)}$ as functions $X_i \times (\mathcal{A} \setminus \{\alpha\}) \to \mathcal{P}(\Theta_{\iota(i)})$ with $\sigma_{\iota(i)}(x, a) = \emptyset$ iff $\sigma_{\iota(i)}(x, a) = \emptyset$ under the original interpretation.

2. $|\Theta_{\iota(i)}| \leq |X| \cdot (|\mathcal{A}| - 1)$: If there were more than $|X| \cdot |\mathcal{A}|$ chips, one would have to be covered on $(\Theta_{\iota(i)-2}, \sigma_{\iota(i)-2})$ (the board resulting from the last reset which is part of $(\Theta_{\iota(i-1)+2}, \sigma_{\iota(i-1)+2}) \overset{\tau_i}{\rightsquigarrow}_g (\Theta_{\iota(i)+2}, \sigma_{\iota(i)+2})$) as there is only one top-most chip on each (x, a), contradicting the fact that the last reset was applied at $(\Theta_{\iota(i)-3}, \sigma_{\iota(i)-3})$.
3. $\sigma_{\iota(i)}(\rho_i) \neq \emptyset$ for any $R \leq i$: We argue per induction on i. First, suppose $R = i$: Then $\rho_i = (X_i, x, 0)$ for some $x \in X_i$. Because any spot $(x, 0)$ on the Safra board without a stack is populated in a greedy run before the nexttrace map is read, it follows that $\sigma_{\iota(i)}(\rho_i) = \sigma_{\iota(i)}(x, 0)$ is not empty. Now, because $\sigma_{\iota(i)}(\rho_i)$ is not empty, it is easily observed that the stack on $\sigma_{\iota(i)}(\rho_i)$ will be moved onto $\sigma_{\iota(i)+1}(\rho_{i+1})$ when computing the τ_i-successor. As any steps that could occur between $(\Theta_{\iota(i)+1}, \sigma_{\iota(i)+1})$ and $(\Theta_{\iota(i+1)}, \sigma_{\iota(i+1)})$ never clear away all stacks on any space on the board which has at least one stack on it, it thus follows that $\sigma_{\iota(i+1)}(\rho_{i+1}) \neq \emptyset$ (although the unique stack on it may not be the one moved over from $\sigma_{\iota(i)}(\rho_i)$ because of a thinning step).
4. There must be a maximal height $1 \leq k \leq |X| \cdot (|\mathcal{A}| - 1)$ such that from some $R < K$ onwards, $|\sigma_{\iota(i)}(\rho_i)| \geq k$ of the height of the stack on ρ_i for every $\iota(i) > K$: This follows from the fact that $|\sigma_{\iota(i)}(\rho_i)| \leq |X| \cdot |\mathcal{A}|$ (as a consequence of 2.) and $|\sigma_{\iota(i)}(\rho_i)| \geq 1$ (as a consequence of 3.).
5. From some $K < N$ onwards, the kth chip of all $\sigma_{\iota(i)}(\rho_i)$ with $N \leq i$ needs to remain the same: As $|\sigma_{\iota(i)}(\rho_i)|$ is never less than k again, meaning the kth chip is never cleared as part of a reset, the only way that the colour of the kth chip could change would be if the stack on $\sigma_{\iota(i)}(\rho_i)$ was 'switched' for some $<_\Theta$-smaller stack with a different kth chip as part of a thinning. Such a stack will also always be smaller according to the lexicographic ordering on the first k elements. But this lexicographic ordering is well-founded on arbitrary finite linear orders, as it is always embeddable into the well-founded ω^k. Thus, such replacements can only take place finitely often.

Thus the kth value of $\sigma_{\iota(i)}(\rho_i)$, call it γ, stays constant for any $N \leq i$, meaning also $\gamma \in \Theta_i$ for all i with $N \leq i$. It suffices to to prove that infinitely many γ-resets take place to conclude the run $(\Theta_i, \sigma_i)_{i \in \omega}$ accepting. As ρ is an accepting run, it passes through states $(x, 0^*) \in F$ infinitely many times. Observe that whenever the run enters $(x, 0^*)$, the trace it follows has attained α, meaning a new chip is placed on top of the stack on $\sigma_{\iota(i)}(\rho_i)$ which is the stack containing γ from N onwards. As k was chosen as the greatest infinitely recurring stack height, it also follows that $|\sigma_{\iota(i)}(\rho_i)| = k$, and thus max $\sigma_{\iota(i)}(\rho_i) = \gamma$, infinitely often. After N, this can only happen if the new chips added by the trace tracked by ρ attaining α are removed from above γ via an γ-reset. Thus, infinitely many γ-resets have to take place along $(\Theta_i, \sigma_i)_{i\in\omega}$, making it an accepting run on τ. □

Lemma 5.2.15. Fix a finite set M of trace maps over $\mathcal{A}$. Now suppose some $\tau \in M^\omega$ had an accepting run $(\Theta_i, \sigma_i)_{i\in\omega}$. Then τ describes a path over $\mathcal{A}$ which satisfies the trace condition of $\mathcal{A}$.

Proof. We prove this by showing that there must exist an accepting run of τ on $\mathfrak{B}(\mathcal{A}, M)$. As the run $(\Theta_i, \sigma_i)_{i\in\omega}$ is accepting, there exist $N \in \omega$ and $\gamma \in \bigcap_{N\leq i} \Theta_i$ such that infinitely many γ-resets take place along the run. Now denote by $\overline{\tau}[i, j] \in M^\omega$ the letters of τ read between the indexes i and j i.e., if $\iota(k-1) < i \leq \iota(k) < \iota(k+n) < j \leq \iota(k+n+1)$ then $\overline{\tau}[i, j] = \tau_k\tau_{k+1}\ldots\tau_{k+n}$. We begin by proving a crucial fact: For $N \leq i \leq j$ if $\gamma \in \bigcup \sigma_j(x, a)$ then there must exist $x' \in X_i$ and $b \in \mathcal{A}$ such that $\gamma \in \bigcup \sigma_i(x', b)$ and $(X_i, x', b) \xrightarrow{\overline{\tau}[i,j]} (X_j, x, a)$ on $\mathfrak{B}(\mathcal{A}, M)$ (for this, we identify 0 and 0^*). We prove this per induction on j. Clearly, if $i = j$ then one may choose $x' := x$ and $b := a$ as $(X_i, x, a) \xrightarrow{\varepsilon} (X_i, x, a)$ in $\mathfrak{B}(\mathcal{A}, M)$. For the inductive step, we proceed per case distinction on the transition step between (Θ_j, σ_j) and $(\Theta_{j+1}, \sigma_{j+1})$:

- $(\Theta_j, \sigma_j) \overset{\tau_k}{\rightsquigarrow} (\Theta_{j+1}, \sigma_{j+1})$: Suppose $\gamma \in \bigcup \sigma_{j+i}(x, a)$. As $\gamma \in \Theta_j$, it is easily observed that a stack containing γ can only have arrived on γ if it was 'moved' there by the previous transition. More formally, that means there have to be a $x' \in X_j$ and $b \in \mathcal{A}$ with $\gamma \in \bigcup \sigma_j(x', b)$ and $(X_j, x', b) \xrightarrow{\tau_k} (X_{j+1}, x, a)$. Per inductive hypothesis, there furthermore have to be $x'' \in X_i$ and $c \in \mathcal{A}$ such that $\gamma \in \bigcup \sigma_i(x'', c)$ and $(X_i, x'', c) \xrightarrow{\overline{\tau}[i,j]} (X_j, x', b)$. As $\overline{\tau}[i, j+1] = \overline{\tau}[i, j]\tau_k$, this yields $(X_i, x'', c) \xrightarrow{\overline{\tau}[i,j+1]} (X_{j+1}, x, a)$ as desired.
- $(\Theta_j, \sigma_j) \overset{W}{\rightsquigarrow} (\Theta_{j+1}, \sigma_{j+1})$: If $\gamma \in \bigcup \sigma_{j+1}(x, a)$ then also $\gamma \in \bigcup \sigma_j(x, a)$ because weakening may only remove stacks. Then the claim readily follows from the inductive hypothesis because $\overline{\tau}[i, j+1] = \overline{\tau}[i, j]$.
- $(\Theta_j, \sigma_j) \overset{R_{\gamma'}}{\rightsquigarrow} (\Theta_{j+1}, \sigma_{j+1})$: As such a reset only removes chips from some stacks, $\gamma \in \bigcup \sigma_{j+1}(x, a)$ means that also $\gamma \in \bigcup \sigma_j(x, a)$. Thus simply proceed per inductive hypothesis.
- $(\Theta_j, \sigma_j) \overset{P}{\rightsquigarrow} (\Theta_{j+1}, \sigma_{j+1})$: Again, $\gamma \in \bigcup \sigma_{j+1}(x, a)$ entails $\gamma \in \bigcup \sigma_j(x, a)$ because σ_{j+1} differs from σ_j only by the addition of some empty stacks (which is thus cannot contain γ). Proceed per inductive hypothesis.

Now let $(r_n)_{n\in\omega}$ be a sequence of indexes of γ-resets after N i.e., a monotone increasing sequence with $N < r_0$ and $(\Theta_{r_n}, \sigma_{r_n}) \overset{R_\gamma}{\rightsquigarrow} (\Theta_{r_n+1}, \sigma_{r_n+1})$. Define the sets $S_n := \{(x, a) \mid \gamma \in \bigcup \sigma_{r_n}(x, a)\}$. The previous result means that for any $(x, a) \in S_{n+1}$ there exist $(x', b) \in S_n$ such that $(X_{r_n}, x', b) \xrightarrow{\overline{\tau}[r_n, r_{n+1}]} (X_{r_{n+1}}, x, a)$ in $\mathfrak{B}(\mathcal{A}, M)$ (in which we identify $0 \in \mathcal{A}$ with 0^* in the automata states). An application of König's Lemma yields a sequence $((x_n, a_n) \in S_n)_{n\in\omega}$ such that $(X_{r_n}, x_n, a_n) \xrightarrow{\overline{\tau}[r_n, r_{n+1}]} (X_{r_{n+1}}, x_{n+1}, a_{n+1})$ for every $n \in \omega$. Notably, each run segment $(X_{r_n}, x_n, a_n) \xrightarrow{\overline{\tau}[r_n, r_{n+1}]} (X_{r_{n+1}}, x_{n+1}, a_{n+1})$ crosses the set F of accepting states of $\mathfrak{B}(\mathcal{A}, M)$ at least once: In $(\Theta_{r_n+1}, \sigma_{n+1})$, each instance of γ is the top-most chip on its respective stack. In $(\Theta_{r_{n+1}}, \sigma_{r_{n+1}})$, on the other hand, every instance of γ is covered. This means that each stack $\gamma \in S \in \sigma(x, a)$ with $(x, a) \in S_{n+1}$ must have 'attained α' at least once between r_n and r_{n+1}. In $\mathfrak{B}(\mathcal{A}, M)$, this corresponds to crossing F. The run segments $(X_{r_n}, x_n, a_n) \xrightarrow{\overline{\tau}[r_n, r_{n+1}]} (X_{r_{n+1}}, x_{n+1}, a_{n+1})$ thus already provide the suffix of an accepting run on τ as F is crossed infinitely often. All that remains is to show that there is a run segment $X_0 \xrightarrow{\overline{\tau}[0, r_0]} (X_{r_0}, x_0, a_0)$ to assemble an accepting run of τ on $\mathfrak{B}(\mathcal{A}, M)$. It follows from another application of the previous result that there has to be an $x \in X_N$ and an $a \in \mathcal{A}$ such that $(X_N, x, a) \xrightarrow{\overline{\tau}[N, r_0]} (X_{r_0}, x_0, a_0)$. Now examine the step $(\Theta_{N-1}, \sigma_{N-1}) \rightsquigarrow (\Theta_N, \sigma_N)$ which one may assume, without loss of generality, introduces the chip γ to Θ_N i.e., $\gamma \notin \Theta_{N-1}$. New chips can only be introduced by the covering phase of a τ_k-step. Thus, new chips can only appear on $(x, 0)$, meaning the run segment above is actually $(X_n, x, 0) \xrightarrow{\overline{\tau}[N, r_0]} (X_{r_0}, x_0, a_0)$. Lastly, observe that the existence of the run $(\Theta_i, \sigma_i)_{i\in\omega}$ already guarantees that $\mathrm{cod}(\tau_k) = \mathrm{dom}(\tau_{k+1})$ as $(\Theta_{\iota(k+1)}, \sigma_{\iota(k+1)}) \overset{\tau_{k+1}}{\rightsquigarrow} (\Theta_{\iota(k+1)+1}, \sigma_{\iota(k+1)+1})$ for each $k \in \omega$. That means that $X_0 \xrightarrow{\overline{\tau}[0, N]} (X_n, x, 0)$ is a run segment on $\mathfrak{B}(\mathcal{A}, M)$. Thus, one may assemble the accepting run of τ on $\mathfrak{B}(\mathcal{A}, M)$ pictured below and an conclude that τ indeed describes a path over $\mathcal{A}$ which satisfies the trace condition.

$$X_0 \xrightarrow{\overline{\tau}[0,N]} (X_n, x, 0) \xrightarrow{\overline{\tau}[N,r_0]} (X_{r_0}, x_0, a_0) \xrightarrow{\overline{\tau}[r_0,r_1]} (X_{r_1}, x_1, a_1) \xrightarrow{\overline{\tau}[r_1,r_2]} \dots$$

□

Theorem 5.2.16. Fix a finite set M of trace maps over $\mathcal{A}$. Then there exists an accepting Safra board run on $\tau \in M^\omega$ if and only if τ describes a path over $\mathcal{A}$ which satisfies the trace condition.

To close the section, we illustrate the connection between Safra boards and the determinisation of Büchi automata (more concretely of $\mathfrak{B}(\mathcal{A}, M)$) via Safra's construction (Safra 1988). We do this by defining a determinised variant of $\mathfrak{B}(\mathcal{A}, M)$ in terms of Safra boards.

To ensure that the automaton we construct has a finite state space, we first prove that one can 'make do' with a finite supply of chips when carrying out greedy transition steps. The last condition asserted in lemma 5.2.18 is crucial for the acceptance condition of the constructed automaton.

Definition 5.2.17. Fix a $\mathcal{A}$, a finite set X and a number $K \geq |X|$. Fixing the supply of chips $\overline{K} = \{n \in \omega \mid n < K \cdot (|\mathcal{A}|+1)\}$, write $\textsc{Sb}(\mathcal{A}, X, K) \subseteq \textsc{Sb}(\mathcal{A}, X)$ for the set of K*-sparse* Safra boards. A board (Θ, σ) is K-sparse if

- $\Theta \subseteq \overline{K}$ and $|\Theta| \leq K \cdot |\mathcal{A}|$
- There is at most one stack on each board position in (Θ, σ)
- There are no stacks on any position (x, α) on (Θ, σ)

Lemma 5.2.18. For finite X, Y and a $K \geq |X|, |Y|$ let $(\Theta_0, \sigma_0) \in \textsc{Sb}(\mathcal{A}, X, K)$. Then there exists, for any $\tau \colon X \to Y$, a Safra board $(\Theta_n, \sigma_n) \in \textsc{Sb}(\mathcal{A}, Y, K)$ and a transition sequence $(\Theta_0, \sigma_0) \rightsquigarrow \ldots \rightsquigarrow (\Theta_n, \sigma_n)$ such that $(\Theta_0, \sigma_0) \overset{\tau}{\rightsquigarrow}_g (\Theta_n, \sigma_n)$. Furthermore, $\Theta_n \cap \Theta_0 = \bigcap_{i \leq n} \Theta_i$.

Proof. For a board (Θ, σ) define $|\sigma| := |\{(x, a) \in \mathrm{dom}(\sigma) \mid \sigma(x, a) \neq \emptyset\}|$ i.e., the number of board positions with stacks on them. We shall argue that the steps 1. through 4. from definition 5.2.11 can be taken from (Θ_0, σ_0) in such a way that the resulting sequence $(\Theta_i, \sigma_i)_{i \leq n}$ is such that $\Theta_i \subseteq \overline{K}$. To ensure that $\Theta_n \cap \Theta_0 = \bigcap_{i \leq n} \Theta_i$, one requires the transitions will only introduce chips from $\Theta' := \overline{K} \setminus \Theta_0$. Observe that from the first sparseness condition on (Θ_0, σ_0), it follows that $|\Theta_0| \leq K \cdot |\mathcal{A}|$ and thus $K \leq |\Theta'|$.

1. Suppose $\gamma_1 < \ldots < \gamma_n \in \Theta_0$ were covered in (Θ_0, σ_0). We begin by showing that each transition of the sequence

$$(\Theta_0, \sigma_0) \overset{R_{\gamma_k}}{\rightsquigarrow} (\Theta_1, \sigma_1) \overset{R_{\gamma_{k-1}}}{\rightsquigarrow} \ldots \overset{R_{\gamma_1}}{\rightsquigarrow} (\Theta_k, \sigma_k)$$

can be taken. The only thing which could prevent a transition $(\Theta_i, \sigma_i) \overset{R_{\gamma_{k-i}}}{\rightsquigarrow} (\Theta_{i+1}, \sigma_{i+1})$ along this sequence from being legal would be that γ_{k-i} was not covered in (Θ_i, σ_i). This can only happen if some earlier reset along this sequence 'uncovered' γ_{k-i}. But this cannot happen: Suppose $\gamma > \gamma'$ and consider some stack with $S \ni \gamma'$ and γ' covered i.e., $\max S \neq \gamma'$. Then there are two possibilities for $S \restriction \gamma$: If $\gamma \notin S$ then $S \restriction \gamma = S$ and γ' thus remains covered. If $\gamma \in S$ then $\gamma \in S \restriction \gamma$, meaning a chip $\gamma > \gamma'$ remains in S and γ' remains covered. Thus, a reset on some $\gamma_j > \gamma_{k-i}$ cannot uncover γ_{k-i}, meaning the γ_{k-i}-reset may be carried out.

As resets only ever remove chips, it is easily observed that $\Theta_k \subseteq \Theta_0 \subseteq \overline{K}$. Because resets never add any new stacks, it follows that $|\sigma_k| = |\sigma_0|$. Now observe the following: If a chip is covered in (Θ_k, σ_k) it is also covered in (Θ_0, σ_0), as resets only ever remove chips from the tops of stacks. Thus, every chip $\gamma \in \Theta_k$ must be at the top of at least one stack on (Θ_k, σ_k): Suppose, towards contradiction, that there was a covered chip in (Θ_k, σ_k). But then it would also have been covered in (Θ_0, σ_0), meaning it would have been among the γ_i and would have been reset by the sequence of resets. But then it cannot be covered in (Θ_k, σ_k) as any reset chip is uncovered by the reset. If each chip of Θ_k must occur on top of at least one stack, it is easily observed that $|\Theta_k| \leq |\sigma_k|$.

2. Carry out the population $(\Theta_k, \sigma_k) \overset{P}{\rightsquigarrow} (\Theta_{k+1}, \sigma_{k+1})$. That is, if $\sigma_k(x, 0) = \emptyset$ then $\sigma_{k+1}(x, 0) = \{\emptyset\}$. As $\Theta_{k+1} = \Theta_k$, it follows $\Theta_{k+1} \subseteq \overline{K}$.
3. Now carry out $(\Theta_{k+1}, \sigma_{k+1}) \overset{\tau}{\rightsquigarrow} (\Theta_{k+2}, \sigma_{k+2})$ with $\tau\colon X \to Y$. During the transformation, the stacks are moved from positions $(x, a) \in X \times \mathcal{A}$ to positions $(y, a') \in Y \times \mathcal{A}$ according to τ. If a stack has landed on some (y, α), it is then moved to $(y, 0)$ and has a new chip added to its top. If multiple stacks landed on (y, α), the same chip is added to all of them. During this step, only at most $|\{(y, \alpha) \mid y \in Y\}| = |Y|$ new chips are introduced to Θ_{k+2}, meaning the supply of chips Θ' is sufficient. Thus, $\Theta_{l+1} \subseteq \overline{K}$.
4. In this step $(\Theta_{k+2}, \sigma_{k+2}) \overset{T}{\rightsquigarrow} (\Theta_{k+3}, \sigma_{k+3})$, some of the stacks, and possibly some of the chips that used to be in them, are removed and no new chips are added. Thus $\Theta_{k+3} \subseteq \overline{K}$.

It is easily observed that after step 4. there is at most one stack on each position of $(\Theta_{k+3}, \sigma_{k+3})$. Furthermore, after step 3. there are no stacks on positions (y, α), a fact which remains unchanged by step 4. It remains to show that $|\Theta_{k+3}| \subseteq K \cdot |\mathcal{A}|$. We have shown that after step 1. $|\Theta_k| = |\sigma_0|$. As there are no stacks on (x, α) in (Θ_0, σ_0), that means that $|\Theta_k| \leq |X| \cdot (|\mathcal{A}| - 1) \leq K \cdot (|\mathcal{A}| - 1)$. It is easily observed that after step 2. $|\Theta_{k+1}| = |\Theta_k|$. As already argued, step 3. adds at most $|Y| \leq K$ new chips, meaning $|\Theta_{k+2}| \leq |\Theta_{k+1}| + K \leq K \cdot |\mathcal{A}|$. As step 4. only removes chips, this means $|\Theta_{k+3}| \leq |\Theta_{k+2}| \leq K \cdot |\mathcal{A}|$ as desired. □

We can thus construct a deterministic Rabin automaton which recognises the trace condition of $\mathcal{A}$, similarly to the non-deterministic Büchi-automaton

$\mathfrak{B}(\mathcal{A}, M)$. For the construction of the automaton, as well as some of the arguments in section 5.3.2, it would be helpful if for each $(\Theta, \sigma) \in \textsc{Sb}(\mathcal{A}, X, K)$ and $\tau \colon X \to Y$ there was some unique $(\Theta', \sigma') \in \textsc{Sb}(\mathcal{A}, Y, K)$ such that $(\Theta, \sigma) \overset{\tau}{\rightsquigarrow}_g (\Theta', \sigma')$. We thus simply assume that for each such (Θ, σ) and τ where this is applicable, such a choice has been made, for example via an application of the axiom of choice or some other means, and treat $\overset{\tau}{\rightsquigarrow}_g$ as an injective function on K-sparse (Θ, σ) for suitable K such that $(\Theta, \sigma) \overset{\tau}{\rightsquigarrow}_g (\Theta', \sigma')$ was always derived according to lemma 5.2.18.

Definition 5.2.19. Let $\mathcal{A}$ be finite and fix a finite set of finite sets O and a set of trace maps $M \subseteq \bigcup_{X,Y \in O} \textsc{Hom}(X, Y)$ over $\mathcal{A}$. Furthermore, set $K := \max_{X \in O} |X|$. The *Safra automaton* $\mathfrak{S}(\mathcal{A}, S, M)$ for a starting object $S \in O$ is the Rabin automaton $(M, Q, \delta, (S, (\Theta_0, \sigma_0)), R)$ where

$$Q := \Sigma X \in O.\ \textsc{Sb}(\mathcal{A}, X, K)$$
$$\delta := ((X, (\Theta, \sigma)), \tau \colon X \to Y) \mapsto (Y, (\Theta', \sigma')) \text{ s.t. } (X, (\Theta, \sigma)) \overset{\tau}{\rightsquigarrow}_g (Y, (\Theta', \sigma'))$$
$$R := \{(\{(X, (\Theta, \sigma)) \in Q \mid \gamma \in \Theta, \gamma \text{ covered}\}, \{(X, (\Theta, \sigma)) \in Q \mid \gamma \notin \Theta\}) \mid \gamma \in \overline{K}\}$$

and $(\Theta_0, \sigma_0) = (\emptyset, (s, a) \mapsto \emptyset)$.

Lemma 5.2.20. For an $\mathcal{A}$, $M \subseteq \bigcup_{X,Y \in O} \textsc{Hom}(X, Y)$ for a finite set of finite sets O, the Safra automaton $\mathfrak{S}(\mathcal{A}, S, M)$ accepts a sequence $\tau \in M^\omega$ if and only if $\mathrm{dom}(\tau_0) = S$ and τ describes a path over $\mathcal{A}$ which satisfies the trace condition.

Proof. First, consider any run $(S, (\Theta_0, \sigma_0)) \xrightarrow{\tau_0} (X_1, (\Theta_1, \sigma_1)) \xrightarrow{\tau_1} \cdots$ of $\tau \in M^\omega$ on $\mathfrak{S}(\mathcal{A}, S, M)$. Each state transition $(X_i, (\Theta_i, \sigma_i)) \xrightarrow{\tau_i} (X_{i+1}, (\Theta_{i+1}, \sigma_{i+1}))$ corresponds to a greedy transition $(\Theta_i, \sigma_i) \overset{\tau_i}{\rightsquigarrow}_g (\Theta_{i+1}, \sigma_{i+1})$ obtained via lemma 5.2.18. Thus, one may 'expand' the run into a Safra board run of the following shape

$$(\Theta_0, \sigma_0) \rightsquigarrow (\Theta_0^1, \sigma_0^1) \rightsquigarrow \cdots \rightsquigarrow (\Theta_0^{n_0}, \sigma_0^{n_0}) \rightsquigarrow (\Theta_1, \sigma_1) \rightsquigarrow \cdots \rightsquigarrow (\Theta_1^{n_1}, \sigma_1^{n_1}) \cdots$$

By comparing definitions 5.2.11 and 5.2.12, it is easy to see that the run above must be a greedy run. Analogously to fact 5.2.13, this greedy Safra board run, and by extension the run on $\mathfrak{S}(\mathcal{A}, S, M)$, exists if and only if $\tau \in M^\omega$ describes a path over $\mathcal{A}$ with $\mathrm{dom}(\tau_0) = S$. The latter condition is caused by the fact that (Θ_0, σ_0) always is a Safra board on S. From lemmas 5.2.14 and 5.2.15 it follows that the Safra board run above is accepting if and only if τ satisfies the trace condition. It thus suffices to argue that the Rabin condition R holds on a run on $\mathfrak{S}(\mathcal{A}, S, M)$ iff the expanded Safra board run is accepting. For this, observe that R holds on a run if there exists some chip $\gamma \in \overline{K}$ and some $N \in \omega$

such that $\gamma \in \Theta_i$ for ever $N \leq i$ and γ is covered in infinitely many (Θ_i, σ_i). By scrutinising lemma 5.2.18, one can see that the former condition means that γ is present on every Safra board along the greedy Safra board run from some point onwards. Definition 5.2.11 dictates that whenever γ is covered in (Θ_i, σ_i), a γ-reset takes place between (Θ_i, σ_i) and $(\Theta_{i+1}, \sigma_{i+1})$ in the greedy run. The Rabin condition is thus completely analogous to the acceptance condition on Safra board runs: It holds if and only if there is some chip γ which is eventually never removed again and reset infinitely often. □

We close this section by deriving some bounds on the sizes of various components of $\mathfrak{S}(\mathcal{A}, S, M)$. Note that these are not optimal bounds for Safra constructions. The reader may consult Piterman (2007), for example, for a more space-efficient construction.

Lemma 5.2.21. Fix finite $\mathcal{A}$ and $M \subseteq \bigcup_{X,Y \in O} \textsc{Hom}(X, Y)$ where O is a finite set of finite sets . Denote the Safra automaton $\mathfrak{S}(\mathcal{A}, S, M) = (M, Q, \delta, s, R)$ and let $K := \max_{X \in O} |X|$.

1. For any $X \in O$ one has

$$|\textsc{Sb}(\mathcal{A}, X, K)| \leq \sum_{C=1}^{K \cdot |A|} \binom{K \cdot (|\mathcal{A}| + 1)}{C} \cdot C! \cdot 2^{C \cdot |X| \cdot (|\mathcal{A}|-1)} = O(K!).$$

2. $|Q| \leq \sum_{X \in O} |\textsc{Sb}(\mathcal{A}, X, K)| = O(|O| \cdot K!)$.
3. $|R| = K \cdot (|\mathcal{A}| + 1) = O(K)$.

Proof. 2. readily follows from 1. Furthermore, 3. holds as $|R| = |\overline{K}| = K \cdot (|\mathcal{A}| + 1)$. To understand the bound in 1., observe that a board $(\Theta, \sigma) \in \textsc{Sb}(\mathcal{A}, X, K)$ with $|\Theta| = C$ consists of three components: A choice $\Theta \subseteq \overline{K}$ (of which there are $\binom{\overline{K}}{C} = \binom{K \cdot (|\mathcal{A}|+1)}{C}$ different ones if $|\Theta| = C$), a linear order imposed on Θ (of which there are $|\Theta|! = C!$ many) and one stack $S \subseteq \mathbf{P}(\Theta)$ on each position (x, a) with $a \neq \alpha$ i.e., a function $\sigma \colon X \times (\mathcal{A} \setminus \{\alpha\}) \to \mathbf{P}(\Theta)$ (of which there are $(2^C)^{|X| \cdot (|\mathcal{A}|-1)} = 2^{C \cdot |X| \cdot (|\mathcal{A}|-1)}$ many). Furthermore taking into account that $1 \leq |\Theta| = C \leq K \cdot |\mathcal{A}|$, one observes the bound stated in 1. □

5.3 Reset proof systems

Fix an activation algebra $\mathcal{A}$. In this section, we show that every cyclic proof system induced by a trace interpretation over $\mathcal{A}$ gives rise to a cyclic proof system whose soundness condition is based on Safra boards. It serves as a starting point for deriving concrete Reset-based proof systems based on concrete cyclic proof systems, as we do in section 5.4.

Given a cyclic proof system $\mathcal{R}$ induced by a trace interpretation $\iota\colon \mathcal{R} \to \mathcal{A}$, the reset proof system $\mathrm{R}(\mathcal{R})$ is obtained by annotating the sequents of $\mathcal{R}$ with $\mathcal{A}$-Safra boards. More specifically, an $\mathcal{R}$-sequent Γ is annotated with a Safra board $(\Theta, \sigma) \in \mathrm{Sb}(\mathcal{A}, \iota(\Gamma))$. Each derivation rule $R \in \mathcal{R}$ is 'lifted' to a corresponding derivation rule in $\mathrm{R}(\mathcal{R})$, the Safra boards annotating the ith premise being the result of the transition of the trace interpretation map r_i on the conclusion's Safra board. Furthermore, the system $\mathrm{R}(\mathcal{R})$ also contains structural rules corresponding to the three types 'bookkeeping transitions' on Safra boards: Weakening, reset and population. The soundness condition of $\mathrm{R}(\mathcal{R})$ requires each local cycle $\gamma(t)$ i.e., path between a bud t and its companion $\beta(t)$, to contain an application of the Reset-rule which leaves an invariant θ, a prefix of the control which remains unchanged along the whole local cycle.

Definition 5.3.1. Fix a cyclic proof system $(\mathrm{Seq}, \mathcal{R}, \rho, \mathrm{Pfs})$ induced by a trace interpretation ι into $\mathcal{A}$. The *reset proof system for* $\mathcal{R}$ is the cyclic proof system $\mathrm{R}(\mathcal{R}) := (\mathrm{R}(\mathrm{Seq}), \mathrm{R}(\mathcal{R}), \mathrm{R}(\rho), \mathrm{R}(\mathrm{Pfs}))$ specified as follows. Sequents in $\mathrm{R}(\mathcal{R})$ are expressions $\Gamma; (\Theta, \sigma)$ where $\Gamma \in \mathrm{Seq}$ is an $\mathcal{R}$-sequent and $(\Theta, \sigma) \in \mathrm{Sb}(\mathcal{A}, \iota(\Gamma))$ is a Safra board on Γ's trace object $\iota(\Gamma)$. The order Θ is called the *control* of $\Gamma; (\Theta, \sigma)$. The derivation rules of $\mathrm{R}(\mathcal{R})$ consist of annotated versions of rules from $\mathcal{R}$ and additional *structural rules*. The structural rules are given by the following three rule schemata.

$$\textsc{Weak}\,\frac{\Gamma; (\Theta', \sigma') \text{ where } (\Theta, \sigma) \overset{W}{\rightsquigarrow} (\Theta', \sigma')}{\Gamma; (\Theta, \sigma)}$$

$$\textsc{Reset}_{\gamma}\,\frac{\Gamma; (\Theta', \sigma') \text{ where } (\Theta, \sigma) \overset{R_\gamma}{\rightsquigarrow} (\Theta', \sigma')}{\Gamma; (\Theta, \sigma)}$$

$$\textsc{Pop}\,\frac{\Gamma; (\Theta', \sigma') \text{ where } (\Theta, \sigma) \overset{P}{\rightsquigarrow} (\Theta', \sigma')}{\Gamma; (\Theta, \sigma)}$$

For each rule $r \in \mathcal{R}$ with $\rho(r) = (\Gamma, \Delta_1, \ldots, \Delta_n)$ and maps $r_i\colon \iota(\Gamma) \to \iota(\Delta_i)$ given by the trace interpretation, the following schema gives rules for each $(\Theta, \sigma) \in \mathrm{Sb}(\mathcal{A}, \iota(\Gamma))$ where $(\Theta, \sigma) \overset{r_i}{\rightsquigarrow} (\Theta_i, \sigma_i)$ for $1 \le i \le n$:

$$r\,\frac{\Delta_1; (\Theta_1, \sigma_1) \quad \cdots \quad \Delta_n; (\Theta_n, \sigma_n)}{\Gamma; (\Theta, \sigma)}$$

Let $D = (C, \lambda, \delta)$ be a preproof of $\mathrm{R}(\mathcal{R})$. Pick some $t \in \mathrm{dom}(\beta)$ and let $\pi(t) = (\Gamma_i; (\Theta_i, \sigma_i))_{i<n}$ be sequents along the path from $\beta(t)$ to t. Let Θ be the longest common prefix of all of the Θ_i. An *invariant* of $\pi(t)$ is any prefix θ of Θ such that an application of a $\max(\theta)$-reset occurs between $\beta(t)$ and t. Sometimes we speak of *the* invariant of $\pi(t)$, in which case we refer to the longest such. An $\mathrm{R}(\mathcal{R})$-preproof satisfies the soundness condition $\mathrm{R}(\textsc{Pfs})$ iff for every $t \in \mathrm{dom}(\beta)$ the path $\pi(t)$ between $\beta(t)$ and t has an invariant.

5.3.1 Soundness

A *reset proof* for $\mathcal{R}$ is a cyclic proof in $\mathrm{R}(\mathcal{R})$. This is essentially a cyclic proof in $\mathcal{R}$ with additional structure in the form of annotations. Any application of a rule corresponding to $R \in \mathcal{R}$ directly impacts the traces running through a preproof while the structural rules perform 'bookkeeping' for the control (Θ, σ). This intuition can be made more formal: There exists a proof morphism from $\mathrm{R}(\mathcal{R})$ into $\mathcal{R}$ arising from stripping away the annotations (Θ, σ).

Fix a cyclic proof system $\mathcal{R}$ induced by a trace interpretation over $\mathcal{A}$. The function $erase\colon \mathrm{R}(\textsc{Seq}) \to \textsc{Seq}$ is defined by $erase(\Gamma; (\Theta, \sigma)) := \Gamma$ on sequents.

Lemma 5.3.2. For a cyclic proof system $\mathcal{R}$ induced by a trace interpretation over $\mathcal{A}$, the function $erase\colon \mathrm{R}(\textsc{Seq}) \to \textsc{Seq}$ can be extended to a preproof morphism $erase\colon \mathrm{R}(\mathcal{R}) \to \mathcal{R}$.

Proof. We need to assign to every rule $\hat{R} \in \mathrm{R}(\mathcal{R})$ a corresponding preproof $erase(\hat{R})$ in $\mathcal{R}$. There are only two cases to consider:

- *$\hat{R}$ is a structural rule:* Then $\hat{R}$ is of shape

$$\hat{R}\,\frac{\Gamma; (\Theta', \sigma') \text{ where } (\Theta, \sigma) \overset{X}{\rightsquigarrow} (\Theta', \sigma')}{\Gamma; (\Theta, \sigma)}$$

  where $X \in \{W, P\} \cup \{R_\gamma \mid \gamma \in \Theta\}$. In any case, we need to find a preproof with assumption $erase(\Gamma; (\Theta, \sigma)) = \Gamma$ and premise $erase(\Gamma; (\Theta, \sigma)) = \Gamma$. Such a preproof is given by the identity preproof of Γ, i.e. the triple $(\{\varepsilon\}, \varepsilon \mapsto \Gamma, \emptyset)$.

- *$\hat{R}$ corresponds to a rule $R \in \mathcal{R}$:* That is, there is $R \in \mathcal{R}$ with $\rho(R) = (\Gamma, \Delta_1, \ldots, \Delta_n)$ and maps $r_i\colon \iota(\Gamma) \to \iota(\Delta_i)$ given by the trace interpretation and $\hat{R}$ is of the form

$$\hat{R}\,\frac{\Delta_1; (\Theta_1, \sigma_1) \quad \ldots \quad \Delta_n; (\Theta_n, \sigma_n)}{\Gamma; (\Theta, \sigma)}$$

where $(\Theta, \sigma) \overset{r_i}{\rightsquigarrow} (\Theta_i, \sigma_i)$ for $1 \leq i \leq n$. Then, analogously to the first case, we need to find a preproof of Γ with open leaves $\Delta_1, \ldots, \Delta_n$ in $\mathcal{R}$. The preproof consisting of exactly one application of R is as desired. $\square$

Lemma 5.3.2 merely establishes that *erase* is a preproof morphism not a proof morphism. Showing the latter is the subject of the remainder of this section. That *erase* constitutes a proof morphism between $\mathrm{R}(\mathcal{R})$ and $\mathcal{R}$ can be understood as a relative soundness result: Suppose $\mathcal{R}$ is sound i.e.„ the system proves only true sequents. As *erase* is a proof morphism, if a sequent $\Gamma; (\Theta, \sigma)$ is provable in $\mathrm{R}(\mathcal{R})$, then there is a cyclic proof of $erase(\Gamma; (\Theta, \sigma))$ in $\mathcal{R}$ obtained via the morphism, and so $\mathrm{R}(\mathcal{R})$ is sound.

The idea behind the soundness argument is rather simple: For every strongly connected component η of an $\mathrm{R}(\mathcal{R})$-proof, one can find a 'shared invariant' which is common to all cycles in η. The properties of such invariants allow one to conclude that reading off the controls (Θ, σ) off any infinite path through the proof which visits precisely $C[\eta]$ infinitely often must be an accepting Safra board run and the underlying trace thus must satisfy the trace condition. The most complicated step of the argument is establishing the existence of such shared invariants.

For the remainder of this section, fix some cyclic proof system $\mathcal{R}$ induced by a trace interpretation $\iota\colon \mathcal{R} \to \mathcal{A}$.

Proposition 5.3.3. Let (C, λ, δ) be an $\mathrm{R}(\mathcal{R})$-proof and let η be a strongly connected component of C. Then there exists some $t \in \eta$ such that the invariant θ of $\beta(t) < t$ is a prefix of the invariant of each $\beta(s) < s$ with $s \in \eta$.

Proof. Observe that one can impose a linear order $\sqsubset$ on η such that for any $s_0 \in \eta$ condition (ii) of the definition of strongly connected component can be fulfilled by taking $s_1, \ldots, s_n$ such that they are $\sqsubset$-less than s_0. Clearly, every downset $\mathrm{Down}_{\sqsubseteq}(s)$ for $s \in \eta$ is a strongly connected component. We prove per induction on the $\sqsubset$-order that for every $s \in \eta$, the strongly connected component $\mathrm{Down}_{\sqsubseteq}(s)$ contains a local cycle $\beta(t) < t$ whose invariant is a prefix of all invariants in $\mathrm{Down}_{\sqsubseteq}(s)$. The claim then follows as $\eta = \mathrm{Down}_{\sqsubseteq}(\max_{\sqsubseteq} \eta)$. The case of the $\sqsubset$-least element is trivial. Thus pick some $s \neq b(\eta)$ and consider $\eta' := \mathrm{Down}_{\sqsubseteq}(s) \setminus \{s\}$. Clearly $\eta' = \mathrm{Down}_{\sqsubseteq}(s')$ for some $s' \in \eta'$ and thus has an element $t' \in \eta'$ with invariant θ' which is a prefix of all invariants of cycles in η'. We first prove that the path $\beta(s) \in C[\eta']$: As $\mathrm{Down}_{\sqsubseteq}(s)$ is a strongly connected component, there needs to be a shortest possible sequence $s_1, \ldots, s_n \in \eta'$ with $0 < n$ such that $\beta(s_i) \leq s_{i+1}$ and $s_n = b(\eta)$. Then $\beta(s_{i+1}) < \beta(s_i)$ always as otherwise the 'detour' through s_{i+1} could be avoided, shortening the sequence. This means that $\beta(s_1) < \beta(s) \leq s_1$, meaning that $\beta(s)$ occurs on the path $\beta(s_1) \leq s_1$. As θ' is a prefix of the invariant of $\beta(s_1) \leq s_1$, it must also be a prefix of the control at $\beta(s)$. There are thus only two possibilities for the invariant θ of $\beta(s) \leq s$: Either θ' is a prefix of it or it is a prefix of θ'. In the former case, t' remains the element in $\mathrm{Down}_{\sqsubseteq}(s)$ whose invariant is a prefix of all other invariants, in the latter s is the new such element by the transitivity of the prefix relation. □

Theorem 5.3.4. The function *erase*: $\mathrm{R}(\textsc{Seq}) \to \textsc{Seq}$ is a cyclic proof system morphism.

Proof. Part of the claim has already been proven in lemma 5.3.2. It only remains to show that if Π is a proof in $\mathrm{R}(\mathcal{R})$ then $erase(\Pi)$ is a proof in $\mathcal{R}$.

Let $\Pi = (C, \lambda, \delta)$ be a cyclic proof of $\Gamma; (\Theta, \sigma)$ in $\mathrm{R}(\mathcal{R})$. For this, it suffices to show that every path $\widehat{\pi'} : \omega \to \mathcal{A}$ induced by a branch π' through $erase(\Pi) = (C', \lambda', \delta')$ satisfies the trace condition of $\mathcal{A}$. There must exist a path π through Π 'following' π'. Let $(\Theta_i, \sigma_i)_{i\in\omega}$ be such that $\lambda(\pi_i') = \Gamma_i; (\Theta_i, \sigma_i)$. Clearly, $(\Theta_i, \sigma_i)_{i\in\omega}$ is a Safra board run of τ with $\tau_i := \widehat{\pi'}(i < i + 1)$. By lemma 5.2.15, it thus suffices to show that $(\Theta_i, \sigma_i)_{i\in\omega}$ is accepting to prove that P satisfies the global trace condition. By lemma 4.1.4, there exists a strongly connected component η of C such that $\mathrm{Inf}(\pi) = C[\eta]$. Thus π remains within $C[\eta]$ from some point onwards, say from index N onwards. By proposition 5.3.3, there furthermore exists $t \in \eta$ such that the invariant θ of $\beta(t) < t$ is a prefix of all controls in the annotations in $C[\eta]$. This means that $\theta \leq \Theta_i$ for all $i \geq N$. Consider the chip $\gamma := \max(\theta)$: From the previous observation follows that $\gamma \in \Theta_i$ for all $i \geq N$. Furthermore, γ is reset on the path $\beta(t) < t$ as θ is an invariant of that path. Thus, infinitely many γ-resets take place along $(\Theta_i, \sigma_i)_{i\in\omega}$, making it an accepting run as desired. □

Corollary 5.3.5 (Soundness)**.** If Π is a proof of $\Gamma; (\Theta, \sigma)$ in $\mathrm{R}(\mathcal{R})$ then $erase(\Pi)$ is a proof of Γ in $\mathcal{R}$.

5.3.2 Completeness and proof search

In this section prove completeness of $\mathrm{R}(\mathcal{R})$ relative to $\mathcal{R}$ i.e., any sequent provable in $\mathcal{R}$ can be provable in $\mathrm{R}(\mathcal{R})$. We do this by showing that proof search can be performed in $\mathrm{R}(\mathcal{R})$ if $\mathcal{R}$ has a finite amount of derivation rules. Thus let $\mathcal{R}$ be such that the set $\mathcal{F}$ of derivation rules is finite and let its soundness condition be induced by a trace interpretation over $\mathcal{A}$.

We begin by constructing a proof search system $\mathrm{S}(\mathcal{R})$ for $\mathcal{R}$. Similarly to $\mathrm{R}(\mathcal{R})$ the sequents of $\mathrm{S}(\mathcal{R})$ are $\mathcal{R}$-sequents annotated with Safra boards. However, the annotations of $\mathrm{S}(\mathcal{R})$ are restricted to be K-sparse for a suitable K. Crucially, the system $\mathrm{S}(\mathcal{R})$ has a finite number of derivation rules if $\mathcal{R}$ does, a difference from $\mathrm{R}(\mathcal{R})$ which eases proof search. More specifically, each rule of $\mathrm{S}(\mathcal{R})$ is formed by taking a rule $R \in \mathcal{R}$, annotating its conclusion with a K-sparse Safra board and annotating the ith premise with the K-sparse Safra board resulting from the *greedy* transition via the trace interpretation map r_i. The soundness condition is a 'global variant' of the acceptance condition of the Safra automata in definition 5.2.19.

Definition 5.3.6. Fix $K := \max\{|\iota(\Gamma)| \mid \Gamma \in \textsc{Seq}\}$. The *proof search system* of $\mathcal{R}$ is the system $\mathrm{S}(\mathcal{R}) = (\mathrm{S}(\textsc{Seq}), \mathrm{S}(\mathcal{R}), \mathrm{S}(\rho), \mathrm{S}(\textsc{Pfs}))$ defined below. The sequents of $\mathrm{S}(\mathcal{R})$ are expressions $\Gamma; (\Theta, \sigma)$ with $\Gamma \in \textsc{Seq}$ a $\mathcal{R}$-sequent and $(\Theta, \sigma) \in \textsc{Sb}(\mathcal{A}, \iota(\Gamma), K)$ a K-sparse Safra board on $\iota(\Gamma)$. The rules of $\mathrm{S}(\mathcal{R})$ comprise, for each $R \in \mathcal{R}$ with $\rho(R) = (\Gamma, \Delta_1, \dots, \Delta_n)$ and maps $r_i \colon \iota(\Gamma) \to \iota(\Delta_i)$, and each $(\Theta, \sigma) \in \textsc{Sb}(\mathcal{A}, \iota(\Gamma), K)$ the rule:

$$\mathrm{R}(\Theta,\sigma)\,\frac{\Delta_1; (\Theta_1, \sigma_1) \quad \dots \quad \Delta_n; (\Theta_n, \sigma_n)}{\Gamma; (\Theta, \sigma)}$$

where $(\Theta, \sigma) \overset{r_i}{\rightsquigarrow}_g (\Theta_i, \sigma_i)$ for $1 \leq i \leq n$.

A $\mathrm{S}(\mathcal{R})$-preproof Π satisfies the soundness condition $\mathrm{S}(\textsc{Pfs})$ if along every infinite path $(\Gamma_i; (\Theta_i, \sigma_i))_{i \in \omega}$ through Π there exists some $N \in \omega$ and $\gamma \in \bigcap_{N \leq i} \Theta_i$ such that γ is covered infinitely often.

As we fixed $\overset{\tau}{\rightsquigarrow}_g$ to be an injective function on K-sparse Safra boards, the choice of R and (Θ, σ) specifies the rule $\mathrm{R}(\Theta, \sigma)$ uniquely.

Lemma 5.3.7. The function $\mathit{expand} \colon \mathrm{S}(\textsc{Seq}) \to \mathrm{R}(\textsc{Seq})$ with

$$\mathit{expand}(\Gamma; (\Theta, \sigma)) := \Gamma; (\Theta, \sigma)$$

can be extended to a proof morphism.

Proof. Towards this claim, first pick some $\mathrm{R}(\Theta, \sigma) \in \mathrm{S}(\mathcal{R})$ arranged as follows

$$\mathrm{R}(\Theta,\sigma)\,\frac{\Delta_1; (\Theta_1, \sigma_1) \quad \dots \quad \Delta_n; (\Theta_n, \sigma_n)}{\Gamma; (\Theta, \sigma)}$$

Then there is $R \in \mathcal{R}$ with $\rho(R) = (\Gamma, \Delta_1, \dots, \Delta_n)$ andtrace maps $r_i \colon \iota(\Gamma) \to \iota(\Delta_i)$ given by the trace interpretation. We have to find a corresponding preproof

$expand(\mathrm{R}(\Theta, \sigma))$ in $\mathrm{R}(\mathcal{R})$. Note that for each $i \leq n$ there is $(\Theta, \sigma) \overset{r_i}{\rightsquigarrow}_g (\Theta_i, \sigma_i)$ with the expanded sequence the expanded sequence

$$(\Theta, \sigma) \overset{R_{\gamma_1}}{\rightsquigarrow} (\Theta_r^1, \sigma_r^1) \ldots \overset{R_{\gamma_k}}{\rightsquigarrow} (\Theta_r^k, \sigma_r^k) \overset{P}{\rightsquigarrow} (\Theta_p, \sigma_p) \overset{\tau_i^{\Delta_i}}{\rightsquigarrow} (\Theta_i^*, \sigma_i^*) \overset{T}{\rightsquigarrow} (\Theta_i, \sigma_i)$$

in which the initial R_γ- and P-steps are shared between all $i \leq n$ (see lemma 5.2.18). Then we may derive $expand(\mathrm{R}(\Theta, \sigma))$ as follows:

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\Delta_1; (\Theta_1, \sigma_1)}{\Delta_1; (\Theta_1^*, \sigma_1^*)}\,\textsc{Weak} \qquad \ldots \qquad \dfrac{\Delta_n; (\Theta_n, \sigma_n)}{\Delta_n; (\Theta_n^*, \sigma_n^*)}\,\textsc{Weak}}{\Gamma; (\Theta_p, \sigma_p)}\,\mathrm{R}}{\Gamma; (\Theta_r^k, \sigma_r^k)}\,\textsc{Pop}}{\vdots}}{\dfrac{\Gamma; (\Theta_r^1, \sigma_r^1)}{\Gamma; (\Theta_0, \sigma_0)}\,\textsc{Reset}_{\gamma_1}}\,\textsc{Reset}_{\gamma_2}$$

To prove that $expand$ preserves the soundness condition, let $\Pi = (C, \lambda, \delta)$ be a $\mathrm{S}(\mathcal{R})$-proof and let $expand(\Pi) = (C', \lambda', \delta')$ be its $expand$-translation. Now consider some $t' \in \mathrm{dom}(\beta')$ and the associated path $(\Gamma_i'; (\Theta_i', \sigma_i'))_{i \leq n}$ between $\beta'(t')$ and t' in $expand(\Pi)$. There is a corresponding path $(\Gamma_i; (\Theta_i, \sigma_i))_{i \leq m}$ through Π. By the soundness condition of $\mathrm{S}(\mathcal{R})$, the path through Π which starts at the root and then cycles infinitely on $(\Gamma_i; (\Theta_i, \sigma_i))_{i \leq m}$ must have some $\gamma \in \bigcap_{i \leq m} \Theta_i$ which is covered somewhere along $(\Gamma_i; (\Theta_i, \sigma_i))_{i \leq m}$. By lemma 5.2.18 that means that $\gamma \in \bigcap_{i \leq n} \Theta_i'$ as well and the fact that γ is covered somewhere along $(\Gamma_i; (\Theta_i, \sigma_i))_{i \leq m}$ means a γ-reset must take place somewhere along $(\Gamma_i'; (\Theta_i', \sigma_i'))_{i \leq n}$. It remains to show that the cycle has an invariant θ with $\max(\theta) = \gamma$. Taking $\theta := \{\gamma' \in \Theta \mid \gamma' \leq \gamma\}$, it remains to show that no chip within θ disappears somewhere along the cycle. But this cannot happen, as it cannot be 'replaced underneath γ' before the bud is reached, as this would require removing γ from the control first. Thus, every cycle in $expand(\Pi)$ has an invariant θ with an accompanying $\max(\theta)$-reset, meaning $expand(\Pi)$ is a $\mathrm{R}(\mathcal{R})$-proof. □

We employ notion of proof search systems to prove completeness of $\mathrm{R}(\mathcal{R})$ relative $\mathcal{R}$: Every sequent provable in $\mathcal{R}$ can also be proven in $\mathrm{R}(\mathcal{R})$. Because the proof is based on a proof search procedure, the result we obtain is even stronger: The $\mathrm{R}(\mathcal{R})$-proof will essentially be an unfolding of the $\mathcal{R}$-proof. Fix a preproof (of any derivation system $\mathcal{S}$) $\Pi = ((T, \beta), \lambda, \delta)$, *unfolding* Π *at bud* $t \in \mathrm{dom}(\beta)$

yields the preproof $\Pi' := ((T', \beta'), \lambda', \delta')$ with

$$T' := T \cup \{tu \mid \beta(t)u \in T\} \qquad \lambda'(s) := \begin{cases} \lambda(s) & s \in T \\ \lambda(\beta(t)u) & s = tu \end{cases}$$

δ' defined analogously to λ' and $\beta' := \beta \setminus \{(t, \beta(t))\} \cup \{(tu, c)\}$ where $t = \beta(t)u$ and either $c = t$ or $c = \beta(t)$. A preproof Π' of Π is an *unfolding* of Π if Π' can be arrived at by repeatedly unfolding Π.

Theorem 5.3.8 (Completeness)**.** Let $(\textsc{Seq}, \mathcal{R}, \rho, \textsc{Pfs})$ be a cyclic proof system induced by a trace interpretation on $\mathcal{A}$. If there is a cyclic proof Π of $\Gamma \in \textsc{Seq}$ in $\mathcal{R}$ then there is a proof Π' of $\Gamma; (\emptyset, (x, a) \mapsto \emptyset)$ in $\mathrm{R}(\mathcal{R})$. Furthermore, $erase(\Pi')$ is an unfolding of Π.

Proof. Let $\Pi = (C, \lambda, \delta)$ be a proof of Γ in $\mathcal{R}$. Consider the subsystem $\mathcal{R}' := (\mathrm{im}(\lambda), \mathrm{im}(\delta), \rho \restriction \mathcal{R}', C \cap \textsc{Pp}(\mathcal{R}'))$ of $\mathcal{R}$. As Π is finite, so is $\mathcal{R}'$. Now fix $M := \{r_i \colon \iota(\Gamma) \to \iota(\Delta_i) \mid R \in \mathcal{R}', \rho(R) = (\Gamma, \Delta_1, \ldots, \Delta_n)\}$, where the r_i are given by the trace interpretation, and construct the Safra automaton $\mathfrak{S}(\mathcal{A}, i(\Gamma), M) = (M, Q, s, \delta_{\mathfrak{S}}, R_{\mathfrak{S}})$ according to definition 5.2.19. We construct the Rabin tree automaton $\mathfrak{A} = (\textsc{Seq}', Q', \Delta, s', R')$ with

$$\begin{aligned}
Q' &:= \{(s, (\Theta, \sigma)) \mid s \in C \setminus \mathrm{dom}(\beta), \Gamma := \lambda(s), (\iota(\Gamma), (\Theta, \sigma)) \in Q\} \\
\Delta' &:= \{((s, (\Theta, \sigma)), (\Delta_1, \ldots, \Delta_n), ((t_1, (\Theta_1, \sigma_n)), \ldots, (t_n, (\Theta_n, \sigma_n)))) \mid \\
&\qquad \text{if } s \in T \setminus \mathrm{Leaf}(T), \mathrm{Chld}(s) = \{t_1, \ldots, t_n\}, \rho(\delta(s)) = (\Gamma, \Delta_1, \ldots, \Delta_n), \\
&\qquad r_i \colon \iota(\Gamma) \to \iota(\Delta_i) \text{ given by the trace interpretation} \\
&\qquad \text{and } \delta_{\mathfrak{S}}((\iota(\Gamma), (\Theta, \sigma)), r_i) = (\iota(\Delta_i), (\Theta_i, \sigma_i))\} \\
s' &:= (\varepsilon, (\Theta_0, \sigma_0)) \text{ where } s = (\iota(\lambda(\varepsilon)), (\Theta_0, \sigma_0))
\end{aligned}$$

The Rabin condition R' is defined by taking $R' := \{(G', B') \mid (G, B) \in R_{\mathfrak{S}}\}$ in which $X' := \{(\{(s, (\Theta, \sigma)) \mid (\iota(\lambda(s)), (\Theta, \sigma)) \in X\}$. It is easily observed that $L(\mathfrak{A})$ contains precisely the 'infinite unfolding' of Π which has a successful run of $\textsc{Sb}(\mathcal{A}, i(\Gamma), K)$ along the paths $P \colon \omega \to \mathcal{A}$ of each of their branches. In other words, the only tree in $L(\mathfrak{A})$ corresponds to the unfolding of Π. By proposition 5.1.5, there exists a regular tree in $L(\mathfrak{A})$ which has a regular run on $\mathfrak{A}$. This run may be turned into an $\mathrm{S}(\mathcal{R}')$-preproof Π' by replacing each step $((s, (\Theta, \sigma)), \langle \Delta_1, \ldots, \Delta_n \rangle, ((t_1, (\Theta_1, \sigma_n)), \ldots, (t_n, (\Theta_n, \sigma_n))))$ corresponding to the rule $R := \delta(s)$ with the corresponding $\mathrm{S}(\mathcal{R}')$-rule $\mathrm{R}(\Theta, \sigma)$:

$$\mathrm{R}(\Theta, \sigma)\frac{\Delta_1; (\Theta_1, \sigma_1) \quad \ldots \quad \Delta_n; (\Theta_n, \sigma_n)}{\Gamma; (\Theta, \sigma)}$$

As the run satisfies the Rabin condition R', the preproof Π' satisfies the soundness condition of $\mathrm{S}(\mathcal{R}')$. The conclusion of Π' is $\Gamma; (\emptyset, (x, a) \mapsto \emptyset)$ as this corresponds to the initial state s' of $\mathcal{A}$. Now, by lemma 5.3.7, $expand(\Pi')$ is an $\mathrm{R}(\mathcal{R}')$-proof (and thus an $\mathrm{R}(\mathcal{R})$-proof). As the states of $\mathfrak{A}$ are labelled by the nodes of C, the regular run on $\mathfrak{A}$ must correspond to an unfolding of Π. Thus, $erase(\Pi')$ must be an unfolding of Π as well. $\square$

5.4 Examples

In this section, we apply the results do derive reset proof systems for the two running examples of this dissertation: $\mathrm{CHA}_<$ (section 5.4.1) and $\mathrm{CHA}_<$ (section 5.4.2). Leigh and Wehr (2024) presents two more examples: cyclic Gödel's T and the modal μ-calculus, which we have chosen to omit from this dissertation. For each system, the abstract reset system $\mathrm{R}(\mathcal{R})$ will serve as a starting point. However, the reset systems we derive all differ from the 'naïve' system $\mathrm{R}(\mathcal{R})$ by a few 'ergonomic' adjustments and a more syntactic annotation mechanism.

5.4.1 Simpson's $\mathrm{CHA}_<$

We propose a reset proof system for Simpson-style cyclic Heyting arithmetic (see section 2.3). We call the system *reset Heyting arithmetic* $\mathrm{RHA}_<$. It is based on the reset system $\mathrm{R}(\mathrm{CHA}_<)$ induced by $\mathrm{CHA}_<$ with some slight modifications.

Sequents of $\mathrm{RHA}_<$ are expressions $\Theta; \sigma \mid \Gamma \Rightarrow \delta$ where $\Gamma \Rightarrow \delta$ is a $\mathrm{CHA}_<$-sequent, Θ is a sequence of distinct characters called the *control* and σ is a finite set of *assignments* $t \mapsto u$ where t is a term in Γ, δ and u is a subsequence $u \sqsubseteq \Theta$. The set of $\mathrm{RHA}_<$-sequents is denoted $\textsc{Seq}_{\mathrm{RHA}_<}$. For an assignment $t \mapsto u$, write $(t \mapsto u)[s/x]$ for $t[s/x] \mapsto u$. This notation extends to sets of assignments $\sigma[s/x]$. Sequents $\varepsilon; \emptyset \mid \Gamma \Rightarrow \delta$ with empty control are identified with $\mathrm{CHA}_<$-sequents $\Gamma \Rightarrow \delta$.

Definition 5.4.1. The *derivation rules* of Simpson's reset Heyting arithmetic are listed below. In each rule, $\Theta'; \sigma'$ denotes the result of first removing all assignments to terms not occurring in the premise from σ and then removing all letters of Θ which are not assigned to at least one term. The rules of $\mathrm{RHA}_<$ contain the rules of $\mathrm{CHA}_<$ adjusted to 'properly treat' the control $\Theta; \sigma$. Observe that the $\textsc{Wk}$ also allows for the 'weakening' of the assignments σ.

$$\text{Ax}\frac{}{\Theta; \sigma \mid \Gamma, \delta \Rightarrow \delta} \qquad \to\text{L}\frac{\Theta'; \sigma' \mid \Gamma \Rightarrow \varphi \quad \Theta'; \sigma' \mid \Gamma, \psi \Rightarrow \delta}{\Theta; \sigma \mid \Gamma, \varphi \to \psi \Rightarrow \delta}$$

$$\to\mathrm{R}\frac{\Theta;\sigma \mid \Gamma,\varphi \Rightarrow \psi}{\Theta;\sigma \mid \Gamma \Rightarrow \varphi \to \psi} \qquad \wedge\mathrm{L}\frac{\Theta;\sigma \mid \Gamma,\varphi,\psi \Rightarrow \delta}{\Theta;\sigma \mid \Gamma,\varphi \wedge \psi \Rightarrow \delta}$$

$$\wedge\mathrm{R}\frac{\Theta';\sigma' \mid \Gamma \Rightarrow \varphi \quad \Theta';\sigma' \mid \Gamma \Rightarrow \psi}{\Theta;\sigma \mid \Gamma \Rightarrow \varphi \wedge \psi}$$

$$\vee\mathrm{L}\frac{\Theta';\sigma' \mid \Gamma,\varphi \Rightarrow \delta \quad \Theta';\sigma' \mid \Gamma,\psi \Rightarrow \delta}{\Theta;\sigma \mid \Gamma,\varphi \vee \psi \Rightarrow \delta} \qquad \vee\mathrm{R}\frac{\Theta;\sigma \mid \Gamma \Rightarrow \varphi_i}{\Theta;\sigma \mid \Gamma \Rightarrow \varphi_1 \vee \varphi_2}$$

$$\forall\mathrm{L}\frac{\Theta;\sigma \mid \Gamma,\varphi[t/x] \Rightarrow \delta}{\Theta;\sigma \mid \Gamma,\forall x.\varphi \Rightarrow \delta} \qquad \forall\mathrm{R}\frac{\Theta;\sigma \mid \Gamma \Rightarrow \varphi \quad x \notin \mathrm{FV}(\Gamma,\forall x.\varphi)}{\Theta;\sigma \mid \Gamma \Rightarrow \forall x.\varphi}$$

$$\exists\mathrm{L}\frac{\Theta;\sigma \mid \Gamma,\varphi \Rightarrow \delta \quad x \notin \mathrm{FV}(\Gamma,\exists x.\varphi,\delta)}{\Theta;\sigma \mid \Gamma,\exists x.\varphi \Rightarrow \delta} \qquad \exists\mathrm{R}\frac{\Theta;\sigma \mid \Gamma \Rightarrow \varphi[t/x]}{\Theta;\sigma \mid \Gamma \Rightarrow \exists x.\varphi}$$

$$\bot\mathrm{L}\frac{}{\Theta;\sigma \mid \Gamma,\bot \Rightarrow \delta} \qquad =\!\mathrm{R}\frac{}{\Theta;\sigma \mid \Gamma \Rightarrow t = t}$$

$$\textsc{Wk}\frac{\Theta';\sigma' \mid \Gamma \Rightarrow \delta}{\Theta;\sigma,\sigma^* \mid \Gamma,\Gamma' \Rightarrow \delta} \qquad \textsc{Cut}\frac{\Theta;\sigma \mid \Gamma \Rightarrow \varphi \quad \Theta;\sigma \mid \Gamma,\varphi \Rightarrow \delta}{\Theta;\sigma \mid \Gamma \Rightarrow \delta}$$

$$\mathrm{S}\frac{\Theta;\sigma \mid \Gamma, t = Sx \Rightarrow \delta \quad x \text{ fresh}}{\Theta;\sigma \mid \Gamma, 0 < t \Rightarrow \delta}$$

The rules =L and Sub 'imported' from $\mathrm{CHA}_<$ need more careful treatment because of their effects on traces. The rule Sub is defined as below:

$$\textsc{Sub}\frac{\Theta';\sigma^* \mid \Gamma \Rightarrow \delta}{\Theta;\sigma[s/x] \mid \Gamma[s/x] \Rightarrow \delta[s/x]}$$

where $\sigma^* := \{t \mapsto u \mid t \in \textsc{Term}(\Gamma \Rightarrow \delta) \text{ and } (t[s/x] \mapsto u) \in \sigma\}$. The rule =L can similarly be defined as

$$=\!\mathrm{L}\frac{\Theta;\sigma^* \mid \Gamma[t/x,s/y] \Rightarrow \delta[t/x,s/y] \quad x,y \notin \mathrm{FV}(s,t)}{\Theta;\sigma \mid \Gamma[s/x,t/y], s = t \Rightarrow \delta[s/x,t/y]}$$

where

$$\begin{aligned}\sigma^* := \{b[t/x,s/y] \mapsto u \mid (b[s/x,t/y] \mapsto u) \in \sigma, b \in \textsc{Term} \text{ and}\\ b[t/x,s/y] \in \textsc{Term}(\Gamma[t/x,s/y],\delta[t/x,s/y])\}.\end{aligned}$$

The axioms of $\mathrm{RHA}_<$ are the arithmetical *axioms* of $\mathrm{CHA}_<$ listed below. This means $\Theta;\sigma \mid \Gamma \Rightarrow \delta$ for any $\mathrm{CHA}_<$-sequent $\Gamma \Rightarrow \delta$ below and any control $\Theta;\sigma$

is an axiom of RA.

$$
\begin{array}{rlrl}
s < t, t < u & \Rightarrow s < u & s < t & \Rightarrow Ss < St \\
& \Rightarrow s + St = S(s+t) & s < t, t < s & \Rightarrow \bot \\
& \Rightarrow s < t \lor s = t \lor t < s & & \Rightarrow t \cdot 0 = 0 \\
s < t, t < Ss & \Rightarrow \bot & & \Rightarrow t < St \\
& \Rightarrow s \cdot St = (s \cdot t) + s & t < 0 & \Rightarrow \bot \\
& \Rightarrow t + 0 = t & &
\end{array}
$$

Lastly, $\mathrm{RHA}_<$ features three derivation rules which have no corresponding rule in $\mathrm{CHA}_<$.

$$
\textsc{Focus}\,\frac{\Theta; \sigma, (t \mapsto \varepsilon) \mid \Gamma \Rightarrow \delta \quad t \in \textsc{Term}(\Gamma, \delta)}{\Theta; \sigma \mid \Gamma \Rightarrow \delta}
$$

$$
\textsc{Reset}_a\,\frac{\Theta'; \sigma, (t_1 \mapsto ua), \ldots, (t_n \mapsto ua) \mid \Gamma \Rightarrow \delta}{\Theta; \sigma, (t_1 \mapsto uau_1), \ldots, (t_n \mapsto uau_n) \mid \Gamma \Rightarrow \delta}
$$

$$
{<}\mathrm{L}\,\frac{\Theta a; \sigma, (s \mapsto ua) \mid \Gamma \Rightarrow \delta \quad a \text{ fresh}}{\Theta; \sigma, (t \mapsto u) \mid \Gamma, s < t \Rightarrow \delta}
$$

where in $\textsc{Reset}_a$, a may not occur in annotations in σ and the u_i must be non-empty.

An $\mathrm{RHA}_<$-preproof satisfies the soundness condition of $\mathrm{RHA}_<$ if every pair of bud $t \in \mathrm{dom}(\beta)$ and companion $\beta(t)$ has an *invariant*: there exists a letter a such that a occurs in all of the controls between t and $\beta(t)$, the prefix of a is constant across these controls, and the $\textsc{Reset}_a$ rule is applied between t and $\beta(t)$. An $\mathrm{RHA}_<$-proof is a *proof of* $\mathrm{CHA}_<$*-sequent* $\Gamma \Rightarrow \delta$ if its root is labelled $\varepsilon; \emptyset \mid \Gamma \Rightarrow \delta$. If there exists an $\mathrm{RHA}_<$-proof of $\Gamma \Rightarrow \delta$ write $\mathrm{RA} \vdash \Gamma \Rightarrow \delta$.

The proof system $\mathrm{RHA}_<$ features one 'ergonomic adjustment' differentiating it from the 'naïve' reset system $\mathrm{R}(\mathrm{CHA}_<)$. In $\mathrm{R}(\mathrm{CHA}_<)$, the rules corresponding to $\mathrm{CHA}_<$-rules add new chips to the control (Θ, σ) whenever progress takes place i.e., whenever there are inequalities $s < t$ in Γ of the conclusion. This can quickly get out hand, making the handling of the control quite unwieldy. In $\mathrm{RHA}_<$, the $\mathrm{CHA}_<$-correspondents never add chips to the control, only remove them if they are no longer used. Instead, the <L rule allows the prover to add chips corresponding to the progress embodied by an inequality $s < t$ in Γ.

Before we continue proving soundness and completeness of $\mathrm{RHA}_<$, we make a general observation about $(\Theta, \sigma) \in \textsc{Sb}(\mathbb{B}, X)$: Because $\mathbb{B} = \{0, 1\}$, we have

$\mathrm{dom}(\sigma) = X \times (\mathbb{B} \setminus \{1\}) = X \times \{0\}$. From now on, we thus treat σ as a function with $\mathrm{dom}(\sigma) = X$ instead to reduce clutter.

We prove soundness of $\mathrm{RHA}_<$ relative to HA by constructing a proof morphism $\mathit{embed}\colon \mathrm{RHA}_< \to \mathrm{R}(\mathrm{CHA}_<)$.

Lemma 5.4.2. There is a function $\mathit{embed}\colon \textsc{Seq}_{\mathrm{RHA}_<} \to \textsc{Seq}_{\mathrm{R}(\mathrm{CHA}_<)}$ which is defined by

$$\mathit{embed}(\Theta; \sigma \mid \Gamma \Rightarrow \delta) := \Gamma \Rightarrow \delta; (\overline{\Theta}, \overline{\sigma})$$

where $\overline{\Theta}$ is the set $\{a \mid a \in \Theta\}$ ordered according to the letters' positions in Θ and $\overline{\sigma}(t) := \{\{x \mid x \in u\} \mid t \mapsto u \in \sigma\}$. It can be extended into a proof morphism $\mathit{embed}\colon \mathrm{RHA}_< \to \mathrm{R}(\mathrm{CHA}_<)$.

Proof. To translate rules corresponding to $\mathrm{CHA}_<$-rules, we need to account for the difference in how $\mathrm{RHA}_<$ and $\mathrm{R}(\mathrm{CHA}_<)$ treat inequalities left of $\Rightarrow$ explained above. Pick a rule $R \in \mathrm{RHA}_<$ with a corresponding $\mathrm{CHA}_<$-rule (i.e. R is not an instance of <L, Wk, Focus or Reset). It is translated as the $\mathrm{R}(\mathrm{CHA}_<)$-preproof as below. Here, we denote the chips that were 'erroneously' added by R by u_i and by σ'_i the 'erroneous' stacks. For this, we employ the notation $\sigma \cup \sigma'$ to denote the function $(x, a) \mapsto \sigma(x, a) \cup \sigma'(x, a)$. Observe that for every $s < t \in \Gamma_i$, the trace map r_i dictating the Safra board transition $(\overline{\Theta}, \overline{\sigma}) \overset{r_i}{\rightsquigarrow} (\overline{\Theta_i} \oplus u_i, \overline{\sigma_i} \cup \sigma'_i)$ contains two transitions concerning the predecessor $t' \xleftarrow{i}_R t$ of t: $(t, 0, t')$ and $(t, 1, s)$. The latter is the cause of a chip being added 'erroneously'. The former ensures that the stacks on t are not removed from $\overline{\sigma_i}$ but simply 'reassigned' to t', just as is done in $\mathrm{RHA}_<$. This guarantees that $\overline{\sigma_i}$ is part of the control in each of the premises.

$$\dfrac{\Theta_1; \sigma_1 \mid \Gamma_1 \Rightarrow \delta_1 \quad \cdots \quad \Theta_n; \sigma_n \mid \Gamma_n \Rightarrow \delta_n}{\Theta; \sigma \mid \Gamma \Rightarrow \delta}\,\mathrm{R} \qquad \overset{\mathit{embed}}{\rightsquigarrow}$$

$$\dfrac{\dfrac{\Gamma_1 \Rightarrow \delta_1; (\overline{\Theta_1}, \overline{\sigma_1})}{\Gamma_1 \Rightarrow \delta_1; (\overline{\Theta_1} \oplus u_1, \overline{\sigma_1} \cup \sigma'_1)}\,\textsc{Weak} \qquad \cdots \qquad \dfrac{\Gamma_n \Rightarrow \delta_n; (\overline{\Theta_n}, \overline{\sigma_n})}{\Gamma_n \Rightarrow \delta_n; (\overline{\Theta_n} \oplus u_n, \overline{\sigma_n} \cup \sigma'_n)}\,\textsc{Weak}}{\Gamma \Rightarrow \delta; (\overline{\Theta}, \overline{\sigma})}\,\mathrm{R}$$

The Reset and Focus rules correspond to applications of the Reset and Pop rules in $\mathrm{R}(\mathrm{CHA}_<)$. The Wk is translated to a combination of the Wk-rule of CA to 'weaken' in $\Gamma \Rightarrow \delta$ (taking care of 'erroneous' chips as above) and the Weak-rule of $\mathrm{R}(\mathrm{CHA}_<)$ to 'weaken' in σ. The only rule for which the translation via *embed* remain open is <L. This translation is achieved by a (possibly vacuous)

application of the Wk-rule from $\mathrm{CHA}_<$, 'simulating' the removal of the inequality $s < t$, followed by the Weak-rule of $\mathrm{R}(\mathrm{CHA}_<)$ to remove all 'erroneously' added chips and stacks i.e., all except the one induced by the inequality $s < t$.

$$\dfrac{\Theta a; \sigma, (s \mapsto ua) \mid \Gamma \Rightarrow \delta \quad a \text{ fresh}}{\Theta; \sigma, (t \mapsto u) \mid \Gamma, s < t \Rightarrow \delta}{<}\mathrm{L} \qquad \overset{embed}{\rightsquigarrow}$$

$$\dfrac{\dfrac{\Gamma \Rightarrow \delta; (\overline{\Theta a}, \overline{\sigma, (s,0) \mapsto ua})}{\Gamma \Rightarrow \delta; (\overline{\Theta a v}, \overline{\sigma, (t \mapsto u)} \cup \overline{(s,0) \mapsto ua} \cup \sigma')}\textsc{Weak}}{\Gamma, s < t \Rightarrow \delta; (\overline{\Theta}, \overline{\sigma, (t \mapsto u)})}\textsc{Wk}$$

By comparing the soundness conditions of $\mathrm{RHA}_<$ and $\mathrm{R}(\mathrm{CHA}_<)$, it is easily observed that *embed* maintains the soundness condition of $\mathrm{RHA}_<$. □

Corollary 5.4.3 (Soundness)**.** If $\mathrm{RHA}_< \vdash \Gamma \Rightarrow \delta$ then $\mathrm{HA} \vdash \Gamma \Rightarrow \delta$.

Proof. If Π is a proof of $\varepsilon; \emptyset \mid \Gamma \Rightarrow \delta$ in $\mathrm{RHA}_<$ then $embed(\Pi)$ is a proof of $\Gamma \Rightarrow \delta; (\emptyset, ((x,a) \mapsto \emptyset))$ in $\mathrm{R}(\mathrm{CHA}_<)$ and thus $erase(embed(\Pi))$ a proof of $\Gamma \Rightarrow \delta$ in $\mathrm{CHA}_<$. As $\mathrm{CHA}_<$ proves the same sequents as HA (see theorem 7.1.3) there must also be a proof of $\Gamma \Rightarrow \delta$ in HA. □

Let $\mathcal{F}$ be a finite fragment of $\mathrm{CHA}_<$. To conclude completeness of $\mathrm{RHA}_<$ relative to HA, we construct a proof morphism $search\colon \mathrm{S}(\mathcal{F}) \to \mathrm{RHA}_<$. For this, recall that $\mathrm{S}(\mathcal{F})$ is the 'proof search system' induced by the derivation rules in $\mathcal{F}$ and the trace interpretation of $\mathrm{CHA}_<$.

Lemma 5.4.4. There is a function $search\colon \mathrm{S}(\textsc{Seq}_{\mathcal{F}}) \to \textsc{Seq}_{\mathrm{RHA}_<}$ with

$$search(\Gamma \Rightarrow \delta; (\Theta, \sigma)) := \widehat{\Theta}; \widehat{\sigma} \vdash \Gamma \Rightarrow \delta$$

where $\widehat{S} \in \Theta^*$ for $S \subseteq \Theta$ is the duplicate-free sequence of length $|\Theta|$ which is strictly sorted according to Θ and, if (Θ, σ) is a K-sparse Safra board on some $\iota(\Gamma \Rightarrow \delta)$, then

$$\begin{aligned} \widehat{\sigma} := \{s \mapsto \widehat{S} \mid s \in \textsc{Term}(\Gamma, \delta) \text{ with } \sigma(s,0) = \{S\}\} \cup \\ \{s \mapsto \varepsilon \mid s \in \textsc{Term}(\Gamma, \Delta) \text{ with } \sigma(s,0) = \emptyset\} \end{aligned}$$

The function can be extended into a proof morphism $search\colon \mathrm{S}(\mathcal{F}) \to \mathrm{RHA}_<$.

Proof. Towards this claim, first pick some $\mathrm{R}(\Theta, \sigma) \in \mathrm{S}(\mathcal{F})$ arranged as follows

$$\dfrac{\Gamma_1 \Rightarrow \delta_1; (\Theta_1, \sigma_1) \quad \dots \quad \Gamma_n \Rightarrow \delta_n; (\Theta_n, \sigma_n)}{\Gamma \Rightarrow \delta; (\Theta, \sigma)}\mathrm{R}(\Theta, \sigma)$$

Then there is $R \in \mathcal{F}$ with $\rho(R) = (\Gamma \Rightarrow \delta, \Gamma_1 \Rightarrow \delta_1, \ldots, \Gamma_n \Rightarrow \delta_n)$ andtrace maps $r_i\colon \iota(\Gamma \Rightarrow \delta) \to \iota(\Gamma_i \Rightarrow \delta_i)$ given by the trace interpretation. Then for each $i \leq n$ there is $(\Theta, \sigma) \overset{r_i}{\rightsquigarrow}_g (\Theta_i, \sigma_i)$ with the expanded sequence the expanded sequence

$$(\Theta, \sigma) \overset{R_{\gamma_1}}{\rightsquigarrow} (\Theta^1_r, \sigma^1_r) \ldots \overset{R_{\gamma_k}}{\rightsquigarrow} (\Theta^k_r, \sigma^k_r) \overset{P}{\rightsquigarrow} (\Theta_p, \sigma_p) \overset{r_i}{\rightsquigarrow} (\Theta^*_i, \sigma^*_i) \overset{T}{\rightsquigarrow} (\Theta_i, \sigma_i)$$

in which the initial R_γ- and P-steps are shared between all $i \leq n$ (see lemma 5.2.18). Similarly to lemma 5.3.7, we may derive the following in $\mathrm{RHA}_<$:

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\widehat{\Theta_1}; \widehat{\sigma_1} \mid \Gamma_1 \Rightarrow \delta_1}{\widehat{\Theta'_1}; \widehat{\sigma'_1} \mid \Gamma_1 \Rightarrow \delta_1}\,\textsc{Wk}}{\vdots}}{\widehat{\Theta_p}; \widehat{\sigma_p} \mid \Gamma_1 \Rightarrow \delta_1}\,{<}\mathrm{L} \qquad \ldots \qquad \dfrac{\dfrac{\dfrac{\widehat{\Theta_n}; \widehat{\sigma_n} \mid \Gamma_n \Rightarrow \delta_n}{\widehat{\Theta'_n}; \widehat{\sigma'_n} \mid \Gamma_n \Rightarrow \delta_n}\,\textsc{Wk}}{\vdots}}{\widehat{\Theta_p}; \widehat{\sigma_p} \mid \Gamma_n \Rightarrow \delta_n}\,{<}\mathrm{L}}{\widehat{\Theta_p}; \widehat{\sigma_p} \mid \Gamma \Rightarrow \delta}\,\mathrm{R}}{\vdots}}{\widehat{\Theta^k_r}; \widehat{\sigma^k_r} \mid \Gamma \Rightarrow \delta}\,\textsc{Focus}}{\vdots}}{\widehat{\Theta_0}; \widehat{\sigma_0} \mid \Gamma \Rightarrow \delta}\,\textsc{Reset}_a$$

That is, first apply all possible $\textsc{Reset}_a$-rules, starting at the Θ_0-greatest a. Then $\textsc{Focus}$ on each $t \in \textsc{Term}(\Gamma \Rightarrow \delta)$ with no $t \mapsto u \in \widehat{\sigma^k_r}$. After applying the RA-rule corresponding to $R \in \mathcal{F}$, apply various instances of <L carefully, as described in the next paragraph. Close each branch of the preproof with an application of $\textsc{Wk}$ which removes all superfluous annotations from the $\widehat{\sigma_i}$.

The application of the <L-instances requires a little more consideration: If $s < t, s < t' \in \Gamma_i$ with $t \neq t'$ then $\sigma^*_i(s, 0)$ will contain the annotations $\sigma_p(t, 0), \sigma_p(t', 0)$ extended by the same $\gamma \in \Theta^*_i$. In RA, on the other hand, the annotations of t and t' can only be extended with separate applications of the <L-rule, meaning the annotations will be extended with two different chips γ, γ'. Observe, however, that after the thinning step $(\Theta^*_i, \sigma^*_i) \overset{T}{\rightsquigarrow} (\Theta_i, \sigma_i)$ only one annotation remains in $\sigma_1(s, 0)$. Thus, the preproof pictured above only applies the <L-instance yielding this 'surviving' annotation of s with the 'correct' chip γ. As visible in the preproof above, this results in a sequent $\widehat{\Theta'_1}; \widehat{\sigma'_1} \mid \Gamma_1 \Rightarrow \delta_1$ rather than the 'naïve' sequent $\widehat{\Theta^*_1}; \widehat{\sigma^*_1} \mid \Gamma_1 \Rightarrow \delta_1$. However, the application of $\textsc{Wk}$ then yields the desired sequent at the leaf.

An argument analogous to that given for *expand* in lemma 5.3.7 shows that *search* maintains the soundness condition. □

Corollary 5.4.5 (Completeness)**.** If $\mathrm{HA} \vdash \Gamma \Rightarrow \delta$ then $\mathrm{RHA}_< \vdash \Gamma \Rightarrow \delta$.

Proof. Suppose there is a proof of $\Gamma \Rightarrow \delta$ in HA. By proposition 2.3.5 there is a proof of the same sequent in $\mathrm{CHA}_<$. Indeed, as cyclic proofs are finite trees, this proof is made in some finite fragment $\mathcal{F}$ of $\mathrm{CHA}_<$. By theorem 5.3.8, there is a proof Π of $\Gamma \Rightarrow \delta; (\emptyset, (x, a) \mapsto \emptyset)$ in $\mathrm{S}(\mathcal{F})$ and thus $search(\Pi)$ is a proof of $\varepsilon; \emptyset \mid \Gamma \Rightarrow \delta$ in $\mathrm{RHA}_<$. □

5.4.2 Simple CHA

Definition 5.4.6 (Reset Heyting arithmetic)**.** The *sequents* of *reset Heyting arithmetic* (RHA) are expressions $\Theta; \sigma \mid \Gamma \Rightarrow \delta$ consisting of a sequent $\Gamma \Rightarrow \delta$ of first-order arithmetic, a finite, repetition-free sequence Θ of distinct letters called the *control* and an *assignment* $\sigma : \mathrm{FV}(\Gamma, \delta) \to \textsc{Sub}(\Theta)$ mapping the free variables of $\Gamma \Rightarrow \delta$ to subsequences of Θ. The sequence $\sigma(x)$ is called the *annotation* of x.

The derivation rules of RHA are given below. In rules in which free variables in the conclusion might not be present in the premises, σ' denotes the restriction of σ to the remaining free variables and Θ' the subsequence of Θ which only contains letters which occur in at least one annotation $\sigma'(x)$. Conversely, in rules in which new free variables might be introduced in the premise, σ^+ denotes the extension of σ which annotates each new variable with the empty sequence ε.

$$\text{Ax}\frac{}{\Theta; \sigma \mid \Gamma, \delta \Rightarrow \delta} \qquad \to\text{L}\frac{\Theta'; \sigma' \mid \Gamma \Rightarrow \varphi \quad \Theta'; \sigma' \mid \Gamma, \psi \Rightarrow \delta}{\Theta; \sigma \mid \Gamma, \varphi \to \psi \Rightarrow \delta}$$

$$\to\text{R}\frac{\Theta; \sigma \mid \Gamma, \varphi \Rightarrow \psi}{\Theta; \sigma \mid \Gamma \Rightarrow \varphi \to \psi} \qquad \wedge\text{L}\frac{\Theta; \sigma \mid \Gamma, \varphi, \psi \Rightarrow \delta}{\Theta; \sigma \mid \Gamma, \varphi \wedge \psi \Rightarrow \delta}$$

$$\wedge\text{R}\frac{\Theta'; \sigma' \mid \Gamma \Rightarrow \varphi \quad \Theta'; \sigma' \mid \Gamma \Rightarrow \psi}{\Theta; \sigma \mid \Gamma \Rightarrow \varphi \wedge \psi}$$

$$\vee\text{L}\frac{\Theta'; \sigma' \mid \Gamma, \varphi \Rightarrow \delta \quad \Theta'; \sigma' \mid \Gamma, \psi \Rightarrow \delta}{\Theta; \sigma \mid \Gamma, \varphi \vee \psi \Rightarrow \delta} \qquad \vee\text{R}\frac{\Theta; \sigma \mid \Gamma \Rightarrow \varphi_i}{\Theta; \sigma \mid \Gamma \Rightarrow \varphi_1 \vee \varphi_2}$$

$$\forall\text{L}\frac{\Theta; \sigma^+ \mid \Gamma, \varphi[t/x] \Rightarrow \delta}{\Theta; \sigma \mid \Gamma, \forall x.\varphi \Rightarrow \delta} \qquad \forall\text{R}\frac{\begin{array}{c} x \notin \mathrm{FV}(\Gamma, \forall x.\varphi) \\ \Theta; \sigma^+ \mid \Gamma \Rightarrow \varphi \end{array}}{\Theta; \sigma \mid \Gamma \Rightarrow \forall x.\varphi}$$

$$\exists\text{L}\frac{\begin{array}{c} x \notin \mathrm{FV}(\Gamma, \exists x.\varphi, \delta) \\ \Theta; \sigma^+ \mid \Gamma, \varphi \Rightarrow \delta \end{array}}{\Theta; \sigma \mid \Gamma, \exists x.\varphi \Rightarrow \delta} \qquad \exists\text{R}\frac{\Theta; \sigma^+ \mid \Gamma \Rightarrow \varphi[t/x]}{\Theta; \sigma \mid \Gamma \Rightarrow \exists x.\varphi}$$

$$\bot\text{L}\,\frac{}{\Theta; \sigma \mid \Gamma, \bot \Rightarrow \delta}$$

$$=\text{L}\,\frac{\Theta; \sigma[t/x, s/y] \mid \Gamma[t/x, s/y] \Rightarrow \delta[t/x, s/y] \quad x, y \notin \mathrm{FV}(s, t)}{\Theta; \sigma[s/x, t/y] \mid \Gamma[s/x, t/y], s = t \Rightarrow \delta[s/x, t/y]}$$

$$=\text{R}\,\frac{}{\Theta; \sigma \mid \Gamma \Rightarrow t = t} \qquad \textsc{Wk}\,\frac{\Theta'; \sigma' \mid \Gamma \Rightarrow \delta}{\Theta; \sigma \mid \Gamma, \Gamma' \Rightarrow \delta}$$

$$\textsc{Sub}\,\frac{x \notin \mathrm{FV}(t) \quad \Theta'; \sigma'^{+} \mid \Gamma \Rightarrow \delta}{\Theta; \sigma \mid \Gamma[t/x] \Rightarrow \delta[t/x]}$$

$$\textsc{Cut}\,\frac{\Theta'; \sigma'^{+} \mid \Gamma \Rightarrow \varphi \quad \Theta; \sigma^{+} \mid \Gamma, \varphi \Rightarrow \delta}{\Theta; \sigma \mid \Gamma \Rightarrow \delta}$$

$$\textsc{Case}_x\,\frac{\Theta'; \sigma' \mid \Gamma(0) \Rightarrow \delta(0) \quad \overset{a \notin \Theta}{\Theta a; \sigma[x \mapsto ua] \mid \Gamma(Sx) \Rightarrow \delta(Sx)}}{\Theta; \sigma[x \mapsto u] \mid \Gamma(x) \Rightarrow \delta(x)}$$

$$\textsc{Clear}\,\frac{\Theta'; \sigma[x \mapsto \varepsilon] \mid \Gamma \Rightarrow \delta}{\Theta; \sigma[x \mapsto u] \mid \Gamma \Rightarrow \delta} \qquad \textsc{Reset}_a\,\frac{\overset{u' \text{ non-empty}}{\Theta'; \sigma[x \mapsto ua] \mid \Gamma \Rightarrow \delta}}{\Theta; \sigma[x \mapsto uau'] \mid \Gamma \Rightarrow \delta}$$

Furthermore, the sequents $\Theta; \sigma \mid \Gamma \Rightarrow \delta$, in which $\Gamma \Rightarrow \delta$ is taken from the sequents below, are axioms.

$$\Rightarrow 0 \neq St \qquad Ss = St \Rightarrow s = t \qquad \Rightarrow s + 0 = 0$$
$$\Rightarrow s + St = S(s + t) \qquad \Rightarrow s \cdot 0 = 0 \qquad \Rightarrow s \cdot St = (s \cdot t) + s$$

The preproofs of RHA are defined analogously to those of CHA. That is, they are a pair $\Pi = (C, \lambda)$ consisting a cyclic tree C in cycle normal form and a labelling function λ. The labelling λ must be according to the derivation rules given above, labelling non-bud leaves with axioms and satisfying $\lambda(\beta(s)) = \lambda(s)$ for $s \in \mathrm{dom}(\beta)$.

An RHA-preproof is an RHA-proof if every local cycle $\gamma(t)$ has an *invariant:* a sequence θa which is a prefix to all controls along the local cycle $\gamma(t)$ of t and $\beta(t)$ and the $\textsc{Reset}_a$ rule is applied along $\gamma(t)$. We often refer to the longest invariant of a local cycle as *the* invariant of said cycle. Write RHA $\vdash \Gamma \Rightarrow \delta$ if there is an RHA-proof of $\varepsilon; \{x \mapsto \varepsilon\} \mid \Gamma \Rightarrow \delta$ i.e., a proof whose endsequent has an empty control and in which all variables are assigned the empty sequence.

The main ergonomic 'adjustment' of RHA is restricting σ to always be a function i.e., every variable is always assigned exactly one stack. Because CHA-traces can never merge i.e., they only every follow a single, fixed variable, there is no need to account for the possibility of a variable being assigned more than one stack. Thus, the σ of R(CHA), and thus of RHA, naturally already is a partial function. Again, to ease usability, we thus 'complete' the σ of RHA into a function, assigning every variable the empty stack by default.

We prove soundness of RHA relative to HA by constructing a proof morphism $\mathit{embed}\colon \mathrm{RHA} \to \mathrm{R(CHA)}$.

Lemma 5.4.7. There is a function $\mathit{embed}\colon \textsc{Seq}_{\mathrm{RHA}} \to \textsc{Seq}_{\mathrm{R(CHA)}}$ which is defined by

$$\mathit{embed}(\Theta; \sigma \mid \Gamma \Rightarrow \delta) := \Gamma \Rightarrow \delta; (\overline{\Theta}, \overline{\sigma})$$

where $\overline{\Theta}$ is the set $\{a \mid a \in \Theta\}$ ordered according to the letters' positions in Θ and $\overline{\sigma}(x) := \{\{a \mid a \in \sigma(x)\} \mid x \in \mathrm{FV}(\Gamma, \delta)\}$. It can be extended into a proof morphism $\mathit{embed}\colon \mathrm{RHA} \to \mathrm{R(CHA)}$.

Proof. First of all, observe that for any trace map $\tau : X \to Y$ used to specify the trace condition in CHA in example 4.3.4 has the property that for any $(x, a, y), (x, b, z) \in \tau$ we have $a = b$ and $x = y = z$. Thus, any of the RHA rules except for Clear and Reset can be translated quite directly to R(CHA): Let r be such a rule, then there exists rule $r' \in$ CHA directly corresponding to it. Then r can be translated as follows:

$$\dfrac{\Theta_1; \sigma_1 \mid \Gamma_1 \Rightarrow \delta_1 \quad \cdots \quad \Theta_n; \sigma_n \mid \Gamma_n \Rightarrow \delta_n}{\Theta; \sigma \mid \Gamma \Rightarrow \delta}\,\mathrm{r} \qquad \overset{\mathit{embed}}{\rightsquigarrow}$$

$$\dfrac{\dfrac{\Gamma_1 \Rightarrow \delta_1; (\overline{\Theta_1}, \overline{\sigma_1})}{\Gamma_1 \Rightarrow \delta_1; (\overline{\Theta_1}, \sigma_1^*)}\,\textsc{Pop} \quad \cdots \quad \dfrac{\Gamma_n \Rightarrow \delta_n; (\overline{\Theta_n}, \overline{\sigma_n})}{\Gamma_n \Rightarrow \delta_n; (\overline{\Theta_n}, \sigma_n^*)}\,\textsc{Pop}}{\Gamma \Rightarrow \delta; (\overline{\Theta}, \overline{\sigma})}\,\mathrm{R}$$

By the previously noted property of trace maps, it is easily observed that always $\Theta_i^* = \overline{\Theta_i}$ as desired. Furthermore, the only way for σ_i^* and $\overline{\sigma_i}$ can differ is if new variables were introduced by $\Gamma_i \Rightarrow \delta_i$ which do not yet have empty stacks (denoted by the $-^+$-notation in the rules of RHA). This can be simulated with an application of Pop.

For the Reset-rule, observe that the Reset-rule of R(CHA) on its own is *precisely* the rule which the Reset-rule of RHA should be translated to. Lastly, the Clear-rule can be translated as a combination of Weak and Pop as follows:

$$\textsc{Clear}\,\frac{\Theta'; \sigma[x \mapsto \varepsilon] \mid \Gamma \Rightarrow \delta}{\Theta; \sigma[x \mapsto u] \mid \Gamma \Rightarrow \delta} \quad \overset{embed}{\rightsquigarrow}$$

$$\textsc{Weak}\,\frac{\textsc{Pop}\,\dfrac{\Gamma \Rightarrow \delta; (\overline{\Theta'}, \overline{\sigma}[x \mapsto \{\emptyset\}])}{\Gamma \Rightarrow \delta; (\overline{\Theta'}, \overline{\sigma}[x \mapsto \emptyset])}}{\Gamma \Rightarrow \delta; (\overline{\Theta}, \overline{\sigma}[x \mapsto \{\{a \in u\}\}])}$$

By comparing the soundness conditions of RHA and R(CHA), it is easily observed that *embed* maintains the soundness condition of RHA. □

Corollary 5.4.8 (Soundness)**.** If RHA $\vdash \Gamma \Rightarrow \delta$ then HA $\vdash \Gamma \Rightarrow \delta$.

Proof. If Π is a proof of $\varepsilon; \{x \mapsto \varepsilon\} \mid \Gamma \Rightarrow \delta$ in RHA then $embed(\Pi)$ is a proof of $\Gamma \Rightarrow \delta; (\emptyset, (x \mapsto \{\emptyset\}))$ in R(CHA) and thus $erase(embed(\Pi))$ a proof of $\Gamma \Rightarrow \delta$ in CHA. As CHA proves the same sequents as HA (see theorem 3.4.9) there must also be a proof of $\Gamma \Rightarrow \delta$ in HA. □

Let $\mathcal{F}$ be a finite fragment of CHA. To conclude completeness of RHA relative to HA, we construct a proof morphism $search\colon \mathrm{S}(\mathcal{F}) \to$ RHA. For this, recall that $\mathrm{S}(\mathcal{F})$ is the 'proof search system' induced by the derivation rules in $\mathcal{F}$ and the trace interpretation of CHA. The only slight complication is introduced by the fact that not all spaces on the Safra boards of $\mathrm{S}(\mathcal{F})$ are populated.

Lemma 5.4.9. There is a function $search\colon \mathrm{S}(\textsc{Seq}_{\mathcal{F}}) \to \textsc{Seq}_{\mathrm{RHA}}$ with

$$search(\Gamma \Rightarrow \delta; (\Theta, \sigma)) := \widehat{\Theta}; \widehat{\sigma} \vdash \Gamma \Rightarrow \delta$$

where $\widehat{S} \in \Theta^*$ for $S \subseteq \Theta$ is the duplicate-free sequence of length $|\Theta|$ which is strictly sorted according to Θ and, if (Θ, σ) is a K-sparse Safra board on some $\iota(\Gamma \Rightarrow \delta)$, then

$$\widehat{\sigma} := x \mapsto \begin{cases} \widehat{S} & \text{if } \sigma(s) = \{S\} \\ \varepsilon & \text{if } \sigma(s) = \emptyset \end{cases}$$

The function can be extended into a proof morphism $search\colon \mathrm{S}(\mathcal{F}) \to$ RHA.

Proof. Towards this claim, first pick some $\mathrm{R}(\Theta, \sigma) \in \mathrm{S}(\mathcal{F})$ arranged as follows

$$r(\Theta,\sigma)\,\frac{\Gamma_1 \Rightarrow \delta_1; (\Theta_1, \sigma_1) \quad \dots \quad \Gamma_n \Rightarrow \delta_n; (\Theta_n, \sigma_n)}{\Gamma \Rightarrow \delta; (\Theta, \sigma)}$$

Then there is $r \in \mathcal{F}$ with $\rho(r) = (\Gamma \Rightarrow \delta, \Gamma_1 \Rightarrow \delta_1, \ldots, \Gamma_n \Rightarrow \delta_n)$ andtrace maps $r_i \colon \iota(\Gamma \Rightarrow \delta) \to \iota(\Gamma_i \Rightarrow \delta_i)$ given by the trace interpretation. Then for each $i \leq n$ there is $(\Theta, \sigma) \overset{r_i}{\rightsquigarrow}_g (\Theta_i, \sigma_i)$ with the expanded sequence

$$(\Theta, \sigma) \overset{R_{\gamma_1}}{\rightsquigarrow} (\Theta^1_r, \sigma^1_r) \ldots \overset{R_{\gamma_k}}{\rightsquigarrow} (\Theta^k_r, \sigma^k_r) \overset{P}{\rightsquigarrow} (\Theta_p, \sigma_p) \overset{r_i}{\rightsquigarrow} (\Theta_i, \sigma_i) \overset{T}{\rightsquigarrow} (\Theta_i, \sigma_i)$$

in which the initial R_γ- and P-steps are shared between all $i \leq n$ (see lemma 5.2.18). Note that the thinning step leaves the board unchanged: As for all r_i we have $(x, a, y), (x, b, z) \in r_i$ entailing $x = y = z$ and $a = b$, the r_i-step can never result in more than one stack at the same board position, as there was no more than one before it. We may derive the following in RHA, similarly to lemma 5.3.7:

$$\cfrac{\cfrac{\widehat{\Theta_1}; \widehat{\sigma_1} \mid \Gamma_1 \Rightarrow \delta_1 \qquad \ldots \qquad \widehat{\Theta_n}; \widehat{\sigma_n} \mid \Gamma_n \Rightarrow \delta_n}{\widehat{\Theta^k_r}; \widehat{\sigma^k_r} \mid \Gamma \Rightarrow \delta}\,\mathrm{R}}{\vdots \atop \cfrac{}{\widehat{\Theta_0}; \widehat{\sigma_0} \mid \Gamma \Rightarrow \delta}\,\textsc{Reset}_a}$$

That is, first apply all possible $\textsc{Reset}_a$-rules, starting at the Θ_0-greatest a. Then observe that the operation $\widehat{\ }$ of the definition of *search* always populates all positions on he board, meaning $\widehat{\sigma^k_r} = \widehat{\sigma_p}$, and the P-steps thus need not be simulated. Thus, the desired premises can be obtained by applying the RHA-rule corresponding to $r \in \mathcal{F}$.

An argument analogous to that given for *expand* in lemma 5.3.7 shows that *search* maintains the soundness condition. □

Corollary 5.4.10 (Completeness)**.** If HA $\vdash \Gamma \Rightarrow \delta$ then RHA $\vdash \Gamma \Rightarrow \delta$.

Proof. Suppose there is a proof of $\Gamma \Rightarrow \delta$ in HA. By theorem 2.2.1 there is a proof of the same sequent in CHA. Indeed, as cyclic proofs are finite trees, this proof is made in some finite fragment $\mathcal{F}$ of CHA. By theorem 5.3.8, there is a proof Π of $\Gamma \Rightarrow \delta; (\emptyset, (x, a) \mapsto \emptyset)$ in $\mathrm{S}(\mathcal{F})$ and thus *search*(Π) is a proof of $\varepsilon; (x \mapsto \varepsilon) \mid \Gamma \Rightarrow \delta$ in RHA. □

5.5 Discussion

The operation $\iota(\mathcal{R}) \mapsto \mathrm{R}(\mathcal{R})$ described in this chapter captures just *one* way of designing a reset proof system. The reset proof system of Afshari, Enqvist, et al. (2022) for first-order μ-calculus demonstrates the potential for variation. The reset rule $\mathrm{RS}(\kappa)$ utilised in that system corresponds to the following transition on Safra boards: Fix a board $(\Theta, \sigma) \in \mathrm{SB}(\mathcal{A}, X)$ and pick a covered $\kappa \in \Theta$. Let $C(\kappa)$ be the set of children of κ:

$$C(\kappa) := \bigl\{\min_{\Theta}\{\gamma \in S \mid \kappa < \gamma\} \bigm| x \in X, a \in \mathcal{A}, S \in \sigma(a, x) \text{ and } \kappa \in S\bigr\}.$$

If every $\gamma \in C(\kappa)$ is also covered, one may perform a 'reset operation' yielding the board $(\Theta \setminus C(\kappa), \sigma \setminus C(\kappa))$ where $(\sigma \setminus C(\kappa))(x, a) := \{S \setminus C(\kappa) \mid S \in \sigma(x, a)\}$. Replacing the reset transitions of definition 5.2.6 with this variant of the reset condition would yield an abstract reset proof system enjoying the same soundness and completeness properties as our chosen form of $\mathrm{R}(\mathcal{R})$, albeit requiring slight modifications to their proofs. Most likely there are multiple permissible alternatives to the 'reset machinery' we present in this chapter. We chose to only cover one, namely the one the closest to the traditional Safra construction for Rabin automata.

Related work

We are aware of very few articles wherein 'pure' reset calculi for cyclic proof systems have been designed: Jungteerapanich (2010) and Stirling (2013) for the modal μ-calculus and Afshari, Enqvist, et al. (2022) for first-order μ-calculus.

The first reset proof systems in the literature were annotated calculi for the modal μ-calculus introduced by Jungteerapanich (2010) and Stirling (2013), respectively capturing satisfiability and validity. Section 6.3.4 of (Leigh and Wehr 2024) gives a more detailed comparison between the Jungteerapanich–Stirling system and one created 'naïvely' using methods similar to those we employ in section 5.4. In sum, while also inspired by the Safra construction, the Jungteerapanich–Stirling system incorporates multiple insights into the semantics of the μ-calculus on which our method cannot make use of. These 'optimisations' lead to smaller proofs (although not in a 'complexity theoretically relevant' manner).

Afshari, Enqvist, et al. (2022) present a reset proof system for the first-order μ-calculus. It is based on the cyclic proof system for the first-order μ-calculus with ordinal approximations put forward by Sprenger and Dam (2003a). The crucial insight underpinning its design is that the mechanism of ordering ordinal variables in the Sprenger–Dam system is already very similar to the control of the Jungteerapanich–Stirling system. Thus, the Sprenger–Dam system is extended into a reset system in a very natural manner.

Both of the aforementioned reset systems were designed by combining automata theoretic considerations, namely the Safra construction, and semantic insights about the particular logic. In either case, the resulting systems are more elegant than the systems generated in this chapter. It should however be noted that proving soundness and completeness of such elegant systems requires considerable effort. We conjecture that for many purposes of cyclic proof theory, the naïve reset proof systems derived in this chapter prove sufficient.

Kloibhofer and Venema (2025) give a reset proof system for the two-way modal μ-calculus. Similar to this chapter, their reset mechanism is motivated by an automaton construction. They employ a generalisation of Safra automata to parity automata, rather than the 'traditional' Safra automata for Büchi-automata as in this chapter.

Focus-style proof systems are another common class of annotated cyclic proof systems in the literature. Instead of annotating each trace value with stacks of chips, focus-style proof systems instead only mark trace values as 'in focus' or 'out of focus'. Similarly to reset proof systems, the soundness condition of focus-style proof systems is a path condition: it only needs to be verified on each local cycle of a preproof. The first focus-style proof system was presented by Marti and Venema (2021a) of the Jungteerapanich–Stirling system to the alternation free fragment of the modal μ-calculus. Since then, focus-style proof systems have been given for modal logics with the master modality (Rooduijn 2021), full computation tree logic CTL* (Afshari, Leigh, and Menéndez Turata 2023), common knowledge over S5 modal logic (Rooduijn and Zenger 2022), the alternation-free fragment of the two-way modal μ-calculus (Rooduijn and Venema 2023) and propositional dynamic logic PDL (Kloibhofer, Trucco Dalmas, et al. 2026). It should be noted that all positive properties of reset proof systems we have mentioned in this dissertation, such as their suitability for proof theoretic investigations and proof search, extend to focus-style proof systems. However, it seems like not all cyclic proof systems admit focus-style proof systems. For example, once quantifier alternation is allowed, the modal μ-calculus requires reset annotations, rather than just focus-style annotations, for completeness.

A work directly related to this chapter is Dekker et al. (2023). In it, the authors present a cyclic proof system for the modal μ-calculus. The proof system relies on an annotation scheme, inspired by a novel construction for determinising parity automata. As presented, the system's soundness condition is a global trace condition, rather than a path condition. The authors state that this is because their annotation scheme does not rely on a control mechanism (similar to Θ in the setting of reset proofs) and by introducing such a control mechanism, their system could be presented in terms of a path condition.

Such as reset and focus-style proof systems have proven well-suited to proof theoretic investigations. So far, the results which employ them are in the areas of interpolation (Afshari, Leigh, and Menéndez Turata 2021; Afshari and Leigh 2022; Marti and Venema 2021b; Kloibhofer and Venema 2025; Kloibhofer, Trucco Dalmas, et al. 2026), translations of cyclic proofs into non-cyclic proofs with suitable induction axioms (Afshari, Leigh, and Menéndez Turata 2024) and cut-elimination results (Afshari and Kloibhofer 2024; Afshari and Kloibhofer 2025).

6 Stack-controlled proof systems

This chapter introduces stack-controlled proofs, a novel representation of cyclic proofs for which the physical and logical ordering of local cycles align. Intuitively, the physical ordering records along which cycle a 'detour' through which other cycle can occur. The logical ordering records which cycle 'preserves the progress' of which other cycle. Their alignment of the physical and logical ordering means that stack-controlled proofs are closely related to finite proofs with induction rules. We rely on this relationship in chapter 7 to derive multiple more concrete results of cyclic proof theory. For a more approachable introduction to stack-controlled proofs and their benefits, we refer readers to section 3.4.

It should be noted that cyclic proofs with the aforementioned alignment have previously been studied. Sprenger and Dam (2003b) employ this normal form, expressed in terms of induction orders, to translate from cyclic proofs for μFOL, first-order logic with least and greatest fixed points, to finite μFOL-proofs with induction rules. Afshari and Leigh (2016) employ an analogous normal form for reset proofs for the modal μ-calculus in a bid[20] to prove cut-free completeness of finite proofs with induction rules.

In both aforementioned works, the alignment of the physical and logical ordering of local cycles is expressed as a combinatorial property of cyclic proofs. The stack-controlled proof systems we introduce in this chapter instead enforce this alignment through their derivation rules. Crucially, this means that stack-controlled proof systems do not feature an additional soundness condition: *well-formedness suffices to ensure soundness*. The name "stack-controlled" derives from the mechanism which achieves this: each sequent is annotated by a stack which tracks both the available 'inductive hypotheses' and the nodes which the proof may 'cycle back to' at any given point.

Similarly to our treatment of reset proofs in chapter 5, this chapter gives a general rendition of stack-controlled proofs in terms of abstract cyclic proof systems. Note that unlike chapter 5, we only study trace conditions specified in terms of $\mathbb{B}$ (see the discussion in section 6.5 for context). We begin by defining

[20] Their argument was found to contain a mistake by Kloibhofer (2023). They have since obtained the same result with a rather different proof strategy in (Afshari, Leigh, and Menéndez Turata 2024).

stack-controlled proofs in section 6.1. We then prove their soundness (the underlying cyclic proof of a stack-controlled proof is a $\iota(\mathcal{R})$-proof) in section 6.2. We prove their completeness (every $\iota(\mathcal{R})$-proof can be unfolded and annotated into a stack-controlled proof) in section 6.3. To illustrate the abstract construction, we present two concrete stack-controlled proof systems: one corresponding to Simpson's $\mathrm{CHA}_<$ (section 6.4.1) and one corresponding to simple CHA (section 6.4.2). We close with a brief discussion in section 6.5.

6.1 Definition

For the remainder of the chapter, fix a derivation system $\mathcal{R} = (\textsc{Seq}, \mathcal{R}, \rho)$ and a trace interpretation $\iota : \mathcal{R} \to \mathbb{B}$ as defined in chapter 4. We call a pair (x, σ) with $\sigma \in \{+, -\}$ a *progress annotated value*, often denoting it by x^σ, and write $\mathrm{Prg}(X) := X \times \{+, -\}$ for set of progress annotated X-values.

We define $\Lambda(\mathcal{R})$, the stack-controlled proof system corresponding to $\iota(\mathcal{R})$. A sequent of $\Lambda(\mathcal{R})$ has the form "$\Gamma \mid \Lambda$" where $\Gamma \in \textsc{Seq}$ is an $\mathcal{R}$-sequent and Λ is a *companion label stack* for Γ. That means Λ is a sequence of *companion labels* of the form $P; \vec{G} \mapsto \Delta$ where Δ is an $\mathcal{R}$-sequent, P is a finite set of progress annotated values and $\vec{G}$ is a sequence of finite sets of progress annotated values. To be able to characterise the nature of P and $\vec{G}$ further, denote the ith companion label in Λ (starting at 0) by $P_i; \vec{G^i} \mapsto \Delta_i$. Λ is a well-formed stack of companion labels for Γ if for every $i < |\Lambda|$

(i) $|\vec{G^i}| = i + 1$ and every $G^i_j \subseteq \mathrm{Prg}(\iota(\Delta_j))$ for $j \leq i$ and
(ii) $P_i \subseteq \mathrm{Prg}(\iota(\Gamma))$.

Remark 6.1.1. A companion label $P_i; \vec{G^i} \mapsto \Delta_i$ represents the 'data associated to a cycle'. The set P_i tracks the *current state of progress* of the trace values the cycle aims to progress. A value x^+ indicates that x has already progressed along the cycle while x^- indicates that must yet be progressed for the cycle to close. The $\vec{G^i}$ store the *guards* which dictate when a cycle may be closed with a bud. Unlike the P_i, which change throughout a proof, reflecting the state of progress, the $\vec{G^i}$ of a companion label remain fixed. As we later define, a companion 'initialises' its associated P_i to X^-, for some set X of trace values, and G^i_i is set to X^+, representing the requirement that every trace value in the initial P_i must be progressed along the cycle. The G^i_j for $j < i$ 'store' the state of the P_j at the companion i.e., the intermediate state of progress of the local cycles which are 'physically greater', whose trace values must be preserved. At the bud, the current state of P_j is then compared to G^i_j to ensure the traces of these cycles have not been disturbed.

Remark 6.1.2. Recall that in section 3.4, we motivated stack-controlled proofs as embodying "$\sqsubseteq^*$ induction orders". Observe also that the notion of companion label diverges quite drastically from those given for SHA, the stack-controlled proofs for CHA presented in said section. This is because the above definitions account for one more complication that is not captured by the intuition of "$\sqsubseteq^*$ induction orders". There are cyclic proofs over $\mathbb{B}$, and thus for example $\mathrm{CHA}_<$-proofs, which cannot be assigned an induction order in the following sense: It is not possible to choose only a single value for each cycle which must be progressed and preserved.

Consider, for example, the cyclic proof sketched in fig. 15 below. It has two buds (at the top) which share a companion (at the bottom). The trace values at each node are $\{x, y\}$. Thick edges indicate progressing connections and dashed edges non-progressing connections. That is, on the left cycle, the trace value x progresses to both x and y but y is not preserved. The right cycle is dual, progressing y to x and y but not preserving x.

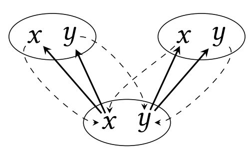


Figure 15: An illustration of cyclic proof over $\mathbb{B}$.

The proof in fig. 15 satisfies the global trace condition: Represent branches through the proof by infinite sequences $\pi \in \{l, r\}^\omega$ of *l*eft and *r*ight. Then $t \in \{x, y\}^\omega$, as defined below, is an infinitely progressing trace:

$$t_{2i} \coloneqq \begin{cases} x & \text{if } \pi_i = l \\ x & \text{if } \pi_i = r \end{cases} \qquad t_{2i-1} \coloneqq t_{2i}$$

On the other hand, no induction order can be assigned to the proof above: Consider w.l.o.g. the induction order where the left cycle is given precedence over the right cycle by the ordering. The left cycle's associated trace value must be x, as this is the only value which progresses along it. If this was a proper induction order, then the right cycle would preserve x but we know that it does not. The dual order, giving the right cycle precedence over the left, can be ruled out analogously. But then there can be no induction order. In fact, even unfolding the proof would not change this. None of its unfoldings can be assigned an induction ordering.

Thus, the naïve induction order approach of identifying each cycle with a single progressing trace value is insufficient. Instead, in general, every cycle must be assigned a *set of trace values* which it must progress and which $\sqsubseteq^*$-lower cycles must preserve. This is accounted for by taking P_i and G^i_j to be sets of trace values.

When we prove the soundness and completeness of SHA in section 6.4.2, those arguments will highlight the fact that the much simpler structure of the companion labels of SHA is only possible because of the various 'simplicities' of simple CHA.

Let $\Gamma, \Delta \in \textsc{Seq}$ and $\tau : \iota(\Gamma) \to \iota(\Delta)$ and Λ be a companion label stack for Γ. We define $\tau(\Lambda)$ as the companion label stack for Δ with $|\tau(\Lambda)| = |\Lambda|$ such that for $i < |\Lambda|$, if the ith label of Λ is $(P_i; \vec{G^i} \mapsto \Xi_i)$ then the ith label of $\tau(\Lambda)$ is $(\tau(P_i); \vec{G^i} \mapsto \Xi_i)$ where

$$\tau(P_i) := \{y^+ \mid x^\sigma \in P_i \wedge (x, 1, y) \in \tau\} \cup \{y^\sigma \mid x^\sigma \in P_i \wedge (x, 0, y) \in \tau\}.$$

Given two sets $X, Y \subseteq \mathrm{Prg}(Z)$ we write

$$X \preceq Y := \forall t^\sigma \in X. \exists \sigma' \geq \sigma.\ t^{\sigma'} \in Y$$

imposing on $\{+, -\}$ the ordering of $- < +$.

The rules of $\Lambda(\mathcal{R})$ consist of annotated $\mathcal{R}$-rules and additional *structural* rules. The structural rules are given below

$$\textsc{Pop}\frac{\Gamma \mid \Lambda}{\Gamma \mid \Lambda, \Lambda'} \qquad \textsc{Drop}\frac{\Gamma \mid \Lambda^*}{\Gamma \mid \Lambda} \qquad \textsc{Comp}\frac{\Gamma \mid \Lambda, (X^-; \pi(\Lambda), X^+ \mapsto \Gamma)}{\Gamma \mid \Lambda}$$

$$\textsc{Bud}\frac{}{\Gamma \mid \Lambda, (X^+; \pi(\Lambda), X^+ \mapsto \Gamma)}$$

where $\pi(\Lambda) = (P_j)_{j<|\Lambda|}$ and $X^\sigma = X \times \{\sigma\}$ for $\sigma \in \{+, -\}$. For the Drop rule, we have that Λ and Λ^* are identical except that $\pi(\Lambda^*) \preceq \pi(\Lambda)$ i.e., a label $(P; \vec{G} \mapsto \Delta) \in \Lambda$ may be replaced by $(P'; \vec{G} \mapsto \Delta) \in \Lambda^*$ with $P' \preceq P$.

For each rule $r \in \mathcal{R}$ with $\rho(r) = (\Gamma, \Delta_1, \ldots, \Delta_n)$ and maps $r_i : \iota(\Gamma) \to \iota(\Delta_i)$ given by the trace interpretation, the following schema gives annotated rules for each companion label stack Λ for Γ:

$$r_\Lambda\frac{\Delta_1 \mid r_1(\Lambda) \quad \ldots \quad \Delta_n \mid r_n(\Lambda)}{\Gamma \mid \Lambda}$$

Note that $\Lambda(\mathcal{R})$ is a derivation system, not a cyclic proof system. A $\Lambda(\mathcal{R})$-*proof* of an $\mathcal{R}$-sequent Γ is a derivation of $\Gamma \mid \varepsilon$ using the rules specified above.

We write $\Lambda(\mathcal{R}) \vdash \Gamma$ to denote the existence of such a proof.

Remark 6.1.3. A different way of obtaining an essentially equivalent system is taking the Bud-rule to be defined in terms of $\preceq$:

$$\textsc{Bud}\frac{\vec{Y} \preceq \pi(\Lambda), X}{\Gamma \mid \Lambda, (X; \vec{Y} \mapsto \Gamma)}.$$

Then Drop can be removed from the rules of $\Lambda(\mathcal{R})$, becoming an admissible rule instead.

Observe that, while $\Lambda(\mathcal{R})$ technically is a derivation system, it still represents a cyclic proof system. Previous cyclic proof systems we have considered in this dissertation impose a bud-companion structure on a proof tree T by means of a partial map $\beta : \mathrm{Leaf}(T) \rightharpoonup \mathrm{Inner}(T)$. In $\Lambda(\mathcal{R})$, this is instead achieved using the Comp- and Bud-rules. The Comp-rule 'stores' the current sequent on the companion label stack, 'remembering' it as an available companion further up in the proof. The Bud-rule, exhibits that the current sequent matches that of a previously 'stored' companion (often first 'digging it out of the stack' using an instance of Pop), closing the branch of the proof. Such an application of the Bud-rule corresponds, in the presentation of cyclic trees via $\beta : \mathrm{Leaf}(T) \rightharpoonup \mathrm{Inner}(T)$, to mapping the conclusion of the Bud-rule to the conclusion of the conclusion corresponding Comp-rule. This idea is made formal by the definition of $erase(\Pi)$, the 'underlying' $\iota(\mathcal{R})$-preproof of a $\Lambda(\mathcal{R})$-proof, below.

Definition 6.1.1 ($erase(\Pi)$)**.** Let $\Pi = (T, \rho)$ be a $\Lambda(\mathcal{R})$-proof. The preproof $erase(\Pi)$ is constructed recursively along its underlying tree T. That is, each $v \in T$ with $\lambda(v) = \Gamma \mid \Lambda$ and $|\Lambda| = n$ will be assigned a function f_v with $\mathrm{dom}(f_v) = (\omega^*)^{n+1}$ such that $f_v(u_1, \ldots, u_n, w)$ is a $\iota(\mathcal{R})$-preproof Π' rooted in w whose buds either have companions within Π' or among the $u_1, \ldots, u_n$. The definition of f_v depends on $\rho(\delta(v))$:

Pop: Let $\mathrm{Chld}(v) = \{v'\}$ and $|\Lambda| = n, |\Lambda'| = m$. Then

$$f_v(u_1, \ldots, u_n, u_{n+1}, \ldots, u_{n+1}, w) = f_{v'}(u_1, \ldots, u_n, w).$$

That is, the u_i corresponding to Λ' are simply discarded.

Drop: Let $\mathrm{Chld}(v) = \{v'\}$. Then $f_v = f_{v'}$ i.e., all arguments are simply passed on.

Comp: Let $\mathrm{Chld}(v) = \{v'\}$. Then $f_v(\vec{u}, w) = f_{v'}(\vec{u}, v, w)$ i.e., v is added as the companion corresponding to the newly added companion label.

Bud: Then $f_v(\vec{u}, u', w)$ consists only of the node w, labelled by $\lambda_\Pi(w)$ with $\beta(w) = u'$. That is, the preproof is a bud pointing at the companion node indicated by the arguments to f_v.

$\mathcal{R}$-rule: Let $\rho(\delta(v)) = r_\Lambda$ corresponding to some $r \in \mathcal{R}$ as below

$$\hat{r}\frac{\Delta_1 \mid r_1(\Lambda) \quad \dots \quad \Delta_n \mid r_n(\Lambda)}{\Gamma \mid \Lambda}$$

and let $\mathrm{Chld}(v) = \{v0, \dots, v(n-1)\}$. Then derive $f_v(\vec{u}, w)$ as follows

$$r\frac{\begin{matrix} f_{v0}(\vec{u}, w0) \\ \Delta_1 \end{matrix} \qquad \dots \qquad \begin{matrix} f_{v(n-1)}(\vec{u}, w(n-1)) \\ \Delta_n \end{matrix}}{\Gamma}$$

where the root of the derivation above is w. That is, simply apply the corresponding $\mathcal{R}$-rule and recurse.

Remark 6.1.4. Note that the idea of representing the cyclic structure of a cyclic proof via Bud- and Comp-rules is, in principle, independent from the notion of stack controlled proofs. Fixing some derivation system $\mathcal{R}$, one can consider a proof system $S(\mathcal{R})$ whose sequents are expressions $\Gamma \mid \Lambda$ where Γ is an $\mathcal{R}$-sequent and is Λ a list of $\mathcal{R}$-sequents. The rules of $S(\mathcal{R})$ are

$$r_\Lambda\frac{\Delta_1 \mid \Lambda \quad \dots \quad \Delta_n \mid \Lambda}{\Gamma \mid \Lambda}$$

for every $r \in \mathcal{R}$ with $\rho(r) = (\Gamma, \Delta_1, \dots, \Delta_n)$ and list Λ of $\mathcal{R}$-sequents, together with the two rules

$$\textsc{Comp}\frac{\Gamma \mid \Lambda, \Gamma}{\Gamma \mid \Lambda} \qquad \textsc{Bud}\frac{}{\Gamma \mid \Lambda, \Gamma, \Lambda'}.$$

Then an $S(\mathcal{R})$-derivation Π can be transformed into an $\mathcal{R}$-preproof $erase(\Pi)$ completely analogously to the procedure described in definition 6.1.1. Conversely, every $\mathcal{R}$-preproof Π can be transformed into a $S(\mathcal{R})$-derivation Π' such that $erase(\Pi') = \Pi$. To ensure the latter property, the set Λ must be managed as a list of sequents, rather than a set, to disambiguate between multiple Comp-instances with the same sequent along a branch.

In the system $S(\mathcal{R})$, derivations correspond merely to $\mathcal{R}$-preproofs, not generally $\mathcal{R}$-proofs for some soundness condition imposed on $\mathcal{R}$. As we prove in the next section, $\Lambda(\mathcal{R})$-derivations correspond to $\iota(\mathcal{R})$-proofs. This is ensured by the further information $P; \vec{G}$ recorded in the companion labels and the additional constraints, in terms of the companion label stacks Λ, imposed on the rules of $\Lambda(\mathcal{R})$.

6.2 Soundness

We prove soundness in the same sense as for reset systems in section 5.3.1: The underlying $\iota(\mathcal{R})$-preproof of a $\Lambda(\mathcal{R})$-proof is a proof. The argument itself also is quite similar to theorem 5.3.4, its RESET-counterpart: Identify an infinite branch with its strongly connected component, identify the $\preceq^*$-greatest cycle and 'read a progressing trace off the annotations'.

Theorem 6.2.1 (Soundness)**.** For any $\Lambda(\mathcal{R})$-proof Π, $erase(\Pi)$ is an $\iota(\mathcal{R})$-proof.

Proof. We have to prove that every infinite branch π through $erase(\Pi)$ has a progressing trace. Observe that any such branch has a corresponding branch π' 'through' Π which at BUD-rules 'jumps back' to the premise of the corresponding COMP-instance. Essentially by lemma 4.1.4, we know that $\mathrm{Inf}(\pi')$ is a 'strongly connected component'.

Consider $r = \min_{<} \mathrm{Inf}(\pi')$, the 'root' of the strongly connected $\mathrm{Inf}(\pi')$: it must be the premise of a COMP-rule. Let $\lambda(r) = \Gamma \mid \Lambda$ and $n = |\Lambda|$. Then for any $v \in \mathrm{Inf}(\pi')$ with $\lambda(v) = \Gamma_v \mid \Lambda_v$ we claim that $|\Lambda'| \geq n$ i.e., the companion label introduced at r is preserved throughout $\mathrm{Inf}(\pi')$: Suppose the companion label was erased via POP at some node $v \in \mathrm{Inf}(\pi')$. As $\mathrm{Inf}(\pi')$ is strongly connected, there should be a path from v to r which does not leave $\mathrm{Inf}(\pi')$. But any bud above v can only have companions below r (whose companion labels remain on the stack) or above v (introduced by later COMP-applications), meaning no such path can exist. Thus, any node $v \in \mathrm{Inf}(\pi')$ has an nth companion label, originating in r, which we shall denote by $P_v; \vec{G}^v \mapsto \Gamma$.

For any $u \in \mathrm{Inf}(\pi')^+$ with final node v such that ru is a valid path, write $\tau_u : \iota(\Gamma_r) \longrightarrow \iota(\Gamma_v)$ for the trace map arising from the composition of the $\iota(\mathcal{R})$-trace maps along it. We claim that for such a ru and any $x^\sigma \in P_v$ there is $y^- \in P_r$ such that $(y, b, x) \in \tau_u$ and furthermore if $\sigma = +$ then $b = 1$. We prove this by induction along u (from left to right) with a case-distinction on the $\Lambda(\mathcal{R})$-rule applied. We only cover the cases which impact the definition of τ_u or change the set P_v:

DROP: Let $u = u'v'v$ (where $v' = r$ if $u = v$). Let $x^\sigma \in P_v$, we know that $P_v \preceq P_{v'}$ and thus that there is $x^{\sigma'} \in P_{v'}$ with $\sigma \leq \sigma'$. Per IH, there is $y^- \in P_r$ with $(y, b, x) \in \tau_{u'v'}$. Further, if $\sigma = +$ then $\sigma' = +$ and thus $b = 1$, as desired.

r_Λ: Then $u = u'v'v$ as above and v is the ith premise of r_Λ applied at v' and thus $P_v = r_i(P_{v'})$. Let $z^\sigma \in P_v$, it could have been included in P_v via either of the clauses of $r_i(P_{v'})$:

- $x^{\sigma'} \in P_{v'}$ and $(x, 1, z) \in r_i$ and $\sigma = +$: Then per IH, there is $y^- \in P_r$ such that $(y, b, x) \in \tau_{u'v'}$ and thus $(y, 1, z) \in \tau_u$ as desired.
- $x^{\sigma} \in P_{v'}$ and $(x, 0, z) \in r_i$: Then per IH, there is $y^- \in P_r$ such that $(y, b, x) \in \tau_{u'v'}$ with $b = 1$ if $\sigma = +$. This extends to $(y, b, z) \in \tau_u$.

To conclude, we know that $\pi' = wru_0ru_1ru_2\ldots$ i.e., starting with some prefix w and then infinitely visiting r with some nodes $u_i \in (\mathrm{Inf}(\pi') \setminus \{r\})^*$ in between. Write v_i for the last node of u_i, which must always be the conclusion of Bud. By the previous result, we know that for every u_i and $x^\bullet \in P_r$ there is $y^\bullet \in P_r$ such that $(y, 1, x) \in \tau_{u_i}$ because $P_{v_i} = P_r^+$. With König's lemma, this yields an infinitely progressing trace following $ru_0ru_1\ldots$. As this corresponds to an infinitely progressing trace along π, the original infinite branch through $erase(\Pi)$, we are done. □

6.3 Completeness

In this section, we show completeness of $\Lambda(\mathcal{R})$ with regards to $\iota(\mathcal{R})$: if $\iota(\mathcal{R}) \vdash \Gamma$ then $\Lambda(\mathcal{R}) \vdash \Gamma$. We do this by demonstrating that a $\Lambda(\mathcal{R})$-proof corresponds to a $R(\mathcal{R})$-proof in a certain normal form.

Recall the cyclic dependency ordering $\sqsubset$ on the cycles $\mathrm{dom}(\beta)$ of a cyclic tree: For $s, t \in \mathrm{dom}(\beta)$, t depends on s, denoted $s \sqsubset t$, if $\beta(t) < \beta(s) < t$. Intuitively, $s \sqsubseteq^* t$ expresses that a cycle $t \to t$ may pass through $\gamma(s)$ any number of times.

In the following, denote the invariant of a cycle $t \in \mathrm{dom}(\beta)$ of a $R(\mathcal{R})$-proof by θ_t. Further, observing that for Safra boards $(\Theta, \sigma) \in \textsc{Sb}(\mathbb{B}, X)$ we always have $\mathrm{dom}(\sigma) = X \times \{0\}$, we write $\sigma(x) := \sigma(x, 0)$.

We begin by defining the various structural properties of reset proofs which constitute the aforementioned normal form. These definitions stem from Afshari and Leigh (2016).

Definition 6.3.1. Fix a $R(\mathcal{R})$-proof Π. It is

(i) a *strictly progressing* proof if, for every bud $s \in \mathrm{dom}(\beta)$ with $\theta_s = \theta a$, two $\textsc{Reset}_a$-applications take place along $\gamma(s)$.

(ii) an *invariant* proof if, whenever two buds $s, t \in \mathrm{dom}(\beta)$ have the same companion $\beta(s) = \beta(t)$ then they have the same invariant i.e., $\theta_s = \theta_t$.

(iii) a *monotone* proof if for every $s, t \in \mathrm{dom}(\beta)$ if $s \sqsubset t$ then $\theta_t \leq \theta_s$ and further if $\theta_t = \theta_s = \theta a$ then a $\textsc{Reset}_a$ must occur along $[t, \beta(s)]$.

(iv) an *injective* proof if β is an injection.

Remark 6.3.1. We have stated before that before stack-controlled proofs represent a normal form of cyclic proofs in which the physical and logical ordering of local cycles align. This condition corresponds exactly to the first half of monotonicity in definition 6.3.1, which directly implies that for every $s, t \in \mathrm{dom}(\beta)$ if $s \sqsubseteq^* t$ then $\theta_t \leq \theta_s$. Recall, for example from the proof of theorem 5.3.4, that the trace values 'progressed along' a cycle $t \in \mathrm{dom}(\beta)$ with $\theta_t = \theta' a$ are precisely those whose annotations contain a. If $s, t \in \mathrm{dom}(\beta)$ share a strongly connected component and $\theta_t \leq \theta_s$, this means that s does not erase all a-annotated trace values and thus preserves the progress of t. Thus, the first half of monotonicity can be rephrased as: if $s \sqsubseteq^* t$ then s preserves the progress of t. In that sense, the physical (i.e. $\sqsubseteq^*$) and logical (i.e. prefix ordering on invariants) ordering on the local cycles aligns.

Remark 6.3.2. Note that that the condition about positions of RESET-rules which is part of monotonicity in definition 6.3.1 does not play a role in the constructions and arguments of this section. We chose to include it nonetheless to faithfully represent the notions as presented by Afshari and Leigh (2016).

The following is Afshari and Leigh (2016, theorem 4.9). While the argument is carried out for a specific reset proof system, it easily generalises to any abstract cyclic proof system $R(\mathcal{R})$. Note that while a mistake was found in the aforementioned article by Kloibhofer (2023), it concerns a later theorem.

Proposition 6.3.2. Every $R(\mathcal{R})$-proof can be unfolded into an invariant, monotone $R(\mathcal{R})$-proof.

Remark 6.3.3. For readers familiar with Sprenger and Dam (2003b), it should be noted that the notion of invariant, monotone reset proofs employed by Afshari and Leigh (2016) is analogous to the notion of injective proofs with tree-compatible induction orders employed by Sprenger and Dam. Specifically, when viewing reset proofs as a form of induction orders, as illustrated in propositions 3.3.2 and 5.3.3, then monotonicity is precisely tree-compatibility with the added condition on the ordering of RESET-applications.

Call a cycle $\gamma(t)$ of an $R(\mathcal{R})$-proof strictly progressive if $\theta_t = \theta a$ and two RESET$_a$-applications take place along t. We first show a technical lemma which will help with obtaining strictly progressive proofs.

Lemma 6.3.3. Let Π be an $R(\mathcal{R})$-proof and $s \in T$ such that for every $t \in \mathrm{dom}(\beta)$ with $s < \beta(t)$, $\gamma(t)$ is strictly progressive. It can be unfolded into an $R(\mathcal{R})$-proof Π' which is identical on every node $s' \not\geq s$ and for every $t \in \mathrm{dom}(\beta')$ with $s \leq \beta'(t)$, $\gamma(t)$ is strictly progressive.

Proof. This result is only interesting if s is a companion. Thus, let $B = \beta^{-1}(s)$ be its companions. Because $s \in \gamma(t)$ for every $t \in B$, we know that there is a strict linear order $\prec$ on B such that $t \prec t'$ entails $\theta_{t'} \leq \theta_t$ i.e., the invariant of t' is maintained by t. Now we construct the new subproof above s using the following procedure: To begin, mark s as open and call it $\varepsilon \in B^*$. Now pick some open node called $u \in B^*$. Mark it as closed and apply one of the following two steps:

(a) if there exist $v, v' \in B^*$ and $t \in B$ such that $u = vtv't$ and $\max_\prec v' \prec t$ then insert a cycle from the node called u to the one called v.

(b) otherwise, graft the subproof Π_s above of s in Π on top of this node u. For each leaf l above u which corresponds to a bud t in Π_s, do the following:

  (i) if $s < \beta(t)$ i.e., the companion of t lies within Π_s but above s, set $\beta(l)$ to be the node corresponding to $\beta(t)$ in the Π_s copy above u.

  (ii) if $\beta(t) < s$ i.e., the companion of t does not lie within Π_s, set $\beta(l) = \beta(t)$.

  (iii) if $\beta(t) = s$, mark l as open and call it $ut \in B^*$.

First, we must argue that the procedure above terminates, yielding a finite $R(\mathcal{R})$-preproof. Suppose not, then there must be an infinite sequence $u \in B^*$ such that u every prefix of u is considered open at some point during the procedure. That means that every prefix v of u cannot satisfy the condition guarding (a) above, as otherwise no nodes whose name is a strict extension of v would ever be considered open. Clearly, this means that $t := \max_\prec B$ may occur at most once in u: If $u = vtv'tu'$ for some $v, v' \in B^*$ and $u \in B^\omega$ then $vtv't$ would satisfy (a). Repeating this argument along $\prec$ lets us conclude that every $t \in B$ may occur in u at most once. But this is not possible as B is finite and u infinite.

Thus, the procedure above terminates, yielding an $R(\mathcal{R})$-preproof Π'. As Π' is the unfolding of a $R(\mathcal{R})$-proof, it remains an $R(\mathcal{R})$-proof. Thus, all that remains to be proven is that for any $t \in \mathrm{dom}(\beta')$ with $s \leq \beta'(t)$ the cycle $\gamma(t)$ is strictly progressing. There are two ways in which the procedure above can create such a cycle above s:

- By following step (a): That means $\beta'(t)$ was called v and t was called $vt'v't'$ for $v, v' \in B^*$ and $t' \in B$. Writing $v' = t'_1 \dots t'_n$, observe that this must mean that $\gamma(t)$ crosses precisely the same rules as the path $\gamma(t')\gamma(t'_1)\dots\gamma(t'_n)\gamma(t')$ with $t'_i \prec t'$. Because thus every $\gamma(t'_i)$ maintains $\theta_{t'}$, $\theta_{t'}$ is an invariant along $\gamma(t)$ and as each $\gamma(t')$ must contain at least one $\textsc{Reset}_a$ for $\theta_{t'} = \theta a$, $\gamma(t)$ contains two and thus is strictly progressing.

- By following step (bi): Then $\gamma(t)$ is a copy of some cycle $\gamma(t')$ in Π_s and thus strictly progressing per assumption. □

Remark 6.3.4. Observe that the termination proof we employ above is highly classical. We believe a more careful combinatorial analysis of the property guarding (a) should yield a constructive bound $k(|B|)$ such that any sequence $u \in B^*$ with $|u| > k(|B|)$ satisfies the condition, yielding a constructive termination proof.

Lemma 6.3.4. Every $R(\mathcal{R})$-proof Π can be unfolded into a strictly progressing proof.

Proof. Simply recursively apply lemma 6.3.3 along the tree T underlying Π. More concretely, fix $R_0 := T$ and name $\Pi_0 := \Pi$. To obtain a Π_{i+1}, if $R_i \neq \emptyset$, pick some $s \in T$ satisfying the precondition of lemma 6.3.3, thus starting at the leaves of Π, and take Π_{i+1} to be the proof produced by lemma 6.3.3 applied at s and $R_{i+1} := R_i \setminus \{s\}$. As T is finite, this must result in a final $\Pi_{|T|}$ which is strictly progressing. □

Lemma 6.3.5. Every $R(\mathcal{R})$-proof Π can be transformed into an injective proof Π' such that $erase(\Pi) = erase(\Pi')$.

Proof. Suppose w.l.o.g. that $s \in T$ with $\beta^{-1}(s) = \{t, t'\}$ and $\lambda(s) = \Gamma; (\Theta, \sigma)$. As depicted below, replace s with an application of Weak which leaves the annotation unchanged, calling the new premise node s'. Then redirect $\beta(t') := s'$.

$$\begin{array}{c} t \qquad t' \\ \ldots \\ \hline s : \Gamma; (\Theta, \sigma) \end{array} \longrightarrow \begin{array}{c} t \qquad t' \\ \ldots \\ \hline s' : \Gamma; (\Theta, \sigma) \\ \hline s : \Gamma; (\Theta, \sigma) \end{array}\,\textsc{Weak}$$

Carry out the analogous modification for every node $s \in T$ with $|\beta(s)| > 1$ to obtain an injective $R(\mathcal{R})$-proof Π'. As $erase$ removes all instances of Weak, we have $erase(\Pi') = erase(\Pi)$. □

Corollary 6.3.6. Every $R(\mathcal{R})$-proof can be unfolded into an injective, monotone and strictly progressing proof.

Proof. First apply lemma 6.3.4 and subsequently proposition 6.3.2, observing that strict progressiveness is preserved under unfoldings. Lastly, apply lemma 6.3.5, observing that it maintains all of the aforementioned properties. □

The translation from $R(\mathcal{R})$-proofs to $\Lambda(\mathcal{R})$-proofs is verbatim for $\mathcal{R}$-rules. The only difficulty of the translation is assigning each sequent a suitable stack Λ of companion labels and inserting the 'Λ-bookkeeping rules' Drop, Pop and Comp at the right locations. Towards the assignment of the Λ, consider a $R(\mathcal{R})$-proof $\Pi = ((T, \beta), \rho)$ and recall that $\gamma(t)$ denotes the local cycle from $[\beta(t), t]$ for $t \in \mathrm{dom}(\beta)$. For the translation from $R(\mathcal{R})$ to $\Lambda(\mathcal{R})$, it is crucial to characterise which cycle's trace values a given node $s \in T$ must preserve. We consider the set $S(s) \subseteq \mathrm{dom}(\beta)$ which collects these:

$$S(s) = \{u \in \mathrm{dom}(\beta) \mid \exists t \in \mathrm{dom}(\beta).\, s \in \gamma(t) \wedge t \sqsubseteq^* u\}$$

and begin by proving that, when imposing a suitable ordering on them, the $S(s)$ can indeed be treated as a stack. In the following, we denote by $<_\beta$ the order on $\mathrm{dom}(\beta)$ induced by β, given by $s <_\beta t$ iff $\beta(s) < \beta(t)$. Observe that strict progressiveness is not required for this part of the argument.

Lemma 6.3.7. Let $\pi = ((T, \beta), \lambda)$ be an injective, monotone $R(\mathcal{R})$. The set $S(s)$ is linearly ordered by $<_\beta$ for every $s \in T$. Moreover, the function $\Sigma : T \longrightarrow \mathrm{dom}(\beta)^*$ assigning to $s \in T$ the enumeration of $S(s)$ according to $<_\beta$ satisfies:

1. If $\varepsilon \notin \mathrm{im}(\beta)$ then $\Sigma(\varepsilon) = \varepsilon$; otherwise $\Sigma(\varepsilon) = \beta^{-1}(\varepsilon)$.
2. If $t \in \mathrm{Chld}(s)$ and $t \notin \mathrm{im}(\beta)$, then $\Sigma(t) \leq \Sigma(s)$.
3. If $t \in \mathrm{Chld}(s)$ and $t \in \mathrm{im}(\beta)$, then there is $u \leq \Sigma(s)$ such that $\Sigma(t) = u \cdot \beta^{-1}(t)$.

Proof. First, notice that per construction, if $t \in S(s)$ then $\beta(t) \leq s$. Hence, $S(s)$ is indeed linearly ordered by $<_\beta$ and the function Σ given above well-defined. Furthermore, $|\beta^{-1}(t)| \leq 1$ by the injectivity of β.

Define $\mathrm{Down}(t) := \{s \in \mathrm{dom}(\beta) \mid s <_\beta t\}$ for $t \in \mathrm{dom}(\beta)$.

1. We know that $S(\varepsilon) = \{\beta^{-1}(\varepsilon) \mid \text{if } \varepsilon \in \mathrm{im}(\beta)\}$ because $\varepsilon \in \gamma(t)$ entails $t = \beta^{-1}(\varepsilon)$.

2, 3. We first prove that if $t \in \mathrm{Chld}(s)$ then $S(t) \subseteq S(s) \cup \beta^{-1}(t)$: Fix some $v \in S(t)$ i.e., such that there is u with $t \in \gamma(u)$ and $u \sqsubseteq^* v$. There are two cases:

   $\beta(u) \neq t$: Then $s \in \gamma(u)$, as $t \in \mathrm{Chld}(s)$, and thus $v \in S(s)$.

   $\beta(u) = t$: Then we consider two cases arising from $u \sqsubseteq^* v$:

   $u = v$: Then $u \in \beta^{-1}(t)$.

   $u \sqsubset v_1 \sqsubseteq^* v$: Then $\beta(v_1) < t$ and thus $s \in \gamma(v_1)$, meaning $v \in S(s)$.

What remains to be shown towards 2. and 3. is that buds are only removed in an $\leq_\beta$-upwards closed manner: if $t \in \mathrm{Chld}(s)$ and $u \in S(s) \setminus S(t)$ i.e., u was removed from $S(-)$ in stepping from s to t, then there is no $u' \in S(t)$ such that $u \leq_\beta u' \neq \beta^{-1}(t)$.

We instead prove an equivalent statement: For $t \in \mathrm{Chld}(s)$, if $u \in S(t)$ with $u \neq \beta^{-1}(t)$ then $S(s) \cap \mathrm{Down}(u) \subseteq S(t)$. If $u \in S(t)$, there must be v such that $t \in \gamma(v)$ and $v \sqsubseteq^* u$. As $\beta^{-1}(u) \neq t$, $s \in \gamma(u)$ as well. Now fix $u' \in S(s) \cap \mathrm{Down}(u)$, meaning there is v' with $s \in \gamma(v')$ and $v' \sqsubseteq^* u'$, as $u' \in S(s)$, and $u' <_\beta u$, as $u' \in \mathrm{Down}(u)$. We distinguish two cases:

$v = v'$: Then $u' \in S(s)$ as $v = v' \sqsubseteq^* u'$.

$v \neq v'$: Because of $\beta(v), \beta(v') < s < v, v'$, we may distinguish two further cases:

$v \sqsubset v'$: Then $v \sqsubset v' \sqsubseteq^* u'$ and thus $u' \in S(t)$.

$v' \sqsubset v$: This case is resolved with a further case distinction: For any x, if $v <_\beta x$ and $x \sqsubseteq^* u'$ then either $v \sqsubseteq^* u'$ or $v <_\beta u'$. From $v' \sqsubset v$ it then follows that $v <_\beta v'$ and thus:

$v \sqsubseteq^* u'$: Then $u' \in S(t)$ as $t \in \gamma(v)$.

$v <_\beta u'$: But $u' <_\beta u$ together with $v \sqsubseteq^* u$ yield $u' <_\beta u \leq_\beta v$, ruling out this option.

It remains to prove the aforementioned disjunction, which we do per induction on the $\sqsubset$-chain witnessing $x \sqsubseteq^* u'$: If $x = u'$ then $v <_\beta x = u'$. Thus suppose $x \sqsubset x' \sqsubseteq^* u'$. Because $v, x' \leq_\beta x$, we may distinguish two cases:

$x' \leq_\beta v$: Then $x' \leq_\beta v <_\beta x \leq x'$ and thus $v \sqsubset x' \sqsubseteq^* u'$.

$v <_\beta x'$: Continue per IH with $x := x'$. □

Remark 6.3.5. Intuitively, lemma 6.3.7 can be read as stating that the alignment of the physical and logical orderings means the sets of preserved cycles $S(s)$ 'behave as stacks' when ordered by $<_\beta$ and the injectivity of π means that these 'stacks' are extended by at most one bud between two nodes, mirroring the Comp-rule.

We can now prove completeness.

Theorem 6.3.8 (Completeness)**.** If $\iota(\mathcal{R}) \vdash \Gamma$ then $\Lambda(\mathcal{R}) \vdash \Gamma$.

Proof. By theorem 5.3.8 and corollary 6.3.6, we may assume an $R(\mathcal{R})$-proof Π of Γ which is injective, monotone and strictly progressing. We translate every subproof $\Pi[s] \vdash \Gamma_s; (\Theta_s, \sigma_s)$, rooted at $s \in T$, to a $\Lambda(\mathcal{R})$-proof $\Pi'[s] \vdash \Gamma_s \mid \Lambda_s$.

The stack Λ_s is computed as follows: For $t \in \mathrm{dom}(\beta)$ let $\theta_t := \theta a_t$. Now let $s \in T$ and $t \in S(s)$ then we define, for every $x \in \iota(\Gamma_s)$ and every $ua_tv \in \sigma_s(x)$ the operation below:

$$(x, ua_tv)^t_s := \begin{cases} x^+ & \text{if } v \neq \varepsilon \text{ and } \textit{precisely one } a_t\text{-}\textsc{Reset} \\ & \quad \text{has taken place along } [\beta(t), s] \\ x^+ & \text{if } \textit{at least two } a_t\text{-}\textsc{Resets} \text{ have} \\ & \quad \text{taken place along } [\beta(t), s] \\ x^- & \text{otherwise} \end{cases} \tag{6.1}$$

We then define the state of progress of cycle $t \in S(s)$ at node $s \in T$ by $P^t_s := \{(x, ua_tv)^t_s \mid x \in \iota(\Gamma_s), ua_tv \in \sigma_s(x)\}$. Now let $\Sigma(s) = t_1 \dots t_n$ then $\Lambda_s := l_1; \dots; l_n$ where

$$l_i := (P^{t_i}_s; P^{t_1}_{\beta(t_i)} \dots P^{t_{i-1}}_{\beta(t_i)} (P^{t_i}_{\beta(t_i)})^+) \mapsto \Gamma_{t_i}$$

The proof proceeds via an induction on the subproofs of Π. Let $\Pi[s]$ be such a subproof. We perform a case distinction:

s IS A BUD: Then $\Lambda_s = \Lambda; (P^s_s; \vec{Y}; (P^s_{\beta(s)})^+ \mapsto \Gamma_s)$ because $s \in S(s)$. An application of DROP followed by BUD proves $\Gamma_s \mid \Lambda_s$ as desired if $P^s_s = (P^s_{\beta(s)})^+$ and $\vec{Y} \preccurlyeq \pi(\Lambda)$:

1. $P^s_s = (P^s_{\beta(s)})^+$: We know that $\Gamma_s = \Gamma_{\beta(s)}$ and further, because they are bud and companion, that $\Gamma_s; (\Theta_s, \sigma_s) = \Gamma_{\beta(s)}; (\Theta_{\beta(s)}, \sigma_{\beta(s)})$. Furthermore, for every $x^\sigma \in P^s_s$ we know that $\sigma = +$ because, by strict progressivity of Π, at least two a_t-RESETS have must have occurred along $[\beta(s), s] = \gamma(s)$. The claim directly follows from these two observations.
2. $P^t_{\beta(s)} \preccurlyeq P^t_s$ for $s \neq t \in S(s)$: As $\Gamma_s; (\Theta_s, \sigma_s) = \Gamma_{\beta(s)}; (\Theta_{\beta(s)}, \sigma_{\beta(s)})$, we again know that both sets contain the same signed trace values. Now consider $x \in \iota(\Gamma_s)$ with $ua_tv \in \sigma_s(x)$ and let $(x, ua_tv)^t_{\beta(s)} = x^{\sigma_{\beta(s)}}$ and $(x, ua_tv)^t_s = x^{\sigma_s}$. It remains to show that $\sigma_{\beta(s)} \leq \sigma_s$. We distinguish two cases:

   NO RESET$_{a_t}$ APPLIED ON $\gamma(s)$: Then $\sigma_{\beta(s)} = \sigma_s$ because the same clause of eq. (6.1) must apply for both.

   RESET$_{a_t}$ APPLIED ON $\gamma(s)$: If $\sigma_{\beta(s)} = +$ because of one of the first two clauses of eq. (6.1) then $\sigma_s = + = \sigma_{\beta(s)}$. If $\sigma_{\beta(s)} = -$ but $v \neq \varepsilon$ then $\sigma_s = + > \sigma_{\beta(s)}$. Otherwise $\sigma_s = - = \sigma_{\beta(s)}$.

s IS AN AXIOM: Then $\Gamma_s \mid \Lambda_s$ is an axiom of $\Lambda(\mathcal{R})$.

s IS AN INNER NODE: Then s was derived from $\mathrm{Chld}(s) = \{t_1, \ldots, t_n\}$ by an application of an $R(\mathcal{R})$-rule r and there are $\Lambda(\mathcal{R})$-proofs of $\Gamma_{t_i} \mid \Lambda_{t_i}$. There are various cases to consider:

$r \in \mathcal{R}$: That is, r is an $\mathcal{R}$-rule 'lifted into' $R(\mathcal{R})$. The rule R has a corresponding $\Lambda(\mathcal{R})$-rule of the same name, To derive $\Gamma_s \mid \Lambda_s$ from $\{(\Gamma_{t_i} \mid \Lambda_{t_i}) \mid 1 \leq i \leq n\}$ we must distinguish two cases for each t_i: either $t_i \in \mathrm{im}(\beta)$ or not. For the purpose of illustration, let $t_1 \notin \mathrm{im}(\beta)$ and $t_2 \in \mathrm{im}(\beta)$. The following derivation, in which $t_3, \ldots, t_n$ are treated analogously to t_1 or t_2 based on the aforementioned case-distinction, is as desired.

$$
\dfrac{
\dfrac{\dfrac{\Gamma_{t_1} \mid \Lambda_{t_1}}{\Gamma_{t_1} \mid \Lambda'_1}\,\textsc{Drop}}{\Gamma_{t_1} \mid r_1(\Lambda_s)}\,\textsc{Pop}
\qquad
\dfrac{\dfrac{\dfrac{\Gamma_{t_2} \mid \Lambda^*_2; (\Gamma^{t_2}_{t_2}; \pi(\Lambda^*_2) \mapsto \Gamma_{t_2})}{\Gamma_{t_2} \mid \Lambda^*_2}\,\textsc{Comp}}{\Gamma_{t_2} \mid \Lambda'_2}\,\textsc{Drop}}{\Gamma_{t_2} \mid r_2(\Lambda_s)}\,\textsc{Pop}
\qquad \cdots \qquad
\begin{array}{c}\Gamma_{t_i} \mid \Lambda_{t_i} \\ \vdots \\ \Gamma_{t_i} \mid r_i(\Lambda_s)\end{array}
\qquad \cdots
}{\Gamma_s \mid \Lambda_s}\,r_{\Lambda_s}
$$

The rule r might transform its labels Λ_s into various $r_i(\Lambda_s)$ in its premises. These might differ from the desired Λ_{t_i}. By lemma 6.3.7 $\Sigma(t_1) \leq \Sigma(s)$ and there exists $u \leq \Sigma(s)$ such that $\Sigma(t_2) = u \cdot s'$ for $\beta(s') = t_2$. Thus, the $r_i(\Lambda_s)$ might contain 'too many labels' (the difference between $\Sigma(s)$ and $\Sigma(t_1)$ or u, respectively), which can be corrected for with an application of POP, resulting in labels Λ'_i. Further, some progress set in $\pi(\Lambda'_i)$ may contain some x^+ while the corresponding set in $\pi(\Lambda_{t_i})$ only contains x^- (but not the other way around), because progress is only tracked by the Λ_{t_i} after the first RESET along the respective cycles. Such 'spurious' values can be erased in the same DROP-application. If $t_i \notin \mathrm{im}(\beta)$ then the labels in the premise of DROP will be Λ_{t_i} as desired. Otherwise, an application of the COMP-rule will extend the Λ^*_i resulting from the DROP-application to the desired Λ_{t_i}.

$r = \textsc{Reset}_a$: Let $\mathrm{Chld}(s) = \{t\}$. We claim that $\Lambda_s = \Lambda_t$ and thus the assumed proof of $\Lambda_t \mid \Gamma_t$ already closes the claim. To conclude $\Lambda_s = \Lambda_t$, it suffices to prove $\pi(\Lambda_s) = \pi(\Lambda_t)$. Thus, let $t' \in S(s) = S(t)$. If $a \neq a_{t'}$ then trivially $P^{t'}_s = P^{t'}_t$. Suppose $a = a_{t'}$, then for $x \in \iota(\Gamma_s)$ we have $ua_{t'} \in \sigma_t(x)$ iff $ua_{t'}v \in \sigma_s(x)$ for some non-empty v. It suffices to prove $(x, ua_{t'}v)^{t'}_s = (x, ua_{t'})^{t'}_t$. We must distinguish two cases: If no $\textsc{Reset}_{a_{t'}}$s have taken place along $[\beta(t'), s]$ then $(x, ua_{t'}v)^{t'}_s = x^- = (x, ua_{t'})^{t'}_t$ as neither the first nor the second clause apply for either. If at least one $\textsc{Reset}_{a_{t'}}$ has taken place along $[\beta(t'), s]$ then $(x, ua_{t'}v)_s^{t'} = x^+ = (x, ua_{t'})_t^{t'}$ because either the first or second clause apply for either.

$r = \textsc{Weak}$: Let $\mathrm{Chld}(s) = \{t\}$. The only difference between Λ_t and Λ_s is that possibly $P_t^{t'} \subseteq P_s^{t'}$ for each $t' \in S(s) = S(t)$. This can be accounted for with an application of $\textsc{Drop}$.

$r = \textsc{Pop}$: Let $\mathrm{Chld}(s) = \{t\}$. We know that σ_s and σ_t may only differ by the presence of more empty stacks in σ_t. As the definition of $\Lambda_\bullet$ only considers non-empty stacks, we have $\Lambda_s = \Lambda_t$ and thus the assumed proof of $\Lambda_t \mid \Gamma_t$ closes the proof.

To close the proof, simply observe that $\Pi'[\varepsilon] \vdash \Gamma \mid \varepsilon$ in $\Lambda(\mathcal{R})$ as desired. □

Example 6.3.9. The argument above requires strict progressivity because it ensures that every trace value tracked along a cycle progresses at least once. This in turn is required for soundness (see theorem 6.2.1). Observe that if only one reset takes place along the cycle of a reset proof, this need not be the case. For this, consider the following trace maps $\tau, \tau' : X \to X$ with $X := \{x, y\}$:

$$\tau := \{(x, 0, y), (y, 1, x)\} \quad \tau' := \{(x, 1, x), (y, 0, y)\}.$$

We give a sketch of a cycle of a reset proof. Here, the underlying set of the Safra boards is always X. We write $(\Theta; x^\theta, y^{\theta'}) := (\Theta; \{(x, \theta), (y, \theta')\})$.

$$(ab; x^{ab}, y^a) \overset{\tau}{\rightsquigarrow} (abc; x^{ac}, y^{ab}) \overset{R}{\rightsquigarrow} (a; x^a, y^a) \overset{\tau'}{\rightsquigarrow} (ab; x^{ab}, y^a)$$

First observe that this is a successful cycle, as an a-reset takes place at the R-step and a in an invariant. However, only the $y \to x$ and not the $x \to y$ trace progresses along one 'round' of the cycle.

6.4 Examples

We present two examples of stack-controlled proof systems for concrete cyclic proof systems. As for the reset proof systems in section 5.4, we make some 'ergonomic adjustments'. For the system $\mathrm{SHA}_<$ corresponding to Simpson's $\mathrm{CHA}_<$, given in section 6.4.1, these adjustments are quite minor. In section 6.4.2, we define a system SHA corresponding to simple CHA. It differs from $\Lambda(\mathrm{CHA})$ quite substantially as SHA incorporates various simplifications to the control stack afforded by the simplicity of traces of CHA.

6.4.1 Simpson's $\mathrm{CHA}_<$

Definition 6.4.1. The *sequents* of $\mathrm{SHA}_<$ are expressions $\Lambda \mid \Gamma \Rightarrow \delta$ where $\Gamma \Rightarrow \delta$ is a $\mathrm{CHA}_<$-sequent. The component Λ is a stack of *companion labels*. A companion label is an expression $X; \vec{Y} \mapsto \Gamma' \Rightarrow \delta'$ where $\Gamma' \Rightarrow \delta'$ is a $\mathrm{CHA}_<$-sequent, and further X and $Y \in \vec{Y}$ are non-empty sets of *progress annotated terms* i.e., expressions t^σ with $\sigma \in \{-, +\}$. For a $\mathrm{SHA}_<$-sequent $\Lambda \mid \Gamma \Rightarrow \delta$ to be well-formed, we further require two conditions of Λ:

(i) all $X; \vec{Y} \mapsto \Gamma' \Rightarrow \delta' \in \Lambda$ are such that $\{t \mid t^\sigma \in X\} \subseteq \textsc{Term}(\Gamma, \delta)$, and

(ii) if $\Lambda = \Lambda_1; (X; \vec{Y} \mapsto \Gamma' \Rightarrow \delta'); \Lambda_2$ then $|\vec{Y}| = |\Lambda_1| + 1$ and $\{t \mid t^\sigma \in Y_i\} \subseteq \textsc{Term}(\Gamma', \delta')$.

For a set X of progress annotated terms, we define $X^\sigma := \{t^\sigma \mid t^{\sigma'} \in X\}$. For two such sets X, Y we define

$$X \preceq Y := \forall t^\sigma \in X \exists \sigma' \geq \sigma.\, t^{\sigma'} \in Y$$

where $- < +$. We also define

$$\pi(\varepsilon) := \varepsilon \quad \pi(\Lambda; (X; \vec{Y} \mapsto \ldots)) := \pi(\Lambda), X.$$

Most rules of $\mathrm{SHA}_<$ are unsurprising first-order or arithmetical rules which do not interact with Λ, except mirroring the behaviour of $\textsc{Drop}$ in the case of rules which remove formulas, and thus possibly annotated term, from the sequent, e.g. $\textsc{Wk}$ or $\vee$R. That is, the Λ^* in $\Lambda^* \mid \Gamma \Rightarrow \delta$ denotes the result of replacing every $(X; \vec{Y} \mapsto \ldots)$ in Λ by $(\{t^\sigma \in X \mid t \in \textsc{Term}(\Gamma, \delta)\}, \vec{Y})$.

$$\text{Ax}\frac{}{\Lambda \mid \Gamma, \delta \Rightarrow \delta} \qquad \to\text{L}\frac{\Lambda^* \mid \Gamma \Rightarrow \varphi \quad \Lambda^* \mid \Gamma, \psi \Rightarrow \delta}{\Lambda \mid \Gamma, \varphi \to \psi \Rightarrow \delta}$$

$$\to\text{R}\frac{\Lambda \mid \Gamma, \varphi \Rightarrow \psi}{\Lambda \mid \Gamma \Rightarrow \varphi \to \psi} \qquad \wedge\text{L}\frac{\Lambda \mid \Gamma, \varphi, \psi \Rightarrow \delta}{\Lambda \mid \Gamma, \varphi \wedge \psi \Rightarrow \delta}$$

$$\wedge\text{R}\frac{\Lambda^* \mid \Gamma \Rightarrow \varphi \quad \Lambda^* \mid \Gamma \Rightarrow \psi}{\Lambda \mid \Gamma \Rightarrow \varphi \wedge \psi} \qquad \vee\text{L}\frac{\Lambda^* \mid \Gamma, \varphi \Rightarrow \delta \quad \Lambda^* \mid \Gamma, \psi \Rightarrow \delta}{\Lambda \mid \Gamma, \varphi \vee \psi \Rightarrow \delta}$$

$$\vee\text{R}\frac{\Lambda \mid \Gamma \Rightarrow \varphi_i}{\Lambda \mid \Gamma \Rightarrow \varphi_1 \vee \varphi_2} \qquad \forall\text{L}\frac{\Lambda \mid \Gamma, \varphi[t/x] \Rightarrow \delta}{\Lambda \mid \Gamma, \forall x.\varphi \Rightarrow \delta}$$

$$\forall\text{R}\frac{\Lambda \mid \Gamma \Rightarrow \varphi \quad x \notin \mathrm{FV}(\Gamma)}{\Lambda \mid \Gamma \Rightarrow \forall x.\varphi} \qquad \exists\text{L}\frac{\Lambda \mid \Gamma, \varphi \Rightarrow \delta \quad x \notin \mathrm{FV}(\Gamma, \delta)}{\Lambda \mid \Gamma, \exists x.\varphi \Rightarrow \delta}$$

$$\exists\text{R}\frac{\Lambda \mid \Gamma \Rightarrow \varphi[t/x]}{\Lambda \mid \Gamma \Rightarrow \exists x.\varphi} \qquad \bot\text{L}\frac{}{\Lambda \mid \Gamma, \bot \Rightarrow \delta} \qquad =\text{R}\frac{}{\Lambda \mid \Gamma \Rightarrow t = t}$$

$$\textsc{Wk}\frac{\Lambda^* \mid \Gamma \Rightarrow \delta}{\Lambda \mid \Gamma, \Gamma' \Rightarrow \delta} \qquad \textsc{Cut}\frac{\Lambda^* \mid \Gamma \Rightarrow \varphi \quad \Lambda \mid \Gamma, \varphi \Rightarrow \delta}{\Lambda \mid \Gamma \Rightarrow \delta}$$

the following arithmetic-specific axioms $\Lambda \mid \Gamma \Rightarrow \delta$ for the sequents $\Gamma \Rightarrow \delta$ listed below

$$\begin{aligned}
s < t, t < u &\Rightarrow s < u & s < t &\Rightarrow Ss < St \\
&\Rightarrow s + St = S(s + t) & s < t, t < s &\Rightarrow \bot \\
&\Rightarrow s < t \vee s = t \vee t < s & &\Rightarrow t \cdot 0 = 0 \\
s < t, t < Ss &\Rightarrow \bot & &\Rightarrow t < St \\
&\Rightarrow s \cdot St = (s \cdot t) + s & t < 0 &\Rightarrow \bot \\
&\Rightarrow t + 0 = t
\end{aligned}$$

and the arithmetic-specific derivation rule

$$\text{S}\frac{\Lambda \mid \Gamma, t = Sx \Rightarrow \delta \quad x \notin \text{FV}(\Gamma, t, \delta)}{\Lambda \mid \Gamma, 0 < t \Rightarrow \delta}.$$

The following are rules of arithmetic which interact with Λ in some way:

$$<\text{L}\frac{\Lambda^+ \mid \Gamma \Rightarrow \delta}{\Lambda \mid \Gamma, s < t \Rightarrow \delta} \qquad \textsc{Sub}\frac{\Lambda' \mid \Gamma \Rightarrow \delta}{\Lambda \mid \Gamma[t/x] \Rightarrow \delta[t/x]}$$

$$=\text{L}\frac{\Lambda^= \mid \Gamma[t/x, s/y] \Rightarrow \delta[t/x, s/y] \quad x, y \notin \text{FV}(s, t)}{\Lambda \mid \Gamma[s/x, t/y], s = t \Rightarrow \delta[s/x, t/y]}$$

For $<$L we replace every $(X; \vec{Y} \mapsto \ldots)$ in Λ by $(X[s \rightsquigarrow^+ s]; \vec{Y} \mapsto \ldots)$ in Λ^+ where

$$X[t \rightsquigarrow^+ s] := \begin{cases} X \cup \{s^+\} & t^\sigma \in X \text{ for some } \sigma \in \{-, +\} \\ X & \text{otherwise} \end{cases}.$$

For Sub we replace every $(X; \vec{Y} \mapsto \ldots)$ in Λ by $(X'; \vec{Y} \mapsto \ldots)$ in Λ' where

$$X' := \bigcup_{u^\sigma \in X} \{v^\sigma \mid v \in \textsc{Term}(\Gamma, \delta) \wedge v[t/x] = u\},$$

in other words: X' contains the same terms as X (if they are preserved in Γ by the substitution) and all substitution instances, which originate from terms in X.

For =L we replace every $(X; \vec{Y} \mapsto \dots)$ in Λ by $(X^=; \vec{Y} \mapsto \dots)$ in $\Lambda^=$ where

$$X^= := \bigcup_{u^\sigma \in X} \{v^\sigma \mid w \in \textsc{Term}(\Gamma, \delta) \wedge u = w[s/x, t/y] \wedge v = w[t/x, s/y]\}.$$

Below are the rules for manipulating the stack and building cycles, the last of the rules of $\mathrm{SHA}_<$:

$$\textsc{Pop}\frac{\Lambda_0 \mid \Gamma \Rightarrow \delta}{\Lambda_0; \Lambda_1 \mid \Gamma \Rightarrow \delta} \qquad \textsc{Drop}\frac{\Lambda' \mid \Gamma \Rightarrow \delta}{\Lambda \mid \Gamma \Rightarrow \delta}$$

$$\textsc{Comp}\frac{\Lambda, (X^-; \pi(\Lambda), X^+ \mapsto \Gamma \Rightarrow \delta) \mid \Gamma \Rightarrow \delta}{\Lambda \mid \Gamma \Rightarrow \delta}$$

$$\textsc{Bud}\frac{}{\Lambda; (X^+; \pi(\Lambda), X^+ \mapsto \Gamma \Rightarrow \delta) \mid \Gamma \Rightarrow \delta}$$

where in $\textsc{Drop}$ Λ' is identical to Λ except that $\pi(\Lambda') \preceq \pi(\Lambda)$ i.e., a label $(X; \vec{Y} \mapsto \dots) \in \Lambda$ may be replaced by $(X'; \vec{Y} \mapsto \dots)$ with $X' \preceq X$.

A proof in $\mathrm{SHA}_<$ is any well-formed derivation in the proof system described above. A sequent $\Gamma \Rightarrow \delta$ of $\mathrm{CHA}_<$-formulas is provable in $\mathrm{SHA}_<$, denoted $\mathrm{SHA}_< \vdash \Gamma \Rightarrow \delta$, if there is a $\mathrm{SHA}_<$-proof of $\varepsilon \mid \Gamma \Rightarrow \delta$.

Completeness of $\mathrm{SHA}_<$ follows directly from the completeness of $\Lambda(\mathrm{CHA}_<)$.

Theorem 6.4.2 (Completeness)**.** If $\mathrm{CHA}_< \vdash \Gamma \Rightarrow \delta$ then $\mathrm{SHA}_< \vdash \Gamma \Rightarrow \delta$.

Proof. By theorem 6.3.8, there exists a $\Lambda(\mathrm{CHA}_<)$-proof Π of $\Gamma \Rightarrow \delta \mid \varepsilon$. It suffices to prove that if $\Lambda(\mathrm{CHA}_<)$ proves $\Gamma \Rightarrow \delta \mid \Lambda$ then $\mathrm{SHA}_<$ proves $\Lambda \mid \Gamma \Rightarrow \delta$, noting that the structure of stack labels coincides between $\Lambda(\mathrm{CHA}_<)$ and $\mathrm{SHA}_<$. We prove this per induction on the derivation and the only interesting case is the rules r_Λ where $r \in \mathrm{CHA}_<$: Similarly to the system $\mathrm{RHA}_<$ of section 5.4.1, $\mathrm{SHA}_<$ features an explicit <L-rule, rather than progressing terms along inequalities $s < t$ in every rule. Thus, consider a rule

$$r_\Lambda\frac{\Gamma_1 \Rightarrow \delta_1 \mid r_1(\Lambda) \quad \dots \quad \Gamma_n \Rightarrow \delta_n \mid r_n(\Lambda)}{\Gamma \Rightarrow \delta \mid \Lambda}$$

and its 'corresponding' $\mathrm{SHA}_<$-rule

$$r'\frac{\Lambda_1 \mid \Gamma_1 \Rightarrow \delta_1 \quad \dots \quad \Lambda_n \mid \Gamma_n \Rightarrow \delta_n}{\Lambda \mid \Gamma \Rightarrow \delta}.$$

Write $P^i; \vec{G}^i \mapsto \Xi_i \Rightarrow \varphi_i$ for the companion labels in Λ_i and $P^i_*; \vec{G}^i \mapsto \Xi_i \Rightarrow \varphi_i$ for those is $r_i(\Lambda)$. We know that $P_i \subseteq P^i_*$ and further that $P^i_* \setminus P_i = \{t^+_1, \dots, t^+_m\}$

where $t_j < s_j \in \Gamma_i$ and $s_j^\bullet \in P_i$ for all $j \leq m$. Then we may simulate the effects of r_Λ in $\mathrm{SHA}_<$ as follows

$$\dfrac{\cdots \quad \dfrac{r_i(\Lambda) \mid \Gamma_i \Rightarrow \delta_i}{\Lambda_i \mid \Gamma_i \Rightarrow \delta_i}{<}\mathrm{L}^* \quad \cdots}{\Lambda \mid \Gamma \Rightarrow \delta}\, r' \qquad \square$$

Similarly, soundness of $\mathrm{SHA}_<$ is quite straight-forward using the result from section 6.2.

Lemma 6.4.3 (Soundness)**.** If $\mathrm{SHA}_< \vdash \Gamma \Rightarrow \delta$ then $\mathrm{CHA}_< \vdash \Gamma \Rightarrow \delta$.

Proof. By theorem 6.2.1 it suffices to prove that $\Lambda(\mathrm{CHA}_<) \vdash \Gamma \Rightarrow \delta$. For this, we prove that if $\mathrm{SHA}_<$ derives $\Lambda \vdash \Gamma \Rightarrow \delta$ then $\Lambda(\mathrm{CHA}_<)$ derives $\Gamma \Rightarrow \delta \mid \Lambda$ per induction on the derivation. As in the proof of theorem 6.4.2, the only important difference between the two systems is the handling of progress: In $\Lambda(\mathrm{CHA}_<)$, every rule r_Λ progresses all possible terms in accordance with the trace interpretation of $\mathrm{CHA}_<$. In $\mathrm{SHA}_<$, there is a dedicated <L-rule which progresses terms along an inequality. In the translation from $\mathrm{SHA}_<$ to $\Lambda(\mathrm{CHA}_<)$ this matters in two ways: First, we must 'simulate' <L in $\Lambda(\mathrm{CHA}_<)$. This can be done by first applying an appropriate $\mathrm{W{\scriptsize K}}_\Lambda$ which does not actually discard any formulas and then removing all spuriously progressed terms with Drop. Second, when translating rules other than $< L$, Pop, Drop, Comp and Bud i.e., when applying some r_Λ with $r \in \mathrm{CHA}_<$, then this r_Λ might spuriously progress terms which must be corrected via Drop in the premises of the r_Λ-application. $\square$

6.4.2 Simple CHA★

The system SHA we present here corresponds to simple CHA and takes advantage of the simplicity of CHA-traces: It suffices to track only a single variable per cycle whose identity does not change.

The sequents of SHA are expressions $\Lambda \mid \Gamma \Rightarrow \delta$, consisting of an HA sequent $\Gamma \Rightarrow \delta$ and an ordered list Λ of *companion labels*, which is treated as a stack. We denote the empty list by ε. A companion label is an expression of the shape $x^\bullet \mapsto \Gamma \Rightarrow \delta$ for a HA-sequent $\Gamma \Rightarrow \delta$, a variable $x \in \mathrm{FV}(\Gamma, \delta)$ and where $x^\bullet \in \{x^+, x^-\}$. We write $\mathrm{Var}(\Lambda)$ for the set $\{x \mid (x^\bullet \mapsto \Gamma \Rightarrow \delta) \in \Lambda\}$ of variables with associated companion labels in Λ.

★ The system SHA presented in this section stems from joint work with Graham E. Leigh, a preprint being available at (Leigh and Wehr 2025). In the aforementioned article, SHA is obtained from CHA via induction orders, rather than reset proofs as presented in this chapter.

A SHA-sequent $\Lambda \mid \Gamma \Rightarrow \delta$ is *well-formed* if for every $x^\bullet \mapsto \Delta \Rightarrow \gamma \in \Lambda$ we have $x \in \mathrm{FV}(\Gamma, \delta)$. Going forward, we assume that every SHA-sequent $\Lambda \mid \Gamma \Rightarrow \delta$ under consideration is well-formed, unless stated otherwise.

Before presenting the rules of the calculus SHA, we fix further operations on stacks of companion labels. Given a variable x, we define Λ^{+x} to be the result of replacing in Λ all occurrences of x^- by x^+:

$$\varepsilon^{+x} := \varepsilon \qquad (y^\bullet \mapsto \Gamma \Rightarrow \delta; \Lambda)^{+x} := \begin{cases} x^- \mapsto \Gamma \Rightarrow \delta; \Lambda^{+x}, & \text{if } y = x, \\ y^\bullet \mapsto \Gamma \Rightarrow \delta; \Lambda^{+x}, & \text{otherwise.} \end{cases}$$

Definition 6.4.4. The *derivation rules* of SHA are of split into three categories:

LOGICAL RULES, which do not interact with companion labels:

$$\textsc{Ax}\,\frac{}{\varepsilon \mid \Gamma, \delta \Rightarrow \delta} \qquad {\to}\mathrm{L}\,\frac{\Lambda^* \mid \Gamma \Rightarrow \varphi \quad \Lambda^* \mid \Gamma, \psi \Rightarrow \delta}{\Lambda \mid \Gamma, \varphi \to \psi \Rightarrow \delta}$$

$${\to}\mathrm{R}\,\frac{\Lambda \mid \Gamma, \varphi \Rightarrow \psi}{\Lambda \mid \Gamma \Rightarrow \varphi \to \psi} \qquad {\wedge}\mathrm{L}\,\frac{\Lambda \mid \Gamma, \varphi, \psi \Rightarrow \delta}{\Lambda \mid \Gamma, \varphi \wedge \psi \Rightarrow \delta}$$

$${\wedge}\mathrm{R}\,\frac{\Lambda^* \mid \Gamma \Rightarrow \varphi \quad \Lambda^* \mid \Gamma \Rightarrow \psi}{\Lambda \mid \Gamma \Rightarrow \varphi \wedge \psi}$$

$${\vee}\mathrm{L}\,\frac{\Lambda^* \mid \Gamma, \varphi \Rightarrow \delta \quad \Lambda^* \mid \Gamma, \psi \Rightarrow \delta}{\Lambda \mid \Gamma, \varphi \vee \psi \Rightarrow \delta} \qquad {\vee}\mathrm{R}\,\frac{\Lambda \mid \Gamma \Rightarrow \varphi_i}{\Lambda \mid \Gamma \Rightarrow \varphi_1 \vee \varphi_2}$$

$${\forall}\mathrm{L}\,\frac{\Lambda \mid \Gamma, \varphi[t/x] \Rightarrow \delta}{\Lambda \mid \Gamma, \forall x.\varphi \Rightarrow \delta} \qquad {\forall}\mathrm{R}\,\frac{\Lambda \mid \Gamma \Rightarrow \varphi \quad x \notin \mathrm{FV}(\Gamma)}{\Lambda \mid \Gamma \Rightarrow \forall x.\varphi}$$

$${\exists}\mathrm{L}\,\frac{\Lambda \mid \Gamma, \varphi \Rightarrow \delta \quad x \notin \mathrm{FV}(\Gamma, \delta)}{\Lambda \mid \Gamma, \exists x.\varphi \Rightarrow \delta} \qquad {\exists}\mathrm{R}\,\frac{\Lambda \mid \Gamma \Rightarrow \varphi[t/x]}{\Lambda \mid \Gamma \Rightarrow \exists x.\varphi}$$

$${=}\mathrm{L}\,\frac{\Lambda^* \mid \Gamma[x/y] \Rightarrow \delta[x/y] \quad x, y \notin \mathrm{FV}(s, t)}{\Lambda \mid \Gamma[s/x, t/y], s = t \Rightarrow \delta[s/x, t/y]}$$

$${=}\mathrm{R}\,\frac{}{\Lambda \mid \Gamma \Rightarrow t = t} \qquad {\perp}\mathrm{L}\,\frac{}{\Lambda \mid \Gamma, \perp \Rightarrow \delta} \qquad \textsc{Wk}\,\frac{\Lambda \mid \Gamma \Rightarrow \delta}{\Lambda \mid \Gamma, \Gamma' \Rightarrow \delta}$$

$$\textsc{Cut}\,\frac{\Lambda^* \mid \Gamma \Rightarrow \varphi \quad \Lambda \mid \Gamma, \varphi \Rightarrow \delta}{\Lambda \mid \Gamma \Rightarrow \delta} \qquad \textsc{Sub}\,\frac{\Lambda^* \mid \Gamma \Rightarrow \delta}{\Lambda \mid \Gamma[t/x] \Rightarrow \delta[t/x]}$$

where Λ^* are adjustments of Λ which ensure well-formedness of the sequent.

That is, let $\Gamma \Rightarrow \delta$ be some HA-sequent and Λ some companion label stack, not necessarily such that $\Lambda \mid \Gamma \Rightarrow \delta$ is a well-formed sequent. Then we define $\Lambda^* \mid \Gamma \Rightarrow \delta$ recursively with $\varepsilon^* := \varepsilon$ and

$$(x^\bullet \mapsto \Gamma' \Rightarrow \delta'; \Lambda)^* := \begin{cases} x^\bullet \mapsto \Gamma' \Rightarrow \delta'; \Lambda^* & \text{if } x \in \mathrm{FV}(\Gamma, \delta) \\ \varepsilon & \text{otherwise} \end{cases}.$$

Arithmetical axioms of the form:

$$\begin{array}{ll} \varepsilon \mid \Gamma \Rightarrow 0 \neq St & \varepsilon \mid \Gamma, Ss = St \Rightarrow s = t \\ \varepsilon \mid \Gamma \Rightarrow s + 0 = 0 & \varepsilon \mid \Gamma \Rightarrow s + St = S(s + t) \\ \varepsilon \mid \Gamma \Rightarrow s \cdot 0 = 0 & \varepsilon \mid \Gamma \Rightarrow s \cdot St = (s \cdot t) + s \end{array}$$

Label rules which manipulate companion labels:

$$\textsc{Comp}\,\frac{\Lambda; (x^- \mapsto \Gamma \Rightarrow \delta) \mid \Gamma \Rightarrow \delta}{\Lambda \mid \Gamma \Rightarrow \delta}$$

$$\textsc{Bud}\,\frac{}{\Lambda; (x^+ \mapsto \Gamma \Rightarrow \delta) \mid \Gamma \Rightarrow \delta} \qquad \textsc{Pop}\,\frac{\Lambda_0 \mid \Gamma \Rightarrow \delta}{\Lambda_0; \Lambda_1 \mid \Gamma \Rightarrow \delta}$$

$$\textsc{Case}_x\,\frac{\Lambda^* \mid \Gamma(0) \Rightarrow \delta(0) \qquad \Lambda^{+x} \mid \Gamma(Sx) \Rightarrow \delta(Sx)}{\Lambda \mid \Gamma(x) \Rightarrow \delta(x)}$$

A proof in SHA is any well-formed derivation in the proof system described above. A sequent $\Gamma \Rightarrow \delta$ of CHA-formulas is provable in SHA, denoted SHA $\vdash \Gamma \Rightarrow \delta$, if there is a SHA-proof of $\varepsilon \mid \Gamma \Rightarrow \delta$.

The following amounts to the admissibility of the Drop-rule in SHA.

Lemma 6.4.5. Let SHA derive $\Lambda \mid \Gamma \Rightarrow \delta$ then it also derives $\Lambda' \mid \Gamma \Rightarrow \delta$ where in Λ', any label $x^- \mapsto \Gamma' \Rightarrow \delta'$ in Λ may be replaced by $x^+ \mapsto \Gamma' \Rightarrow \delta'$.

Proof. Easy induction on the derivation of $\Lambda \mid \Gamma \Rightarrow \delta$. □

A key observation behind the simplification SHA makes over Λ(CHA) is the fact that the trace interpretation τ giving rise to CHA only employs trace maps which are partial injections.

Definition 6.4.6. Let $\mathcal{A}$ be an activation algebra. A trace map τ over $\mathbb{B}$ is a partial injection if

1. it is functional: for any $x \in X$ if $(x, b, y), (x, b', y') \in \tau$ then $y = y'$, and

2. it is injective: for any $y \in Y$ if $(x, b, y), (x', b', y) \in \tau$ then $x = x'$.

Fix a trace interpretation $\iota : \mathcal{R} \longrightarrow \mathbb{B}$ for some derivation system $\mathcal{R}$. We say ι *only employs partial injections* if for every $r \in \mathcal{R}$ with $\rho(r) = (\Gamma, \Delta_1, \dots, \Delta_n)$ every $r_i : \iota(\Gamma) \longrightarrow \iota(\Delta_i)$ is a partial injection.

Towards completeness, first observe the following key result about $R(\mathcal{R})$-proofs induced by trace interpretations which employ only partial injections.

Lemma 6.4.7. Let $R(\mathcal{R})$ be induced by $\iota : \mathcal{R} \longrightarrow \mathbb{B}$ which only employs partial injections. Further, let Π be an $R(\mathcal{R})$-proof with end-sequent $\Gamma; (\emptyset; \_ \mapsto \emptyset)$ and $\Delta; (\Theta, \sigma)$ be one of its sequents. For any $a \in \Theta$ there exists precisely one $x \in \iota(\Delta)$ such that $a \in^2 \sigma(x)$ i.e., there is some S such that $a \in S \in \sigma(x)$.

Proof. We prove the claim for any $\Gamma_v; (\Theta_v, \sigma_v) = \lambda(v)$ and $v \in T$ per induction on the length of v.

$v = \varepsilon$: Trivial as $\Theta = \emptyset$.

$v = v'n$: We perform a case-distinction on the rule applied at v':

Pop: Then $n = 0$. If $a \in \Theta_{v'0} = \Theta_{v'}$ then there exists precisely one $x \in \iota(\Gamma_{v'}) = \iota(\Gamma_{v'0})$ such that $a \in^2 \sigma_{v'}(x)$. As $\sigma_{v'}(x) \subseteq \sigma_{v'0}(x)$ we know $a \in^2 \sigma_{v'0}(x)$. It can be in no other $\sigma_{v'0}(y)$ because the symmetric difference $\sigma_{v'}(y) \Delta \sigma_{v'0}(y) \subseteq \{\emptyset\}$.

Weak: Then $n = 0$. If $a \in \Theta_{v'0} \subseteq \Theta_{v'}$ and there exists precisely one x with $a \in^2 \sigma_{v'}(x)$. We know there must be $y \in \iota(\Gamma_{v'0})$ with $a \in^2 \sigma_{v'0}(y)$ as $a \notin \Theta$ otherwise. As for every $y \in \iota(\Gamma_{v'0})$ we have $\sigma_{v'0}(y) \subseteq \sigma_{v'}(y)$, only $a \in^2 \sigma_{v'0}(x)$ is possible.

Reset$_b$: Then $n = 0$. If $a \in \Theta_{v'0} \subseteq \Theta_{v'}$ there exists precisely one x such that $a \in^2 \sigma_{v'}(x)$. As in the Weak-case, $a \in^2 \sigma_{v'0}(y)$ for some y and $y \neq x$ is impossible as for any $S \in \sigma_{v'0}(y)$ there is $S' \in \sigma_{v'}(y)$ with $S' \subseteq S$ and $a \in^2 \sigma_{v'}(x)$ precisely.

$r \in \mathcal{R}$: That means $(\Theta_{v'}, \sigma_{v'}) \overset{r_i}{\rightsquigarrow} (\Theta_{v'n}, \sigma_{v'n})$. There are two cases to consider. If $a \in \Theta_{v'n} \setminus \Theta_{v'}$ then a was added as part of the Cover step, meaning per definition $a \in^2 \sigma_{v'n}(x)$ for precisely one x. Otherwise, if $a \in \Theta_{v'}$ then there exists a unique $x \in \iota(\Gamma_{v'})$ with $a \in^2 \sigma_{v'}(x)$. Because r_n is a partial injection, a can only be moved to at most one $y \in \iota(\Gamma_{v'n})$ in the Move step and will not be moved further during the Cover step. $\square$

We can now prove completeness. With the previous result, we can still rely on theorem 6.3.8.

Theorem 6.4.8 (Completeness)**.** If CHA $\vdash \Gamma \Rightarrow \delta$ then SHA $\vdash \Gamma \Rightarrow \delta$.

Proof. By theorem 6.3.8, there is a Λ(CHA)-proof $\Pi \vdash \Gamma \Rightarrow \delta \mid \varepsilon$. We make a few observations: First, note that lemma 6.4.7 applies to the R(CHA)-proof Π_R used by the proof of theorem 6.3.8 to construct Π. That means that for any sequent $\Gamma' \Rightarrow \delta' \mid \Lambda$ and any companion label $P_i; \vec{G^i} \mapsto \Gamma_i \Rightarrow \delta_i$ in Λ we have that $|P_i| \leq 1$ and $|G^i_j| = 1$ for every $j \leq i$. For the latter fact, observe that if $G^i_j = \emptyset$ at any point, this means that at the companion v in Π_R which corresponds to the Comp-instance which inserted this label, the invariant of the cycle $\Sigma(v)_j$ is not preserved anymore, contradicting monotonicity of Π_R.

We use the following result to translate Π to a SHA-proof: Let v be a node of Π such that $\lambda(v) = \Gamma_v \Rightarrow \delta_v \mid \Lambda_v$. If Λ_v is such that for every label $P_i; \vec{G^i} \mapsto \Gamma_i \Rightarrow \delta_i$ of Λ_v we have $P_i = \{x_i^{\sigma_1}\}$ for some $x_i \in \textsc{Var}$ then there is a derivation of $\Lambda'_v \mid \Gamma_v \Rightarrow \delta_v$ with $\Lambda'_v = (x_1^{\sigma_1} \mapsto \Gamma_1 \Rightarrow \delta_1; \ldots)$. We prove it per induction on the underlying tree of Π. This claim is completely trivial if $\delta(v) \in \{\textsc{Comp}, \textsc{Bud}, \textsc{Pop}\}$. If $\delta(v) = \textsc{Drop}$, simply observe that if the conclusion of an Drop-instance meets the preconditions laid out above then theorem 6.3.8 will never apply Drop in Π in a way that cannot be mirrored by an application of lemma 6.4.5, also meaning the IH may be applied to the premise of Drop to complete the proof. Lastly, let $\delta(v) = r_{\Lambda_v}$ for some $r \in$ CHA: Then we know, looking at the argument of theorem 6.3.8, the subproof above v in Π must look as follows:

$$\dfrac{\textsc{Pop}\,\dfrac{\Gamma_1 \Rightarrow \delta_1 \mid \Lambda_1}{\Gamma_1 \Rightarrow \delta_1 \mid r_1(\Lambda_v)} \quad \cdots \quad \textsc{Pop}\,\dfrac{\Gamma_i \Rightarrow \delta_i \mid \Lambda_i}{\Gamma_i \Rightarrow \delta_i \mid r_i(\Lambda_v)} \quad \cdots}{\Gamma_v \Rightarrow \delta_v \mid \Lambda_v}\, r_{\Lambda_v}$$

Where crucially, if any such Pop-instance actually removes any labels from $r_i(\Lambda_v)$, it does so exactly in the manner mirroring the operation $\Lambda*$ used to define the binary rules of SHA: If $x_j \notin \mathrm{FV}(\Gamma_i, \delta_i)$ for the variable x_j from the companion labels of Λ_v then the Pop-instance will remove its companion label and all those above it from $r_i(\Lambda_v)$. Thus, the SHA-rule corresponding to r may be applied to derive:

$$\dfrac{\Lambda'_1 \mid \Gamma_1 \Rightarrow \delta_1 \quad \ldots \quad \Lambda'_i \mid \Gamma_i \Rightarrow \delta_i \quad \ldots}{\Lambda'_v \mid \Gamma_v \Rightarrow \delta_v}\, r$$

Because of the observation above, the Λ_i satisfy the precondition, meaning the IH may be applied.

Applying the result above to ε of Π then yields a SHA-proof of $\varepsilon \mid \Gamma \Rightarrow \delta$ as desired. $\square$

To obtain soundness, it suffices to translate SHA-proofs into Λ(CHA)-proofs. As the latter system is less strict, this is rather straight-forward.

Theorem 6.4.9 (Soundness)**.** If SHA $\vdash \Gamma \Rightarrow \delta$ then CHA $\vdash \Gamma \Rightarrow \delta$.

Proof. It suffices to prove $\Lambda(\mathrm{CHA}) \vdash \Gamma \Rightarrow \delta$. We prove a slightly more general result: Let SHA derive $\Lambda \mid \Gamma \Rightarrow \delta$ with $\Lambda = (x_i^{\sigma_i} \mapsto \Gamma_i \Rightarrow \delta_i)_{i \leq n}$. Then $\Lambda(\mathrm{CHA})$ derives $\Gamma \Rightarrow \delta \mid \Lambda'$ for any Λ' with $|\Lambda'| = n$ and where the ith label $P_i; \vec{G}^i \mapsto \Gamma_i \Rightarrow \delta_i$ of Λ' is such that $P_i = \{x_i^{\sigma_i}\}$, $G_i^i = \{x_i^+\}$ and for every $j < i$ we have $G_j^i = \{x_j^{\sigma_j^i}\}$ where $\sigma_j^i \leq \sigma_j$.

We prove this per induction, only handling the cases which are of interest. Observe that the Λ^* operation in the binary rules of SHA may be handled analogously to its instance in the $\textsc{Case}_x$-rule below.

Bud: Then $\Lambda' = \Lambda_0; \{x_n^+\}; (G_j^n)_{j<n}, \{x_n^+\} \mapsto \Gamma \Rightarrow \delta$ with $G^n \preceq \pi(\Lambda_0)$. Then derive

$$\dfrac{\dfrac{}{\Gamma \Rightarrow \delta \mid \Lambda_0; \{x_n^+\}; \pi(\Lambda_0), \{x_n^+\} \mapsto \Gamma \Rightarrow \delta}\,\textsc{Bud}}{\Gamma \Rightarrow \delta \mid \Lambda_0; \{x_n^+\}; (G_j^n)_{j<n}, \{x_n^+\} \mapsto \Gamma \Rightarrow \delta}\,\textsc{Drop}$$

Comp: Let $x_{n+1}^- \mapsto \Gamma \Rightarrow \delta$ be the new SHA-companion label. Then derive:

$$\dfrac{\Gamma \Rightarrow \delta \mid \Lambda'; \{x_{n+1}^-\}; \pi(\Lambda'), \{x_{n+1}^+\} \mapsto \Gamma \Rightarrow \delta}{\Gamma \Rightarrow \delta \mid \Lambda'}$$

noting that $\Lambda'; \{x_{n+1}^-\}; \pi(\Lambda'), \{x_{n+1}^+\} \mapsto \Gamma \Rightarrow \delta$ satisfies the preconditions to apply the IH.

$\textsc{Case}_x$: This is only interesting if $x = x_i$ for some $i \leq n$. Then the last rule applied is as follows:

$$\dfrac{\Lambda^* \mid \Gamma(0) \Rightarrow \delta(0) \qquad \Lambda^{+x_i} \mid \Gamma(S\,x_i) \Rightarrow \delta(S\,x_i)}{\Lambda \mid \Gamma(x_i) \Rightarrow \delta(x_i)}\,\textsc{Case}_{x_i}$$

then derive

$$\dfrac{\dfrac{\Gamma(0) \Rightarrow \delta(0) \mid \Lambda'_*}{\Gamma(0) \Rightarrow \delta(0) \mid \textsc{Case}_{x_i,1}(\Lambda')}\,\textsc{Pop} \qquad \Gamma(S\,x_i) \Rightarrow \delta(S\,x_i) \mid \textsc{Case}_{x_i,2}(\Lambda')}{\Gamma(x_i) \Rightarrow \delta(x_i) \mid \Lambda'}\,\textsc{Case}_{x_i,\Lambda'}$$

where Λ'_* is simply the first $|\Lambda^*|$ companion labels from Λ' which we observe is a prefix of $\textsc{Case}_{x_i,1}(\Lambda')$ as these labels do not interact with x_i. Further note that both Λ'_* and $\textsc{Case}_{x_i,2}(\Lambda')$ are such that the inductive hypothesis may be applied. ☐

6.5 Discussion

Whereas our definition of abstract reset systems in chapter 5 covers trace conditions over any finite $\mathcal{A}$, the construction in this chapter is restricted to trace conditions over $\mathbb{B}$. This is, plainly, because we do not know how to extend this construction to arbitrary $\mathcal{A}$. One of the core difficulties is how to define a generalised variant of the Drop-rule (or, equivalently, how to give a generalised account of preservation). Naïvely, reading the annotations x^- and x^+ as an annotation with $\mathbb{B}$, one could consider companion labels collecting trace values x^a annotated with elements $a \in \mathcal{A}$. The progress rule over $\mathbb{B}$ we present here allows replacing a progress set P with a set P' over the same set of trace values as long as

$$\forall x^a \in P'. \exists b \geq a.\ x^b \in P.$$

Intuitively, this just reduces the level of recorded progress from P to P' in a harmless manner: any trace following the annotations through x ending in some y^+ will progress, even if it is 'forgotten' that x has already progressed previously. This definition does not work for general activation algebras. Recall the three point failure algebra $\mathbb{F} = \{0, 1, 2\}$ with $\alpha = 1$ in which the maximal value 2 represents an 'unrecoverable failure' (modelled by the fact that there is no $a \in \mathcal{A}$ with $2 \vee a$). By the original definition of Drop, replacing $P = \{x^2\}$ by $P' = \{x^0\}$ would be allowed, but this is not harmless anymore: A trace running through x with accumulated activation algebra element 2 will never progress anymore and 'forgetting' this fact would break the notion of 'preservation' embodied by the Drop-rule. It is possible that a subtler definition, such as

$$\forall x^a \in P'.\ a \leq \alpha \rightarrow \exists \alpha \geq b \geq a.\ x^b \in P$$

could work but we have not yet had time to investigate this further.

We know of one other construction in the literature which follow the same goal of aligning the physical and logical ordering of cycles of a cyclic proof. The ranked proofs of ranked proofs from Kuperberg et al. (2021) rely on a ranking function which dictates both the physical and logical ordering of cycles (and thus forces their alignment). However, these ranked proofs only seem applicable to the contraction-free fragment of the type theory they consider, and do not seem to easily generalise to unrestricted traces as our stack-controlled proofs.

As discussed in section 3.5, another way of ensuring the alignment of the physical and logical ordering of cycles is to simply restrict one's notion of 'proof' to only such aligned cyclic proofs in the first place. Examples of this in the literature are Curzi and Das (2021) and Beklemishev et al. (2025). A shortcoming of this method is that one might be excluding certain cyclic proofs which one is interested in for other reasons, such as proof theoretic content. No such problems arise for stack-controlled proofs because, by simply unfolding it, any cyclic proof can be represented as a stack-controlled proof.

Lastly, Grotenhuis and Otten are currently working on a construction similar to stack-controlled proofs, aiming to derive results similar to those in chapter 7 for type theories. Similar to us, they start by considering reset proofs. Recall that each cycle in a reset proof has a highest chip a in its invariant, the trace values which the cycle progresses and other cycles must preserve being exactly those annotated by a. They observe that in most cyclic proof systems, there exists, between the companion of the cycle and the root, an individual 'original' trace value o at whose progress point a is newly introduced. There must then be a trace between o and any a-annotated trace value in the cycle, meaning that o acts as a bound on all such trace values. Thus, instead of tracking the progress of each individual a-annotated trace value, it suffices to identify a decrease of this common bound along the cycle (or some unfolding thereof). Such cycles then quite directly correspond to induction on the common bound. This would lead to their translation method being applicable in more settings, as it does away with the reliance on a maximum-operation on trace values of the method we present here.

7 Cyclic proof theory

We have noted before that stack-controlled proof systems are structurally similar to proof systems with induction rules. This chapter we relies on this similarity to obtain various more concrete results in cyclic proof theory. In section 7.1, we give a translation from $\mathrm{SHA}_<$ into HA and thus show that $\mathrm{CHA}_<$ and HA prove the same theorems. In section 7.2 we refine this argument to identify fragments of $\mathrm{CPA}_<$ which correspond to the inductive fragments $\mathrm{I}\Pi_n$ for $n > 0$. Section 7.3 adapts Gentzen's proof of the consistency of PA to SPA. Lastly, we give an abstract presentation of a direct[22] soundness argument for cyclic proof systems in section 7.4, which relies on the completeness of stack-controlled proofs as an intermediate step.

7.1 Unravelling Simpson's $\mathrm{CHA}_<$

This section we gives a translation from $\mathrm{SHA}_<$ into HA. It can be seen as an extension of the translation from SHA to HA we give in section 3.4. Like said simpler results, the translation of $\mathrm{SHA}_<$ illustrates what we mean by the 'structural similarities' between stack-controlled proof systems and those with induction rules.

We introduce some useful notation to speak about the control stack Λ of a $\mathrm{CHA}_<$-sequent $\Lambda \mid \Gamma \Rightarrow \delta$. We shall name the components of the ith companion label, counting from the left and starting at 1, as follows: $(P_i; \vec{G}^i \mapsto \Gamma_i \Rightarrow \delta_i)$ and we shall denote by $\vec{x}_i$ some fixed sequentialisation of $\mathrm{FV}(\Gamma_i, \delta_i)$. By $\vec{y}_i, \vec{z}_i$, etc. we denote sequences of fresh variables of the same length as $\vec{x}_i$. Note that $\vec{x}_i$ and $\vec{x}_j$ need not be disjoint but we require that any $\vec{y}_i, \vec{z}_i, \vec{y}_j, \vec{z}_j$ are. Observe: Given a sequent $\Lambda \mid \Gamma \Rightarrow \delta$, consider a companion label $(P_i; \vec{G}^i \mapsto \Gamma_i \Rightarrow \delta_i)$: Then $\mathrm{FV}(P_i) \subseteq \mathrm{FV}(\Gamma, \delta)$ and $\mathrm{FV}(G^i_j) \subseteq \vec{x}_i$.

To translate from $\mathrm{SHA}_<$ to HA, a key step is to formulate the *inductive invariants* induced by the labels Λ of a sequent $\Lambda \mid \Gamma \Rightarrow \delta$. Each label in Λ gives rise to one invariant i.e., we define the set of invariants for a given Λ as $\mathcal{H}_\Lambda :=$

[22] That is, it avoids classical, 'indirect' reasoning. We are tempted to call it constructive. However, there are some details regarding ordinal representation which make this classification a little more subtle.

$\{\varphi_i^\Lambda \mid 0 < i \leq |\Lambda|\}$ where the invariant of the ith label $(P_i; \vec{G}^i \mapsto \Gamma_i \Rightarrow \delta_i)$ is defined as:

$$\varphi_i^\Lambda := \forall \vec{y}_i.\, \gamma^{\Lambda,i} \to (\Gamma_i \Rightarrow \delta_i)[\vec{y}_i/\vec{x}_i]$$

where we suppress the Λ superscript if it is clear from the context. Here we write $\Gamma_i \Rightarrow \delta_i := \bigwedge \Gamma_i \to \delta_i$. The *guards* are given by $\gamma^{\Lambda,i} := \bigwedge_{j \leq i} \gamma_j^{\Lambda,i}$ with

$$\gamma_j^{\Lambda,i} := \bigwedge_{t^\sigma \in G_j^i} \bigvee_{s^{\sigma'} \in P_j} t[\vec{y}_i/\vec{x}_i] <^{\sigma,\sigma'} s$$

where $<^{+,-} := <$ and $<^{\sigma,\sigma'} := \leq$ otherwise.

Before we prove the main embedding theorem, we observe some facts about syntax. All of these are easily verified.

Fact 7.1.1.

(i) Whenever $a <^{\sigma_a,\sigma_b} b <^{\sigma_b,\sigma_c} c$ then $a <^{\sigma_a,\sigma_c} c$.

(ii) $\sigma \leq \sigma'$ iff $<^{\sigma,\sigma'} = \leq$.

(iii) $\mathrm{FV}(\varphi_i^\Lambda) = \mathrm{FV}(\gamma^{\Lambda,i}) = \mathrm{FV}(\pi(\Lambda)) \subseteq \mathrm{FV}(\Gamma, \delta)$.

With the definitions fixed, we can prove the main theorem of this section: $\mathrm{SHA}_<$ embeds into HA, the control stack Λ being transformed into a set of inductive invariants $\mathcal{H}_\Lambda$.

Theorem 7.1.2. If $\Lambda \mid \Gamma \Rightarrow \delta$ is derivable in $\mathrm{SHA}_<$ then $\mathrm{HA} \vdash \Gamma, \mathcal{H}_\Lambda \Rightarrow \delta$.

Proof. The proof proceeds by induction on the structure of the $\mathrm{SHA}_<$-proof. Only the rules interacting with Λ, and thus the inductive invariant directly, are of interest and we shall restrict our attention to them. The rules in which Λ is 'cleared' via Λ^*, such as $\to$L or $\wedge$R can be handled analogously to Pop.

Comp: Let $\Lambda = \Lambda'; (X^-; \pi(\Lambda), X^+ \mapsto \Gamma \Rightarrow \delta)$ with $i := |\Lambda|$. Observe that in this case, $\Gamma_i = \Gamma, \delta_i = \delta$ and further $\vec{x}_i = \mathrm{FV}(\Gamma, \delta)$, $G_i^i = X^+$ and $G_{<j}^i = \pi(\Lambda)$. We may assume $\Gamma, \mathcal{H}_{\Lambda'}, \varphi_i \Rightarrow \delta$ to derive $\Gamma, \mathcal{H}_{\Lambda'} \Rightarrow \delta$. Define $\max X := \max\{t \mid t^\sigma \in X\}$ and consider the formula

$$\iota_m := \forall \vec{y}_i.\, m = \max(X[\vec{y}_i/\vec{x}_i]) \to \gamma_{<i}^i \to (\Gamma \Rightarrow \delta)[\vec{y}_i/\vec{x}_i]$$

where $\gamma_{<i}^i := \bigwedge_{j<i} \gamma_j^{\Lambda,i}$. We derive the following:

$$
\begin{prooftree}
\AxiomC{$\Gamma, \mathcal{H}_{\Lambda'}, \varphi_i \Rightarrow \delta$}
\UnaryInfC{$\mathcal{H}_{\Lambda'}, \varphi_i \Rightarrow \Sigma$}
\LeftLabel{\textsc{Sub}}
\UnaryInfC{$\mathcal{H}_{\Lambda'}[\vec{y}_i/\vec{x}_i], \varphi_i[\vec{y}_i/\vec{x}_i] \Rightarrow \Sigma[\vec{y}_i/\vec{x}_i]$}
\LeftLabel{$\dagger$}
\UnaryInfC{$\mathcal{H}_{\Lambda'}, \gamma^i_{j<i}, \varphi_i[\vec{y}_i/\vec{x}_i] \Rightarrow \Sigma[\vec{y}_i/\vec{x}_i]$}
\LeftLabel{$\star$}
\UnaryInfC{$\mathcal{H}_{\Lambda'}, \gamma^i_{j<i}, \forall m' < \max(X[\vec{y}_i/\vec{x}_i]).\ \iota_{m'} \Rightarrow \Sigma[\vec{y}_i/\vec{x}_i]$}
\UnaryInfC{$\mathcal{H}_{\Lambda'}, \forall m' < m.\ \iota_{m'} \Rightarrow \iota_m$}
\LeftLabel{\textsc{Ind}}
\UnaryInfC{$\mathcal{H}_{\Lambda'} \Rightarrow \forall m.\iota_m$}
\AxiomC{}
\RightLabel{$*$}
\UnaryInfC{$\Gamma, \forall m.\iota_m \Rightarrow \delta$}
\LeftLabel{\textsc{Cut}}
\BinaryInfC{$\Gamma, \mathcal{H}_{\Lambda'} \Rightarrow \delta$}
\end{prooftree}
$$

where $\Sigma := \bigwedge \Gamma \to \delta$ and $\star$ uses

$$\gamma^i_{j<i}, \forall m' < \max(X[\vec{y}_i/\vec{x}_i]).\ \iota_{m'}, \Rightarrow \varphi_i[\vec{y}_i/\vec{x}_i].$$

We prove this entailment in prose: Let $\vec{z}_i$ be such that $\gamma^i[\vec{z}_i/\vec{y}_i, \vec{y}_i/\vec{x}_i]$. To prove $(\Gamma \Rightarrow \delta)[\vec{z}_i/\vec{x}_i]$, we apply $\forall m' < \max(X[\vec{y}_i/\vec{x}_i]).\ \iota_{m'}$ with $m' := \max(X[\vec{z}_i/\vec{x}_i])$ and $\vec{y}_i := \vec{z}_i$. A few claims must be proven to do so:

- $\max(X[\vec{z}_i/\vec{x}_i]) < \max(X[\vec{y}_i/\vec{x}_i])$: Let $t^+ \in X^+$ be such that $t[\vec{z}_i/\vec{x}_i] = \max(X[\vec{z}_i/\vec{x}_i])$. By $\gamma^i_i[\vec{z}_i/\vec{y}_i, \vec{y}_i/\vec{x}_i]$ there is $s^- \in X^-$ such that

$$\max(X[\vec{z}_i/\vec{x}_i]) = t[\vec{z}_i/\vec{x}_i] < s[\vec{y}_i/\vec{x}_i] \le \max(X[\vec{y}_i/\vec{x}_i]).$$

- $\gamma^i_{j<i}[\vec{z}_i/\vec{y}_i]$: Let $j < i$ and let $t_1^{\sigma_1} \in G^i_j = P_j$. By $\gamma^i_j[\vec{z}_i/\vec{y}_i, \vec{y}_i/\vec{x}_i]$, there exists $t_2^{\sigma_2} \in P_j$ such that $t_1[\vec{z}_i/\vec{x}_i] <^{\sigma_1,\sigma_2} t_2[\vec{y}_i/\vec{x}_i]$. By γ^i_j, there then exists $t_3^{\sigma_3} \in P_j$ such that $t_2[\vec{y}_i/\vec{x}_i] <^{\sigma_2,\sigma_3} t_3$. By transitivity, thus $t_1[\vec{z}_i/u] <^{\sigma_1,\sigma_3} t_3$ as desired.

Further, $\dagger$ uses

$$\mathcal{H}_{\Lambda'}, \gamma^i_{j<i} \Rightarrow \bigwedge \mathcal{H}_{\Lambda}[\vec{y}_i/\vec{x}_i]$$

which we, too, prove in prose, more concretely $\varphi_j, \gamma^i_{j<i} \Rightarrow \varphi_j[\vec{y}_i/\vec{x}_i]$: Let $\vec{y}_j$ be given such that $\gamma^j[\vec{y}_i/\vec{x}_i]$ and (note the differing subscripts!). We prove $(\Gamma_j \Rightarrow \delta_j)[\vec{y}_j/\vec{x}_j]$ by applying φ_j with $\vec{y}_j := \vec{y}_j$. Only γ^j remains to be proven: Let $k \le j$ and $t_1^{\sigma_1} \in G^j_k$, by $\gamma^j_k[\vec{y}_i/\vec{x}_i]$ there is $t_2^{\sigma_2} \in P_k$ with $t_1[\vec{y}_j/\vec{x}_j] <^{\sigma_1,\sigma_2} t_2[\vec{y}_i/\vec{x}_i]$. Because $G^i_k = P_k$, γ^i_k furthermore yields $t_3^{\sigma_3} \in P_k$ such that $t_2[\vec{y}_i/\vec{x}_i] <^{\sigma_2,\sigma_3} t_3$. By transitivity, this yields $t_1[\vec{y}_j/\vec{x}_j] <^{\sigma_1,\sigma_3} t_3$.

Lastly, $*$ is derived by instantiating $\forall m.\ \iota_m$ with $m := \max X$ and $\vec{y}_i := \vec{x}_i$. Observe that for this, $\gamma^i_{j<i}[\vec{x}_i/\vec{y}_i]$ is trivially true.

Bud: Let $\Lambda = \Lambda', (X^+; \pi(\Lambda'), X \mapsto \Gamma \Rightarrow \delta)$ with $|\Lambda| = i$. We want to derive $\Gamma, \mathcal{H}_\Lambda \Rightarrow \delta$. For this, we use

$$\varphi_i := (\forall \vec{y}_i.\ \gamma^i \to (\Gamma \Rightarrow \delta)[\vec{y}_i/\vec{x}_i]) \in \mathcal{H}_\Lambda$$

instantiated with $\vec{y}_i \coloneqq \vec{x}_i$. It remains to prove the two hypothesis of this instance of φ_i:

- $\gamma_i^i[\vec{x}_i/\vec{y}_i]$: For any $t^\sigma \in X$ we know $t^+ \in X^+$ and clearly $t <^{\sigma,+} t$ because $<^{\sigma,+}=\leq$.
- $\gamma_{<j}^i[\vec{x}_i/\vec{y}_i]$: This is trivial: For any $t^\sigma \in P_j$ with $j < i$ we have $t^\sigma \in P_j$ and $t <^{\sigma\sigma} t$ because $<^{\sigma\sigma}= \leq$ always.

<L: We may thus assume $\Gamma, \mathcal{H}_{\Lambda^+} \Rightarrow \delta$ to derive $\Gamma, s < t, \mathcal{H}_\Lambda \Rightarrow \delta$. For this, it suffices to show that $\varphi_i^\Lambda, s < t \Rightarrow \varphi_i^{\Lambda^+}$ for every $i \leq |\Lambda|$. For this, in turn, it suffices to prove $\gamma^{\Lambda^+,i} \Rightarrow \gamma^{\Lambda,i}$: Let $j \leq i$ and consider $u^\sigma \in G_j^i$. Per assumption, we know that $v^{\sigma'} \in P_j[t \rightsquigarrow^+ s]$ such that $u[\vec{y}_i/\vec{x}_i] <^{\sigma,\sigma'} v$ and $\sigma \leq \sigma'$. The only interesting case is if $v^\sigma = s^+$ but $s^+ \notin P_j$. That means $t^{\sigma''} \in P_j$ and, because $u <^{\sigma,\sigma'} s < t$ we have $u <^{\sigma,\sigma''} t$ regardless of the nature of σ''.

Sub: We may assume that $\Gamma, \mathcal{H}_{\Lambda'} \Rightarrow \delta$ to derive that $\Gamma[t/z], \mathcal{H}_\Lambda \Rightarrow \delta[t/z]$. It suffices to prove that $\mathcal{H}_\Lambda \Rightarrow \bigwedge \mathcal{H}_{\Lambda'}[t/z]$ for the following construction:

$$\dfrac{\mathcal{H}_\Lambda \Rightarrow \bigwedge \mathcal{H}_{\Lambda'}[t/z] \qquad \dfrac{\Gamma, \mathcal{H}_{\Lambda'} \Rightarrow \delta}{\Gamma[t/z], \mathcal{H}_{\Lambda'}[t/z] \Rightarrow \delta[t/z]}\,\textsc{Sub}}{\Gamma[t/z], \mathcal{H}_\Lambda \Rightarrow \delta[t/z]}\,\textsc{Cut}$$

As in the <L-case, we need to derive $\gamma^{\Lambda',i}[t/z] \Rightarrow \gamma^{\Lambda,i}$. Observe that, writing P'_j for the jth set in $\pi(\Lambda')$, $P'_j[t/z] = P_j$ for all $j \leq i$, trivialising the claim.

=L: We may assume that $\Gamma[t/x, s/y], \mathcal{H}_{\Lambda^=} \Rightarrow \delta[t/x, s/y]$ to derive that $\Gamma[s/x, t/y], s = t, \mathcal{H}_\Lambda \Rightarrow \delta[s/x, t/y]$. Similarly to the Sub-case, it suffices to prove that $s = t, \mathcal{H}_\Lambda \Rightarrow \bigwedge \mathcal{H}_{\Lambda^=}$ for the following construction:

$$\dfrac{s = t, \mathcal{H}_\Lambda \Rightarrow \bigwedge \mathcal{H}_{\Lambda^=} \qquad \dfrac{\Gamma[t/x, s/y], \mathcal{H}_{\Lambda^=} \Rightarrow \delta[t/x, s/y]}{\Gamma[s/x, t/y], s = t, \mathcal{H}_{\Lambda^=} \Rightarrow \delta[s/x, t/y]}\,{=}\mathrm{L}}{\Gamma[s/x, t/y], s = t, \mathcal{H}_\Lambda \Rightarrow \delta[s/x, t/y]}\,\textsc{Cut}$$

Thus, we need to derive $s = t, \gamma^{\Lambda^=,i} \Rightarrow \gamma^{\Lambda,i}$. Write $P_j^=$ for the jth set of $\pi(\Lambda^=)$ and fix some $a^\sigma \in G_j^i$. By $\gamma^{\Lambda^=,i}$ there is $b^{\sigma'} \in P_j^=$ such that $a[\vec{y}_i/\vec{x}_i] <^{\sigma,\sigma'} b$. Per definition of $\Lambda^=$ that means there is $w \in \textsc{Term}(\Gamma, \delta)$ with $b = w[t/x, s/y]$ and further $w[s/x, t/y]^{\sigma'} \in P_j$. Then $a[\vec{y}_i/\vec{x}_i] <^{\sigma,\sigma'} w[s/x, t/y]^{\sigma'}$ follows from $s = t$ as desired.

Drop: Again it suffices to show that $\mathcal{H}_\Lambda \Rightarrow \bigwedge \mathcal{H}_{\Lambda'}$ and thus that $\gamma^{\Lambda',i} \Rightarrow \gamma^{\Lambda,i}$. Write P'_j for the jth set in $\pi(\Lambda')$, then we know $P'_j \preceq P_j$ for all $j \leq i$. Thus, if $t^\sigma \in G^i_j$ then there is $s^{\sigma'} \in P'_j$ with $t[\vec{y}_i/\vec{x}_i] <^{\sigma,\sigma'} s$. By $P'_j \preceq P_j$ we know there is $s^{\sigma''} \in P_j$ with $\sigma' \leq \sigma''$ and thus $t[\vec{y}_i/\vec{x}_i] <^{\sigma,\sigma''} s$ as desired.

Pop: From $\Gamma, \mathcal{H}_{\Lambda_0} \Rightarrow \delta$ we can conclude $\Gamma, \mathcal{H}_{\Lambda_0,\Lambda_1} \Rightarrow \delta$ via weakening. □

The proof above again illustrates the 'structural similarity' between stack-controlled and inductive proof systems, generally, and $\mathrm{SHA}_<$ and HA, specifically. Observe that the translation from $\mathrm{SHA}_<$ to HA is carried out in one 'recursive pass' over the $\mathrm{SHA}_<$-prooftree. Furthermore, each Comp-rule is translated into one Ind-instance and this is (except for reasoning about $<$ in HA) the only source of inductions in the resulting HA-proof. In this manner, the cyclic structure of $\mathrm{SHA}_<$ and the inductive structure of HA correspond to each other.

Theorem 7.1.2 of course directly entails that $\mathrm{CHA}_<$ and HA prove the same theorems.

Theorem 7.1.3. $\mathrm{CHA}_< \vdash \Gamma \Rightarrow \delta$ iff $\mathrm{HA} \vdash \Gamma \Rightarrow \delta$.

Proof. We have shown in proposition 2.3.5 that $\mathrm{HA} \vdash \Gamma \Rightarrow \delta$ entails $\mathrm{CHA}_< \vdash \Gamma \Rightarrow \delta$. Conversely, if $\mathrm{CHA}_< \vdash \Gamma \Rightarrow \delta$ then $\mathrm{SHA}_< \vdash \varepsilon \mid \Gamma \Rightarrow \delta$ by theorem 6.4.2. By theorem 7.1.2, this means $\mathrm{HA} \vdash \Gamma \Rightarrow \delta$ as $\mathcal{H}_\varepsilon = \emptyset$. □

Example 7.1.4. For the sake of the example, let us extend the language with two unary predicates A, B. Consider the following proof:

$$
\begin{prooftree}
\AxiomC{}
\LeftLabel{\textsc{Bud}}
\UnaryInfC{$\Lambda[\{x^+,y^+\}] \mid H, A(x), B(y) \Rightarrow \bot$}
\LeftLabel{\textsc{Sub}}
\UnaryInfC{$\Lambda[\{a^+,b^+\}] \mid H, A(a), B(b) \Rightarrow \bot$}
\LeftLabel{\textsc{Drop, Wk}}
\UnaryInfC{$\Lambda[\{x^-,a^+,b^+,y^-\}] \mid \Gamma, A(a), B(b) \Rightarrow \bot$}
\LeftLabel{$<$L}
\UnaryInfC{$\Lambda[\{x^-,y^-\}] \mid \Gamma, A(a), B(b), a < x, b < x \Rightarrow \bot$}
\AxiomC{}
\RightLabel{\textsc{Bud}}
\UnaryInfC{$\Lambda[\{x^+,y^+\}] \mid H, A(x), B(y) \Rightarrow \bot$}
\RightLabel{\textsc{Sub}}
\UnaryInfC{$\Lambda[\{a^+,b^+\}] \mid H, A(a), B(b) \Rightarrow \bot$}
\RightLabel{\textsc{Drop, Wk}}
\UnaryInfC{$\Lambda[\{x^-,y^-,a^+,b^+\}] \mid \Gamma, A(a), B(b) \Rightarrow \bot$}
\RightLabel{$<$L}
\UnaryInfC{$\Lambda[\{x^-,y^-\}] \mid \Gamma, A(a), B(b), a < y, b < y \Rightarrow \bot$}
\LeftLabel{H}
\BinaryInfC{$\{x^-,y^-\}; \{x^+,y^+\} \mapsto H, A(x), B(y) \Rightarrow \bot \mid H, A(x), B(y) \Rightarrow \bot$}
\LeftLabel{\textsc{Comp}}
\UnaryInfC{$\varepsilon \mid H, A(x), B(y) \Rightarrow \bot$}
\UnaryInfC{$\varepsilon \mid H \Rightarrow \forall xy.\, \neg(A(x) \wedge B(y))$}
\end{prooftree}
$$

where

$$
\begin{aligned}
H &:= \forall xy.\, A(x) \wedge B(y) \rightarrow \exists ab.\, I(x) \vee I(y), \\
I(z) &:= a < z \wedge b < z \wedge A(a) \wedge B(a), \\
\Gamma &:= H, A(x), B(y), \\
\Lambda[X] &:= X; \{x^+, y^+\} \mapsto H, A(x), B(y) \Rightarrow \bot.
\end{aligned}
$$

We shall now apply theorem 7.1.2 to it. The invariant is as follows:

$$
\begin{aligned}
\varphi^X := \forall x'y'.\, &\Big(\bigvee t^\sigma \in X.\, x' <^{+,\sigma} t\Big) \wedge \Big(\bigvee t^\sigma \in X.\, y' <^{+,\sigma} t\Big) \\
&\rightarrow H, A(x'), B(x') \Rightarrow \bot.
\end{aligned}
$$

The translated proof is:

$$
\begin{prooftree}
\AxiomC{}
\UnaryInfC{$\varphi^{\{x^+,y^+\}}, H, A(x), B(y) \Rightarrow \bot$}
\LeftLabel{Sub}
\UnaryInfC{$\varphi^{\{a^+,b^+\}}, H, A(a), B(b) \Rightarrow \bot$}
\LeftLabel{“Drop”, Wk}
\UnaryInfC{$\varphi^{\{x^-,a^+,b^+,y^-\}}, \Gamma, A(a), B(b) \Rightarrow \bot$}
\LeftLabel{“<L”}
\RightLabel{2}
\UnaryInfC{$\varphi^{\{x^-,y^-\}}, \Gamma, A(a), B(b), a, b < x \Rightarrow \bot$}
\LeftLabel{H}
\UnaryInfC{$\varphi^{\{x^-,y^-\}}, H, A(x), B(y) \Rightarrow \bot$}
\LeftLabel{Sub}
\UnaryInfC{$\varphi^{\{x_m^-,y_m^-\}} \Rightarrow \Sigma_m$}
\LeftLabel{$\star$}
\RightLabel{1}
\UnaryInfC{$\forall m' < \max\{x_m, y_m\}.\ \iota_{m'} \Rightarrow \Sigma_m$}
\LeftLabel{Ind}
\UnaryInfC{$\Rightarrow \forall m.\ \iota_m$}
\AxiomC{$\iota_m[\max\{x,y\}/m], H, A(x), B(y) \Rightarrow \bot$}
\UnaryInfC{$\forall m.\ \iota_m, H, A(x), B(y) \Rightarrow \bot$}
\LeftLabel{Cut}
\BinaryInfC{$H, A(x), B(y) \Rightarrow \bot$}
\UnaryInfC{$H \Rightarrow \forall xy.\ \neg(A(x) \wedge B(y))$}
\end{prooftree}
$$

where we only treat the x-premise of H (the other one is analogous) and we define:

$$
\begin{aligned}
\iota_m &:= \forall x_m y_m.\ \max\{x_m, y_m\} = m \to \Sigma_m, \\
\Sigma_m &:= H, A(x_m), B(y_m) \Rightarrow \bot.
\end{aligned}
$$

To make it easier to follow the φ^X, we explicitly write out the guards on x' (the bound on y' is analogous):

$$
\begin{aligned}
\varphi^{\{x_m^-,y_m^-\}} &\rightsquigarrow x' < x_m \vee x' < y_m \\
\varphi^{\{x^-,y^-\}} &\rightsquigarrow x' < x \vee x' < y \\
\varphi^{\{x^-,a^+,b^+,y^-\}} &\rightsquigarrow x' < x \vee x' \leq a \vee x' \leq b \vee x' < y \\
\varphi^{\{a^+,b^+\}} &\rightsquigarrow x' \leq a \vee x' \leq b \\
\varphi^{\{x^+,y^+\}} &\rightsquigarrow x' \leq x \vee x' \leq y
\end{aligned}
$$

To justify the proof above, observe that at the spot marked with 1, the guard of $\varphi^{x_m^-,y_m^-}$ clearly entails that $\max\{x', y'\} < \max x_m, y_m$ and at the spot marked 2, the guard of $\varphi^{x^-,a^+,b^+,y^-}$ entails that of φ^{x^-,y^-} because $a < x$ and $b < x$.

7.2 Fragments of Simpson's $\mathrm{CPA}_<$

Simpson (2017), when introducing $\mathrm{CPA}_<$, gives as an example a cyclic proof of the totality of the Ackermann-function which only uses Σ_1-formulas. This is surprising because this totality result can famously only be proven over at least $\mathrm{I}\Sigma_2$ i.e., PA with induction restricted to Σ_2-formulas. Das (2020) strengthens this observation into a result: He defines a proof Π of $\mathrm{CPA}_<$ to be in the class $\mathrm{C}\Sigma_n$ if the sequent of every companion only mentions Σ_n-formulas.[23] He then proves in (Das 2020, theorem 4.1) that $\mathrm{C}\Sigma_n$ and $\mathrm{I}\Sigma_{n+1}$ have the same Π_{n+1}-consequences for any $n \in \omega$. This can be explained by the fact, because of the Sub-rule and the mechanism of cycles, the free variables of a sequent are implicitly quantified (see remark 7.2.7 for more details).

While Das' result clarifies the discrepancy first observed by Simpson, it only identifies classes of $\mathrm{CPA}_<$-proofs with fragments $\mathrm{I}\Sigma_n$ of PA over limited classes of theorems. Beklemishev et al. (2025) closes this gap: They identify $\mathrm{I}\Sigma_n$ for $n > 0$ with classes of proofs in a cyclic proof system for PA. Instead of taking into account the complexity of all formulas in sequents, they divide the formulas into 'induction formulas' and 'side formulas', counting only the complexity of induction towards the induction complexity of a cyclic proof. However, the cyclic proof system they consider has a rather restrictive soundness condition, as discussed in section 3.5. This seems to have been done to simplify the translation of cyclic proofs to $\mathrm{I}\Sigma_n$.

In this section, we extend the result of (Beklemishev et al. 2025) to the more common cyclic proof system $\mathrm{CPA}_<$: We define fragments $\mathrm{CI}\Pi_n$ of $\mathrm{CPA}_<$ which prove the same theorems as $\mathrm{I}\Pi_n$ for $n > 0$. Similar to the approach of Beklemishev et. al, our classification distinguishes between *induction formulas* and *side formulas*, indicated by a chosen class of *induction variables*. The translation for $\mathrm{CI}\Pi_n$ to $\mathrm{I}\Pi_n$ then reduces to a more careful rendition of the unravelling result of section 7.1.

We begin by defining by defining a variant CPA_V of $\mathrm{CPA}_<$ with syntactically distinguished induction variables.

Definition 7.2.1 (CPA_V)**.** The sequents of CPA_V are of the form $\Gamma \Rightarrow_V \Delta$ where Γ, Δ are finite sets of arithmetic formulas (just as in $\mathrm{CHA}_<$) and $V \subseteq \mathrm{Var}$ is a finite set of *induction variables*. The derivation rules of CPA_V are as those of $\mathrm{CHA}_<$ adapted to Peano arithmetic in the usual manner: allowing for a finite set of formulas on the right-hand side of $\Rightarrow_V$ and using classical, rather than

[23] Technically, this is not quite correct. He considers the universal closures of cyclic proofs which only mention Σ_n-formulas, but by partial cut-elimination, these two notions coincide when only considering Π_{n+1}-theorems.

intuitionistic, first-order logical rules. For all rules obtained in such a way, the choice of V must be identical in premises and conclusion. CPA_V furthermore features another rule for manipulating the variable set V:

$$\textsc{Var}\frac{\Gamma \Rightarrow_{V'} \Delta}{\Gamma \Rightarrow_V \Delta}$$

where $V, V' \subseteq \textsc{Var}$.

We specify the trace condition of CPA_V as a trace interpretation over $\mathbb{B}$. For a sequent $\Gamma \Rightarrow_V \Delta$ we fix $\iota(\Gamma \Rightarrow_V \Delta) = \textsc{Term}_V(\Gamma, \Delta)$ where

$$\textsc{Term}_V(\Phi) := \{t \in \textsc{Term}(\Phi) \mid \mathrm{FV}(t) \subseteq V\}$$

that is, only terms whose free variables stem exclusively from V can be followed by traces. The trace maps for the rules except $\textsc{Var}$ are then analogous to those of $\mathrm{CHA}_<$, except for their domains being restricted to V-terms as described above. Lastly, let $r = \textsc{Var}$ as above, then

$$r_1 := \{(t, 0, t) \mid t \in \textsc{Term}(\Gamma, \Delta) \text{ and } \mathrm{FV}(t) \subseteq V \cap V'\}.$$

We say $\mathrm{CPA}_V \vdash \Gamma \Rightarrow \Delta$ if there is a CPA_V-proof of $\Gamma \Rightarrow_\emptyset \Delta$.

For the following, we introduce some notation. Let $V \subseteq \textsc{Var}$ and $\Gamma \subseteq \textsc{Form}$ then $\Gamma^{\ni V} := \{\varphi \in \Gamma \mid \mathrm{FV}(\varphi) \cap V \neq \emptyset\}$. The *induction formulas* of a sequent $\Gamma \Rightarrow_V \Delta$ are $\Gamma^{\ni V} \cup \Delta^{\ni V}$ i.e., those mentioning the induction variables V. The *side formulas* thus are the remaining formulas i.e., $\Gamma \setminus \Gamma^{\ni V}$ and $\Delta \setminus \Delta^{\ni V}$, which we denote by $\Gamma^{\not\ni V}$ and $\Delta^{\not\ni V}$, respectively. For a CPA_V-proof Π and $s \in \mathrm{dom}(\beta)$ we define its set of *stable variables* $\mathrm{SV}(s) := (\mathrm{FV}(\Gamma^{\ni V}, \Delta^{\ni V}) \setminus V)$ if $\lambda(s) = \Gamma \Rightarrow_V \Delta$. Intuitively, stable variables are those appearing free in induction formulas without being induction variables. Their value must remain constant within cycles, as expressed more formally below.

Definition 7.2.2 (Stable variables)**.** Let Π be a CPA_V-proof. We say that it has *stable variables* if the following holds for every $s \in \beta$: Let η be the maximal strongly connected component with $s \in \eta$ then we require:

(i) For every $t \in C[\eta]$ with $\lambda(t) = \Gamma \Rightarrow_V \Delta$ the sets V and $\mathrm{SV}(s)$ are disjoint i.e., $V \cap \mathrm{SV}(s) = \emptyset$ and

(ii) For every instance of the rules $\textsc{Sub}$, $\forall$R and $\exists$L in $C[\eta]$ which introduce a fresh variable x we have $x \notin \mathrm{SV}(s)$.

Any $\mathrm{CPA}_<$-proof can be annotated with a naïve set of induction variables which ensures the proof has stable variables.

Proposition 7.2.3. If $\mathrm{CPA}_< \vdash \Gamma \Rightarrow \Delta$ then $\mathrm{CPA}_V \vdash \Gamma \Rightarrow \Delta$ with stable variables.

Proof. Let Π be the $\mathrm{CPA}_<$-proof and fix

$$V := \{x \in \textsc{Var} \mid x \text{ free in some sequent of } \Pi\}.$$

Denote by Π_V the CPA_V-preproof in which every sequent is annotated with variable set V. This clearly remains a CPA_V-proof because all traces of Π may be considered towards its proofhood. For $\mathrm{CPA}_V \vdash \Gamma \Rightarrow_\emptyset \Delta$ simply derive

$$\textsc{Var}\,\dfrac{\dfrac{\Pi_V}{\Gamma \Rightarrow_V \Delta}}{\Gamma \Rightarrow_\emptyset \Delta}$$

□

Stable CPA_V-proofs can be assigned an be assigned an induction complexity based on the complexity of their induction formulas, depending on their position relative to the $\Rightarrow$-symbol in the sequent.

Definition 7.2.4 ($\mathrm{CI\Pi}_n$)**.** A CPA_V-proof Π is a $\mathrm{CI\Pi}_n$-proof for $n > 0$ if it has stable variables and the following holds: For every $s \in \mathrm{dom}(\beta)$ with $\lambda(s) = \Gamma \Rightarrow_V \Delta$ we have $\Gamma^{\ni V} \subseteq \Sigma_n$ and $\Delta^{\ni V} \subseteq \Pi_n$.

As usual with cyclic proof systems, embedding $\mathrm{I\Pi}_n$ into $\mathrm{CI\Pi}_n$ is not very complicated.

Lemma 7.2.5. If $\mathrm{I\Pi}_n \vdash \Gamma \Rightarrow \Delta$ then $\mathrm{CI\Pi}_n \vdash \Gamma \Rightarrow \Delta$.

Proof. The $\mathrm{I\Pi}_n$-proof can mostly be translated to a non-cyclic CPA_V-proof (and thus $\mathrm{CI\Pi}_n$-proof) with $V = \emptyset$ at every node. The only exception from this statement is the induction axiom. Thus, let $\varphi \in \Pi_n$ then we derive the proof in fig. 16. Note that we implicitly apply $\textsc{Wk}$ in some places to aid readability. The $\star$ marks the cycle. In fig. 16, $\varphi^* := \forall x.\ \varphi(x) \to \varphi(Sx)$ and x, y are chosen fresh. First, we must argue that the above is a CPA_V-proof. For this observe that the following is a progressing trace along the $\star$-cycle, where the progression happens at the "$\textsc{Cut}$ + axiom"-step.

$$x \xrightarrow{=\mathrm{L}} Sy \xrightarrow{\textsc{Cut}\text{ + axiom}} y \xrightarrow{\textsc{Sub}} x$$

It is easily verified that the trace only ever mentions free variables in the V-set of the current sequent. Now to verify that the above is a $\mathrm{CI\Pi}_n$-proof, observe that at the bud/companion, x only occurs freely in $\varphi(x)$ because of its freshness and

$$
\begin{prooftree}
\AxiomC{$\varphi(0), \varphi^* \Rightarrow_{\{x\}} \varphi(x) \quad (\star)$}
\LeftLabel{\textsc{Var}}
\UnaryInfC{$\varphi(0), \varphi^* \Rightarrow_{\{x,y\}} \varphi(x)$}
\LeftLabel{\textsc{Sub}}
\UnaryInfC{$\varphi(0), \varphi^* \Rightarrow_{\{x,y\}} \varphi(y)$}
\AxiomC{$\varphi(Sy) \Rightarrow_{\{x,y\}} \varphi(Sy)$}
\LeftLabel{$\rightarrow$L}
\BinaryInfC{$\varphi(0), \varphi^*, \varphi(y) \rightarrow \varphi(Sy) \Rightarrow_{\{x,y\}} \varphi(Sy)$}
\LeftLabel{$\forall$L}
\UnaryInfC{$\varphi(0), \varphi^* \Rightarrow_{\{x,y\}} \varphi(Sy)$}
\LeftLabel{\textsc{Wk}}
\UnaryInfC{$y < Sy, \varphi(0), \varphi^* \Rightarrow_{\{x,y\}} \varphi(Sy)$}
\LeftLabel{\textsc{Cut} + axiom}
\UnaryInfC{$\varphi(0), \varphi^* \Rightarrow_{\{x,y\}} \varphi(Sy)$}
\LeftLabel{\textsc{Var}}
\UnaryInfC{$\varphi(0), \varphi^* \Rightarrow_{\{x\}} \varphi(Sy)$}
\LeftLabel{=L}
\UnaryInfC{$x = Sy, \varphi(0), \varphi^* \Rightarrow_{\{x\}} \varphi(x)$}
\LeftLabel{S}
\UnaryInfC{$0 < x, \varphi(0), \varphi^* \Rightarrow_{\{x\}} \varphi(x)$}
\AxiomC{$\varphi(0) \Rightarrow_{\{x\}} \varphi(0)$}
\LeftLabel{=L}
\UnaryInfC{$0 = x, \varphi(0) \Rightarrow_{\{x\}} \varphi(x)$}
\AxiomC{}
\UnaryInfC{$x < 0 \Rightarrow_{\{x\}}$}
\LeftLabel{$\vee$L}
\TrinaryInfC{$0 < x \vee 0 = x \vee x < 0, \varphi(0), \varphi^* \Rightarrow_{\{x\}} \varphi(x)$}
\LeftLabel{\textsc{Cut} + axiom}
\UnaryInfC{$\varphi(0), \varphi^* \Rightarrow_{\{x\}} \varphi(x) \quad (\star)$}
\LeftLabel{$\forall$R}
\UnaryInfC{$\varphi(0), \varphi^* \Rightarrow_{\{x\}} \forall x.\, \varphi(x)$}
\LeftLabel{$\rightarrow$R}
\UnaryInfC{$\Rightarrow_{\{x\}} \varphi(0) \rightarrow \varphi^* \rightarrow \forall x.\, \varphi(x)$}
\LeftLabel{\textsc{Var}}
\UnaryInfC{$\Rightarrow_{\emptyset} \varphi(0) \rightarrow \varphi^* \rightarrow \forall x.\, \varphi(x)$}
\end{prooftree}
$$

Figure 16: A proof of the induction axiom for $\varphi \in \Pi_n$ in $\mathrm{C}\mathrm{I}\Pi_n$

$\varphi \in \Pi_n$ per assumption. Towards separation of variables, notice that the maximal strongly connected component containing the $\star$-cycle is just the $\star$-cycle itself. It contains only one instance of SUB, which introduces a fresh variable $x \notin \mathrm{SV}(\star)$, meaning the variables of the proof above are stable. □

Remark 7.2.1. The result above motivates the division of sequents into induction and side formulas. The proof of the induction axiom requires the side formula $\varphi^* \in \Pi_{n+1}$ (for $\varphi \in \Pi_n$) to be present along the $\star$-cycle. Thus, simply restricting the complexity of all formulas in the sequent, as Simpson and Das do, does not yield a notion of cyclic logical complexity which can sensibly capture $\mathrm{I}\Pi_n$.

Remark 7.2.2. Translating $\mathrm{I}\Pi_n$ with the induction rule IND instead of induction axioms as in lemma 7.2.5 is no more complicated. The naïve translation of the IND-rule does not introduce any interleaving cycles to the resulting CPA-proof, leaving the same simple cyclic structure as lemma 7.2.5: Every strongly connected component consists of exactly one cycle.

$$
\textsc{Ind}\,\frac{\begin{array}{c} x \notin \mathrm{FV}(\Gamma, \Delta) \\ \Gamma, \varphi(x) \Rightarrow \varphi(Sx), \Delta \end{array}}{\Gamma, \varphi(0) \Rightarrow \varphi(t), \Delta}
$$

This claim might seem to contradict the comments we make in remark 3.5.1 about the admissibility of the rule above in simplistic sCHA. However, the sCHA-result needs to account for IND appearing inside a simple cycle, whereas lemma 7.2.5 does not.

We continue by defining SPA_V, the the stack-controlled version of CPA_V. It is obtained analogously to $\mathsf{SHA}_<$ as described in section 6.4.1. That is, the sequents of SPA_V are expressions $\Lambda \mid \Gamma \Rightarrow_V \Delta$ where $\Gamma \Rightarrow_V \Delta$ is a CPA_V-sequent. The component Λ is a stack of companion labels, expressions $X; \vec{Y} \mapsto \Gamma' \Rightarrow_{V'} \Delta'$ where $\Gamma' \Rightarrow_{V'} \Delta'$ is a SPA_V-sequent, and further X and $Y \in \vec{Y}$ are non-empty sets of *progress annotated terms* i.e., expressions t^σ with $\sigma \in \{-, +\}$. Analogously to $\mathsf{SHA}_<$, we denote the ith SPA_V-companion label in Λ by $(P_i; \vec{G}^i \mapsto \Gamma_i \Rightarrow_{V_i} \Delta_i)$, from the left and starting at 1. For a SPA_V-sequent $\Lambda \mid \Gamma \Rightarrow_V \Delta$ to be well-formed, we further require that for $P_i; \vec{G}^i \mapsto \Gamma_i \Rightarrow_{V_i} \Delta_i \in \Lambda$:

(i) $\{t \mid t^\sigma \in P_i\} \subseteq \textsc{Term}_V(\Gamma, \Delta)$ and

(ii) $|\vec{G}^i| = i$ and $\{t \mid t^\sigma \in G^i_j\} \subseteq \textsc{Term}_{V_j}(\Gamma_j, \Delta_j)$ for all $j \leq i$.

Most rules of SPA_V are those of SPA_V, restricted to well-formed sequents as defined above. The $\textsc{Var}$-rule of CPA_V transfers to SPA_V as follows:

$$\textsc{Var}\frac{\Lambda \upharpoonright V' \mid \Gamma \Rightarrow_{V'} \Delta}{\Lambda \mid \Gamma \Rightarrow_V \Delta}$$

where $\Lambda \upharpoonright V'$ replaces the label $(P_i; \vec{G}^i \mapsto \Gamma_i \Rightarrow_{V_i} \Delta_i) \in \Lambda$ by $(P_i \upharpoonright V'; \vec{G}^i \mapsto \Gamma_i \Rightarrow_{V_i} \Delta_i)$ where $P_i \upharpoonright V := \{t^\sigma \in P_i \mid \mathrm{FV}(t) \subseteq V'\}$.

As for CPA_V before, we define a notion of stable variables. Observe that, as defined below, it does not directly correspond to the eponymous notion of CPA_V. We define the set of *stable variables* of Λ as $\mathrm{SV}(\Lambda) := \bigcup_{i \leq |\Lambda|} (\mathrm{FV}(\Gamma_i^{\ni V_i}, \Delta_i^{\ni V_i}) \setminus V_i)$.

Definition 7.2.6 (Stable variables)**.** Let Π be an SPA_V-proof. We say that it has *stable variables* if:

(i) For every sequent $\Lambda \mid \Gamma \Rightarrow_V \Delta$ in Π the sets V and $\mathrm{SV}(\Lambda)$ are disjoint i.e., $V \cap \mathrm{SV}(\Lambda) = \emptyset$ and

(ii) For every instance of the rules $\textsc{Sub}$, $\forall$R and $\exists$L in Π with conclusion $\Lambda \mid \Gamma \Rightarrow_V \Delta$ which introduce the fresh variable x we have $x \notin \mathrm{SV}(\Lambda)$.

The definition of $\mathsf{SI\Pi}_n$ is analogous to that of $\mathsf{CI\Pi}_n$.

Definition 7.2.7 ($\mathsf{SI\Pi}_n$)**.** A SPA_V-proof Π is a $\mathsf{SI\Pi}_n$-proof for $n > 0$ if it has stable variables and the following holds: Let $\Lambda \mid \Gamma \Rightarrow_V \Delta$ be a sequent of Π and let $(P_i; \vec{G}^i \mapsto \Gamma_i \Rightarrow_{V_i} \Delta_i) \in \Lambda$ then $\Gamma_i^{\ni V_i} \in \Sigma_n$ and $\Delta_i^{\ni V_i} \in \Pi_n$.

Remark 7.2.3. For a SPA_V-proof Π of $\varepsilon \mid \Gamma \Rightarrow_V \Delta$ with an initially empty stack, this is equivalent to requiring of every conclusion $\Lambda \mid \Gamma' \Rightarrow_{V'} \Delta'$ of a $\textsc{Comp}$-instance in Π that $\Gamma'^{\ni V'} \in \Sigma_n$ and $\Gamma'^{\ni V'} \in \Pi_n$. However, when considering 'partial' SPA_V proofs with non-empty stacks in their conclusion, such as in the generalised inductive hypothesis of lemma 7.2.13, the phrasing above is more convenient.

As noted, the notions of stable variables for CPA_V- and SPA_V-proofs do not directly correspond. Thus, we must first prove a crucial lemma about the CPA_V-notion before we can clarify their relationship.

Lemma 7.2.8. If Π is a CPA_V-preproof with stable variables then so is any unfolding of Π.

Proof. Let Π' be an unfolding of Π. Observe the following:

- Π and Π' unfold into the same ∞-preproof Π^∞.
- Every path from the root through Π uniquely corresponds to such a path through Π^∞ which in turn uniquely corresponds to such a path through Π'.

Suppose, towards contradiction, that Π' did not have stable variables. This means there must be $s' \in \mathrm{dom}(\beta')$ and a node t' in η', the maximal strongly connected component of C' containing s', violating condition (i) or (ii) of definition 7.2.2. Observe that there furthermore must be the following:

- the $\pi_{s'} = [\varepsilon, s']$ from the root of C' to s' and
- a path $\pi_{s't'}$ through C' from s' to t' and
- a path $\pi_{t's'}$ through C' from t' to s'.

As discussed above, $\pi_{s'}$ corresponds to a unique path from the root of C' to some node s. As Π' is an unfolding of Π, $s \in \mathrm{dom}(\beta)$ and $\lambda'(s') = \lambda(s)$ and $\mathrm{SV}(s') = \mathrm{SV}(s)$. The same argument means the path $\pi_{s'}\pi_{s't'}$ leads to some node $t \in T$ with $\lambda'(t') = \lambda(t)$ and, if applicable, $\delta'(t') = \delta(t)$. That means t violates condition (i) or (ii) of definition 7.2.2. If there was a connected component containing s and t, this would mean Π did not have stable variables, leading to a contradiction. Observe that this is indeed the case, as $\pi_{s'}\pi_{s't'}\pi_{t's'}$ must correspond to a path through C ending in s, again because Π' is an unfolding of Π. That means t lies on a cycle from s to s and thus in the maximal strongly connected component containing s. □

The crucial fact relating stable variables in CPA_V and SPA_V is given below: When translating from CPA_V to SPA_V following the procedure given for $\mathrm{CHA}_<$ and $\mathrm{SHA}_<$ in theorem 6.4.2, if the CPA_V-proof had separated variables, so will the resulting SPA_V-proof. Observe that this is not an if and only if: There are CPA_V-proofs which do not have stable variables according to the CPA_V-condition but whose translated SPA_V-version will have them. This is because we chose the notion of separated variables in CPA_V to be pleasant to state and verify, rather than complete, which would have made it much more finicky and ultimately less informative for identifying $\mathsf{CI\Pi}_n$.

Proposition 7.2.9. If $\mathsf{CPA}_V \vdash \Gamma \Rightarrow \Delta$ with stable variables then $\mathsf{SPA}_V \vDash \Gamma \Rightarrow \Delta$ with stable variables.

Proof. We can just follow the argument of theorem 6.4.2 which transforms $\mathsf{CHA}_<$-proofs into $\mathsf{SHA}_<$-proofs to transform a CPA_V-proof Π into an SPA_V-proof Π'. Towards separation of variables, observe that Π' is, ignoring the bookkeeping rules, an unfolding of Π. By lemma 7.2.8, that means Π' must have stable variables 'when regarded as a CPA_V-proof'. Observe that this entails separation of variables in the SPA_V-sense: If s is a node of Π' with $\lambda(s) = \Lambda \mid \Gamma \Rightarrow_V \Delta$ and let t be the bud corresponding to a label in Λ. Then s is on the direct path from '$\beta'(t)$' to t, and thus contained in the maximal strongly connected component of t, meaning the desired properties follow directly from the CPA_V-notion of stable variables. □

This correspondence directly yields an embedding from $\mathsf{CI\Pi}_n$ into $\mathsf{SI\Pi}_n$.

Corollary 7.2.10. If $\mathsf{CI\Pi}_n \vdash \Gamma \Rightarrow \Delta$ then $\mathsf{SI\Pi}_n \vdash \Gamma \Rightarrow \Delta$.

Proof. Translate the CPA_V-proof Π to a SPA_V-proof Π' according to proposition 7.2.9. Then separation of variables is already given. For the sequent condition, simply observe that every companion in Π' is labelled by a sequent which also labels the sequent of some bud in Π. □

Before we translate $\mathsf{SI\Pi}_n$ to $\mathsf{I\Pi}_n$, we make the following simple observation.

Proposition 7.2.11. Any CPA_V- or SPA_V-proof with stable variables in a $\mathsf{CI\Pi}_n$- or $\mathsf{SI\Pi}_n$-proof, respectively, for some $n \in \omega$.

Proof. As such proofs have finitely many companions whose sequents mention only finitely many induction formulas each, there must be a minimal logical complexity class capturing them all. □

To translate from $\mathsf{SI\Pi}_n$ to $\mathsf{I\Pi}_n$, a key step is to formulate the *inductive invariants*. These are a variant of those used in section 7.1 for $\mathsf{CHA}_<$ with one key difference: they only mention induction formulas. That is, each label in Λ gives rise to one invariant i.e., we define the set of invariants for a given Λ as $\mathcal{H}_\Lambda := \{\varphi_i^\Lambda \mid 0 < i \le |\Lambda|\}$ where the invariant of the ith label $(P_i; \vec{G}^i \mapsto \Gamma_i \Rightarrow_{V_i} \Delta_i)$ is defined as:

$$\varphi_i^\Lambda := \forall \vec{y}_i.\ \gamma^{\Lambda,i} \to (\Gamma_i^{\ni V_i} \Rightarrow \Delta_i^{\ni V_i})[\vec{y}_i/\vec{x}_i]$$

where $\vec{x}_i = V_i$ and $\vec{y}_i$ are fresh with $|\vec{y}_i| = |\vec{x}_i|$. As before, we suppress the Λ superscript if it is clear from the context. Here, $\Phi \Rightarrow \Psi := \bigwedge \Pi \to \bigvee \Psi$. The *guards* $\gamma^{\Lambda,i} := \bigwedge_{j \le i} \gamma_j^{\Lambda,i}$ are defined as before with

$$\gamma_j^{\Lambda,i} := \bigwedge_{t^\sigma \in G_j^i} \bigvee_{s^{\sigma'} \in P_j} t[\vec{y}_i/\vec{x}_i] <^{\sigma,\sigma'} s$$

where $<^{+,-} := <$ and $<^{\sigma,\sigma'} := \le$ otherwise.

Before proving the embedding of $\mathrm{SI\Pi}_n$ into $\mathrm{I\Pi}_n$, we make some observations concerning the free variables and logical complexity of the inductive invariants.

Proposition 7.2.12.

(i) For $\Lambda \Rightarrow \Gamma \Rightarrow_V \Delta$ we have $\mathrm{FV}(\mathcal{H}_\Lambda) = \mathrm{SV}(\Lambda) \cup (V \cap \mathrm{FV}(\Gamma, \Delta))$.

(ii) If Λ satisfies the condition of $\mathrm{SI\Pi}_n$ then $\mathcal{H}_\Lambda \subseteq \Pi_n$.

Proof.

(i) For φ_i^Λ we have $\mathrm{FV}((\Gamma_i^{\ni V_i} \Rightarrow \Delta_i^{\ni V_i})[\vec{y}_i/\vec{x}_i]) \subseteq \mathrm{SV}(\Lambda) \cup \vec{y}_i$ and $\mathrm{FV}(\gamma_j^{\Lambda,i}) = \mathrm{FV}(G_j^i[\vec{y}_i/\vec{x}_i]) \cup \mathrm{FV}(P_j) \subseteq \vec{y}_i \cup (V \cap \mathrm{FV}(\Gamma, \Delta))$. As φ_i^Λ closes over $\vec{y}_i$ that means $\mathrm{FV}(\varphi_i^\Lambda) \subseteq \mathrm{SV}(\Lambda) \cup (V \cap \mathrm{FV}(\Gamma, \Delta))$ as desired.

(ii) For φ_i^Λ observe that, because $\Gamma_i^{\ni V_i} \subseteq \Sigma_n$ and $\Delta_i^{\ni V_i} \subseteq \Pi_n$ per assumption, we have $(\Gamma_i^{\ni V_i} \Rightarrow \Delta_i^{\ni V_i}) \in \Pi_n$. As $\gamma^{\Lambda_i} \in \Delta_1 (:= \Pi_1 \cap \Sigma_1)$ and Π_n is closed under universal quantification, that means $\varphi_i^\Lambda \in \Pi_n$ as desired.

□

Lemma 7.2.13. If $\mathrm{SI\Pi}_n \vdash \Gamma \Rightarrow \Delta$ with $n > 0$ then $\mathrm{I\Pi}_n \vdash \Gamma \Rightarrow \Delta$.

Proof. The proof proceeds analogously to the translation of $\mathrm{SHA}_<$ into HA (theorem 7.1.2) i.e., we prove that $\mathrm{SI\Pi}_n \vdash \Gamma \Rightarrow_V \Delta$ then $\mathrm{I\Pi}_n \vdash \Gamma, \mathcal{H}_\Lambda \Rightarrow \Delta$. Notably the rules whose cases are concerned with the 'adjustment' of the $\mathcal{H}_\Lambda$ to mirror a changed stack are completely analogous. We thus only cover the cases which change from those in theorem 7.1.2:

Comp: Let $\Lambda = \Lambda'; (X^-; \pi(\Lambda), X^+ \mapsto \Gamma \Rightarrow_V \Delta)$ with $i := |\Lambda|$. Observe that in this case, $\Gamma_i = \Gamma, \Delta_i = \Delta, V_i = V$ and further $\vec{x}_i = \mathrm{FV}(\Gamma, \Delta)$, $G_i^i = X^+$ and $G_{<j}^i = \pi(\Lambda)$. We may assume $\Gamma, \mathcal{H}_{\Lambda'}, \varphi_i \Rightarrow \Delta$ to derive $\Gamma, \mathcal{H}_{\Lambda'} \Rightarrow \Delta$. Define $\max X := \max\{t \mid t^\sigma \in X\}$ and consider the formula

$$\iota_m := \forall \vec{y}_i.\ m = \max(X[\vec{y}_i/\vec{x}_i]) \to \gamma_{<i}^i \to (\Gamma^{\ni V} \Rightarrow \Delta^{\ni V})[\vec{y}_i/\vec{x}_i]$$

where $\gamma^i_{<i} \coloneqq \bigwedge_{j<i} \gamma^{\Lambda,i}_j$. We derive the following:

$$
\begin{prooftree}
\AxiomC{$\Gamma, \mathcal{H}_{\Lambda'}, \varphi_i \Rightarrow \Delta$}
\UnaryInfC{$\Gamma^{\not\ni V}, \mathcal{H}_{\Lambda'}, \varphi_i \Rightarrow \Sigma, \Delta^{\not\ni V}$}
\LeftLabel{Sub}
\UnaryInfC{$\Gamma^{\not\ni V}, \mathcal{H}_{\Lambda'}[\vec{y}_i/\vec{x}_i], \varphi_i[\vec{y}_i/\vec{x}_i] \Rightarrow \Sigma[\vec{y}_i/\vec{x}_i], \Delta^{\not\ni V}$}
\LeftLabel{$\dagger$}
\UnaryInfC{$\Gamma^{\not\ni V}, \mathcal{H}_{\Lambda'}, \gamma^i_{j<i}, \varphi_i[\vec{y}_i/\vec{x}_i] \Rightarrow \Sigma[\vec{y}_i/\vec{x}_i], \Delta^{\not\ni V}$}
\LeftLabel{$\star$}
\UnaryInfC{$\Gamma^{\not\ni V}, \mathcal{H}_{\Lambda'}, \gamma^i_{j<i}, \forall m' < \max(X[\vec{y}_i/\vec{x}_i]).\ \iota_{m'} \Rightarrow \Sigma[\vec{y}_i/\vec{x}_i], \Delta^{\not\ni V}$}
\UnaryInfC{$\Gamma^{\not\ni V}, \mathcal{H}_{\Lambda'}, \forall m' < m.\ \iota_{m'} \Rightarrow \iota_m, \Delta^{\not\ni V}$}
\LeftLabel{Ind}
\UnaryInfC{$\Gamma^{\not\ni V}, \mathcal{H}_{\Lambda'} \Rightarrow \forall m. \iota_m, \Delta^{\not\ni V}$}
\AxiomC{}
\RightLabel{$*$}
\UnaryInfC{$\Gamma^{\ni V}, \forall m. \iota_m \Rightarrow \Delta^{\ni V}$}
\LeftLabel{Cut}
\BinaryInfC{$\Gamma, \mathcal{H}_{\Lambda'} \Rightarrow \Delta$}
\end{prooftree}
$$

where $\Sigma \coloneqq \bigwedge \Gamma^{\ni V} \to \Delta^{\ni V}$ and $\star$ uses

$$\gamma^i_{j<i}, \forall m' < \max(X[\vec{y}_i/\vec{x}_i]).\ \iota_{m'}, \Rightarrow \varphi_i[\vec{y}_i/\vec{x}_i].$$

and $\dagger$ uses

$$\mathcal{H}_{\Lambda'}, \gamma^i_{j<i} \Rightarrow \bigwedge \mathcal{H}_\Lambda[\vec{y}_i/\vec{x}_i].$$

The proof of both is analogous those in theorem 7.1.2, observe for $\dagger$ that, as $\mathrm{SV}(\Lambda) \cap V = \emptyset$, the substitutions in $\mathcal{H}_\Lambda$ only act upon the guards γ^j. The crucial observation justifying the application of Sub is that $\vec{x}_i, \vec{y}_i \notin \mathrm{FV}(\Gamma^{\not\ni V}, \Delta^{\not\ni V})$.

Lastly, $*$ is derived by instantiating $\forall m.\ \iota_m$ with $m \coloneqq \max X$ and $\vec{y}_i \coloneqq \vec{x}_i$. Observe that for this, $\gamma^i_{j<i}[\vec{x}_i/\vec{y}_i]$ is trivially true.

Bud: Let $\Lambda = \Lambda', (X^+; \pi(\Lambda'), X \mapsto \Gamma \Rightarrow \delta)$ with $|\Lambda| = i$. We want to derive $\Gamma, \mathcal{H}_\Lambda \Rightarrow \delta$. For this, we use

$$\varphi_i \coloneqq \forall \vec{y}_i.\ \gamma^i \to (\Gamma_i^{\ni V_i} \Rightarrow \Delta_i^{\ni V_i})[\vec{y}_i/\vec{x}_i] \in \mathcal{H}_\Lambda$$

instantiated with $\vec{y}_i \coloneqq \vec{x}_i$. It remains to prove $\gamma^i[\vec{x}_i/\vec{y}_i]$ but this is analogous to theorem 7.1.2.

Var: For this, observe that Var acts like an instance of Drop on Λ, so it is analogous to that case.

$\forall$L: Suppose that $\mathcal{H}_\Lambda, \Gamma \Rightarrow \psi(x), \Delta$ where $x \notin \mathrm{FV}(\Gamma, \Delta, \forall x.\psi)$ to deduce that $\mathcal{H}_\Lambda, \Gamma \Rightarrow \forall x.\psi, \Delta$. As the proof has stable variables, we know that $x \notin \mathrm{SV}(\Lambda)$. As $\mathrm{FV}(\mathcal{H}_\Lambda) \subseteq \mathrm{SV}(\Lambda) \cup (V \cap \mathrm{FV}(\Gamma, \forall x.\psi, \Delta))$ this means $x \notin \mathrm{FV}(\mathcal{H}_\Lambda, \Gamma, \forall x.\psi, \Delta)$ and we may simply reapply $\forall$L.

Note that the case for $\exists$R is analogous to that of $\forall$L and the case for Sub will need an analogous justification for the reapplication of Sub. □

Remark 7.2.4. Observe that the proof strategy above does not work for $\mathrm{SI}\Pi_0$. Crucially, the inductive invariants will not be in Π_0 because they contain universal quantifications.

Remark 7.2.5. Differing from both Das (2020) and Beklemishev et al. (2025), which establish correspondences to $\mathrm{I}\Sigma_n$, we focus on $\mathrm{I}\Pi_n$ instead. Of course, this does not really matter as $\mathrm{I}\Sigma_n = \mathrm{I}\Pi_n$. However, measuring the logical complexities involved in our translation in terms of Π, rather than Σ, is natural as the inductive invariants we use are Π_n because of the universal quantification. One could instead follow the formulation of inductive invariants in (Sprenger and Dam 2003b), who give 'dualised' inductive invariants which are naturally Σ_n instead.

Remark 7.2.6. In proposition 7.2.3 we observe that an $\mathrm{CPA}_<$-proof Π can be transformed into a CPA_V-proof Π' with stable variables by simply fixing V to be the set of all free variables of Π. Observe that in this case lemma 7.2.13 would produce exactly the same PA-proof as the PA-analogue of theorem 7.1.2, because the inductive invariants will close over every free variables and include all formulas of the sequent.

We can now directly conclude the identification of $\mathrm{I}\Pi_n$ and $\mathrm{CI}\Pi_n$.

Theorem 7.2.14. For $n > 0$, provability in $\mathrm{I}\Pi_n$ and $\mathrm{CI}\Pi_n$ coincide.

Proof. The direction from $\mathrm{I}\Pi_n$ to $\mathrm{CI}\Pi_n$ is lemma 7.2.5. For the converse direction, observe that if $\mathrm{CI}\Pi_n \vdash \Gamma \Rightarrow \Delta$ then $\mathrm{SI}\Pi_n \vdash \Gamma \Rightarrow \Delta$ by corollary 7.2.10. Then $\mathrm{I}\Pi_n \vdash \Gamma \Rightarrow \Delta$ by lemma 7.2.13 because $n > 0$. □

Remark 7.2.7. We have noted before that the 'discrepancy' indicated by the result of Das (2020) of $\mathrm{C}\Sigma_n$ and $\mathrm{I}\Sigma_{n+1}$ proving the same $\mathrm{I}\Pi_{n+1}$-theorems can be explained as the SUB-rule together with the mechanisms of cycles 'hiding' an implicit universal quantification on the right-hand side. Towards this, consider the 'sketch' of a CPA-proof below, the star marking a cycle as usual:

$$\dfrac{\dfrac{\Rightarrow A(x)\ (\star)}{\Rightarrow A(t)}\,\textsc{Sub} \qquad \dfrac{\vdots}{A(t) \Rightarrow A(x)}}{\Rightarrow A(x)\ (\star)}$$

Along the $\star$-cycle, the value of x is changed to that of some different t. Intuitively, this means that the $\star$-cycle is 'really' proving $\forall x.\ A(x)$, rather than just $A(x)$. Observe that exactly this pattern appears in the cyclic proof of the termination of the Ackermann function we give in fig. 4 in chapter 3. If one takes $A \in \Sigma_1$, as in the Ackermann example observed by Simpson (2017), the proof above 'embodies' $\mathrm{I}\Pi_2 (= \mathrm{I}\Sigma_2)$, rather than $\mathrm{I}\Sigma_1$ as one might naïvely assume.

By the same reasoning, the results in this section cannot be naïvely dualised. That is, consider the fragment $\mathrm{CI}\Sigma_n$ of CPA_V which in analogous to

$\mathrm{C}\Pi$, replacing the condition on companion nodes by the requirement that $\Gamma^{\ni V} \in \Pi_n, \Delta^{\ni V} \in \Sigma_n$. This fragment would not be equivalent to $\mathrm{I}\Sigma_n$ or $\mathrm{I}\Pi_n$. Rather, it would prove some Π_{n+1}-theorems of $\mathrm{I}\Sigma_{n+1}$, as discussed above.

7.3 Consistency of simple CPA*

Gentzen gave the first proof-theoretic argument for the consistency of Peano arithmetic. In this section, we adapt Gentzen's argument to the setting of cyclic PA. While said consistency already follows from 'composing' the translations from CPA to PA presented in this dissertation with any consistency argument for PA, we believe that some insight about cyclic proofs can be obtained by carrying out the argument over cyclic proofs directly.

Because the result we present in this section is rather technical and intricate, we begin by first giving a more informal overview over the main ideas involved in it. We hope that this makes the later parts easier to follow.

Gentzen's argument Because Gentzen's original argument is not that well known anymore, having rightfully been supplanted by that of Schütte (1950), we begin by summarising it.

Gentzen (1938) gives a proof-theoretic argument for the consistency of PA. As with plain first-order logic, consistency is easily established for cut-free proofs. However, PA is not cut-free complete, hence a more sophisticated method must be applied.

Depending on the choice of language, consistency can, for example, be stated as the unprovability of "$\perp$" or "$0 = 1$". Thus, Gentzen observes that it suffices to study proofs with *closed, atomic endsequents* i.e., proofs whose endsequents consist only of equations between closed terms. Intuitively, it is quite clear that proofs of such endsequents only 'need' equational reasoning via the =L and =R rules, using the arithmetical axioms of PA (or even just Robinson's Q). While such proofs still require CUTS to introduce said axioms into the proof, only CUTS on closed equations are needed. A straightforward, proof-theoretic consistency argument can be given for this fragment of PA (atomic, closed endsequent and only atomic, closed CUTS). Thus, the brunt of Gentzen's argument is a cut-elimination procedure which, given an arbitrary PA-proof with atomic, closed endsequent, produces a proof in the aforementioned fragment.

Consider the first part of a PA-proof which is directly above the root, assuming the endsequent is atomic and closed. Intuitively, it cannot contain logical rules for any connectives, as such formulas must 'first' be introduced via CUT. Thus, the only logical rules it can contain are =L and =R. Gentzen calls this prefix of the proof the *end-part* and it is crucial to his argument. Formally, the end-part is

* This section is based on joint work with Graham E. Leigh. We follow the presentation of Gentzen's argument given by Mancosu et al. (2021).

defined as the prefix of the proof 'underneath' the *border*. An application of a logical rule of a connective is considered to be *on the border* if, between it and the root, no logical rule is applied. Thus, the end-part contains exactly the part of the proof 'before' logical rules are applied, crucially already containing CUT-instances introducing complex formulas.

Gentzen's cut-elimination procedure simply repeats the following three steps:

- Step 1. Close the end-part.
- Step 2. Clean the proof.
- Step 3. Reduce a complex CUT on the border.

Step 1 is slightly complicated, so we begin by explaining the other two. Step 2 can be considered a bookkeeping step, importantly removing 'unneeded' instances of CUT i.e., those whose cut-formulas on one side are not used by the proof. Step 3 relies on this 'cleanliness' for the following crucial observation: if the proof still contains a non-atomic CUT, there must be such an CUT whose cut-formulas, both in the left and right premise, are principal in a rule application on the border. The CUT can the be partially reduced while *only working within the end-part*. Thus, the end-part acts as a kind of 'work site' for the cut-elimination procedure.

With the idea of the end-part as a work site in mind, we can make sense of step 1. The main obstacle of cut-elimination in PA, both generally but also in this specific case considered by Gentzen, is induction. For Gentzen's argument, it is easiest to work with the induction rule IND below.

$$\textsc{Ind}\frac{\Gamma, \varphi(x) \Rightarrow \varphi(Sx), \Delta \qquad x \notin \mathrm{FV}(\Gamma, \Delta)}{\Gamma, \varphi(0) \Rightarrow \varphi(t), \Delta}$$

In general, nothing prevents instances of IND from appearing in the end-part, which would hinder step 3. Thus, they must somehow first be removed from the end-part. Gentzen does this by *closing* the end-part. The end-part is considered closed if no sequent appearing in it contain free variables. As the premise of IND contains the free variable x (or can be trivially eliminated otherwise), a closed end-part can be considered IND-free. Thus, step 1 closes the end-part, thereby guaranteeing the absence of IND from the end-part for step 3. For most rules of PA, the closing procedure is simply the usual substitution lemma, replacing free variables with closed terms. The crucial case of the closing procedure is when instances of IND are encountered. For simplicity, assume that, after substitution, t is some numeral $\overline{n}$. Then the IND-instance may be replaced by a sequence of CUTS as depicted below, where each $\Gamma, \varphi(\overline{m}) \Rightarrow \varphi(\overline{m+1}), \Delta$ is a substitution instance of the premise $\Gamma, \varphi(x) \Rightarrow \varphi(Sx), \Delta$.

$$\textsc{Cut}\frac{\Gamma, \varphi(0) \Rightarrow \varphi(\overline{1}), \Delta \qquad \Gamma, \varphi(\overline{1}) \Rightarrow \varphi(\overline{2}), \Delta \qquad \dots \qquad \Gamma, \varphi(\overline{n-1}) \Rightarrow \varphi(\overline{n}), \Delta}{\Gamma, \varphi(0) \Rightarrow \varphi(\overline{n}), \Delta}$$

Naïvely considered, this replacement of Ind by an n-fold Cut increases the size of the proof, somewhat uncontrolledly as the value of n cannot necessarily be anticipated. As usual for proof-theoretical arguments for consistency of arithmetics, termination of the cut-elimination procedure is thus ensured by annotating proofs with an ordinal size bound and proving that each 'run' of steps 1 – 3 reduces said bound.

Moving to CPA The key observation motivating our argument in this section is the following: the notion of end-part directly transfers to simple CPA but more importantly, the structure of a closed end-part is the same in PA and CPA. A closed end-part of a CPA-proof trivially contains no induction axioms or rules, because they are not part of CPA. Similarly to the Ind-rule, instances of the Case-rule are absent, because they rely on free variables. Most importantly, a closed end-part cannot contain any companions of cycles: any successful cycle must have a free variable trace along it, meaning any companion must contain free variables.

Based on this observation, the steps 2 and 3 of the procedure described above directly transfers to the setting of CPA, as they only operate within the end-part. Thus, it 'only' remains to adapt the end-part closing procedure.

Closing the end-part The main focus of this section is closing the end-part of cyclic proofs of PA. This is difficult mostly because, analogously to the elimination of Ind in PA, this requires the elimination of cycles. To simplify this elimination, we work in SPA, rather than CPA directly, thus adding a 'preprocessing step' of translating CPA to SPA to our consistency proof.

The general approach to the end-part closing procedure is similar to that of Gentzen: the end-part of any proof, even with open, non-atomic endsequent, may be closed by applying a closing substitution to it. Most of the cases are the same as the usual 'substitution lemma' with the additional observation that this closes the end-part.

Towards the "elimination of cycles", consider the case in which the closing procedure encounters an instance of Comp as below where $\mathrm{FV}(\Gamma, \Delta) = \{x\}$

$$\textsc{Comp}\frac{x^- \mapsto \Gamma \Rightarrow \Delta \mid \Gamma \Rightarrow \Delta}{\varepsilon \mid \Gamma \Rightarrow \Delta}$$

and a closing substitution θ. For simplicity, assume again that $\theta = \{x \mapsto \overline{n}\}$. Similarly to Ind of PA, we must replace this with a sequence of Cuts to derive $\varepsilon \mid \Gamma[\theta] \Rightarrow \Delta[\theta]$. Unlike Ind, there is no induction formula φ, instead the whole sequent must be turned into an inductive invariant $\Gamma \rightarrow \Delta$. Further, it is not sufficient to just assume $(\Gamma \rightarrow \Delta)[\overline{m}/x]$ to conclude $\Gamma[\overline{m+1}/x] \Rightarrow \Delta[\overline{m+1}/x]$, following the argument embodied by the subproof above Comp, as said proof might 'access' any $k < m + 1$ via 'enough' Case$_x$-applications. Thus, one must Cut with the inductive invariant $\forall x < \overline{n}.\Gamma \rightarrow \Delta$ as below.

$$\dfrac{\dfrac{\dots \quad \dfrac{\Pi_m}{\varepsilon \mid \Gamma[\overline{m}/x] \Rightarrow \Delta[\overline{m}/x]} \quad \dots}{\varepsilon \mid \Rightarrow \forall x < \overline{n}.(\Gamma \rightarrow \Delta)} \qquad \dfrac{\Pi_R}{\varepsilon \mid \forall x < \overline{n}.(\Gamma \rightarrow \Delta), \Gamma[\overline{n}/x] \Rightarrow \Delta[\overline{n}/x]}}{\varepsilon \mid \Gamma[\overline{n}/x] \Rightarrow \Delta[\overline{n}/x]}\ \textsc{Cut}$$

The proof Π_R is obtained by a recursive application of the end-part closing procedure, which is generalised to allow for non-empty Λ, somewhat analogously to the translation procedure from SHA to HA in theorem 3.4.9. On the left-hand side, observe that to prove $\forall x < \overline{n}.(\Gamma \rightarrow \Delta)$, it suffices to prove $\Gamma[\overline{m}/x] \Rightarrow \Delta[\overline{m}/x]$ for all $m < n$, which can be done analogously to $\Gamma[\theta] \Rightarrow \Delta[\theta]$. Of course, this is only the simplest case. If the stack of the encountered Comp-rule is not empty, a yet more sophisticated inductive invariant than the one above must be constructed. We do not discuss these details here.

One could consider the seemingly simpler translation:

$$\dfrac{\dots\ \varepsilon \mid \Gamma[\overline{m}/x] \Rightarrow \Delta[\overline{m}/x]\ \dots \qquad \varepsilon \mid \dots, (\Gamma \rightarrow \Delta)[\overline{m}/x], \dots, \Gamma[\overline{n}/x] \Rightarrow \Delta[\overline{n}/x]}{\varepsilon \mid \Gamma[\overline{n}/x] \Rightarrow \Delta[\overline{n}/x]}\ \textsc{Cut}$$

That is, simply cutting in $(\Gamma \rightarrow \Delta)[\overline{m}/x]$ for all $m < n$. However, this increases the size of the sequent in the right-most premise of the Cuts in an 'uncontrollable' manner, as it depends on the value of $\overline{n}$. As (parts of) this sequent might be part of the inductive invariants of Comp-instances higher in the proof, this means the syntactic complexity of said invariants becomes 'uncontrollable', likely making it impossible to assign sufficient cut-ranks to those Comp-instances.

For a rather technical, but also very important, detail, observe that the case above essentially 'requires' that $\theta(x)$ is some numeral. The same is true for the case of the Case$_x$-rule. Thus, the closing procedure would be easier to carry out if θ was not just a closing substitution but an *assignment* i.e., $\mathrm{im}(\theta) \subseteq \{\overline{n} \mid n \in \omega\}$. The Sub-rule of CPA poses a problem to this approach. Suppose the procedure encountered an Sub-instance such as the one below:

$$\dfrac{\varepsilon \mid \Gamma \Rightarrow \Delta}{\varepsilon \mid \Gamma[t/x] \Rightarrow \Delta[t/x]}\ \textsc{Sub}$$

and suppose the closing substitution θ was an assignment. One might be tempted to use a recursive application of the closing procedure to obtain the desired sequent $\Gamma[\theta[x \mapsto t[\theta]]] \Rightarrow \Delta[\theta[x \mapsto t[\theta]]]$, except for one important detail preventing this: as $t[\theta]$ need only be closed but no numeral, $[\theta[x \mapsto t[\theta]]]$ may not be an assignment anymore. Of course, there always is some n such that $\mathrm{PA} \vdash t[\theta] = \overline{n}$ because $t[\theta]$ is closed. By CUTting in this equation, one can easily transform the claim to the one about the assignment $\theta[x \mapsto \overline{n}]$. However, this is 'at the cost' of a proof size increase. There are two options how to handle this 'incompatibility':

1. Assume that the closing θ is an assignment and make the 'adjustment' described above in the SUB-case or
2. Assume that the closing θ is just closed and make the 'adjustment' in the COMP- and CASE$_x$-cases.

Crucially, the ordinal annotations of the rules during whose cases the adjustment is made must take into account the adjustment's 'costly' proof size increase. As SUB is introduced during the cut-reduction step (see lemma 7.3.8), in which the rules inserted must have 'economical' ordinal annotations, we have opted for option 2.

Another important detail is how to handle the following situation during the closing procedure

$$\textsc{Comp}\,\dfrac{x^- \mapsto \Gamma \Rightarrow \varphi, \Delta \mid \Gamma \Rightarrow \varphi, \Delta}{\forall\mathrm{R}\,\dfrac{\varepsilon \mid \Gamma \Rightarrow \varphi, \Delta}{\varepsilon \mid \Gamma \Rightarrow \forall x.\varphi, \Delta}}$$

with some closing θ with $x \notin \mathrm{dom}(\theta)$. Of course, the substitution θ may simply be 'pulled through' as follows, leaving the following situation:

$$\forall\mathrm{R}\,\dfrac{\varepsilon \mid \Gamma[\theta] \Rightarrow \varphi[\theta], \Delta[\theta]}{\varepsilon \mid \Gamma[\theta] \Rightarrow \forall x.\varphi[\theta], \Delta[\theta]}$$

At this point, the COMP-case we describe cannot be applied as $x \notin \mathrm{dom}(\theta)$. However, observe that the $\forall$R application above is on the border, meaning the procedure need not continue any further to close the end-part. For such situations, we introduce another version of the COMP-rule called COMP$_s$ which represents a 'suspended' or 'paused' state of the closing procedure, whose variant for empty stacks is given below:

$$\textsc{Comp}_s\,\dfrac{x^- \mapsto \Gamma \Rightarrow \Delta \mid \Gamma \Rightarrow \Delta}{\varepsilon \mid \Gamma[\theta] \Rightarrow \Delta[\theta]}$$

where we require that $x \notin \mathrm{dom}(\theta)$.

Extended language To manage the invariants introduced by the closing procedure, we require a rule which lets us 'shift' the bounds of bounded quantifiers, such as the example below.

$$\textsc{Shift}\frac{\varepsilon \mid \Gamma, \forall x < \overline{n}.\ \varphi, \Delta}{\varepsilon \mid \Gamma, \forall x < \overline{n+1}.\ \varphi, \Delta}$$

With the usual definition of $\forall x < t.\varphi := \forall x.(\exists a.x + a = t) \to \varphi$, this rule is admissible by means of Cuts at the 'cost' of some increase to the proof size. As this is somewhat finicky to reason about as part of a cut-reduction procedure, we instead opt to extend the language of first-order arithmetic with connectives

$$\forall x < n.\varphi \quad \text{and} \quad \forall x \leq n.\varphi \quad \text{where } n \in \omega.$$

Crucially, we do not allow arbitrary terms as bounds, but only fixed numbers $n \in \omega$. This means that the =L and Sub rules cannot interact with such bounds, making the cut-reduction arguments for these connectives easier. As these bounded quantifiers are only auxiliary connectives introduced during the closing procedures, this 'inflexibility' is not a problem.

7.3.1 Ordinal-bounded stack-controlled PA: SPA_O

We begin by defining an ordinal annotated version SPA_O of SPA. As in the case of PA, the ordinal annotations are later used to prove termination of the cut-reduction procedure.

This chapter employs a (conservative) extension of Peano Arithmetic which adds bounded quantification as a primitive connective. That is, we employ the following syntax:

$$\begin{aligned} s, t \in \textsc{Term} &::= x \mid 0 \mid Ss \mid s + t \mid s \cdot t && x \in \textsc{Var} \\ \varphi, \psi \in \textsc{Form} &::= s = t \mid \bot \mid \varphi \to \psi \mid \forall x.\varphi \mid \forall x < n.\varphi \mid \forall x \leq n.\varphi && n \in \omega \end{aligned}$$

The sequents of SPA_O consider are of the form $\Lambda \vdash^\alpha_k \Gamma \Rightarrow \Delta$ where $\Gamma \Rightarrow \Delta$ is a PA-sequent in the language above with Γ, Δ *lists of formulas*[25], α is a *size bound* in the usual ordinal notation system for ε_0 and $k < \omega$ is a *cut-rank*. Λ is a list of companion labels $x^\sigma \mapsto \Gamma \Rightarrow \Delta$, with the same restrictions as in SHA.

We begin by specifying the rules of 'arithmetic in' SPA_O. These are mostly taken from (Mancosu et al. 2021). Deviating from them, we introduce an explicit

[25] Note that this differs from all other proof systems in this dissertation, whose sequents are *sets* of formulas. However, for the cut-reduction argument we present here is made somewhat simpler by not having to consider the implicit contractions of set sequents.

Raise-rule for raising cut-ranks. This simplifies the presentation of the results in this section somewhat. Below, $\alpha + \beta$ denotes the usual commutative addition operator defined on the ordinal notation system for ε_0 and $s(\varphi)$ is the usual measure of formula size. Further, for an ordinal $\beta = \omega^{\alpha_1} + \ldots < \varepsilon_0$ in Cantor normal form, we write $\beta^+ := \omega^{\alpha_1+1}$.

Logical rules The following rules which do not interact with companion labels:

$$\to\mathrm{L}\frac{\Lambda \vdash_k^\beta \Pi \Rightarrow \varphi, \Sigma \quad \Lambda \vdash_k^\alpha \Gamma, \psi \Rightarrow \Delta}{\Lambda \vdash_k^{\alpha+\beta} \Gamma, \Pi, \varphi \to \psi \Rightarrow \Delta, \Sigma}$$

$$\to\mathrm{R}\frac{\Lambda \vdash_k^\alpha \Gamma, \varphi \Rightarrow \psi, \Delta}{\Lambda \vdash_k^{\alpha+1} \Gamma \Rightarrow \varphi \to \psi, \Delta} \qquad \forall\mathrm{L}\frac{\Lambda \vdash_k^\alpha \Gamma, \varphi[t/x] \Rightarrow \Delta}{\Lambda \vdash_k^{\alpha+1} \Gamma, \forall x.\varphi \Rightarrow \Delta}$$

$$\forall\mathrm{R}\frac{\Lambda \vdash_k^\alpha \Gamma \Rightarrow \varphi[y/x], \Delta \quad y \notin \mathrm{FV}(\Gamma, \Delta)}{\Lambda \vdash_k^{\alpha+1} \Gamma \Rightarrow \forall x.\varphi, \Delta}$$

$$=\mathrm{L}\frac{\Lambda \vdash_k^\alpha \Gamma[t/x, s/y] \Rightarrow \Delta[t/x, s/y] \quad x, y \notin \mathrm{FV}(s, t)}{\Lambda \vdash_k^{\alpha+1} \Gamma[s/x, t/y], s = t \Rightarrow \Delta[s/x, t/y]}$$

$$=\mathrm{R}\frac{}{\Lambda \vdash_k^1 \Rightarrow t = t} \qquad \bot\mathrm{L}\frac{}{\Lambda \vdash_k^1 \bot \Rightarrow}$$

$$\mathrm{Wk}\frac{\Lambda \vdash_k^\alpha \Gamma \Rightarrow \Delta}{\Lambda \vdash_k^\alpha \Gamma, \Gamma' \Rightarrow \Delta, \Delta'} \qquad \mathrm{ExL}\frac{\Lambda \vdash_k^\alpha \Gamma, \psi, \varphi, \Gamma' \Rightarrow \Delta}{\Lambda \vdash_k^\alpha \Gamma, \varphi, \psi, \Gamma' \Rightarrow \Delta}$$

$$\mathrm{ExR}\frac{\Lambda \vdash_k^\alpha \Gamma \Rightarrow \Delta, \psi, \varphi, \Delta'}{\Lambda \vdash_k^\alpha \Gamma \Rightarrow \Delta, \varphi, \psi, \Delta'} \qquad \mathrm{CtrL}\frac{\Lambda \vdash_k^\alpha \Gamma, \varphi, \varphi \Rightarrow \Delta}{\Lambda \vdash_k^\alpha \Gamma, \varphi \Rightarrow \Delta}$$

$$\mathrm{CtrR}\frac{\Lambda \vdash_k^\alpha \Gamma \Rightarrow \varphi, \varphi, \Delta}{\Lambda \vdash_k^\alpha \Gamma \Rightarrow \varphi, \Delta} \qquad \mathrm{Sub}\frac{\Lambda \vdash_k^\alpha \Gamma \Rightarrow \Delta}{\Lambda \vdash_k^\alpha \Gamma[t/x] \Rightarrow \Delta[t/x]}$$

$$\mathrm{Cut}\frac{\begin{array}{c} s(\varphi) < k \text{ or } \varphi \text{ atomic} \\ \Lambda \vdash_k^\alpha \Gamma \Rightarrow \varphi, \Delta \quad \Lambda \vdash_k^\beta \Pi, \varphi \Rightarrow \Sigma \end{array}}{\Lambda \vdash_k^{\alpha+\beta} \Gamma, \Pi \Rightarrow \Delta, \Sigma} \qquad \mathrm{Raise}\frac{\Lambda \vdash_{k+1}^\alpha \Gamma \Rightarrow \Delta}{\Lambda \vdash_k^{\omega^\alpha} \Gamma \Rightarrow \Delta}$$

ARITHMETICAL AXIOMS These are of the form:

$$\begin{array}{ll} \varepsilon \vdash^1_k \ 0 = St \Rightarrow & \varepsilon \vdash^1_k Ss = St \Rightarrow s = t \\ \varepsilon \vdash^1_k \ \Rightarrow s + 0 = 0 & \varepsilon \vdash^1_k \ \Rightarrow s + St = S(s+t) \\ \varepsilon \vdash^1_k \ \Rightarrow s \cdot 0 = 0 & \varepsilon \vdash^1_k \ \Rightarrow s \cdot St = (s \cdot t) + s \end{array}$$

Further, we give the following size and cut-bounds to the usual 'cyclic structure rules' of SPA:

$$\textsc{Comp}\frac{\begin{array}{c} s(\varphi(\Lambda; (x^- \mapsto \Gamma \Rightarrow \Delta))) < k \\ \Lambda; (x^- \mapsto \Gamma \Rightarrow \Delta) \vdash^\alpha_k \Gamma \Rightarrow \Delta \end{array}}{\Lambda \vdash^{\alpha^+}_k \Gamma \Rightarrow \Delta}$$

$$\textsc{Bud}\frac{}{\Lambda; (x^+ \mapsto \Gamma \Rightarrow \Delta) \vdash^\omega_k \Gamma \Rightarrow \Delta} \qquad \textsc{Drop}\frac{\Lambda \vdash^\alpha_k \Gamma \Rightarrow \Delta}{\Lambda; \Lambda' \vdash^{\alpha+1}_k \Gamma \Rightarrow \Delta}$$

$$\textsc{Case}_x\frac{\begin{array}{c} x \in \mathrm{FV}(\Gamma, \Delta) \\ \Lambda \vdash^\alpha_k \Gamma(0) \Rightarrow \Delta(0) \qquad \Lambda^{+x} \vdash^\beta_k \Gamma(S\,x) \Rightarrow \Delta(S\,x) \end{array}}{\Lambda \vdash^{\alpha+\beta+\omega}_k \Gamma(x) \Rightarrow \Delta(x)}$$

Remark 7.3.1. The bounds and restrictions of COMP are easiest understood by recalling the same for the usual induction rule

$$\textsc{Ind}\frac{\vdash^\alpha_k \Gamma, \varphi(x) \Rightarrow \varphi(Sx), \Delta \quad s(\varphi) < k \quad x \notin \mathrm{FV}(\Gamma, \Delta)}{\vdash^{\alpha^+}_k \Gamma, \varphi(0) \Rightarrow \varphi(t), \Delta}.$$

The IND-rule 'anticipates' its replacement by a sequence of CUTS on $\varphi(\overline{n})$. Thus, it must ensure that the cut-rank is sufficient. Further, the CUTS will be between (substitution instances of) its premise, resulting in a proof of size $\alpha \cdot n + m < \alpha^+$ for some $n, m \in \omega$.

In SPA_O, the COMP-rule takes the place of IND. It, too, will be replaced by a sequence of CUTS. To accommodate for the added complexity of cyclic proofs, the cut-formulas are substitution instances of more complex inductive invariant $\varphi(\Lambda; (x^- \mapsto \Gamma \Rightarrow \Delta))$ derived from the control stack. We defer the definition of said invariants to section 7.3.2, as this detail is not so important to presenting the rules of SPA_O.

While we have not yet specified all rules of SPA_O, we can already embed any CPA-proof into it, using only the rules listed so far.

Proposition 7.3.1. If CPA $\vdash \Gamma \Rightarrow \Delta$ then there exists α such that $\varepsilon \vdash^{\alpha}_{0} \Gamma \Rightarrow \Delta$.

Proof. First, observe that any CPA-proof can be translated into an SPA-proof by following theorem 6.4.8. Next, it may be translated to a variant using list-sequents, as in this section, rather than set-sequents, as in the definition of CPA. Then, size bounds and cut-ranks may be added according to the rules of SPA_{O}, resulting is a derivation $\varepsilon \vdash^{\beta}_{k} \Gamma \Rightarrow \Delta$ for some $\beta < \varepsilon_0$ and some $k < \omega$. Lastly, one may apply k instances of Raise to obtain $\varepsilon \vdash^{\omega_k(\beta)}_{0} \Gamma \Rightarrow \Delta$ as desired. □

Next, we define the rules for the bounded quantifiers. Note that these are chosen not to provide a complete axiomatisation of bounded quantification but rather as exactly those auxiliary rules which we need for our argument.

Let $<^{-}:=<$ and $<^{+}:=\leq$. We write $\forall \vec{x} <^{\vec{\sigma}} \vec{n}$ as a shorthand for a sequence of bounded quantifications $\forall x_0 <^{\sigma_0} n_0 \ldots \forall x_i <^{\sigma_i} n_i \ldots$. Consider two such sequences $\forall \vec{x} <^{\vec{\sigma}} \vec{n}$ and $\forall \vec{x} <^{\vec{\sigma}'} \vec{m}$. We say that $\vec{\sigma}'; \vec{m}$ is a shifting of $\vec{\sigma}; \vec{n}$, writing $\vec{\sigma}'; \vec{m} \preceq \vec{\sigma}; \vec{n}$ if for every $i < |x|$:

- either $\sigma'_i = +$ and $\sigma_i = -$ and $m_i < n_i$,
- or $\sigma'_i = -$ or $\sigma_i = +$ and $m_i \leq n_i$.

Observe that these are exactly the 'shifts' of bounds which make Shift sound.

$$\forall{<}\mathrm{R}\ \frac{\varepsilon \vdash^{\alpha_0}_{k} \Gamma \Rightarrow A(0) \quad \ldots \quad \varepsilon \vdash^{\alpha_{n-1}}_{k} \Gamma \Rightarrow A(\overline{n-1})}{\varepsilon \vdash^{\Sigma\alpha_i}_{k} \Gamma \Rightarrow \forall x < n.\, A(x)}$$

$$\forall{\leq}\mathrm{R}\ \frac{\varepsilon \vdash^{\alpha_0}_{k} \Gamma \Rightarrow A(0) \quad \ldots \quad \varepsilon \vdash^{\alpha_n}_{k} \Gamma \Rightarrow A(\overline{n})}{\varepsilon \vdash^{\Sigma\alpha_i}_{k} \Gamma \Rightarrow \forall x \leq n.\, A(x)}$$

$$\forall{\leq}\mathrm{L}\ \frac{\varepsilon \vdash^{\alpha}_{k} \Gamma, A(\overline{n}) \Rightarrow B}{\varepsilon \vdash^{\alpha+1}_{k} \Gamma, \forall x \leq n.\, A(x) \Rightarrow B}$$

$$\textsc{Shift}\ \frac{\varepsilon \vdash^{\alpha}_{k} \Gamma, \forall \vec{x} <^{\vec{\sigma}'} \vec{m}.\, \varphi \Rightarrow \Delta \quad \text{where } \vec{\sigma}'; \vec{m} \preceq \vec{\sigma}; \vec{n}}{\varepsilon \vdash^{\alpha}_{k} \Gamma, \forall \vec{x} <^{\vec{\sigma}} \vec{n}.\, \varphi \Rightarrow \Delta}$$

We call a substitution $\theta : V \to \textsc{Term}$ with $V \subseteq \textsc{Var}$ *closed* if $\mathrm{im}(\theta)$ are closed terms. If $\mathrm{im}(\theta) \subseteq \{\overline{n} \mid n \in \omega\}$ we call θ an assignment.

Lastly, the system contains one 'bookkeeping' rule, which represents suspended states of the end-part closing procedure. The final, proof which is produced by the consistency argument will contain no instances of this rule. As noted before,

$\varphi(\Lambda; (x^- \mapsto \Gamma \Rightarrow \Delta))$ and $\mathcal{H}_\Lambda$ will be defined in section 7.3.2.

$$\textsc{Comp}_s \frac{\begin{array}{c} s(\varphi(\Lambda; (x^- \mapsto \Gamma \Rightarrow \Delta))) < k \\ \Lambda; x^- \mapsto \Gamma \Rightarrow \Delta \vdash^\beta_k \Gamma \Rightarrow \Delta \end{array}}{\varepsilon \vdash^{\beta^+}_k \mathcal{H}_\Lambda[\theta], \Gamma[\theta] \Rightarrow \Delta[\theta]}$$

We further require of $\textsc{Comp}_s$ that θ must be closed with $\textsc{Var}(\Lambda) \subseteq \mathrm{dom}(\theta)$, $\theta \upharpoonright \textsc{Var}(\Lambda)$ an assignment and $x \notin \mathrm{dom}(\theta)$.

7.3.2 Consistency

In this section, we adapt Gentzen's argument, as presented in (Mancosu et al. 2021), to SPA_O. We present the various steps involved in the argument in the order in which they are applied.

As noted before, we begin by proving that the end-part of a SPA_O-proof can be closed.

Definition 7.3.2 (End-part)**.** Consider a proof in SPA_O. A node of the derivation is said to be *in the end-part* if between it and the root of the proof, there is no instance of a logical connective rule. That is, no instance of →L, →R, ∀L, ∀R, ∀R', ∀≤L, ∀<R or ∀≤R.

An instance of the aforementioned logical rules whose conclusion is in the end-part is called a *boundary rule application*.

We call the end-part of a proof *closed* if all sequents of nodes in the end-part are closed.

Observe that if $\varepsilon \vdash^\alpha_k \Gamma \Rightarrow \Delta$ has a closed end-part, the end-part may not contain instances of $\textsc{Comp}$, $\textsc{Comp}$', $\textsc{Sub}$, $\textsc{Bud}$ or $\textsc{Case}_x$ as each of these rules requires the availability of free variables, either in its premises or consequence.

The following is a somewhat technical lemma which we use throughout the end-part closing argument. Certain key cases require the closing substitution to be an assignment (on certain arguments). The following lemma, allows one to always work with one, at a 'small' proof size cost.

Lemma 7.3.3. Let θ be a closed substitution and θ' the assignment with $\mathrm{dom}(\theta') = \mathrm{dom}(\theta)$ and

$$\theta'(x) := \overline{\theta(x)^{\mathbb{N}}}$$

i.e. the assignment mapping every x to its numeral value. If $\Lambda \vdash^\beta_k \Gamma[\theta'] \Rightarrow \Delta[\theta']$ then there is $n < \omega$ such that $\Lambda \vdash^{\beta+n}_k \Gamma[\theta] \Rightarrow \Delta[\theta]$. Further, if the assumed proof has a closed end-part, so does the produced proof.

Proof. Let w.l.o.g. θ already be an assignment except for one $\theta(x)$ which merely a closed term. Then there is a proof, consisting only of instances of =L, =R, atomic CUTs and arithmetical axioms of $\Lambda \vdash_k^m \Rightarrow \theta(x) = \theta'(x)$ for some $m < \omega$. Then derive

$$\dfrac{\Lambda \vdash_k^m \Rightarrow \theta(x) = \theta'(x) \qquad \dfrac{\Lambda \vdash_k^{\leq\beta} \Gamma[\theta'] \Rightarrow \Delta[\theta']}{\Lambda \vdash_k^{\leq\beta+1} \theta(x) = \theta'(x), \Gamma[\theta] \Rightarrow \Delta[\theta]}\,{=}\mathrm{L}}{\Lambda \vdash_k^{\leq\beta+m+1} \Gamma[\theta] \Rightarrow \Delta[\theta]}\,\textsc{Cut}$$

and fixing $n := m + 1$. The proof of $\Lambda \vdash_k^m \Rightarrow \theta(x) = \theta'(x)$ is fully closed. Further, if the assumed proof of $\Lambda \vdash_k^{\leq\beta} \Gamma[\theta'] \Rightarrow \Delta[\theta']$ has a closed end-part, this must mean $\mathrm{FV}(\Gamma, \Delta) \subseteq \mathrm{dom}(\theta') = \mathrm{dom}(\theta)$ and thus the whole derivation above has a closed end-part. □

The closing procedure roughly proceeds by replacing suitable instances of COMP by a series of CUTs on suitable inductive invariants. We continue by defining said inductive invariants.

As before, companion labels are of the form $x^\sigma \mapsto \Gamma \Rightarrow \Delta$ with the usual side-conditions. Let $x_i^{\sigma_i} \mapsto \Gamma_i \Rightarrow \Delta_i$ be the ith companion label in Λ, counting from the left and starting at 1, and let Λ_i be all the labels *before* the ith label. We define $\mathcal{H}_\Lambda := \{\varphi_i \mid 1 \leq i \leq |\Lambda|\}$ where

$$\varphi_i := \forall x_i' <^{\sigma_i} x_i.\forall \vec{u} \leq \vec{v}.\forall \vec{y}.(\Gamma_i \to \Delta_i)[x_i'/x_i, \vec{u}/\vec{v}]$$

with $\vec{v} = \textsc{Var}(\Lambda_i) \setminus \{x\}$, $\vec{y} = \mathrm{FV}(\Gamma_i, \delta_i) \setminus \{\vec{v}, x\}$ and $x_i', \vec{u}$ fresh and $\Phi \to \Psi$ being a shorthand for $\bigwedge \Phi \to \bigvee \Psi$. Observe that the formula above is not well-formed: The upper bounds of the bounded quantifiers in our language are natural numbers, not terms such as the variables used in the definition of the φ_i. In the rest of this section, we will only ever consider substitution instances $\varphi_i[\theta]$ where $\theta \restriction \textsc{Var}(\Lambda)$ is an assignment. Then, if $\forall y' <^\sigma y.\psi$ occurs as part of such a φ_i and $\theta(y) = \overline{n}$ we set $(\forall y' <^\sigma y.\psi)[\theta] = \forall y' <^\sigma n.\psi[\theta]$. While this is a drastic abuse of notation, it is the easiest way to define such sets $\mathcal{H}_\Lambda[\theta]$. We shall write $\varphi(\Lambda)$ for the last 'formula' $\varphi_{|\Lambda|}$.

The following is essentially 'the right inductive invariant' to prove that the end-part of a SPA_O-proof can always be closed.

Lemma 7.3.4. Let $\Lambda \vdash_k^\alpha \Gamma \Rightarrow \Delta$ and θ be closed with $\textsc{Var}(\Lambda) \subseteq \mathrm{dom}(\theta) \subseteq \mathrm{FV}(\Gamma, \Delta)$ and $\theta \restriction \textsc{Var}(\Lambda)$ an assignment. Then $\varepsilon \vdash_k^{\leq\alpha} \mathcal{H}_\Lambda[\theta], \Gamma[\theta] \Rightarrow \Delta[\theta]$. Furthermore, if $\mathrm{dom}(\theta) = \mathrm{FV}(\Gamma, \Delta)$ then the aforementioned proof has a closed end-part.

Proof. Proof by induction on α. Case-distinction on the last rule applied. In many cases, the claim about the closed end-part follows directly from the inductive hypothesis and by looking at the proof rules applied. If so, we do not further comment on it.

Comp: Then the situation is as follows:

$$\textsc{Comp}\ \dfrac{\dfrac{\Pi}{\Lambda; x^- \mapsto \Gamma \Rightarrow \Delta \vdash_k^{\beta} \Gamma \Rightarrow \Delta}}{\Lambda \vdash_k^{\beta^+} \Gamma \Rightarrow \Delta}$$

There are two cases to consider. First, if $x \notin \mathrm{dom}(\theta)$ then derive

$$\textsc{Comp}_s\ \dfrac{\dfrac{\Pi}{\Lambda; x^- \mapsto \Gamma \Rightarrow \Delta \vdash_k^{\beta} \Gamma \Rightarrow \Delta}}{\varepsilon \vdash_k^{\beta^+} \mathcal{H}_\Lambda[\theta], \Gamma[\theta] \Rightarrow \Delta[\theta]}$$

Otherwise, denote by φ^* the 'formula' $\varphi(\Lambda; x^- \mapsto \Gamma \Rightarrow \Delta)$ i.e.,

$$\varphi^* = \forall x' < x.\forall \vec{u} \leq \vec{v}.\forall \vec{y}.(\Gamma \to \Delta)[x'/x, \vec{u}/\vec{v}]$$

where $\vec{v} = \textsc{Var}(\Lambda) \setminus \{x\}$, $\vec{y} = \mathrm{FV}(\Gamma, \Delta) \setminus \{\vec{v}, x\}$ and $x', \vec{u}$ fresh.
By lemma 7.3.3 it suffices to derive $\varepsilon_k^{<\beta^+} \mathcal{H}_\Lambda[\theta], \Gamma[\theta] \Rightarrow \Delta[\theta]$, assuming that θ is an assignment. We prove this claim per induction on $\theta(x)$. Suppose the claim held for all θ^* with $\theta^*(x) < \theta(x)$. Then we derive the following:

$$\textsc{CtrL}\ \dfrac{\textsc{Cut}\ \dfrac{\forall{<}\mathrm{R}, \forall{\leq}\mathrm{R}\ \dfrac{\dots \quad \textsc{Shift}\ \dfrac{\dfrac{\dfrac{\text{IH: } \theta^*(x) < \theta(x)}{\varepsilon \vdash_k^{\leq \beta\cdot n^* + l^*} \mathcal{H}_\Lambda[\theta^*], \Gamma[\theta^*] \Rightarrow \Delta[\theta^*]}}{\varepsilon \vdash_k \mathcal{H}_\Lambda[\theta^*] \Rightarrow \forall \vec{y}.\ (\Gamma \to \Delta)[\theta^*]}}{\varepsilon \vdash_k^{\leq \beta\cdot n^* + l_2^*} \mathcal{H}_\Lambda[\theta] \Rightarrow \forall \vec{y}.\ (\Gamma \to \Delta)[\theta^*]} \quad \dots}{\varepsilon \vdash_k^{\leq \beta\cdot n + l} \mathcal{H}_\Lambda[\theta] \Rightarrow \varphi^*[\theta]} \qquad \Pi'}{\varepsilon \vdash_k^{\leq \beta\cdot (n+1) + l} \mathcal{H}_\Lambda[\theta], \mathcal{H}_\Lambda[\theta], \Gamma[\theta] \Rightarrow \Delta[\theta]}}{\varepsilon \vdash_k^{\leq \beta\cdot (n+1) + l} \mathcal{H}_\Lambda[\theta], \Gamma[\theta] \Rightarrow \Delta[\theta]}$$

$$\Pi' : \quad \dfrac{\text{IH 2.}}{\varepsilon \vdash_k^{\leq \beta} \mathcal{H}_\Lambda[\theta], \varphi^*[\theta], \Gamma[\theta] \Rightarrow \Delta[\theta]}$$

We first explain the derivation, then how the bounds are arrived at. The Cut-application is allowed as the Comp-rule ensures $s(\varphi^*) < k$. On the right-hand side of the Cut, we apply the inductive hypothesis 2. of the outer induction, noting that $\mathcal{H}_{\Lambda;x^- \mapsto \Gamma \Rightarrow \Delta} = \mathcal{H}_\Lambda, \varphi^*$. On the left-hand side, we apply the $\forall{<}$R and $\forall{\leq}$R rules until all bounded quantifiers have been removed from φ^*, replacing x' and the $\vec{u}$ with numerals. This results in a finite number of open premises, each of the form $\mathcal{H}_\Lambda[\theta] \Rightarrow \forall\vec{y}.(\mathcal{H}_\Lambda, \Gamma \to \Delta)[\theta^*]$, where θ^* is the assignment with $\theta^*(z)$ being the numeral inserted by the right-rule of the bounded quantification on $z \in \textsc{Var}(\Lambda) \cup \{x\}$. Next, the $\mathcal{H}_\Lambda[\theta]$ are 'adjusted' to $\mathcal{H}_\Lambda[\theta^*]$ via a Shift-application, observing that the only difference between them is that some of the upper bounds in $\mathcal{H}_\Lambda[\theta^*]$ may be lower than those in $\mathcal{H}_\Lambda[\theta]$. With applications of $\forall$R, $\to$R, $\wedge$L and $\vee$R, the sequents can be transformed back into the shape $\mathcal{H}_\Lambda[\theta^*], \Gamma[\theta^*] \Rightarrow \Delta[\theta^*]$, to which we may apply the inner inductive hypothesis, observing that $\theta^*(x) < \theta(x)$ by the structure of the $\forall$<R-rule.

To understand the ordinal bound, observe that the inner inductive hypothesis yields derivations of size $\leq \beta \cdot n^* + l^*$ with $n^*, l^* \in \omega$ for every assignment θ^* as described above which, at the cost of increasing l^* to some $l_2^* \in \omega$, can be transformed into derivations of $\mathcal{H}_\Lambda[\theta] \Rightarrow \forall\vec{y}.\ (\mathcal{H}_\Lambda, \Gamma \to \Delta)[\theta^*]$. As the size of $\forall$<R, $\forall\leq$R is simply the finite sum of the size of their premises, these combine into a derivation of size $\leq \beta \cdot n + l$ for some $n, l \in \omega$. The rest of the bound calculation is self-explanatory.

Towards the closed end-part, if θ is closed, then the proof produced by the IH for the right-hand premise of Cut already has a closed end-part. Observe that the application of $\forall$<R on the left-hand premise is part of the border, thus only the premises of the Cut as well as premise and conclusion of CtrL must be closed. It is easily observed they are if $\mathrm{dom}(\theta) = \mathrm{FV}(\Gamma, \Delta)$.

Comp$_s$: Then the situation is as follows:

$$\textsc{Comp}_s\ \dfrac{\dfrac{\Pi}{\Lambda; x^- \mapsto \Gamma \Rightarrow \Delta \vdash_k^\beta \Gamma \Rightarrow \Delta}}{\varepsilon \vdash_k^{\beta+} \mathcal{H}_\Lambda[\theta'], \Gamma[\theta'] \Rightarrow \Delta[\theta']}$$

where θ' is a closed substitution with $\textsc{Var}(\Lambda) \subseteq \mathrm{dom}(\theta')$, $\theta' \upharpoonright \textsc{Var}(\Lambda)$ an assignment and $x \notin \mathrm{dom}(\theta')$. Then the desired $\varepsilon \vdash_k^{\leq\alpha} \mathcal{H}_\Lambda[\theta' \cup \theta], \Gamma[\theta' \cup \theta] \Rightarrow \Delta[\theta' \cup \theta]$ an be obtained completely analogously to the Comp-case, noting that $\mathrm{dom}(\theta) \cap \mathrm{dom}(\theta') = \emptyset$.

Bud: Then the situation is as follows:

$$\textsc{Bud}\ \dfrac{}{\Lambda; (x^+ \mapsto \Gamma \Rightarrow \Delta) \vdash_k^\omega \Gamma \Rightarrow \Delta}$$

Denote by φ^* the 'formula' $\varphi(\Lambda; x^+ \mapsto \Gamma \Rightarrow \Delta)$ i.e.,

$$\varphi^* = \forall x' \leq x.\forall\vec{u} \leq \vec{v}.\forall\vec{y}.(\Gamma \to \Delta)[x'/x, \vec{u}/\vec{v}]$$

where $\vec{v} = \textsc{Var}(\Lambda) \setminus \{x\}$, $\vec{y} = \mathrm{FV}(\Gamma, \Delta) \setminus \{\vec{v}, x\}$ and $x', \vec{u}$ fresh.

Such a derivation is given by:

$$\forall\!\leq \mathrm{L}^*, \forall\mathrm{L}\ \frac{\overline{\varepsilon \vdash_k (\Gamma \to \Delta)[\theta], \mathcal{H}_\Lambda[\theta], \Gamma[\theta] \Rightarrow \Delta[\theta]}}{\varepsilon \vdash_k^{<\omega} \varphi^*[\theta], \mathcal{H}_{\Lambda'}[\theta], \Gamma[\theta] \Rightarrow \Delta[\theta]}$$

Observe that he premise of the $\forall\leq$L-steps is an instance of the tautological sequent $\Phi \to \Psi, \Phi \Rightarrow \Psi$ and thus has a proof of some finite size.

$\textsc{Drop}$: Then the situation is as follows:

$$\textsc{Drop}\ \frac{\Lambda \vdash_k^{\beta} \Gamma \Rightarrow \Delta}{\Lambda; \Lambda' \vdash_k^{\beta+1} \Gamma \Rightarrow \Delta}$$

and we derive

$$\textsc{Wk}\ \frac{\dfrac{\mathrm{IH}}{\varepsilon \vdash_k^{\leq\beta} \mathcal{H}_\Lambda[\theta], \Gamma[\theta] \Rightarrow \Delta[\theta]}}{\varepsilon \vdash_k^{\leq\beta} \mathcal{H}_{\Lambda;\Lambda'}[\theta], \Gamma[\theta] \Rightarrow \Delta[\theta]}$$

$\textsc{Case}_x$: Only the case of $x \in \mathrm{dom}(\theta)$ as below is interesting:

$$\textsc{Case}_x\ \frac{\dfrac{\Pi_0}{\Lambda \restriction x \vdash_k^{\beta} \Gamma[0/x] \Rightarrow \Delta[0/x]} \qquad \dfrac{\Pi_S}{\Lambda^{+x} \vdash_k^{\beta'} \Gamma[Sx/x] \Rightarrow \Delta[Sx/x]}}{\Lambda \vdash_k^{\beta+\beta'+\omega} \Gamma \Rightarrow \Delta}$$

By lemma 7.3.3, it suffices to prove $\varepsilon \vdash_k^{\leq\beta+\beta'} \mathcal{H}_\Lambda[\theta], \Gamma[\theta] \Rightarrow \Delta[\theta]$, assuming that θ is an assignment. We consider two cases.

$\theta(x) = 0$: In this case, simply derive

$$\textsc{Wk}\ \frac{\dfrac{\mathrm{IH}\ 2.\ \text{for}\ \Pi_0}{\varepsilon \vdash_k^{\leq\beta} \mathcal{H}_{\Lambda\restriction x}[\theta], \Gamma[\theta] \Rightarrow \Delta[\theta]}}{\varepsilon \vdash_k^{\leq\beta} \mathcal{H}_\Lambda[\theta], \Gamma[\theta] \Rightarrow \Delta[\theta]}$$

$\theta(x) = \overline{m+1}$: In this case, we define $\theta' := \theta[x \mapsto \overline{m}]$ and derive

$$\textsc{Shift}\ \frac{\dfrac{\mathrm{IH}\ 2.\ \text{for}\ \Pi_s}{\varepsilon \vdash_k^{\leq\beta'} \mathcal{H}_{\Lambda^{+x}}[\theta'], \Gamma[\theta] \Rightarrow \Delta[\theta]}}{\varepsilon \vdash_k^{\leq\beta'} \mathcal{H}_\Lambda[\theta], \Gamma[\theta] \Rightarrow \Delta[\theta]}$$

where we note that $\bullet[Sx/x][\theta'] = \bullet[\theta]$. Observe that the difference between $\mathcal{H}_\Lambda[\theta]$ and $\mathcal{H}_{\Lambda^{+x}}[\theta']$ is that all x-bounds are reduced from $m+1$ to m and that some bounds of the form $\forall x' < x$ are replaced by $\forall x' \leq x$. Both of these changes can be accomplished by $\textsc{Shift}$.

SUB: Then the situation is as follows:

$$\textsc{Sub}\ \frac{\Lambda \vdash_k^{\beta} \Gamma \Rightarrow \Delta}{\Lambda \vdash_k^{\beta} \Gamma[t/x] \Rightarrow \Delta[t/x]}$$

The only interesting case is if $\mathrm{dom}(\theta) = \mathrm{FV}(\Gamma[t/x], \Delta[t/x])$. In this case, extend θ to $\theta' := \theta[x \mapsto t[\theta]]$. Clearly, θ' is still closed and $\mathrm{dom}(\theta') = \mathrm{FV}(\Gamma, \Delta)$. Then, per IH, there is a proof of $\varepsilon \vdash_k^{\leq\beta} \mathcal{H}_\Lambda[\theta'], \Gamma[\theta'] \Rightarrow \Delta[\theta']$ with a closed end-part. This also closes this case, as $\bullet[t/x][\theta] = \bullet[\theta']$.

CUT: Then the situation is as follows:

$$\frac{\dfrac{\Pi_L}{\varepsilon \vdash_k^{\beta} \Gamma \Rightarrow \varphi, \Delta} \qquad \dfrac{\Pi_R}{\varepsilon \vdash_k^{\beta'} \Pi, \varphi \Rightarrow \Sigma}}{\varepsilon \vdash_k^{\beta+\beta'} \Gamma, \Pi \Rightarrow \Delta, \Sigma}$$

The only interesting situation arises if φ introduces new free variables i.e., $\mathrm{FV}(\varphi) \not\subseteq \mathrm{dom}(\theta)$. In this case, derive

$$\frac{\dfrac{\text{IH for } \Pi_L}{\varepsilon \vdash_k^{\leq\beta} \Gamma[\theta_L] \Rightarrow (\varphi, \Delta)[\theta_L]} \qquad \dfrac{\text{IH for } \Pi_R}{\varepsilon \vdash_k^{\leq\beta'} (\Pi, \varphi)[\theta_R] \Rightarrow \Sigma[\theta_R]}}{\varepsilon \vdash_k^{\leq\beta+\beta'} (\Gamma, \Pi)[\theta] \Rightarrow (\Delta, \Sigma)[\theta]}$$

where

$$\theta_\bullet(x) \mapsto \begin{cases} \theta(x) & x \in \mathrm{dom}(\theta) \\ 0 & x \in \mathrm{FV}(\varphi) \setminus \mathrm{dom}(\theta) \end{cases}.$$

Observe that the $\theta_\bullet$ still satisfy the requirements on the substitution.

$\forall$R: Then the situation is as follows:

$$\forall\mathrm{R}\ \frac{\Lambda \vdash_k^{\beta} \Gamma \Rightarrow \varphi[y/x], \Delta}{\Lambda \vdash_k^{\beta+1} \Gamma \Rightarrow \forall x.\varphi, \Delta}$$

where y is fresh. Then derive

$$\forall\mathrm{R}\ \frac{\dfrac{\text{IH}}{\varepsilon \vdash_k^{\leq\beta} \mathcal{H}_\Lambda[\theta], \Gamma[\theta] \Rightarrow \varphi[y/x, \theta], \Delta[\theta]}}{\varepsilon \vdash_k^{\leq\beta+1} \mathcal{H}_\Lambda[\theta], \Gamma[\theta] \Rightarrow \forall x.\varphi[\theta], \Delta[\theta]}$$

noting that $x, y \notin \mathrm{dom}(\theta)$. Towards the closed end-part, observe that the $\forall$R-rule is on the border of the constructed proof. Thus, the end-part thus consists only of the conclusion, which is closed if $\mathrm{dom}(\theta) = \mathrm{FV}(\Gamma, \forall x.\varphi, \Delta)$.

OTHER RULES: Usual substitution + weakening argument. □

Remark 7.3.2. Observe that the size of the proof 'produced' in the Bud-case of lemma 7.3.4 is independent of the choice of θ, only depending on the contents of $\Lambda; x^+ \mapsto \Gamma \Rightarrow \Delta$. Thus, one could define a function h such that $h(\Lambda; x^+ \mapsto \Gamma \Rightarrow \Delta)$ is said size and redefine the Bud-rule as

$$\textsc{Bud}\frac{}{\Lambda; (x^+ \mapsto \Gamma \Rightarrow \Delta) \vdash_k^{h(\Lambda; x^+ \mapsto \Gamma \Rightarrow \Delta)} \Gamma \Rightarrow \Delta}.$$

This would lead to slightly smaller size bounds on some of the proofs. We chose to forgo this 'optimisation' to ease the presentation of SPA_O.

Remark 7.3.3. The argument above is the reason why the size bound of $\textsc{Case}_x$ needs to be $\alpha + \beta + \omega$, rather than the more intuitive $\alpha + \beta + 1$: to be able to 'resolve' $\textsc{Case}_x$ under a substitution θ with $x \in \mathrm{dom}(\theta)$, we require $\theta(x) = \overline{n}$ for some n. However, as x need not be in $\textsc{Var}(\Lambda)$, we cannot assume this, only that $\theta(x)$ is closed. We must thus apply lemma 7.3.3, at some 'cost' $< \omega$, to ensure θ is an assignment.

That the end-parts of SPA_O-proofs of closed endsequents, atomic or otherwise, can be closed follows directly from lemma 7.3.4.

Corollary 7.3.5 (End-part closing)**.** Let $\varepsilon \vdash_k^\alpha \Gamma \Rightarrow \Delta$ with Γ, Δ closed. Then $\varepsilon \vdash_k^{\leq\alpha} \Gamma \Rightarrow \Delta$ with a closed end-part.

Proof. Simply apply lemma 7.3.4 with $\theta = \emptyset$ and observe that $\mathcal{H}_\varepsilon = \emptyset$. □

As noted before, the closed end-parts of SPA_O and the proof system PA considered by Mancosu et al. (2021) are identical. We may thus 'import' some results directly to our setting.

We call an end-part Wk-*free* if it contains no instances of Wk. The following result is (Mancosu et al. 2021, proposition 7.34). This is the step which we described as "cleaning the proof" in the introduction to this section.

Fact 7.3.6. Let $\varepsilon \vdash_k^\alpha \Gamma \Rightarrow \Delta$ with Γ, Δ atomic and a closed end-part. Then $\varepsilon \vdash_k^{<\alpha} \Gamma' \Rightarrow \Delta'$ with $\Gamma' \subseteq \Gamma, \Delta' \subseteq \Delta$ and a closed, Wk-free end-part.

Crucially, a closed, Wk-free end-part of the proof of an atomic end-sequent cannot contain any instances of Cut which have at least one premise such that all of the 'traces' of its cut-formula end in Wk. This means that fact 7.3.6 is the part of the procedure in which Cut-instances are actually removed from the proof. The following is a direct consequence of the observation about Cuts above. It is (Mancosu et al. 2021, proposition 7.36).

Fact 7.3.7. Let Π be a proof of $\varepsilon \vdash^{\alpha}_{k} \Gamma \Rightarrow \Delta$ with a closed, WK-free end-part. Then either all of Π is in the end-part or the end-part contains an instance of a non-atomic CUT with both of its premises containing a boundary rule application to its cut-formula.

We now prove cut-elimination. As the closed end-part does not contain any 'cyclic rules', most of there reductions are standard. We thus only treat those reductions which involve novel rules we introduced here, referring the reader to, for example, (Mancosu et al. 2021) for the remaining cases.

Lemma 7.3.8 (Cut-reduction)**.** Let Π be a proof of $\varepsilon \vdash^{\alpha}_{0} \Gamma \Rightarrow \Delta$ with Γ, Δ atomic and a closed, WK-free end-part. If Π is not its own end-part then $\varepsilon \vdash^{\beta}_{0} \Gamma \Rightarrow \Delta$ with $\beta < \alpha$.

Proof. By fact 7.3.7 there exists a non-atomic CUT-application with cut-formula C in the end-part such that both of its premises contain a boundary rule application whose principal formula is the cut-formula. We reduce this CUT, the reduction strategy depending on the principal connective of the C. Observe that, as all these reductions operate within the closed end-part and no COMP-instances can occur within it, every sequent we mention has stack ε. We omit this information from the derivations below to save some space and reduce clutter.

$\forall x.A$: As $s(\forall x.A) > 0$, there must be a RAISE-application 'underneath' the CUT-application under consideration. We only handle the case of the right-rule being $\forall$R', the case of $\forall$R being analogous. Including the RAISE, the situation can be sketched as follows

$$
\cfrac{
\cfrac{
\cfrac{\cfrac{\Pi'_L}{\vdash^{\alpha'}_{k_L} \Gamma' \Rightarrow \varphi_L[y/x], \Delta'}}{\vdash^{\alpha'+1}_{k_L} \Gamma' \Rightarrow \forall x.\varphi_L, \Delta'}\;\forall\mathrm{R}
\quad\quad
\cfrac{\cfrac{\Pi'_R}{\vdash^{\beta'}_{k_R} \Pi', \varphi_R(t) \Rightarrow \Sigma'}}{\vdash^{\beta'+1}_{k_R} \Pi', \forall x.\varphi_R \Rightarrow \Sigma'}\;\forall\mathrm{L}
}{
\vdots\,\Pi_L \qquad\qquad \vdots\,\Pi_R
}
}{
\cfrac{\vdash^{\alpha}_{k} \Gamma \Rightarrow \forall x.\varphi, \Delta \qquad \vdash^{\beta}_{k} \Pi, \forall x.\varphi \Rightarrow \Sigma}{\vdash^{\alpha+\beta}_{k} \Gamma, \Pi \Rightarrow \Delta, \Sigma}\;\textsc{Cut}
}
$$

$$
\begin{array}{c}
\vdots\,\Pi \\
\cfrac{\vdash^{\gamma}_{k} \Xi \Rightarrow \Theta}{\vdash^{\omega^{\gamma}}_{k-1} \Xi \Rightarrow \Theta}\;\textsc{Raise} \\
\vdots
\end{array}
$$

where t closed and $\varphi, \varphi_L, \varphi_R$ differ only by terms and bounds, which may be caused by instances of =L and SHIFT along Π_L and Π_R. Further, $\alpha'+1 \leq \alpha$, $\beta' + 1 \leq \beta$ and $\alpha + \beta \leq \gamma$.

Replace this with the following derivation. We write $\Pi_\bullet, \varphi(t)$ for the derivation $\Pi_\bullet$ weakened by $\varphi(t)$, in which case all substitutions caused by =L and shifts caused by Shift in $\forall x.\varphi$ are 'replayed' in this additional $\varphi(t)$. Further, observe that the Cut on $\varphi(t)$ satisfies $s(\varphi(t)) < k-1$ as $s(\forall x.\varphi) = s(\varphi) + 1 < k$.

$$
\begin{array}{c}
\text{Cut}\ \dfrac{\Pi_1 \qquad\qquad\qquad \Pi_2}{\text{Ctr}\ \dfrac{\vdash^{\omega^{\gamma_L}+\omega^{\gamma_R}}_{k-1} \Xi, \Xi \Rightarrow \Theta, \Theta}{\vdash^{\omega^{\gamma_L}+\omega^{\gamma_R}}_{k-1} \Xi \Rightarrow \Theta}} \\
\vdots
\end{array}
$$

$$
\Pi_1 : \quad
\begin{array}{c}
\text{Cut}\ \dfrac{
\begin{array}{c}
\text{Wk}\ \dfrac{\text{Sub}\ \dfrac{\dfrac{\Pi'_L}{\vdash^{\alpha'}_{k_L} \Gamma' \Rightarrow \varphi_L[y/x], \Delta'}}{\vdash^{\alpha'}_{k_L} \Gamma' \Rightarrow \varphi_L(t), \Delta'}}{\vdash^{\alpha'}_{k_L} \Gamma' \Rightarrow \forall x.\varphi_L, \varphi_L(t), \Delta'} \\
\vdots\ \Pi_L, \varphi(t) \\
\vdash^{\alpha^*}_{k_L} \Gamma \Rightarrow \forall x.\varphi, \varphi(t)\Delta
\end{array}
\qquad
\begin{array}{c}
\dfrac{\dfrac{\Pi'_R}{\vdash^{\beta'}_{k_R} \Pi', \varphi_R(t) \Rightarrow \Sigma'}}{\vdash^{\beta'+1}_{k_R} \Pi', \forall x.\varphi_R \Rightarrow \Sigma'}\ \forall\text{L} \\
\vdots\ \Pi_R \\
\vdash^{\beta}_{k} \Pi, \forall x.\varphi \Rightarrow \Sigma
\end{array}
}{\vdash^{\alpha^*+\beta}_{k} \Gamma, \Pi \Rightarrow \varphi(t), \Delta, \Sigma} \\
\vdots\ \Pi, \varphi(t) \\
\text{Raise}\ \dfrac{\vdash^{\gamma_L}_{k} \Xi \Rightarrow \varphi(t), \Theta}{\vdash^{\omega^{\gamma_L}}_{k-1} \Xi \Rightarrow \varphi(t), \Theta}
\end{array}
$$

$$
\Pi_2 : \quad
\begin{array}{c}
\text{Cut}\ \dfrac{
\begin{array}{c}
\forall\text{R}\ \dfrac{\dfrac{\Pi'_L}{\vdash^{\alpha'}_{k_L} \Gamma' \Rightarrow \varphi_L[y/x], \Delta'}}{\vdash^{\alpha'+1}_{k_L} \Gamma' \Rightarrow \forall x.\varphi_L, \Delta'} \\
\vdots\ \Pi_L \\
\vdash^{\alpha}_{k} \Gamma \Rightarrow \forall x.\varphi, \Delta
\end{array}
\qquad
\begin{array}{c}
\dfrac{\dfrac{\Pi'_R}{\vdash^{\beta'}_{k_R} \Pi', \varphi_R(t) \Rightarrow \Sigma'}}{\vdash^{\beta'}_{k_R} \Pi', \varphi_R(t), \forall x.\varphi_R \Rightarrow \Sigma'}\ \text{Wk} \\
\vdots\ \Pi_R, \varphi(t) \\
\vdash^{\beta^*}_{k} \Pi, \varphi(t), \forall x.\varphi \Rightarrow \Sigma
\end{array}
}{\vdash^{\alpha+\beta^*}_{k} \Gamma, \Pi, \varphi(t) \Rightarrow \Delta, \Sigma} \\
\vdots\ \Pi, \varphi(t) \\
\text{Raise}\ \dfrac{\vdash^{\gamma_R}_{k} \Xi, \varphi(t) \Rightarrow \Theta}{\vdash^{\omega^{\gamma_R}}_{k-1} \Xi, \varphi(t) \Rightarrow \Theta}
\end{array}
$$

We continue by justifying that the claimed ordinal bounds hold. Observe that for any sub-derivation of the end-part with one open premise, such as

Π_L, Π_R, Π or the remaining derivation underneath the raise rule, decreasing the ordinal bound at its open premise will lead to a strict decrease in the ordinal bound at its root, e.g. $\alpha^* < \alpha$ at the left instance of Π_L above because the bound at its premise is α' instead of $\alpha' + 1$. This is because the only rule which could 'disturb' the propagation of such a decrease is the Comp-rule, which we know does not occur in the closed end-part. For this reason, we have $\alpha^* < a$, $\beta^* < \beta$ and hence $\gamma_L, \gamma_R < \gamma$. Hence $\omega^{\gamma_L} + \omega^{\gamma_R} < \omega^{\gamma}$, meaning the ordinal bound of the resulting derivation is lower than that of the initial derivation over all.

$\forall y <^\sigma n$: As $s(\forall x <^\sigma n.A) > 0$, there must be a Raise-application 'underneath' the Cut-application under consideration. Including the Raise, the situation can be sketched as below. Again, $\varphi, \varphi_L, \varphi_R$ differ only by terms, which may be caused by instances of =L along Π_L and Π_R. Because m must have been arrived at from n using only Shift-applications, we have $m <^\sigma n$ and thus that the $\varphi_L(\overline{m})$-case must be among the premises of the $\forall{<^\sigma}$R-application. Further, $1 + \sum \alpha_i \leq \alpha$, $\beta' + 1 \leq \beta$ and $\alpha + \beta \leq \gamma$.

$$
\begin{array}{c}
\textsc{Cut}\ \dfrac{
\begin{array}{c}
\forall{<^\sigma}\mathrm{R}\ \dfrac{\cdots \quad \dfrac{\Pi_L^m}{\vdash_{k_L}^{\alpha_m} \Gamma' \Rightarrow \varphi_L(\overline{m}), \Delta'} \quad \cdots}{\vdash_{k_L}^{1+\sum \alpha_i} \Gamma' \Rightarrow \forall x <^\sigma n.\varphi_L, \Delta'} \\
\vdots\ \Pi_L \\
\vdash_k^{\alpha} \Gamma \Rightarrow \forall x <^\sigma n.\varphi, \Delta
\end{array}
\qquad
\begin{array}{c}
\dfrac{\dfrac{\Pi_R'}{\vdash_{k_R}^{\beta'} \Omega', \varphi_R(\overline{m}) \Rightarrow \Pi'}}{\vdash_{k_R}^{\beta'+1} \Omega', \forall x \leq m.\varphi_R \Rightarrow \Pi'}\ \forall{\leq}\mathrm{L} \\
\vdots\ \Pi_R \\
\vdash_k^{\beta} \Omega, \forall x <^\sigma n.\varphi \Rightarrow \Pi
\end{array}
}{\vdash_k^{\alpha+\beta} \Gamma, \Omega \Rightarrow \Delta, \Pi} \\
\vdots\ \Pi \\
\textsc{Raise}\ \dfrac{\vdash_k^{\gamma} \Xi \Rightarrow \Theta}{\vdash_{k-1}^{\omega^\gamma} \Xi \Rightarrow \Theta} \\
\vdots
\end{array}
$$

Replace this with the following derivation. We again write $\Pi_\bullet, \varphi(\overline{m})$ for the derivation $\Pi_\bullet$ weakened by $\varphi(\overline{m})$, in which case all substitutions caused by =L and shifts caused by Shift in $\forall x <^\sigma \overline{n}.\varphi$ are 'replayed' in this additional $\varphi(\overline{m})$. Further, observe that the Cut on $\varphi(\overline{m})$ satisfies $s(\varphi(\overline{m})) < k - 1$ as $s(\forall x <^\sigma \overline{n}.\varphi) = s(\varphi) + 1 < k$.

$$
\begin{array}{c}
\textsc{Ctr}\ \dfrac{\textsc{Cut}\ \dfrac{\Pi_1 \qquad\qquad \Pi_2}{\vdash_{k-1}^{\omega^{\gamma_L}+\omega^{\gamma_R}} \Xi, \Xi \Rightarrow \Theta, \Theta}}{\vdash_{k-1}^{\omega^{\gamma_L}+\omega^{\gamma_R}} \Xi \Rightarrow \Theta} \\
\vdots
\end{array}
$$

$$
\Pi_1 :\quad
\begin{array}{c}
\dfrac{
\begin{array}{c}
\dfrac{\dfrac{\Pi_L^m}{\vdash^{\alpha_m}_{k_L} \Gamma' \Rightarrow \varphi_L(\overline{m}), \Delta'}}{\vdash^{\alpha_m}_{k_L} \Gamma' \Rightarrow \forall x <^\sigma n.\varphi_L, \varphi_L(\overline{m}), \Delta'}\,\textsc{Wk} \\
\vdots\ \Pi_L, \varphi(\overline{m}) \\
\vdash^{\alpha^*}_k \Gamma \Rightarrow \forall x <^\sigma n.\varphi, \varphi(\overline{m}), \Delta
\end{array}
\qquad
\begin{array}{c}
\dfrac{\dfrac{\Pi'_R}{\vdash^{\beta'}_{k_R} \Omega', \varphi_R(\overline{m}) \Rightarrow \Pi'}}{\vdash^{\beta'+1}_{k_R} \Omega', \forall x \leq m.\varphi_R \Rightarrow \Pi'}\,\forall{\leq}\mathrm{L} \\
\vdots\ \Pi_R \\
\vdash^{\beta}_k \Omega, \forall x <^\sigma n.\varphi \Rightarrow \Pi
\end{array}
}{\vdash^{\alpha^*+\beta}_k \Gamma, \Omega \Rightarrow \varphi(\overline{m}), \Delta, \Pi}\,\textsc{Cut} \\
\vdots\ \Pi, \varphi(\overline{m}) \\
\dfrac{\vdash^{\gamma_L}_k \Xi \Rightarrow \varphi(\overline{m}), \Theta}{\vdash^{\omega^{\gamma_L}}_{k-1} \Xi \Rightarrow \varphi(\overline{m}), \Theta}\,\textsc{Raise}
\end{array}
$$

$$
\Pi_2 :\quad
\begin{array}{c}
\dfrac{
\begin{array}{c}
\dfrac{\ldots \quad \dfrac{\Pi_L^m}{\vdash^{\alpha_m}_{k_L} \Gamma' \Rightarrow \varphi_L(\overline{m}), \Delta'} \quad \ldots}{\vdash^{\sum \alpha_i}_{k_L} \Gamma' \Rightarrow \forall x <^\sigma n.\varphi_L, \Delta'}\,\forall{<^\sigma}\mathrm{R} \\
\vdots\ \Pi_L \\
\vdash^{\alpha}_k \Gamma \Rightarrow \forall x <^\sigma n.\varphi, \Delta
\end{array}
\qquad
\begin{array}{c}
\dfrac{\dfrac{\Pi'_R}{\vdash^{\beta'}_{k_R} \Omega', \varphi_R(\overline{m}) \Rightarrow \Pi'}}{\vdash^{\beta'}_{k_R} \Omega', \varphi_R(\overline{m}), \forall x \leq m.\varphi_R \Rightarrow \Pi'}\,\textsc{Wk} \\
\vdots\ \Pi_R, \varphi(\overline{m}) \\
\vdash^{\beta^*}_k \Omega, \varphi(\overline{m}), \forall x <^\sigma n.\varphi \Rightarrow \Pi
\end{array}
}{\vdash^{\alpha+\beta^*}_k \Gamma, \Omega, \varphi(\overline{m}) \Rightarrow \Delta, \Pi}\,\textsc{Cut} \\
\vdots\ \Pi, \varphi(\overline{m}) \\
\dfrac{\vdash^{\gamma_R}_k \Xi, \varphi(\overline{m}) \Rightarrow \Theta}{\vdash^{\omega^{\gamma_R}}_{k-1} \Xi, \varphi(\overline{m}) \Rightarrow \Theta, \Theta}\,\textsc{Raise}
\end{array}
$$

The justification of the decrease in the ordinal bound is analogous to the case of $\forall x.\varphi$, noting that $\alpha_m < 1 + \sum \alpha_i$ and $\beta' < \beta' + 1$.

□

Having derived each individual step of the procedure, we can now prove the cut-elimination theorem for SPA_O.

Theorem 7.3.9 (Cut-elimination)**.** If Π is a proof of $\varepsilon \vdash^{\alpha}_0 \Gamma \Rightarrow \Delta$ with Γ, Δ closed atomic then there are $\Gamma' \subseteq \Gamma, \Delta' \subseteq \Delta$ and a proof Π' of $\varepsilon \vdash^{\beta}_0 \Gamma' \Rightarrow \Delta'$ such that Π' is closed and only atomic Cuts.

Proof. We shall use the procedure below to arrive at the proof Π' starting from the proof $\Pi : \varepsilon \vdash^{\alpha}_0 \Gamma \Rightarrow \Delta$:

1. Apply corollary 7.3.5 with $\theta = \emptyset$ to obtain $\Pi_c : \varepsilon \vdash^{\leq\alpha}_0 \Gamma \Rightarrow \Delta$ with a closed end-part.
2. Apply fact 7.3.6 to Π_c to obtain $\Pi_w : \varepsilon \vdash^{\leq\alpha}_0 \Gamma' \Rightarrow \Delta'$ with $\Gamma \subseteq \Gamma', \Delta \subseteq \Delta'$ and a closed, Wk-free end-part.
3. By fact 7.3.7 there are two possibilities:
   (a) Π_w is not its own end-part: Then, by lemma 7.3.8 there exists $\Pi_r : \varepsilon \vdash^{<\alpha}_0 \Gamma' \Rightarrow \Delta'$. Continue from step 1. using $\Pi := \Pi_r$.

(b) Π_w is its own closed end-part: First, this means there are now 'cyclic rules', such as Comp, Bud or Case in Π_w. Further, observe that any trace of a non-atomic cut-formula must end either in Wk or a boundary rule application. As Π_w contains neither, it cannot contain non-atomic Cuts. Thus, we may pick $\Pi' := \Pi_w$ and are done.

If the procedure above always terminates, we can always obtain the desired Π'. Termination is ensured by the fact that whenever the procedure 'jumps' from step 3a. to step 1., the ordinal size of the proof has decreased. Thus, non-termination of the procedure leads an infinitely decreasing sequence in ε_0 which we know cannot exist. □

Corollary 7.3.10. CPA is consistent.

Proof. If no CPA derivation of the empty sequent i.e., $\Rightarrow$, exists then CPA is consistent. Suppose there was such a derivation Π. By proposition 7.3.1, this derivation can be transformed into $\Pi' : \varepsilon \vdash_0^\alpha \Rightarrow$ in SPA_O. By theorem 7.3.9 one can obtain a closed derivation $\Pi'' : \varepsilon \vdash_0^\beta \Rightarrow$ with $\beta \leq \alpha$ without non-atomic Cuts. Now a simple induction proves that a proof of $\Rightarrow$ involving only closed, atomic formulas, like Π'', cannot exist. □

We close this section by deriving the usual ordinal analysis theorem in terms of PRA for SPA.

Theorem 7.3.11. $\mathrm{PRA} + \mathrm{TI}(\varepsilon_0) \vdash$ "SPA is consistent"

Proof. If no SPA derivation of the empty sequent i.e., $\Rightarrow$, exists then SPA is consistent. Working within PRA, suppose there was such a derivation Π. Following proposition 7.3.1, this derivation can be transformed, primitive recursively, into $\Pi' : \varepsilon \vdash_0^\alpha \Rightarrow$ in SPA_O. By theorem 7.3.9 one can obtain a closed derivation $\Pi'' : \varepsilon \vdash_0^\beta \Rightarrow$ with $\beta \leq \alpha$ without non-atomic Cuts. Observe that each 'iteration step' of the procedure used to obtain Π'' from Π' is primitive recursive and the termination of the procedure over all follows from $\mathrm{TI}(\varepsilon_0)$. Now PRA can prove that a proof of $\Rightarrow$ involving only closed, atomic formulas, like Π'', cannot exist. □

Remark 7.3.4. We believe that theorem 7.3.11 can be extended to obtain

$$\mathrm{PRA} + \mathrm{TI}(\varepsilon_0) \vdash \text{"CPA is consistent"}.$$

The core problem in this is that PRA is a quantifier-free theory but the concept of CPA-proof, expressed in terms of the global trace condition, requires some sort of universal quantification ("for all branches ..."). Thus, a different, PRA-friendly definition of "Π is a CPA-proof" must be found. A good candidate are induction orders (see section 3.2). In (Leigh and Wehr 2025) a translation from CPA (with induction orders) to SHA is given, which should readily adapt to PRA, thus extending theorem 7.3.11. Another, arguably more 'natural' or traditional, PRA-friendly phrasing of CPA-proofhood is given by the automata construction for deciding the global trace condition (see e.g. definition 5.1.2). However, as this definition is rather removed from the 'meat and bones' of branches and traces, we are not as confident in 'how useful it would be in PRA', for example if it is equivalent to the existence of an induction orders over PRA. To give a truly 'natural' rendering of CPA-proofhood, some slightly stronger second-order theory, which can at least 'speak of' infinite traces and branches, such as RCA_0, seems a more sensible setting.

7.4 An abstract direct soundness argument

Brotherston (2006, section 4.2.1) presents an abstract rendition of the typical, classical soundness proof for cyclic proof systems (which we have reproduced below in proposition 7.4.2). In this section, we give a similar abstract rendition of a *direct soundness proof*, avoiding the use of 'indirect reasoning',[26] based on the existence of stack-based proof systems for cyclic proof systems whose trace condition is specified over $\mathbb{B}$. This argument also underlies, in some way or another, all applications is this section.

An abstract soundness argument requires a suitable abstract notion of semantics. The following is quite similar to that used in Brotherston (2006). Only, our notion of local soundness is not as directly tailored towards our argument as in Brotherston's case.

Definition 7.4.1. Let $\iota(\mathcal{R})$ be a cyclic proof system for $\iota : \mathcal{R} \to \mathbb{B}$. A *cyclic semantics* consists of

(i) for every sequent $\Gamma \in \textsc{Seq}$ a class $\mathbb{I}_\Gamma$ of *interpretations*
(ii) a *satisfaction (class) relation* $\vDash$ such that $I \vDash \Gamma$ is defined for every $I \in \mathbb{I}_\Gamma$
(iii) for every $I \in \mathbb{I}_\Gamma$ an *ordinal assignment* $\alpha^\Gamma_I : \iota(\Gamma) \to O$ where O is the class of ordinals.

We require the cyclic semantics to be *locally sound*. That is, let $r \in \mathcal{R}$ be a derivation rule with $\rho(r) = (\Gamma, \Delta_1, \dots, \Delta_n)$. For every $I \in \mathbb{I}_\Gamma$ then there must exist sets $\mathcal{I}_i \subseteq \mathbb{I}_{\Delta_i}$ for each $i \leq n$ such that if $I' \vDash \Delta_i$ for every $I' \in \mathcal{I}_i$ and every $i \leq n$ then $I \vDash \Gamma$. Furthermore, we require the interpretations $I' \in \mathcal{I}_i$ to be compatible with the trace interpretation ι in the following way: Let $x \in \iota(\Gamma)$ and $y \in \iota(\Delta_i)$ then

- if $(x, 0, y) \in r_i$ then $\alpha^\Gamma_I(x) \geq \alpha^{\Delta_i}_{I'}(y)$ and
- if $(x, 1, y) \in r_i$ then $\alpha^\Gamma_I(x) > \alpha^{\Delta_i}_{I'}(y)$.

Remark 7.4.1. Above, we try to follow the terminology of Brotherston (2006, section 4.2.1). The only deviation in the data is that we call ordinal assignments α, not σ, as we follow the convention of the rest of this chapter of using σ to denote values from $\{+, -\}$.

For the remainder of this section, we fix a cyclic proof system $\iota(\mathcal{R})$ and a cyclic semantics for it.

[26] Ideally, this argument is fully constructive. However, this depends on the precise notion of ordinals which is employed in the concrete instance. More details below.

We begin by adapting Brotherston's abstract rendition of the usual soundness argument, which he gives for ∞-proofs, to cyclic proofs. Note that this proof is quite indirect i.e., classical, in ways usually uncommon for soundness proofs. In the following, we *italicise* every 'indirect' step we wish to avoid in our proof.

Proposition 7.4.2. If $\iota(\mathcal{R}) \vdash \Gamma$ then $I \vDash \Gamma$ for every $I \in \mathbb{I}_\Gamma$.

Proof. Let Π be a $\iota(\mathcal{R})$-proof of Γ. *Towards contradiction, suppose* Γ *was invalid.* That is, suppose there was $I \in \mathbb{I}_\Gamma$ with $I \nvDash \Gamma$.

As usual, write $\Gamma_v := \lambda(v)$ for $v \in T$. Now we make the following observation: Consider some $v \in T$ such that there is $I' \in \mathbb{I}_{\Gamma_v}$ with $I' \nvDash \Gamma_v$. First, $v \notin \mathrm{Leaf}(T) \setminus \mathrm{dom}(\beta)$ as all axioms of $\mathcal{R}$ must be valid by local soundness. Now, if $v \in \mathrm{Inner}(T)$, *by the converse of local soundness* $\delta(v)$ must have an unsound premise i.e., there must be $v' \in \mathrm{Chld}(v)$ such that there is $I'' \in \mathbb{I}_{\Gamma_{v'}}$ with $I'' \nvDash \Gamma_{v'}$. If $v \in \mathrm{dom}(\beta)$ we can arrive at a similar situation with $v' := \beta(v)$ and $I'' := I'$.

Now 'iterate' the observation above *using König's lemma*, starting at $I_0 := I$, $v_0 := \varepsilon$ and $\Gamma_0 := \Gamma$. This yields an infinite branch $(v_i)_{i\in\omega}$ through Π with $\lambda(v_i) = \Gamma_i$ and $I_i \in \mathbb{I}_{\Gamma_i}$ with $I_i \nvDash \Gamma_i$. This yields an infinite progressing trace $(x_i)_{i\in\omega}$, w.l.o.g. starting at Γ_0 i.e., $x_i \in \iota(\Gamma_i)$ for every $i \in \omega$. Because the branch was obtained by using local soundness,[27] we know that $\alpha^{\Gamma_{i+1}}_{I_{i+1}}(x_{i+1}) \leq \alpha^{\Gamma_i}_{I_i}(x_i)$ at every step with a strict inequality at progression steps. But this means $(\alpha^{\Gamma_i}_{I_i}(x_i))_{i\in\omega}$ is an infinitely decreasing sequence of ordinals, which cannot exist. ☐

We now proceed to give the direct soundness argument. For it, the notion of satisfaction of a cyclic semantics for $\iota(\mathcal{R})$ must be extended to a notion of satisfaction for $\Lambda(\mathcal{R})$-sequents.

Definition 7.4.3. Let $\Gamma \mid \Lambda$ be a $\Lambda(\mathcal{R})$-sequent and $I \in \mathbb{I}_\Gamma$. For a companion label $(P_i; \vec{G}^i \mapsto \Gamma_i) \in \Lambda$, I *satisfies* $(P_i; \vec{G}^i \mapsto \Gamma_i)$, written as $I \vDash (P_i; \vec{G}^i \mapsto \Gamma_i)$ if for every $I' \in \mathbb{I}_{\Gamma_i}$ we may conclude $I' \vDash \Gamma_i$ from

$$\gamma^{I',I}_{\Lambda,j} := \forall x^\sigma \in G^i_j \exists y^{\sigma'} \in P_j.\ \alpha^{\Gamma_i}_{I'}(x) <^{\sigma,\sigma'} \alpha^\Gamma_I(y)$$

for every $j \leq i$ where $<^{+,-} := <$ and $<^{\sigma,\sigma'} := \leq$ otherwise. We say I *satisfies* Λ, writing $I \vDash \Lambda$ if every companion label in Λ is satisfied by I. We say I *satisfies* $\Gamma \mid \Lambda$, writing $I \vDash \Gamma \mid \Lambda$ if $I \vDash \Lambda$ entails $I \vDash \Gamma$.

[27] Technically, this property of the ordinal assignments must be explicitly specified in the property for which König's lemma is applied. As this is rather finicky, we elide this detail here to make the argument slightly easier to follow.

We can now prove the main lemma which sits at the core of the soundness argument. Note that this argument is essentially the translation of $\mathsf{SHA}_<$ to HA in theorem 7.1.2 'lifted to the meta-level'. For the following, also recall the following facts about the $<^{\sigma,\sigma'}$ notation:

(i) Whenever $a <^{\sigma_a,\sigma_b} b <^{\sigma_b,\sigma_c} c$ then $a <^{\sigma_a,\sigma_c} c$.

(ii) If $\sigma \leq \sigma'$ then $<^{\sigma,\sigma'} = \leq$.

Lemma 7.4.4. If $\Lambda(\mathcal{R}) \vdash \Gamma \mid \Lambda$ then for every $I \in \mathbb{I}_\Gamma$ we have $I \vDash \Gamma \mid \Lambda$.

Proof. We prove this by induction along the $\Lambda(\mathcal{R})$-proof.

COMP: Fix $I \in \mathbb{I}_\Gamma$ and suppose that $I \vDash \Lambda$. Fix $n := |\Lambda|$, we will prove

$$\forall I' \in \mathbb{I}_\Gamma.\ (\forall j \leq n.\forall x^\sigma \in P_j \exists y^{\sigma'} \in P_j.\ \alpha^\Gamma_{I'}(x) <^{\sigma,\sigma'} \alpha^\Gamma_I(y)) \to I' \vDash \Gamma. \quad (\star)$$

Before we prove it, first observe that $(\star)$ concludes $I \vDash \Gamma$: Simply pick $I' := I$ and $y^{\sigma'} := x^\sigma$.

We prove $(\star)$ per induction on $\mu_{I'} := \max\{\alpha^\Gamma_{I'}(x) \mid x^- \in X^-\}$. Thus fix $I' \in \mathbb{I}_\Gamma$ and suppose $(\star)$ held for all I'' with $\mu_{I''} < \mu_{I'}$. Furthermore, suppose that

$$\forall j \leq n.\forall x^\sigma \in P_j \exists y^{\sigma'} \in P_j.\ \alpha^\Gamma_{I'}(x) <^{\sigma,\sigma'} \alpha^\Gamma_I(y). \quad (\dagger)$$

Per inductive hypothesis, we know that $I' \vDash \Gamma \mid \Lambda, (X^-; \pi\Lambda, X^+ \mapsto \Gamma)$. Thus, it suffices to prove $I' \vDash \Lambda, (X^-; \pi\Lambda, X^+ \mapsto \Gamma)$ to conclude $I' \vDash \Gamma$.

$I' \vDash \Lambda$: Let $(P_i; \vec{G}^i \mapsto \Gamma_i) \in \Lambda$ with $i \leq n$. We know that $I \vDash (P_i; \vec{G}^i \mapsto \Gamma_i)$. We can conclude from $(\dagger)$ that $I' \vDash (P_i; \vec{G}^i \mapsto \Gamma_i)$ as well: Let $I_i \in \mathbb{I}_{\Gamma_i}$ such that $\gamma^{I_i,I'}_{\Lambda,j}$ for every $j \leq i$ i.e.,

$$x^\sigma \in G^i_j \exists y^{\sigma'} \in P_j.\ \alpha^{\Gamma_i}_{I_i} <^{\sigma,\sigma'} \alpha^\Gamma_{I'}(y).$$

To conclude $I_i \vDash \Gamma_i$, it suffices to prove $\gamma^{I_i,I}_{\Lambda,j}$ for every $j \leq i$ because of $I \vDash (P_i; \vec{G}^i \mapsto \Gamma_i)$. Thus, pick $x^\sigma \in G^i_j$, we know there is $y^{\sigma'} \in P_j$ such that $\alpha^{\Gamma_i}_{I_i} <^{\sigma,\sigma'} \alpha^\Gamma_{I'}(y)$. By $(\dagger)$ we know that there is $z^{\sigma''} \in P_j$ such that $\alpha^\Gamma_{I'}(y) <^{\sigma',\sigma''} \alpha^\Gamma_I(z)$ and thus that $\alpha^{\Gamma_i}_{I_i}(x) <^{\sigma,\sigma''} \alpha^\Gamma_I(z)$ as desired.

$I' \vDash (X^-; \pi\Lambda, X^+ \mapsto \Gamma)$: Let $I'' \in \mathbb{I}_\Gamma$ be such that

$$\forall x^\sigma \in P_j \exists y^{\sigma'} \in P_j.\ \alpha^\Gamma_{I''}(x) <^{\sigma,\sigma'} \alpha^\Gamma_{I'}(y) \quad (*.1)$$

and

$$\forall x^\bullet \in X^+ \exists y^\bullet \in X^-.\ \alpha^\Gamma_{I''}(x) < \alpha^\Gamma_{I'}(y). \quad (*.2)$$

We use our inductive hypothesis for ($\star$) to prove the claim. For this, first note that $\mu_{I''} < \mu_I$: Let $x^\bullet \in X$ be such that $\alpha^\Gamma_{I''}(x) = \mu_{I''}$ then by ($*$.2) we have

$$\mu_{I''} = \alpha^\Gamma_{I''}(x) < \alpha^\Gamma_{I'}(y) \leq \mu_{I'}$$

for some $y^\bullet \in X$. It remains to show that for every $j \leq i$ we have

$$\forall x^\sigma \in P_j \exists y^{\sigma'} \in P_j.\ \alpha^\Gamma_{I''}(x) <^{\sigma,\sigma'} \alpha^\Gamma_I(y)$$

which follows directly from ($\dagger$) and ($*$.1).

Bud: Suppose $I \in \mathbb{I}_\Gamma$ such that $I \vDash (X^+; \pi(\Lambda), X^+ \mapsto \Gamma)$. To conclude $I \vDash \Gamma$ from this, we must prove

$$\forall x^\sigma \in P_j \exists y^{\sigma'} \in P_j.\ \alpha^\Gamma_I(x) <^{\sigma,\sigma'} \alpha^\Gamma_I(y),$$

which follows directly by picking $y^{\sigma'} := x^\sigma$, and

$$\forall x^\bullet \in X^+ \exists y^\bullet \in X^+.\ \alpha^\Gamma_I(x) \leq \alpha^\Gamma_I(y)$$

which is trivial.

Drop: Suppose $I \in \mathbb{I}_\Gamma$ and $I \vDash \Lambda$. Per IH, we know that $I \vDash \Gamma \mid \Lambda^*$ with $\pi(\Lambda^*) \preceq \pi(\Lambda)$. It thus suffices to prove $I \vdash \Lambda^*$. Let $(P^*_i; \vec{G}^i \mapsto \Gamma_i) \in \Lambda^*$ then we know that $I \vDash (P_i; \vec{G}^i \mapsto \Gamma_i)$ for some $P_i \succeq P^*_i$. Towards $I \vDash (P^*_i; \vec{G}^i \mapsto \Gamma_i)$, let $I' \in \mathbb{I}_{\Gamma_i}$ be such that $\gamma^{I',I}_{\Lambda^*,j}$ for every $j \leq i$. To deduce $I' \vDash \Gamma_i$, we apply $I \vDash (P_i; \vec{G}^i \mapsto \Gamma_i)$. Let $x^\sigma \in G^i_j$ then by $\gamma^{I',I}_{\Lambda^*,j}$ there is $y^{\sigma'} \in P^*_j$ such that $\alpha^{\Gamma_i}_{I'}(x) <^{\sigma,\sigma'} \alpha^\Gamma_I(y)$. As $P^*_j \preceq P_j$ we know that $y^{\sigma''} \in P_j$ with $\sigma' \leq \sigma''$ and thus $\alpha^{\Gamma_i}_{I'}(x) <^{\sigma,\sigma''} \alpha^\Gamma_I(x)$ as desired.

Pop: Per IH we know $I \vDash \Gamma \mid \Lambda$ for every $I \in \mathbb{I}_\Gamma$. This directly entails $I \vDash \Gamma \mid \Lambda, \Lambda'$.

r_Λ: Let $r \in \mathcal{R}$ be such that $\rho(r) = (\Gamma, \Delta_1, \ldots, \Delta_n)$. Fix $I \in \mathbb{I}_\Gamma$ and suppose that $I \vDash \Lambda$. By local soundness, there are sets $\mathcal{I}_i \subseteq \mathbb{I}_{\Delta_i}$ such that we can conclude $I \vDash \Gamma$ if $I_i \vDash \Delta_i$ for every $I_i \in \mathcal{I}_i$ and $i \leq n$. Thus, fix some $i \leq n$ and $I_i \in \mathcal{I}_i \subseteq \mathbb{I}_{\Delta_i}$. Per IH, we know that $I_i \vDash \Delta_i \mid r_i(\Lambda)$, meaning it suffices to prove $I_i \vDash r_i(\Lambda)$. Thus, let $(r_i(P_j); \vec{G}^j \mapsto \Gamma_j) \in r(\Lambda)$ and fix $I'_j \in \mathbb{I}_{\Gamma_j}$ with $\gamma^{I'_j,I_i}_{r_i(\Lambda),k}$ for all $k \leq j$ to deduce $I'_j \vDash \Gamma_j$. By $I \vDash \Lambda$ it suffices to prove $\gamma^{I'_j,I}_{\Lambda,k}$ for all $k \leq j$ to conclude this. Thus, let $x^\sigma \in G^j_k$ by $\gamma^{I'_j,I_i}_{r_i(\Lambda),k}$ we know there is $y^{\sigma'} \in r_i(P_k)$ such that $\alpha^{\Gamma_j}_{I'_j}(x) <^{\sigma,\sigma'} \alpha^{\Delta_i}_{I_i}(y)$. Per definition of $r_i(P_k)$, there are two different reasons for which $y^{\sigma'}$ could be in $r_i(P_k)$:

- $\sigma' = +$ and there is $z^{\sigma''} \in P_k$ with $(z, 1, y) \in r_i$: By the local soundness condition, that means $\alpha^{\Delta_i}_{I_i}(y) < \alpha^{\Gamma}_{I}(z)$ and thus $\alpha^{\Gamma_j}_{I'_j}(x) <^{\sigma,\sigma''} \alpha^{\Gamma}_{I}(z)$ by transitivity, regardless of the nature of σ''.
- there is $z^{\sigma'} \in P_k$ with $(z, 0, y) \in r_i$: By the local soundness condition, that means $\alpha^{\Delta_i}_{I_i}(y) \leq \alpha^{\Gamma}_{I}(z)$ and thus $\alpha^{\Gamma_j}_{I'_j}(x) <^{\sigma,\sigma'} \alpha^{\Gamma}_{I}(z)$ by transitivity, regardless of the nature of σ'.

□

Remark 7.4.2. Observe that the proof of lemma 7.4.4 avoids all of the classical steps highlighted in proposition 7.4.2. The only aspect that might be constructively dubious is the notion $\mu_{I'} := \max\{\alpha^{\Gamma}_{I'}(x) \mid x^- \in X^-\}$, which is somewhat contingent on the notion of ordinals involved. Because the set $\{\alpha^{\Gamma}_{I'}(x) \mid x^- \in X^-\}$ is always finite, its maximum can always be computed as long as the ordering on the ordinals involved is decidable. That means as long as the ordinals in the proof above can be represented as computable ordinals (in the sense of Kleene (1938)), for example, this is constructively unproblematic.

Now soundness is a direct consequence of the previous lemma.

Theorem 7.4.5 (Soundness)**.** If $\iota(\mathcal{R}) \vdash \Gamma$ then for all $I \in \mathbb{I}_\Gamma$ we have $I \vDash \Gamma$.

Proof. By theorem 6.3.8 we know that $\Lambda(\mathcal{R}) \vdash \Gamma$, meaning $\Lambda(\mathcal{R}) \vdash \Gamma \mid \varepsilon$. By lemma 7.4.4 this means $I \vDash \Gamma$ for every $I \in \mathbb{I}_\Gamma$ as desired. □

7.4.1 Examples

To demonstrate that this abstract method actually covers most cyclic proof systems from the literature, we describe how to give a cyclic semantics to some concrete cyclic proof systems.

Simple CHA For the semantics of simple CHA given by the standard model i.e., the same one as considered in theorem 2.1.4 fix

$$\mathbb{I} := \{\rho : V \to \omega \mid V \subseteq \textsc{Var}\} \qquad \mathbb{I}_{\Gamma\Rightarrow\delta} := \{\rho : \mathrm{FV}(\Gamma, \delta) \to \omega\}$$

and take $\alpha^{\Gamma\Rightarrow\delta}_{\rho} := \rho : \mathrm{FV}(\Gamma, \delta) \to \omega$. Define $\rho \vDash \Gamma \Rightarrow \delta$ as usual.

Proposition 7.4.6. The above cyclic semantics is locally sound for $\iota(\mathrm{CHA})$.

Proof. We only prove some of the interesting rules. We only comment on the ordinal condition in interesting cases.

Case_x: Let $x \in \mathrm{FV}(\Gamma, \delta)$ and $\rho \in \mathbb{I}_{\Gamma \Rightarrow \delta}$. We perform a case-distinction on $\rho(x)$:

$\rho(x) = 0$: Then we set $\mathcal{I}_1 := \{\rho \setminus \{(x, 0)\}\}$ and $\mathcal{I}_2 := \emptyset$. We now must conclude $\rho \vDash \Gamma \Rightarrow \delta$ from $\rho \setminus \{(x, 0)\} \vDash \Gamma[0/x] \Rightarrow \delta[0/x]$, which is trivial. Lastly, observe that the ordinal condition is maintained because for the left-hand premise, only non-x variables must be preserved.

$\rho(x) = n + 1$: Analogously to the previous case, set $\mathcal{I}_1 := \emptyset$ and $\mathcal{I}_2 := \{\rho[x \mapsto n]\}$. The ordinal condition is maintained as only x progresses in the for the right-hand premise.

$\forall$R: Fix ρ with $\mathrm{dom}(\rho) = \mathrm{FV}(\Gamma, \forall x.\varphi)$. We set $\mathcal{I}_1 := \{\rho[x \mapsto n] \mid n \in \omega\}$. Thus, we must prove that $\rho[x \mapsto n] \vDash \Gamma \Rightarrow \varphi$ entails $\rho \vDash \Gamma \Rightarrow \forall x.\varphi$ with $x \notin \mathrm{FV}(\Gamma, \forall x.\varphi)$. Again, his is trivial.

Wk: The only interesting case is if the weakening removes a free variable. Thus, let w.l.o.g. $\Gamma' = \{\varphi\}$ such that $\mathrm{FV}(\Gamma, \varphi, \delta) \setminus \mathrm{FV}(\Gamma, \delta) = \{x\}$. Then set $\mathcal{I}_1 := \{\rho \restriction \mathrm{FV}(\Gamma, \delta)\}$. Suppose $\rho \restriction \mathrm{FV}(\Gamma, \delta) \vDash \Gamma \Rightarrow \delta$ and $\rho \vDash \Gamma, \varphi$. From this we can conclude $\rho \restriction \mathrm{FV}(\Gamma, \delta) \vDash \Gamma$ and thus $\rho \restriction \mathrm{FV}(\Gamma, \delta) \vDash \delta$ and $\rho \vDash \delta$.

Cut: The only interesting case if the cut-formula introduces new free variables. Thus let cut-formula be φ and, w.l.o.g., $\mathrm{FV}(\delta) \subseteq \mathrm{FV}(\Gamma)$ and $\mathrm{FV}(\varphi) \setminus \mathrm{FV}(\Gamma) = \{x\}$. Let further ρ with $\mathrm{dom}(\rho) = \mathrm{FV}(\Gamma, \delta)$ be given. Then fix $\mathcal{I}_1 = \mathcal{I}_2 := \{\rho[x \mapsto 0]\}$. From $\rho[x \mapsto 0] \vDash \Gamma \Rightarrow \varphi$ and $\rho[x \mapsto 0] \vdash \Gamma, \varphi \Rightarrow \delta$ and $\rho \vDash \Gamma$ we can conclude $\rho[x \mapsto 0] \vDash \delta$ and thus $\rho \vDash \delta$ as $x \notin \mathrm{FV}(\delta)$.

□

Simpsons $\mathrm{CHA}_<$ Again, we only consider the semantics given by the standard model. However, the definition of the interpretation is more subtle than in simple CHA:

$$\mathbb{I} := \{\rho : V \to \omega \mid V \subseteq \text{Var}\} \qquad \mathbb{I}_{\Gamma \Rightarrow \delta} := \{\rho : \mathrm{FV}(\Gamma, \delta) \to \omega\}$$

together with the usual satisfaction relation $\rho \vDash \Gamma \Rightarrow \delta$. For the ordinal assignments, define

$$\alpha_\rho^{\Gamma \Rightarrow \delta}(t \in \text{Term}(\Gamma, \delta)) := t^\rho \in \omega$$

i.e. the usual term interpretation in the standard model.

Proposition 7.4.7. The above cyclic semantics is locally sound for $\iota(\mathrm{CHA}_<)$.

Proof. Generally, the argument is quite straightforward and essentially follows that in proposition 7.4.6 for $\iota(\mathrm{CHA})$. The only complication is the treatment of inequalities to ensure the properties of the ordinal assignments α_I^Γ. We illustrate for the case of $\to$R, analogous precautions must be made to the other cases.

$\to$R: Fix $\rho : \mathrm{FV}(\Gamma, \varphi \to \psi) \to \omega$. Now consider the set of inequalities $\Xi := \{s < t \in \Gamma \cup \{\varphi\}\}$. We distinguish two cases:

- $\rho \vDash \Xi$: Then set $\mathcal{I}_1 := \{\rho\}$. Clearly, we can conclude $\rho \vDash \Gamma \Rightarrow \varphi \to \psi$ from $\rho \vDash \Gamma, \varphi \Rightarrow \psi$. It remains to verify the relationship between $\alpha_\rho^{\Gamma \Rightarrow \varphi \to \psi}$ and $\alpha_\rho^{\Gamma, \varphi \Rightarrow \psi}$. As they are identical by construction, denote them both by α. Further, denote by $\tau : \textsc{Term}(\Gamma, \varphi \to \psi) \to \textsc{Term}(\Gamma, \varphi, \psi)$ the trace map from the conclusion of $\to$R to its premise. Let $t \in \textsc{Term}(\Gamma, \varphi \to \psi)$ and $s \in \textsc{Term}(\Gamma, \varphi, \psi)$. There are two cases we must treat:
  - $(t, 0, s) \in \tau$: That means $t = s$ and thus $\alpha(t) \geq \alpha(s)$ as desired.
  - $(t, 1, s) \in \tau$: That means $s < t \in \Gamma \cup \{\varphi\}$. We know $\rho \vDash s < t$ per assumption and thus that $\alpha(t) > \alpha(s)$ as desired.
- there is $s < t \in \Xi$ such that $\rho \nvDash s < t$: That means $\rho \vDash \Gamma \Rightarrow \varphi \to \psi$. Then we may set $\mathcal{I}_1 := \emptyset$, as no further checks are needed to ensure $\rho \vDash \Gamma \Rightarrow \varphi \to \psi$.

□

Example 7.4.8. To illustrate why the complication above is needed, consider the following $\to$R instance

$$\to\mathrm{R}\,\frac{\Gamma, x < y, \varphi \Rightarrow \psi}{\Gamma, x < y \Rightarrow \varphi \to \psi}$$

and some assignment $\rho : \mathrm{FV}(\Gamma, x < y, \varphi, \psi) \to \omega$ such that $\rho(y) \leq \rho(x)$. Suppose that, towards local soundness, one naïvely chose $\mathcal{I}_1 = \{\rho\}$. Because $\rho \nvDash x < y$, the preservation of satisfaction is trivial. However, the property about the α is violated: If $\tau : \textsc{Term}(\Gamma, x < y, \varphi \to \psi) \to \textsc{Term}(\Gamma, x < y, \varphi, \psi)$ is the trace map of the rule instance above, then $(y, 1, x) \in \tau$ which would then require $\rho(x) = \alpha_\rho(x) < \alpha_\rho(y) = \rho(y)$, which we know to be false. The only way to avoid this problem is to set $\mathcal{I}_1 = \emptyset$ in the first place.

7.5 Discussion

First we must address the obvious: every result in this section is just a different way to 'exploit' the same core idea. Stack-controlled proofs embody a normal from in which inductive invariants are 'easily' read off. Indeed, this observation essentially already goes back to Sprenger and Dam (2003b). The only advancement of the idea we present here is that, as now most cyclic proof systems can be represented via stack-controlled proofs, these insights are much more widely applicable.

The soundness argument in section 7.4 demonstrates that, in principle, stack-controlled proofs uniformly allow for the extraction of inductive invariants from cyclic proofs, regardless of the underlying logic. This does not directly mean that, for example, the translation result from section 7.1 is easily transferred to other cyclic proof systems. As we see it, there are two requirements which restrict such transfers:

- The inductive invariants must be *expressible in the logic*. The core difficulty in this seems to be *expressing the transitive closure of progress*. In first-order arithmetic, this can be done by the easily defined $<$-relation. However, this is more complicated in other logics. For example, the author has been working together with Reuben Rowe on adapting some of the results in this section to the setting of cyclic proofs for transitive closure logic (as first studied by Docherty and Rowe (2019)). While expressing transitive closures in this logic is trivial, expressing the 'single step' progress relation in arbitrary first-order signatures has been one of the main problems in this endeavour so far.
- The *maximum of progress/trace values* must be expressible. That is, in settings which use the 'complex' stack-controlled proofs in the style of $\mathrm{SHA}_{<}$, in which companion labels track multiple trace values, the translation of the COMP-rule requires taking the maximum of those trace values. Indeed, this is also the key 'obstacle' which prevents us from calling the result of section 7.4 constructive. It should be noted that the work of Grotenhuis and Otten we discuss in section 6.5 has the circumvention of the requirement of using max as one of its main motivating factors.

The considerations in the previous paragraph point to good candidates to apply the results of this type to: cyclic 'proof' systems for type systems, such as (Kuperberg et al. 2021; Das 2021; Curzi and Das 2021). As noted before, the translation from 'plain' cyclic system to stack-controlled proofs maintains all proof theoretic, including computational, content, as it only involves proof unfolding and the addition of 'bookkeeping rules'. Thus, studying such cyclic type systems via stack-controlled proofs is a viable approach. This setting also eases the requirement of expressing inductive invariants. For example, if one were to translate from a cyclic type system to one using recursion rules, the inductive invariants take the role of 'semantic conditions' on the translated proof, ensuring the denotations of the cyclic and recursive proof are the same. Thus, the invariant need not be expressed in the type theory directly but rather at the meta-level, similar to the soundness argument of section 7.4.

Another broad setting to which the results of this chapter should transfer are cyclic proof systems with explicit ordinal annotations. Sprenger and Dam (2003b) already work with such a system for first-order logic with least and greatest fixed points, annotating each fixed point quantifier with an abstract unfolding ordinal. Using techniques which inspired both chapters 6 and 7, they translated from a cyclic proof system for such a logic to a 'traditional' finitary proof system with unfolding rules for fixed points and induction rules for the ordinals. The inductive invariants are stated in terms of ordinals, similarly to section 7.4. Such an approach could be generalised, similar to section 7.4, by giving a general method of annotating a cyclic proof system with ordinals and then translating it to a finitary proof system with ordinal induction rules.

All results of this chapter can be seen as variation or extensions of the translation from cyclic to inductive first-order arithmetic via stack-controlled proofs we, for example, present in sections 3.4 and 7.1. Thus, it is worth discussing other translation methods from cyclic to inductive proof systems for Peano or Heyting arithmetic in the literature. When Simpson (2017) introduces $\mathrm{CPA}_{<}$, he proves it equivalent to PA. For this, he formalises the usual soundness proof of cyclic proof systems in ACA_0, a subsystem of second order arithmetic, obtaining PA-proofs via the conservativity of ACA_0 over said system. Das (2020) refines this argument, obtaining bounds on the logical and proof-size complexity of the resulting PA-proofs. Berardi and Tatsuta (2017) take a different approach. They use Ramsey-style order-theoretic principles formalisable in HA to obtain an induction principle mirroring the cyclic structure of a CHA or CPA-proof, using which translating the original proof to HA or PA, respectively, becomes straightforward. Ikebuchi (2025) simplifies this method by removing the intricate proof of the induction principle. Instead, they use a result from the topic of size-change termination from programming language theory by Lee (2009). Paraphrasing, the result states that every cyclic proof has a ranking function, a way of mapping the trace values of every bud and companion into a finite lexicographical ordering, using only max and min, such that the value of this function decreases along every cycle. This means the induction principle derived by Berardi and Tatsuta (2017) can simply be replaced by induction on the result of the ranking function.

The proof methods of Simpson (2017), Das (2020), and Berardi and Tatsuta (2017) are rather 'intimately' tied to the setting of arithmetic and thus likely do not transfer well to other settings. However, we believe the method of Ikebuchi (2025) might transfer quite widely, as the notion of ranking function it relies on is rather generic. Indeed, we believe that the results of sections 7.1, 7.2 and 7.4 in this chapter could just as well be derived using their method. This might well be true for the consistency result in section 7.3 as well, although this would likely require a rather different approach, somehow taking the ranking function into account for the ordinal annotations. We believe that in many situations, the method of Ikebuchi (2025) via ranking functions might be preferable to the method via stack-controlled proofs we use in this section because Ikebuchi's method seems easier to understand and explain. One possible shortcoming of Ikebuchi's translation method is that, like Berardi's and Tatsuta's, the resulting inductive proofs are somewhat structurally dissimilar to the original cyclic proofs, possibly making preservation of certain proof theoretic content more complicated. It would be interesting to see if our translation method by recursion on the proof tree could be combined with the idea of ranking functions, resulting in proofs which were more structurally similar.

8 Constructivity of the global trace condition⋆

In this chapter, we engage in some *constructive reverse mathematics*. Similarly to the more well-known classical reverse mathematics (see e.g. Simpson (2009)), the goal of constructive reverse mathematics is measuring the 'strength', or in this case *non-constructivity*, of theorems of ordinary mathematics by identifying them with certain non-constructive principles. This is achieved by proving the theorem and principles equivalent over a constructive base theory. The eponymous *reverse direction*, from theorem to principles, demonstrates that the principles used in the common direction, from principles to theorem, were 'truly required' and the principles used were 'not too strong'. We recommend Ishihara (2006) or Diener (2020) for a more thorough overview of constructive reverse mathematics.

The usual notions of proof, both finite or at least well-founded, are constructively well-behaved. This is not the case for the usual notion of cyclic proof. Specifically, we demonstrate the non-constructivity of the usual global trace condition by proving its decidability equivalent to the additive Ramsey principle and other non-constructive principles. Complementing this, we propose a constructive alternative notion for which decidability holds constructively and whose equivalence to the standard GTC is again equivalent to the additive Ramsey principle. These two results on their own are not too surprising: Lichter and Smolka (2018) study the constructivity of infinite word automata which, as noted in section 5.1, are closely connected to cyclic proof theory and specifically the global trace condition. The results about the global trace condition directly mirror theirs about infinite word automata, including the role of the additive Ramsey principle and method used to constructivise the global trace condition. Furthering the results of Lichter and Smolka (2018), we extend the equivalence to include two more non-constructive principles, which we call the iterated limited principle of omniscience (ILPO) and thin weak König's lemma (TWKL). In the presence of unique choice (UC), the equivalence further extends to the limited principle of omniscience (LPO).

Warning. Instead of fixing one constructive foundation of mathematics over which to carry out this constructive analysis, we instead present the results of

⋆ The contents of this chapter are joint work with Dominik Kirst.

this chapter in the language of set theory and relying on an intuitive notion of 'constructive mathematics.' Any proposition claimed in this chapter is constructively provable and every proof constructive. We take this approach in large part because, for the concepts studied in this chapter, the choice of foundation does not matter: the results transfer to most of the usual foundations of constructive mathematics. We postpone the discussion of more subtle technical details around the constructivity of our results, including considerations which arise when working with various specific foundations, to section 8.5.

8.1 Overview

We continue by giving an overview of the main results of this chapter.

To motivate the 'project', first observe that the global trace condition is not the right notion of cyclic proofhood from a constructivist perspective: There are relatively simple cyclic proofs which classically satisfy the global trace condition but for which this cannot be proven constructively.

Consider, for example, the schematic depiction of a cyclic proof over $\mathbb{B}$ in fig. 17. It has two buds (at the top) which share a companion (at the bottom). The trace values at each node are $\{x, y\}$. Thick edges indicate progressing connections and dashed edges non-progressing connections. That is, on the left cycle, the trace value x progresses, the trace value y being preserved. On the right cycle, y progresses but x is not preserved.

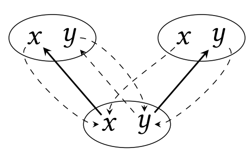


Figure 17: An illustration of an abstract cyclic derivation over $\mathbb{B}$.

Classically, the cyclic proof in fig. 17 satisfies the global trace condition: Consider some branch through it. Then

- if the right cycle is taken infinitely often, y progresses infinitely and is always preserved.
- if the right cycle is taken finitely often, then x progresses infinitely and is only 'disturbed' finitely.

As, classically, one of the two options above always applies, every branch has a progressing trace. However, this argument fails in a constructive setting. Representing a branch through the proof by an infinite sequence $\pi \in \{l, r\}^\omega$ of *l*eft and *r*ight, the above case-distinction reduces to

- there are infinitely many $i \in \omega$ with $\pi_i = r$, or
- there exists some $k \in \omega$ such that $\pi_{k+i} = l$ for all $i \in \omega$.

The intuitive understanding of constructive mathematics as 'computably tractable mathematics' rules out that the above case-distinction could be made constructively: This would, essentially, mean constructing a Turing machine which, given the code of a Turing machine computing π, makes the decision above. It is clear that such a Turing machine cannot be constructed.

Note that, to make some of the proofs in this section more straight-forward, we only consider the global trace condition over $\mathbb{B}$. We believe that the results should readily extend to any reasonable[29] activation algebra $\mathcal{A}$.

Mirroring Lichter and Smolka (2018), we also consider the constructively suitable notion of periodic $\iota(\mathcal{R})$-proofs. We call a branch π of a cyclic tree *periodic* if there exist $\pi_1, \pi_2 \in T^*$ such that $\pi = \pi_1\pi_2^\omega$. That is, a branch is periodic if it eventually keeps repeating some cycle through the tree. We can then distinguish between two variants of the global trace condition: one which considers all branches and one which only considers periodic branches.

Definition 8.1.1. Let Π be a $\mathcal{R}$-preproof.

1. It is a *periodic $\iota(\mathcal{R})$-proof* if every periodic branch through Π is progressing.
2. It is an *unrestricted $\iota(\mathcal{R})$-proof* if every branch through Π is progressing.

We claim that periodic proofhood is the right constructive counterpart to the usual global trace condition, which corresponds to unrestricted proofhood. One way we demonstrate this is proving that decidability of *periodic* proofhood holds constructively.

Theorem 8.2.12 (Informative periodic proof checking)**.** Let Π be an $\mathcal{R}$-preproof. Then either

(i) Π is a periodic $\iota(\mathcal{R})$-proof, or

(ii) Π has a periodic branch which is not progressing.

On the other hand, we also demonstrate that unrestricted proofhood is non-constructive. For this, we prove that various (classical) theorems involving unrestricted proofhood are equivalent to combinations of the additive Ramsey

[29] For example, this will require decidability of the equality of $\mathcal{A}$.

principle (AR), the iterated limited principle of omniscience (ILPO) and the think weak König's lemma (TWKL), as defined below.

For a tree $T \subseteq \omega^*$ we write $T_i := \{u \in T \mid |u| = i\}$. The width of T is *globally bounded by* n if $|T_i| \leq n$ for every $i \in \omega$. T is *thin* if its width is globally bounded by some $n \in \omega$.

Definition 8.1.2. We fix the following three principles:

ILPO: For every $f : \omega \to \mathbb{B}$,

(i) for every $i \in \omega$ there exists $j > i$ such that $f(j) = 1$ or

(ii) there is $k \in \omega$ such that $f(k + i) = 0$ for every $i \in \omega$.

TWKL: Every infinite decidable thin tree $T \subseteq \mathbb{B}^*$ has an infinite branch.

AR: Let $(\Gamma, +)$ be a finite *semi-group* i.e., $+ : \Gamma^2 \to \Gamma$ is associative and Γ finite. Then every sequence $\sigma \in \Gamma^\omega$ has a factorisation i.e., a strictly monotone sequence $\tau : \omega \to \omega$ and some $g \in \Gamma$ such that for every $n \in \omega$

$$\sum_{i=\tau(n)}^{i<\tau(n+1)} \sigma(i) = g.$$

The non-constructivity of unrestricted proofhood is witnessed by the following result. We believe that readers should, at this point, have an intuitive understanding of the various items. Concrete definitions of the notions involved are given in later sections of this chapter. In the following, the statements (ii) – (v) should really be phrased as: "For all cyclic proof systems $\iota(\mathcal{R})$, …". We omit this for brevity.

Theorem 8.1.3. The following are equivalent:

(i) The additive Ramsey principle

(ii) ILPO and TWKL

(iii) *Unrestricted reset soundness:* If Π_R is a sound reset annotation of an $\mathcal{R}$-preproof Π then Π is an unrestricted $\iota(\mathcal{R})$-proof.

(iv) *Informative unrestricted proof checking:* Let Π be an $\mathcal{R}$-preproof. Then it either is an unrestricted $\iota(\mathcal{R})$-proof or has a periodic, non-progressing branch.

(v) *$\mathcal{R}$-proof coincidence:* Let Π be a $\mathcal{R}$-preproof. It is a periodic $\iota(\mathcal{R})$-proof iff it is an unrestricted $\iota(\mathcal{R})$-proof.

Proof.

$(i) \to (ii)$: These are lemmas 8.3.9 and 8.3.10 for ILPO and TWKL, respectively.

$(ii) \to (iii)$: This is theorem 8.4.2.

$(iii) \to (iv)$: Follows directly from lemma 8.2.11.

$(iv) \to (v)$: In the interesting direction, let Π be a periodic $\iota(\mathcal{R})$-proof. By (iv) it is either an unrestricted $\iota(\mathcal{R})$-proof, which means we are done, or has a periodic non-progressing branch, contradicting our assumption.

$(v) \to (i)$: This is theorem 8.4.5. □

Under unique choice, the principle ILPO coincides with the well-known limited principle of omniscience (LPO), both entailing TWKL. Thus, LPO may be added to the list of equivalent statements of theorem 8.1.3 in many constructive foundations of mathematics. We discuss this in more detail in section 8.5.

The non-constructivity results are summed up by the graph in fig. 18. Each arrow denotes an implication, often annotated with a reference to its proof. The dashed lines denote implications which only hold under unique choice, see section 8.5 for details. The nodes are vaguely aligned by non-constructive strength, stronger ones to the left of weaker ones. The node "ILPO + TWKL" describes the conjunction of the two principles. The node "UCPT" describes "unrestricted cyclic proof theory", standing in for the points (iii) – (v) of theorem 8.1.3. The principles LPO, WKL and LLPO are recalled in section 8.3.

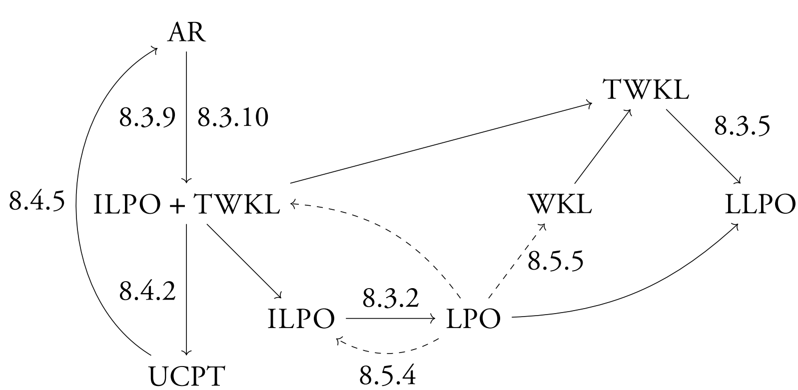


Figure 18: Relating the non-constructive notions of this chapter

The rest of this chapter is concerned with providing the details for the claims made so far. Section 8.2 covers all of the purely constructive results relevant to this chapter i.e., those which do not interact with the non-constructive principles outlined above. In section 8.3 we study the two novel non-constructive principles ILPO and TWKL which we introduce in this thesis. Section 8.4 proves the two main 'reductions', $(ii) \to (iii)$ and $(v) \to (i)$ of theorem 8.1.3. Section 8.5 provides more details on considerations around constructivity which we leave rather vague in the rest of this chapter. We essentially close the chapter with a discussion in section 8.6. The results in this chapter rely on a handful of technical lemmas whose proofs are not very enlightening. We have thus relegated these lemmas to section 8.7.

Remark 8.1.1. Recall that it is admissible to employ classical reasoning in a constructive setting as long as the properties involved are decidable. That is, if one can constructively prove that $P \vee \neg P$ then one can, for example, constructively deduce P from $\neg\neg P$. Examples of decidable properties are $n = m$ and $n < m$ for $n, m \in \omega$.

In the remainder of this chapter, any instance of classical-seeming reasoning will be unproblematic as explained above. Sometimes, this may require slightly more careful considerations about the representation of objects, which we discuss in section 8.5.

8.2 Purely constructive results

This section covers all the main results of this chapter that remain in the 'constructive realm' i.e., those which make no reference to non-constructive principles.

In the following, we redefine some notions around cyclic proofs. All definitions we give here are classically equivalent to those used elsewhere in this thesis. However, the definitions presented below are more suitable to the constructive setting or allow for the proofs in this chapter to be slightly more direct. Fix a cyclic tree $C = (T, \beta)$. We shall always assume that T and $\mathrm{dom}(\beta)$ are decidable. Nodes $s, t \in T$ are *adjacent* if $t = sj$ for some $j \in \omega$ or if $s \in \mathrm{dom}(\beta)$ and $t = \beta(s)$. A sequence $\pi \in T^\omega$ is a *branch* through C if $\pi_0 = \varepsilon$ and π_{i+1} is adjacent to π_i for every $i \in \omega$.

For the following definitions, let $\mathcal{R}$ be a cyclic proof system whose soundness condition is given by a trace-interpretation $\iota : \mathcal{R} \to \mathbb{B}$. Furthermore, we assume that any trace map $\tau : X \to Y$ in $\mathbb{B}$ is decidable, or conversely restrict our attention only to those that are.[30]

Definition 8.2.1. $\Pi = (C, \lambda, \delta)$ be a $\mathcal{R}$-preproof. Let $s, t \in T$ be adjacent in C, then we denote by $\iota(s,t) : \iota(\lambda(s)) \to \iota(\lambda(t))$ the following map over $\mathbb{B}$:

1. if $t = sj$ for some $j \in \omega$ then $\iota(s,t) := (\delta(s))_j$ and
2. if $s \in \mathrm{dom}(\beta)$ and $t = \beta(s)$ then $\iota(s,t) = 1_{\iota(\lambda(s))}$.

We expand the above to $\iota(\pi) : \iota(\pi_0) \to \iota(\pi_{|\pi|-1})$ where $\pi \in T^+$ is a finite path through C as follows:

$$\iota(\pi) := \begin{cases} 1_{\iota(\pi_0)} & \text{if } |\pi| = 1 \\ \iota(\pi_1 \dots \pi_{|\pi|-1}) \circ \iota(\pi_0, \pi_1) & \text{otherwise} \end{cases}.$$

A branch $\pi \in T^\omega$ through Π is called *progressing* if there exists an *offset* $k \in \omega$ and a sequence of *trace values* $x : \Pi i \in \omega.\ \iota(\lambda(\pi_{k+i}))$ such that

(i) for every $i \in \omega$ we have $(x_i, b, x_{i+1}) \in \iota(\pi_{k+i}, \pi_{k+i+1})$ for some $b \in \mathbb{B}$ and
(ii) for every i there is $j > i$ such that $(x_j, 1, x_{j+1}) \in \iota(\pi_{k+j}, \pi_{k+j+1})$.

A key ingredient to our reductions will be reset proofs as discussed in chapter 5. We reproduce slightly modified and simplified definitions for them, focused on greedy runs.

Definition 8.2.2. A Safra board on a finite set A is a tuple (Θ, σ) consisting of a finite linear order Θ and a function $\sigma : A \to \mathcal{P}(\Theta) \setminus \{\emptyset\}$ such that for every $\gamma \in \Theta$ there is $a \in A$ such that $\gamma \in \sigma(a)$. The class $\textsc{Sb}(A)$ contains all Safra boards on A.

Let $\tau : A \to B$ be a trace map over $\mathbb{B}$. The *greedy Safra τ-step* $(\Theta, \sigma) \overset{\tau}{\rightsquigarrow}_g (\Theta^*, \sigma^*)$ for $(\Theta, \sigma) \in \textsc{Sb}(A)$ and $(\Theta^*, \sigma^*) \in \textsc{Sb}(B)$ is computed as follows:

MOVE: Define $\overline{\sigma} : B \times \mathbb{B} \to \mathcal{P}(\mathcal{P}(\Theta))$ via $\overline{\sigma}(b,p) := \{\sigma(a) \mid (a,p,b) \in \tau\}$.

COVER: Fix a linear order $\Theta_c \cap \Theta = \emptyset$ such that $\eta : \Theta_c \approx B'$ where $B' := \{b \in B \mid |\overline{\sigma}(b,1)| > 0\}$ and define $\sigma_c : B \times \mathbb{B} \to \mathcal{P}(\mathcal{P}(\Theta \cup \Theta_c))$ via

$$\sigma_c(b,p) := \begin{cases} \{S \cup \{\eta(b)\} \mid s \in \overline{\sigma}(b,1)\} & \text{if } p = 1 \\ \sigma_c(b,0) & \text{otherwise} \end{cases}.$$

THIN: Define $\sigma_T : B \to \mathcal{P}(\Theta \cup \Theta_c)$ via $\sigma_T(b) := \min_{\sqsubset}(\{\emptyset\} \cup \sigma_c(b,0) \cup \sigma_c(b,1))$ where $S \sqsubset S'$ for $S, S' \subseteq \Theta \cup \Theta_c$ if $\min_{\Theta \oplus \Theta_c}(S \Delta S') \in S$.

[30] This is certainly a reasonable assumption as all 'real' cyclic proof systems from the literature have this property.

Populate: Fix a linear order Θ_P such that $\eta : \Theta_P \approx \{b \in B \mid \sigma_T(b) = \emptyset\}$ and define

$$\sigma_P(b) := \begin{cases} \{\eta(b)\} & \text{if } \sigma_T(b) = \emptyset \\ \sigma_T(b) & \text{otherwise} \end{cases}$$

Reset: Define $R := \{\gamma \in \Theta \mid \forall b \in B.\, \gamma \in \sigma_P(b) \to \exists \gamma' >_{\Theta \oplus \Theta_c} \gamma.\, \gamma' \in \sigma_P(b)\}$. We say that any $\gamma \in R$ was *reset* by $\overset{\tau}{\rightsquigarrow}_g$. Now define

$$\sigma^*(b) := \begin{cases} \{\gamma \in \sigma_P(b) \mid \gamma \leq R\} & \text{if } R \cap \sigma_P(b) \neq \emptyset \\ \sigma_P(b) & \text{otherwise} \end{cases}$$

$$\Theta^* := \{\gamma \in \Theta \cup \Theta_C \cup \Theta_P \mid \exists b \in B.\, \gamma \in \sigma^*(b)\}.$$

Remark 8.2.1. We shall treat $\overset{\tau}{\rightsquigarrow}_g$ as a function. With the right representation of Safra boards, e.g. $A, B \subseteq \omega$ and $\Theta, \Theta_C \subseteq \omega$, this can be ensured. More details on this in section 8.5.

The notion of reset annotation is somewhat analogous to the proof search systems discussed in section 5.3.2.

Definition 8.2.3. Let $(\textsc{Seq}, \mathcal{R}, \rho, \iota)$ be a cyclic proof system and $\Pi = (C, \lambda, \delta)$ a $\mathcal{R}$-preproof. Then $\Pi_R = (C_R, \lambda_R, \theta_R)$ is a *reset annotation of* Π if

(i) $\lambda_R : T_R \longrightarrow T$ such that s is adjacent to t in C_R iff $\lambda_R(s)$ is adjacent to $\lambda_R(t)$ in C and

(ii) if $s \in \mathrm{dom}(\beta_R)$ then $\lambda_R(s) = \lambda_R(\beta_R(s))$ and

(iii) $\theta_R(s) \in \textsc{Sb}(\iota(\lambda(\lambda_R(s))))$ and

(iv) for s adjacent to t in C_R we have $\theta_R(s) \overset{\iota(\lambda_R(s), \lambda_R(t))}{\rightsquigarrow_g} \theta_R(t)$.

Write $\theta_R(s) := (\Theta_s, \sigma_s)$. We call Π_R *sound* if for every $s \in \mathrm{dom}(\beta_R)$ there exists a linear order Θ_p

(i) which is a prefix to Θ_t for every $t \in [\beta_R(s), s]$ and

(ii) such that there exists $t \in [\beta_R(s), s]$ such that $\max \Theta_t$ is *reset* in $\theta_R(t) \rightsquigarrow_g \theta_R(t^+)$ (i.e. the step from t to its successor along the path).

Proposition 8.2.4. For any $\mathcal{R}$-preproof Π there is a reset annotation Π_R.

Proof. Π_R is obtained by performing "annotation search" along the branches of Π, stopping at the first repeat. In the interesting case of a potentially infinite branch π of C, proceed it as follows:

1. Annotate $\pi_0 = \varepsilon$ with $(\Theta, a \mapsto \{\eta(a)\})$ by fixing some $\eta : \Theta \approx \iota(\lambda(\varepsilon))$.
2. ...

i. Suppose π_{i-1} have been annotated by θ_{i-1}. Annotate π_i with the θ such $\theta_{i-1} \overset{\iota(\pi_{i-1},\pi_i)}{\rightsquigarrow_g} \theta$. Now consider the annotations of $[\pi_0, \pi_{i-1}]$:
   (a) if there exists π_j with $j < i$ annotated by θ_j such that $\theta_j = \theta$ then stop the search and insert a 'back edge' to π_j
   (b) otherwise, continue the search.

The technical details of the determinisation of $\overset{\tau}{\rightsquigarrow}_g$, which we discuss in section 8.5, guarantee that there exists a

$$k = f(|T|, \max\{|\iota(\lambda(t))| \mid t \in T\})$$

such that case (a) must have occurred before the search reaches depth k. In other words, we can always obtain a reset annotation Π_R of Π. □

To determine soundness or unsoundness of an $\mathcal{R}$-preproof via its reset annotation Π_R, we need to consider cycles through Π_R.

Definition 8.2.5. Let Π be a $\mathcal{R}$-preproof. A *reset cycle along* Π consists of a sequence $\rho \in T^*$ and a function $\theta : \Pi i < |\rho|.\ \textsc{Sb}(\iota(\lambda(\rho)))$ such that

1. $|\rho| > 1$ and $\rho_0 = \rho_{|\rho|-1}$ and $\theta_0 = \theta_{|\rho|-1}$
2. ρ_i is adjacent to ρ_{i+1} for every $i < |\rho| - 1$ and
3. $\theta(i) \overset{\iota(\rho_i,\rho_{i+1})}{\rightsquigarrow_g} \theta(i+1)$ for every $i < |\rho| - 1$.

We call a reset cycle *sound* if there is a non-empty Θ_p, which is a prefix to every Θ_i, and a $(\max \Theta_p)$-reset takes place along θ.

We call a reset cycle *failed* if it is not sound i.e., for every non-empty Θ_p, which is a prefix to every Θ_i, no $(\max \Theta_p)$-reset takes place along θ.

We begin by proving that an $\mathcal{R}$-preproof with a sound reset annotation is indeed a periodic $\mathcal{R}$-proof. This result requires three lemmas. The proofs of the first two is relegated to section 8.7.

Proposition 8.2.6. Let (ρ, θ) be reset cycle along an $\mathcal{R}$-preproof Π. Let there be $a \in \bigcap_{i<|\rho|} \Theta_i$ and let at least two a-resets take place along the reset cycle. Then for every $x \in \iota(\lambda(\rho_0))$ with $a \in \sigma_0(x)$ there exists $y \in \iota(\lambda(\rho_0))$ with $a \in \sigma_0(y)$ such that $(x, 1, y) \in \iota(\rho)$.

We follow the convention of set theory that $n = \{m \in \omega \mid m < n\}$ for $n \in \omega$.

Proposition 8.2.7. Let $R \subseteq n \times n$ with $n > 0$ be surjective and decidable. Then there exists a cycle $x \in n^{>1}$ through R i.e., $x_0 = x_{|x|-1}$ and $(x_i, x_{i+1}) \in R$.

Proposition 8.2.8. Let Π_R be a sound reset annotation of some Π. For any decidable strongly connected component $\eta \subseteq \mathrm{dom}(\beta_R)$ there exists $s \in \eta$ such that the invariant along $[\beta(s), s]$ is an invariant of all of $C_R[\eta]$.

Proof. If $C_R[\eta]$ is decidable, the proof of proposition 5.3.3 is constructive, as all sets involved become decidable. □

Theorem 8.2.9. Let Π be an $\mathcal{R}$-preproof and Π_R a sound reset annotation of it. Then Π is a periodic $\mathcal{R}$-proof.

Proof. Let Π_R be a sound reset annotation of Π and let $\pi_1\pi_2^\omega$ be a branch through Π. This induces a periodic branch $\rho_1\rho_2^\omega$ through Π_R which 'follows' $\pi_1\pi_2^\omega$. As $\rho_2\rho_{2,0}$ describes a cycle through Π_R, the nodes occurring in ρ_2 must form a strongly connected component $C[\eta]$ of Π_R. By proposition 8.2.8, there exists Θ_p such that Θ_p is a prefix of Θ_t for any $t \in C[\eta]$. Let $a := \max \Theta_p$, we furthermore know that an a-reset must take place along $\rho_2\rho_{2,0}$. Thus, $\rho_2\rho_2\rho_{2,0}$ is a reset cycle such that a occurs in every control and two a-resets take place along it. Writing $\sigma := \sigma_{\rho_{2,0}}$, we know by proposition 8.7.1 that for every $\sigma(y) \ni a$ there is $\sigma(x) \ni a$ with $(x, 1, y) \in \iota(\rho_2\rho_2\rho_{2,0})$. Thus the relation

$$S := \{(x, y) \mid (x, 1, y) \in \iota(\rho_2\rho_2\rho_{2,0}), a \in \sigma(x), \sigma(y)\}$$

is surjective. By proposition 8.7.2 this means there is a sequence $c \in X^{>1}$ such that $(c_i, c_{i+1}) \in S$ for every $i < |c| - 1$ and $c_0 = c_{|c|-1}$. Thus, there is a progressing trace $(x_i \in \iota(\lambda((\pi_1\pi_2^\omega)_{m+i})))$ with $m = |\rho_1|$ and

$$x_i = c_{n \bmod (|c|-1)} \text{ whenever } i = 2|\rho_2\rho_2|.$$

The remaining x_i can then be 'filled in' according to $\iota(\rho_2\rho_2\rho_{2,0})$. □

Conversely, unsound reset annotations witness that a $\mathcal{R}$-preproof is no periodic $\mathcal{R}$-proof. The following has a rather intricate, uninformative proof, which we relegate to section 8.7.

Proposition 8.2.10. If (ρ, θ) is a failed reset cycle of an $\mathcal{R}$-preproof Π then there exists a periodic branch of Π which is not progressing.

Combining the previous results yields constructive reset proof checking.

Lemma 8.2.11 (Reset proof checking)**.** Let Π be an $\mathcal{R}$-preproof. Then either

i there is a sound reset annotation of Π, or

ii Π has a periodic branch which is not progressing.

Proof. By proposition 8.2.4 there exists a reset annotation Π_R of Π. Now either every cycle $[\beta_R(s), s]$ of Π_R either has an appropriate prefix to make Π_R a sound reset annotation or one of the cycles of Π_R forms a failed reset cycle. In the latter case, Π has a periodic, non-progressing branch by proposition 8.7.4. □

Together with the characterisation of sound reset annotations, this yields constructive decidability of periodic $\mathcal{R}$-proofhood.

Theorem 8.2.12 (Informative periodic proof checking)**.** Let Π be an $\mathcal{R}$-preproof. Then either

(i) Π is a periodic $\iota(\mathcal{R})$-proof, or
(ii) Π has a periodic branch which is not progressing.

Proof. By proposition 8.2.4, there exists a sound reset annotation Π_R of Π or Π has a periodic, non-progressing branch by proposition 8.7.4. In the former case, Π is a periodic $\mathcal{R}$-proof by theorem 8.2.9. □

8.3 The principles ILPO and TWKL

In this section, we study the two non-constructive principles ILPO and TWKL which seem novel to the literature. We begin by placing them relative to more well-known non-constructive principles, thereby getting an indication of their strength and witnessing their non-constructivity. We then prove both principles equivalent to 'stronger looking' principles which the reductions in section 8.4 rely on. Lastly, we prove that the additive Ramsey principle (AR) implies both principles.

Recall the limited principle of omniscience (LPO).

Definition 8.3.1 (LPO)**.** For $f : \omega \to \mathbb{B}$

(i) either $f(i) = 0$ for all $i \in \omega$
(ii) or there is n with $f(n) = 1$.

The iterated principle of limited omniscience ILPO can be viewed as a countably iterated version of LPO and is thus at least as strong as LPO.

Proposition 8.3.2. ILPO implies LPO.

Proof. Fix $f : \omega \to \mathbb{B}$ and apply ILPO to f. In case (i) of ILPO, pick (ii) of LPO and provide the first 1. In case (ii) of ILPO, there is k such that always $f(k + i) = 0$. Decide whether $f(i) = 0$ for all $i < k$. If so, pick case (i) of LPO. If there is $f(i) = 1$ with $i < k$, pick case (ii) of LPO and provide i. □

As we discuss in more detail in section 8.5, LPO and ILPO are equivalent under unique choice.

The thin König's lemma TWKL 'sits' between the weak König's lemma (WKL) and the lesser limited principle of omniscience (LLPO).

Definition 8.3.3.

WKL: Every infinite decidable tree $T \subseteq \mathbb{B}^*$ has an infinite branch.

LLPO: Let $f : \omega \to \mathbb{B}$ with at most one $i \in \omega$ such that $f(i) = 1$, then

(i) $f(2i) = 0$ for every $i \in \omega$ or

(ii) $f(2i + 1) = 0$ for every $i \in \omega$.

Proposition 8.3.4. WKL implies TWKL.

Lemma 8.3.5. TWKL implies LLPO.

Proof. Fix $f : \omega \to \mathbb{B}$ with at most one 1. Consider the following tree:

$$\begin{aligned} T \coloneqq \{\varepsilon\} &\cup \{00^n \mid n \in \omega, \forall i \leq n.\ f(2i) = 0\} \\ &\cup \{11^n \mid n \in \omega, \forall i \leq n.\ f(2i+1) = 0\}. \end{aligned}$$

It is clearly decidable and its width is globally bounded by 2. Furthermore, it is infinite as f has at most one 1. Thus, we may apply TWKL, obtaining an infinite branch $b \in \mathbb{B}^\omega$ of T. If $b = 0^\omega$ then case (i) of LLPO applies, if $b = 1^\omega$ then case (ii) applies. □

Under dependent choice, LLPO and WKL (and thus also TWKL) are equivalent (see Diener (2020, proposition 1.4.5)). We do not know if, without dependent choice, TWKL implies WKL. So far, we have not managed to prove this but it may well be that TWKL and WKL coincide.

We have chosen to define ILPO and TWKL as simply as possible, following the usual style of non-constructive principles. However, the reductions in section 8.4 rely on equivalent, although 'stronger looking', formulations of them. We thus continue by establishing said equivalences.

We give one more, equivalent, formulation of ILPO.

Proposition 8.3.6. ILPO is equivalent to the following principle:

ILPO_ω For any $f : \omega \to \omega$ bounded by some $n \in \omega$ there exists $g : n \to \mathbb{B}$:

(i) for any $g(a) = 1$ and $i \in \omega$ there is $j > i$ such that $f(i) = a$ and

(ii) there exists an offset k such that any $g(f(k + i)) = 1$.

Proof.

$\mathrm{ILPO}_\omega \to \mathrm{ILPO}$: We may consider $f : \omega \to \mathbb{B}$ as $f : \omega \to \omega$ bounded by 1. Applying ILPO_ω yields a $g : \omega \to \mathbb{B}$ such that

(i) for any $f(a) = 1$ and $i \in \omega$ there is $j > i$ such that $f(i) = a$ and

(ii) there exists an offset k such that any $g(f(k + i)) = 1$.

Perform a case-distinction on $g(1)$:

$g(1) = 1$: Then (i) means that for every $i \in \omega$ there is $j > i$ with $f(j) = 1$.

$g(1) = 0$: Then $g(0) = 1$ because otherwise $g(f(k)) = 0$. Then from offset k on, $f(k + i) = 0$ for any $i \in \omega$.

$\mathrm{ILPO} \to \mathrm{ILPO}_\omega$: Fix $f : \omega \to \omega$ bounded by some n. We instead prove a slightly different property, by induction on m: There exists $g : m \to \mathbb{B}$ such that:

(i) for any $g(a) = 1$ and $i \in \omega$ there is $j > i$ such that $f(i) = a$ and

(ii) there exists an offset k such that if $f(k+i) < m$ then $g(f(k+i)) = 1$.

Then, by taking $m := n$, the desired property follows.

$m = 0$: Then $g := \emptyset$ works trivially.

$m + 1$: Suppose there already was a suitable set $g' : m \to \mathbb{B}$ and a bound k' for it. Apply ILPO to $f_n(i) := (f(i) = m) : \mathbb{B}$ and consider the outcome:

(i) We choose $g := g'[m \mapsto 1]$. Towards (i), the outcome of ILPO guarantees there exists $j > i$ for every $i \in \omega$ such that $f(i) = m$. Towards (ii), we choose $k := k'$ and observe that if $f(k' + i) < m + 1$, then either $f(k' + i) = m$ and thus $g(f(k' + i)) = 1$ or $f(k' + i) < n$ and thus $g(f(k' + i)) = g'(f(k' + i)) = 1$ per assumption.

(ii) We choose $A := g'[m \mapsto 0]$. Part (i) follows directly from the inductive hypothesis. Towards (ii), observe that ILPO supplies a k_m such that $f(k_m + i) \neq n$ for all $i \in \omega$. Thus $k := \max\{k, k_m\}$ satisfies the claim. □

We also give two more, equivalent, formulations of TWKL:

Definition 8.3.7.

TWKL$_\omega$ Let $T \subseteq \omega^*$ be a decidable infinite tree with a globally bounded width then it has an infinite branch.

TWKL$_R$ Let $R : \omega \to \mathcal{P}(n \times n)$ for some $n > 0$ be such that the R_i for every $i \in \omega$ is surjective and decidable. Then there exists $x : \omega \to n$ such that for every $i \in \omega$ we have $(x_i, x_{i+1}) \in R_i$.

Proposition 8.3.8. TWKL, TWKL$_\omega$ and TWKL$_R$ are equivalent.

Proof.

TWKL $\to$ TWKL$_\omega$: Let $T \subseteq \omega^*$ be globally bounded by n. Observe that this must mean for any $a_1 \dots a_m \in T$ the $a_i < n$. We code T into $T' \subseteq \mathbb{B}^*$ via

$$T' := \{1^{a_1}0^{n-a_1} \dots 1^{a_m}0^{n-a_m} \mid a_1 \dots a_m \in T\}.$$

First observe that if $b' \in \mathbb{B}^\omega$ is a branch through T' then $b \in \omega^\omega$ with

$$b_i := |\{j \in \omega \mid in \leq j < i(n+1) \text{ and } b'_j = 1\}|$$

is a branch through T. To close, we show that T' is globally bounded by n. Thus, let $i \in \omega$ and let $k = \lfloor \frac{i}{n} \rfloor$ and $m = i - kn$. We can observe that

$$T'_i = \{1^{a_1}0^{n-a_1} \dots 1^{a_k}0^{n-a_k}1^{\max(a_{k+1},m)}0^{m \dot{-} a_k} \mid a_1 \dots a_k a_{k+1} \in T_{k+1}\}$$

where $x \dot{-} y = 0$ if $x < y$. As T_{k+1} clearly injects into the right-hand side of the equation above and $|T_{k+1}| \leq n$ per assumption, we can conclude $|T'_i| \leq n$.

TWKL$_\omega$ $\to$ TWKL$_R$: Let the $R : \omega \to \mathcal{P}(n \times n)$ be given. We construct $T \subseteq \omega^*$ recursively via

$$T_0 := \{\varepsilon\} \qquad\qquad T_1 := \{0, \dots, n-1\}$$

$$T_{i+1} := \{ulk \mid k < n \wedge l = \min\{l \mid (l,k) \in R_{i-1}\} \wedge ul \in T_i\}$$

Per induction on i, it easily follows that for every $l < n$ there is a unique $u \in \omega^*$ such that $ul \in T_{i+1}$. Thus, T is infinite and its width is globally bounded by n. Per assumption, that means there is an infinite branch $x \in \omega^\omega$ through T. Per construction of T, we know that $(x_i, x_{i+1}) \in R_i$ for every $i \in \omega$ as desired.

TWKL$_R$ $\to$ TWKL: Let $T \subseteq \mathbb{B}^*$ be globally bounded by n. We can order each T_i lexicographically as $t^i_0, \dots, t^i_{|T_i|-1} \in T_i$. Then define

$$R_i := \{(l,k) \mid t^i_l < t^{i+1}_k \wedge k < |T_{i+1}|\} \cup \{(0,k) \mid |T_{i+1}| \leq k < n\} \subseteq n \times n$$

Observe that the R_i are surjective: For every $k < n$, we distinguish two cases: If $k < |T_{i+1}|$ there is $t_k^{i+1} \in T_{i+1}$ which must have a predecessor $t_l^i \in T_i$ and thus $(l, k) \in R_i$. Otherwise, if $|T_{i+1}| \leq k < n$ then $(0, k) \in R_i$. Hence there is $x \in \omega^\omega$ with $(x_i, x_{i+1}) \in R_i$ for every $i \in \omega$. It suffices to prove that $x_i < |T_i|$ for every $i \in \omega$ as this gives rise to an infinite sequence $(t_{x_i}^i \in T_i)$ with $t_{x_i}^i < t_{x_j}^j$ whenever $i < j$ which induces an infinite branch through T. Thus, suppose there was $|T_i| \leq x_i$ and consider the nature of x_{i+1}. We know that $(x_i, x_{i+1}) \in R_i$ but as $|T_i| > 0$, this cannot be by the second clause of R_i. Thus, that would mean $t_{x_i}^i < t_{x_{i+1}}^{i+1}$ but as $x_i \geq |T_i|$, there is no $t_{x_i}^i$, leading to a contradiction. □

Remark 8.3.1. While the principles ILPO_ω and TWKL_R are phrased in terms of domains $n \in \omega$, they readily generalise to arbitrary finite sets.

We close this section by proving that the additive Ramsey principle implies both ILPO and TWKL to establish implication $(i) \rightarrow (ii)$ of theorem 8.1.3.

Lemma 8.3.9. The additive Ramsey principle implies ILPO.

Proof. Fix $f : \omega \rightarrow \mathbb{B}$. Then the additive Ramsey principle may be applied to it directly, observing that $(\mathbb{B}, \vee)$ is a finite semi-group. Thus, suppose there was a strictly increasing sequence $\tau : \omega \rightarrow \omega$ such that $b = \bigvee_{i=\tau(n)}^{i<\tau(n+1)} f(i)$ for some $b \in \omega$. Then we distinguish two cases:

- $b = 0$: Then $f(\tau(0) + i) = 0$ for every $i \in \omega$: Observe that for such an i there must be $n \leq i$ such that $\tau(n) \leq \tau(0)+i < \tau(n+1)$. As $\bigvee_{j=\tau(n)}^{j<\tau(n+1)} f(j) = 0$ that means that $f(\tau(0) + i) = 0$.
- $b = 1$: Then for every $i \in \omega$ there is $j > i$ with $f(j) = 1$: We know that $i < \tau(i + 1)$ and that, because $\bigvee_{j=\tau(i+1)}^{j<\tau(i+2)} f(j) = 1$ there must be $i < \tau(i + 1) \leq j < \tau(i + 2)$ with $f(j) = 1$. □

Lemma 8.3.10. The additive Ramsey principle implies TWKL

Proof. We prove TWKL_R. $(\{R \subseteq n \times n \mid R \text{ surjective \& decidable}\}, ;)$ with

$$R; R' := \{(a, c) \mid \exists b.\ (a, b) \in R \wedge (b, c) \in R'\}$$

forms a finite semi-group. Applying the additive Ramsey principle to the assumed $R : \omega \rightarrow \mathcal{P}(n \times n)$ yields a strictly monotone $\tau : \omega \rightarrow \omega$ and a surjective $S \subseteq n \times n$ such that $R_{\tau(n)}; R_{\tau(n)+1}; \ldots; R_{\tau(n+1)-1} = S$ for all $n \in \omega$. By proposition 8.7.2, there exists a cycle through S i.e., there are $y_0 y_1 \ldots y_m \in n^+$ with $(y_i, y_{i+1}) \in S$ for every $i < m$ and $(y_m, y_0) \in S$. To construct the sequence $x \in n^\omega$ through the R_i, we begin by fixing $x_{\tau(n)} \coloneqq y_{n \bmod m}$. We then 'fill in the gaps' for x_i with $i \notin \mathrm{im}(\tau)$ as follows:

$i < \tau(0)$: We define, for $i < \tau(0)$ the following

$$x_i \coloneqq \min\{k < n \mid (k, x_{i+1}) \in R_i\}.$$

Observe that this such k must always exist by the surjectivity of the R_i and that the definition terminates as $\tau(0) - i$ decreases in the 'recursive call'. Further, the $(x_i)_{i \leq \tau(0)}$ then clearly form a path through the $(R_i)_{i<\tau(0)}$, ending in $x_{\tau(0)}$.

$\tau(n) < i < \tau(n+1)$: We know that $R_{\tau(n)}; \ldots ; R_{\tau(n+1)-1} = S$ and, per definition, that $(x_{\tau(n)}, x_{\tau(n+1)}) \in S$. Per definition of the ;-operation, there must thus be $x_{\tau(n)+1}, \ldots, x_{\tau(n+1)-1}$ such that $(x_{\tau(n)+j}, x_{\tau(n)+j+1}) \in R_{\tau(n)+j}$ for $j < 1 + \tau(n+1) - \tau(n)$. Then fix these. □

8.4 Main reductions

This section covers the two 'main reductions' of theorem 8.1.3 i.e., those which require the most work, even when relying on the purely constructive results from section 8.2. These are the steps $(ii) \to (iii)$ and $(v) \to (i)$. We call them 'forwards' and 'backwards', respectively, because they lead from non-constructive principles to a theorem of cyclic proof theory, or the converse.

8.4.1 Forwards

Lemma 8.4.1. Let C be a cyclic tree and $\pi \in T^\omega$ a branch through C. If $\mathrm{Inf}(\pi)$ is decidable and there exists an offset k such that $\pi_{k+i} \in \mathrm{Inf}(\pi)$ for every $i \in \omega$, then it is as scc $C[\{s \in \mathrm{Inf}(\pi) \mid s \in \mathrm{dom}(\beta)\}]$.

Proof. If $\mathrm{Inf}(\pi)$ is as described above, the proof of lemma 4.1.4 becomes constructive. □

Theorem 8.4.2. ILPO and TWKL imply *unrestricted reset soundness:* If Π_R is a sound reset annotation of a $\mathcal{R}$-preproof Π then Π is an unrestricted $\mathcal{R}$-proof.

Proof. Let $\pi \in T^\omega$ be a branch through Π. Then there is a corresponding $\pi_R \in T_R^\omega$ through Π_R such that $\lambda_R \circ \pi_R = \pi$. By ILPO_ω, $\mathrm{Inf}(\pi)$ is decidable and thus an scc $C[\eta]$ by lemma 8.4.1. By proposition 8.2.8 there exists some $t \in \eta$ such that the invariant Θ_t of $[\beta(t), t]$ is an invariant for all of $C[\eta]$. Fix $a \coloneqq \max \Theta_t$. By the previous application of ILPO_ω we know that for every i there is $j > i$ such that $\pi_R(j) = t$. Thus we obtain a strictly increasing function $f : \omega \to \omega$

with $\pi_R(f(i)) = t$ for all $i \in \omega$ and $\pi_R[f(0) :]$ remains within $C[\eta]$. Fix the set $X := \{x \in \iota(\lambda(\lambda_R(t))) \mid a \in \sigma_t(x)\}$ of trace values marked by γ at t. Define $R : \omega \to \mathcal{P}(\Gamma \times \Gamma)$ via

$$R_i := \{(x, y) \in \Gamma^2 \mid (x, 1, y) \in \iota(\pi([f(2i) : f(2i + 2)]))\}.$$

Each R_i is decidable as the individual trace maps along Π are decidable. Because a never disappears along $\pi_R[f(2i) : f(2i + 2)]$ and a is reset at least twice along it, we may apply proposition 8.7.1 to conclude that each R_i is a surjection. By TWKL$_R$, this $(x_{f(2i)} \in \iota(\pi_{f(2i)}))_{i\in\omega}$ with $(x_{f(2i)}, 1, x_{f(2i+2)}) \in \iota(\pi_{f(2i)} \dots \pi_{f(2i+2)})$ for every $i \in \omega$. The values x_j for $f(2i) < j < f(2i + 2)$ can then be 'filled in' from the R_i to yield a progressing trace along π. □

8.4.2 Backwards

Lemma 8.4.3. For any finite semi-group $(\Gamma, +)$ there exists a derivation system $\mathcal{R}$, a trace interpretation $\iota : \mathcal{R} \to \mathbb{B}$ and a periodic $\iota(\mathcal{R})$-proof Π such that Π being an unrestricted $\iota(\mathcal{R})$-proof is equivalent to every sequence $\sigma : \omega \to \Gamma$ having a factorisation.

Proof. As Γ is finite, let $g_1 \dots g_n$ be a (repetition-free) enumeration of it. We define $\mathcal{R} := (\text{Seq}, \mathcal{R}, \rho)$ with $\text{Seq} := \{\star\}$ and $\mathcal{R} := \{\dagger\}$ with

$$\rho(\dagger) := (\star, \underbrace{\star, \dots, \star}_{n \text{ times}}).$$

Furthermore, set $\iota(\star) := \Gamma \times (\Gamma \cup \{\bullet\})$ where $\bullet \notin \Gamma$ and

$$\begin{aligned} \dagger_i := &\{((g, g'), 0, (g, g' + g_i)) \mid (g, g') \in \iota(\star)\} \\ &\cup \{((g, g'), 1, (g, \bullet)) \mid (g, g') \in \iota(\star) \land g' + g_i = g\}. \end{aligned}$$

where $\bullet + g = g$ for $g \in \Gamma$. Then Π is the preproof deriving $\star$ consisting of a single application of $\dagger$ all of whose premises 'cycle back' to the root.

First, consider the function $\gamma : \Gamma^\omega \to T^\omega$ given below, which is a bijection between sequences $\sigma \in \Gamma^\omega$ and branches through Π.

$$\gamma(\sigma)_i := \begin{cases} k - 1 \in \omega^1 & \text{if } i = 2j + 1 \text{ and } \sigma_j = g_k \\ \varepsilon & \text{otherwise} \end{cases}$$

We shall now make use of the following claim. Its proof is somewhat technical, so we relegate it to section 8.7. Intuitively, a trace along $\gamma(\sigma)$ 'picks out' a value g in the first component and then computes the 'partial sum,' progressing and resetting the partial sum to an additional neutral element $\bullet$ whenever the partial sum reaches g. Thus, the partial sums must reach g infinitely often, giving rise to a factorisation of σ into g, if the trace progresses infinitely often.

Claim 8.4.4. Every $\sigma \in \Gamma^\omega$ has a factorisation iff $\gamma(\sigma)$ has a progressing trace.

The above result already means that if Π is an unrestricted $\iota(\mathcal{R})$-proof then every $\sigma \in \Gamma^\omega$ has a factorisation. It remains to show that Π is a periodic $\iota(\mathcal{R})$-proof. Thus, let $\pi_1\pi_2^\omega$ be a branch through Π. Let w.l.o.g. $\pi_2 = \varepsilon k_1 \dots \varepsilon k_n$ for some $k_1, \dots, k_n < |\Gamma| - 1$. By the previous result, it suffices to show that $\sigma := \gamma^{-1}(\pi_1\pi_2) \in \Gamma^\omega$ has a factorisation. Let $m = \frac{|\pi_1|}{2}$ then observe that $\sigma_{m+i} = g_{k_{i \bmod n}+1}$. In other words, fixing $g := \sum_{i=1}^{i<n+1} g_{k_i+1}$ we have $g = \sum_{i=m+ln}^{i<m+(l+1)n} \sigma_i$ for every $l \in \omega$ and thus that $\tau(l) := m + ln$ is a factorisation of σ. $\square$

Theorem 8.4.5. Suppose that for every cyclic proof system $\mathcal{R}$, every $\mathcal{R}$-preproof is a periodic $\mathcal{R}$-proof iff it is an unrestricted $\mathcal{R}$-proof. Then the additive Ramsey principle holds.

Proof. Towards the additive Ramsey principle, let $(\Gamma, +)$ be a finite semi-group. By lemma 8.4.3 there exists a cyclic proof system $\mathcal{R}$ and a periodic $\mathcal{R}$-proof Π such that $(\Gamma, +)$ satisfies the additive Ramsey principle iff Π is also an unrestricted $\mathcal{R}$-proof. But this holds per assumption. $\square$

8.5 Clarifying constructivity

In previous sections, we have been rather light on the technical details necessary to fully convince a reader familiar with constructive mathematics of the constructivity of the results. In this section, we mainly kinds of such details: considerations around the coding/representation of the objects involved and the adaptability of the results to various settings of constructive mathematics.

Compatible foundations As stated before, the results in this section can be derived over most constructive foundations of mathematics. The arguments, as presented so far, can be transferred directly to set-theoretic or second order settings, such as constructive ZF (Aczel 1984) and Bishop-style constructive mathematics (Bishop 1967). With a few small adjustments, which we discuss later in this section, our results can also be adapted to type theoretic foundations, such as Martin-Löf type theory (MLTT) (Martin-Löf 1975; Martin-Löf 1982), the Calculus of Inductive Constructions (CIC) (Coquand and Huet 1986; Paulin-

Mohring 1993) or univalent mathematics (HoTT) (The Univalent Foundations Program 2013).

The results of this chapter do not transfer to first-order settings, such as Heyting arithmetic. This is unsurprising as the global trace condition cannot even be stated without quantification over branches and traces, which are 'second-order concepts'. Indeed, the additive Ramsey principle has been proving equivalent to Σ^0_3-LLPO over HA by Berardi and Steila (2017), diverging from our results in this chapter.

Finiteness We take a set X to be finite if there is a bijection $c : n \to X$ for $n \in \omega$. Note that this immediately entails discreteness of X (decidability of equality between X-elements).

Determinisation of $\overset{\tau}{\rightsquigarrow}_g$: In remark 8.2.1 we claim that one may always treat the relation $\overset{\tau}{\rightsquigarrow}_g$ as a function. To justify this claim, first observe that Safra boards are only ever used 'in the context of' some cyclic proof system $\mathcal{R}$ and *a singular proof* Π. Thus, we may first observe that the following set is finite because T underlying Π is finite

$$X := \{\iota(\lambda(t)) \mid t \in T\}$$

and we may restrict our attention to Safra boards on $A \in X$. Further, as X is finite and every $A \in X$ is finite, there is a function $c' : \Pi A \in X.\Sigma n \in \omega.\, n \to A$ such that every $c_A := \pi_2(c'(A)) : n_A \to A$ with $n_A := \pi_1(c'(A))$ is a bijection.

Now we can define a more concrete set $\textsc{Sb}(A)$ of Safra boards on $A \in X$. We say $(\Theta, \sigma) \in \textsc{Sb}(A)$ if $\Theta = (B, <_\Theta)$ where $B \subseteq 3K$, where $K := \max\{|A| \mid A \in X\}$, and $<_\Theta \subseteq B \times B$ is a linear order and $\sigma : A \to \mathcal{P}(B) \setminus \{\emptyset\}$. Furthermore, we require that for every $b \in B$ there is $a \in A$ with $b = \max \sigma(a)$. Observe that the latter must mean $|B| \leq |A|$. Now we can, for $\tau : A \to A'$ for $A' \in X$, define a deterministic $(\Theta, \sigma) \overset{\tau}{\rightsquigarrow}_g (\Theta^*, \sigma^*) \in \textsc{Sb}(A')$.

Move: Define $\overline{\sigma} : A' \times \mathbb{B} \to \mathcal{P}(\mathcal{P}(\Theta))$ via

$$\overline{\sigma}(a', p) := \{\sigma(a) \mid (a, p, a') \in \tau\}.$$

Cover: Let $A'_C := \{a' \in A' \mid |\overline{\sigma}(a', 1)| > 0\}$. Now fix $B_c \subseteq 3K \setminus B$ with $|B_c| = |A'_c|$ such that $\sum B_c$ is minimal, noting that there are enough 'free elements' in $3K \setminus B$ as $|B| \leq |A| \leq K$ and similarly $|A'_c| \leq |A'| \leq K$. Further, let $\eta_c : B_c \to A'_c$ be the bijection such that the ith element $b \in B_c$ is mapped to $\eta_c(b) \in A'_c$ such that $c^{-1}_{A'}(\eta_c(b))$ is the ith element of

$c_{A'}^{-1}(A'_c)$. Fix $\Theta_c := (B_c, <)$ and define $\sigma_c : A' \times \mathbb{B} \longrightarrow \mathcal{P}(\mathcal{P}(B \cup B_c))$ via

$$\sigma_c(a', 1) := \{S \cup \{\eta_c(a')\} \mid s \in \overline{\sigma}(a', 1)\} \qquad \sigma_c(a', 0) := \overline{\sigma}(a', 0)$$

THIN: Define $\sigma_T : B \longrightarrow \mathcal{P}(\Theta \cup \Theta_c)$ via $\sigma_T(b) := \min_{\sqsubset}(\{\emptyset\} \cup \sigma_c(b, 0) \cup \sigma_c(b, 1))$ where $S \sqsubset S'$ for $S, S' \subseteq \Theta \cup \Theta_c$ if $\min_{\Theta \oplus \Theta_c}(S \Delta S') \in S$.

POPULATE: Define $A'_P := \{a' \in A' \mid \sigma_T(b) = \emptyset\}$. Now fix $B_P \subseteq 3K \setminus (B \cup B_c)$ with $|B_P| = |A'_P|$, again observing that $|B| + |B_c| \leq |A| + |A'| \leq 2K$ and thus there are enough 'free elements' in $3K \setminus (B \cup B_c)$. Fix a bijection $\eta_P : B_P \longrightarrow A'_P$ analogously to η_c above and $\Theta_P := (B_p, <)$ and define

$$\sigma_P(b) := \begin{cases} \{\eta(b)\} & \text{if } \sigma_T(b) = \emptyset \\ \sigma_T(b) & \text{otherwise} \end{cases}$$

RESET: Define $R := \{b \in B \mid \forall a' \in A'.\ b \in \sigma_P(a') \rightarrow \exists b' >_{\Theta \oplus \Theta_c} b.\ b' \in \sigma_P(a')\}$. We say that any $b \in R$ was *reset* by $\overset{\tau}{\rightsquigarrow}_g$. Now define

$$\sigma^*(a') := \begin{cases} \{b \in \sigma_P(a') \mid b \leq R\} & \text{if } R \cap \sigma_P(a') \neq \emptyset \\ \sigma_P(a') & \text{otherwise} \end{cases}$$

$$\Theta^* := \{b \in \Theta \cup \Theta_C \cup \Theta_P \mid \exists a' \in a'.\ b \in \sigma^*(a')\}$$

and observe that for and $b \in \Theta^*$ there is a' with $\max \sigma^*(a') = b$ (otherwise $b \in R$ above). That means $(\Theta^*, \sigma^*) \in \mathrm{SB}(A')$ as desired.

Depth bound for proposition 8.2.4 In the proof of proposition 8.2.4 we claim a depth bound

$$f(|T|, \max\{|A| \mid A \in X\})$$

on the search of reset annotations for branches of a proof Π, writing $X := \{\iota(\lambda(t)) \mid t \in T\}$ again. Simply observe that any search state is a pair (t, θ) where $t \in T$ and $\theta \in \mathrm{SB}(\iota(\lambda(t)))$. Thus, any such search branch containing at least $|\Sigma t \in T.\ \mathrm{SB}(\iota(\lambda(t)))| + 1$ different search states must contain a duplicate, terminating the search by inserting a 'back edge'.

Now observe that, if we employ the Safra board representations discussed above and setting $K := \max\{|A| \mid A \in X\}$, we can observe that any $\theta \in \mathrm{SB}(A)$ with $A \in X$ consists of a $B \subseteq 3K$ with $|B| \leq K$, of which there are $\binom{3K}{K}$, a linear order on B, of which there are $!|B| \leq !K$, and a function $\sigma : A \longrightarrow \mathcal{P}(B) \setminus \{\emptyset\}$, of which there are $(2^{|B|})^{|A|} \leq 2^{K^2}$. Thus

$$|\mathrm{SB}(A)| \leq \binom{3K}{K} \cdot !K \cdot 2^{K^2}$$

and we may thus set

$$f(|T|, K) := 2 + |T| \cdot \binom{3K}{K} \cdot !K \cdot 2^{K^2} \geq |\Sigma t \in T.\ \mathrm{Sb}(\iota(\lambda(t)))| + 2.$$

Adapting to type theories The presentation of the definitions and results in this chapter employs the language of set theory. Nonetheless, the results adapt quite readily to type-theoretical foundations, such as MLTT, CIC and HoTT. Sets such as ω, $\mathbb{B}$, $A \times B$, A^* or $A \to B$ may simply be represented by their usual corresponding types. The finite sets used by $\iota(\Gamma)$ may be represented by finite types. Decidable subsets $X \subseteq Y$ of a set Y with type representation A may be represented by functions $A \to \mathbb{B}$. With these considerations, every notion present in this chapter can be represented rather naturally.

Some results in this chapter, lemma 8.3.10 core among them, rely on functional extensionality, for example in the form of extensionality for decidable subsets of $n \times n$. This can of course be accommodated for by extending one's type theory with functional extensionality. Alternatively, one can observe that these results are only ever concerned with finite relations, which can for example be represented as ordered sequences of tuples. The results about these representations then reflect back on the proper functions up-to observational equivalence ("homotopy") of functions. This suffices to obtain all the main results i.e., theorems 8.1.3 and 8.2.12, because they do not mention equality between functions.

Choice-like principles In theorem 8.4.2 and proposition 8.7.5, we 'convert' results of the form $\forall n \in \omega \exists m \in \omega.\ P(n, m)$ into functions $f : \omega \to \omega$ with $P(n, f(n))$ for all $n \in \omega$. While this may look like an application of the axiom of choice, it is an instance of the following principle which is true in constructive set theory, constructive type theory and Bishop-style constructive mathematics:

Proposition 8.5.1. Let$P(n, m)$ be decidable for every $n, m \in \omega$ and suppose $\forall n \exists m.\ P(n, m)$. Then there is $f : \omega \to \omega$ such that $P(n, f(n))$ for all $n \in \omega$.

Proof. While the details of the arguments differ, one essentially constructs

$$f(n) := \min\{m \in \omega \mid P(n, m)\}. \qquad \square$$

Interpretation of $\vee$ and $\exists$ in type theories CIC distinguishes between computational disjunction + and existence Σ and their logical counterparts $\vee$ and $\exists$. In MLTT, the 'standard versions' of disjunction and existence are computational. However, logical variants $\vee$ and $\exists$ can be defined by means of propositional truncation, as is common in HoTT. The computational version of a connective implies its logical counterpart but, generally, not the other way around. This raises the question how instances of 'or' and 'exists' in this chapter should be interpreted.

First of all, we note that theorem 8.2.12 is provable as the 'strongest' variant of its statement: computationally interpreting the positively occurring connectives and logically interpreting the negatively occurring ones. This means that the 'weaker' interpretations of the two clauses of theorem 8.2.12 are equivalent to their aforementioned strongest counterparts. We demonstrate this below for clause (ii) of theorem 8.2.12, the case for clause (i) being analogous. Thus, the choice of connective interpretation for these two properties does not matter.

Proposition 8.5.2. Let Π be an $\mathcal{R}$-preproof. The following are equivalent:

(i) $\exists\pi$ periodic branch of Π. $\neg(\Sigma(x_i)_{i\in\omega}$ π-trace. $(x_i)_{i\in\omega}$ progressing)

(ii) $\Sigma\pi$ periodic branch of Π. $\neg(\exists(x_i)_{i\in\omega}$ π-trace. $(x_i)_{i\in\omega}$ progressing).

Proof. Only the 'top-to-bottom' direction is interesting. Assume the premise, then by theorem 8.2.12, either the desired conclusion holds or

$$\forall\pi \text{ periodic branch through } \Pi. \Sigma(x_i)_{i\in\omega} \text{ trace along } \pi.\ (x_i)_{i\in\omega} \text{ progressing.}$$

Suppose the latter as the former closes the proof. Then we derive a contradiction. As we are now proving the proposition $\perp$, we may eliminate on the assumed $\exists$, obtaining a periodic branch π through Π for which there do not computably exist progressing traces. But by the outcome of theorem 8.2.12, such a trace can computably be obtained, a contradiction. □

The main remaining consideration is the interpretation of connectives for the statements of theorem 8.1.3. First, note that because of the observation about theorem 8.2.12 above, the disjunction in (iv) may always be chosen computationally. Similarly, by proposition 8.5.1, the choice of interpretation of the existential in (i) of ILPO does not matter. The choice of the remaining connectives i.e., the disjunction in ILPO, existential in (ii) of ILPO, existential in TWKL, existential in the additive Ramsey principle as well as the existential in unrestricted $\mathcal{R}$-proofhood, remains to be fixed. There are two sensible choices: all computational or all logical. Each choice gives rise to a provable version of the main result theorem 8.1.3. With other choices, the equivalences between statements with 'incompatible choices' will break down.

Collapse into LPO We have claimed in section 8.1 that under unique choice, ILPO and LPO coincide, entailing TWKL. Unique choice (UC) is defined below.

Definition 8.5.3 (Unique choice)**.** Let $\forall x \in X. \exists! y \in Y.\ P(x, y)$ then there is $f : X \to Y$ such that $P(x, f(x))$ for every $x \in X$.

Unique choice holds in CZF, IZF, MLTT, HoTT and Bishop-style CM. Furthermore, in CIC unique choice holds when interpreting all existentials computationally. Thus, the only non-collapsing setting which we have mentioned so far is the CIC under the 'logical versions' of ILPO and TWKL.

The following is based on proposition 1.2.5 of Diener (2020), modified to make the role of unique choice more explicit.

Proposition 8.5.4. Under UC, LPO also implies ILPO.

Proof. Consider $f : \omega \to \mathbb{B}$. Observe that, either by unique choice or by the computational interpretation of existence in LPO, the following function can be defined

$$g(i) := \begin{cases} 1 & (j \mapsto f(i + j)) \text{ is all zeros} \\ 0 & \text{otherwise} \end{cases}.$$

Now apply LPO to g.

(i) Then case (i) of ILPO applies. Let $i \in \omega$ be given and apply LPO to $(n \mapsto f(i + n + 1))$. In case of (ii) of LPO, we obtain the $j > i$ with $f(j) = 1$ as desired. In case of (i) of LPO, this means that $g(j) = 1$ for some $j > i$, a contradiction.

(ii) There is an $k \in \omega$ such that $g(k) = 1$. Then case (ii) of ILPO applies with the same k. □

Recall that, by proposition 8.3.2, ILPO implies LPO, making them equivalent under UC. Under UC, LPO and ILPO furthermore entail TWKL, making them equivalent to the additive Ramsey principle and unrestricted cyclic proof theory. The following is a version of Diener (2020, proposition 1.4.5) using LPO instead of LLPO which does not rely on dependent choice.[31] Dependent choice can be removed from the aforementioned argument because we only consider decidable trees.

[31] Note that we believe that the choice between LPO and LLPO does not affect the removal of dependent choice from the argument. We chose to use LPO because it is already present in the context and makes the argument simpler to present.

Proposition 8.5.5. Under UC, LPO implies WKL.

Proof. Observe the following: Using LPO, one can decide whether a binary tree T is infinite or not. For this, apply LPO to the following function

$$f_T(n) := \neg \bigvee_{v \in 2^n} v \in T.$$

Each $f_T(n)$ is essentially a computable, finite disjunction. If $f_T(n) = 1$ that means $T \cap 2^n = \emptyset$, meaning T is finite. If f_T is all zeroes, then there exists branches of arbitrary length, making T infinite.

As in proposition 8.3.2, the setting allows this to be used to define, for a fixed binary tree $T \subseteq \mathbb{B}^*$, a function $g : \mathbb{B}^* \to \mathbb{B}$ with

$$g_T(v) := \begin{cases} 1 & v \in T \text{ and the subtree of } T \text{ at } v \text{ is infinite} \\ 0 & \text{otherwise} \end{cases}.$$

Now fix an infinite tree $T \subseteq 2^*$ and consider the following function:

$$b(0) := \varepsilon \qquad b(n+1) := \begin{cases} b(n)0 & g(p(n)0) = 1 \\ b(n)1 & \text{otherwise} \end{cases}.$$

Then an inductive argument shows that $b'(i) := b(i+1)_i$ is an infinite branch through T. □

Corollary 8.5.6. Under UC, LPO implies TWKL.

8.6 Discussion

Recall informative unrestricted proof checking, principle (iv) of theorem 8.1.3, which states that every $\mathcal{R}$-preproof is an $\iota(\mathcal{R})$ or has a *periodic* non-progressing branch. The aspect which makes this "informative" is the periodicity of the counterexample. Readers familiar with cyclic proof theory will have noticed that this phrasing of proof checking is different from the way it is usually stated in the literature, given below.

Definition 8.6.1 (Uninformative unrestricted proof checking)**.** Let Π be an $\mathcal{R}$-preproof. Then it either is an unrestricted $\iota(\mathcal{R})$-proof or has a non-progressing branch.

There are two reasons why we study the informative, rather than the seemingly more common uninformative version of the theorem. First, we would like to note that while proof checking is usually stated in the uninformative phrasing, what is actually proven is the informative variant. All proof methods, be it via automata constructions akin to definition 5.1.2, directly via the additive Ramsey principle as in (Afshari and Wehr 2024, theorem 5.3) or via some other form of cyclic proof representation, yield a periodic counterexample branch. Thus, informative proof checking is arguably widespread in the literature.

Second, and maybe more importantly, uninformative proof checking seems to be very weak. Clearly, informative proof checking entails uninformative proof checking. The opposite direction seems, at least intuitively, false: if uninformative proof checking deems some $\mathcal{R}$-preproof Π a non-proof, yielding an arbitrary counterexample branch, there is little to be done constructively. Because the counterexample is arbitrary, there seems to be little hope of 'periodising' it or otherwise extracting a periodic counterexample from it. Similarly, the constructive informative periodic proof checking is of no help: constructively, it seems consistent that Π may be a periodic $\iota(\mathcal{R})$-proof with an aperiodic counterexample branch.[32]

Taken together, these points motivate our exclusion of uninformative unrestricted proof checking from our study in this chapter: it seems much weaker than all other principles, supplying constructively 'worthless' evidence of non-proofhood, and there seems to be no clear proof strategy for obtaining it which does not rely on the stronger principles studied in this chapter. These factors make it seem rather unlikely that there exists an 'insightful' non-constructive principle to measure its difficulty with.

Infinite Ramsey principles, such as the additive Ramsey principle we consider here, are well-studied in classical reverse mathematics. Classically, such Ramsey principles occupy a special position because they at first eluded the traditional classification in terms of the 'big five'. Hirschfeldt (2015) is a textbook essentially entirely dedicated to the reverse mathematics of Ramsey principles. The additive Ramsey principle, specifically, has been studied by Kołodziejczyk et al. (2019), as part of an investigation into the reverse mathematics of infinite word automata. They prove it equivalent to Σ^0_2-induction over RCA_0. These results of classical reverse mathematics have been used in the study of cyclic proofs by Das (2020) to obtain a result similar to section 7.2.

Ramsey principles have also been studied from the constructive perspective. The non-constructivity of the Ramsey principle for pairs had already been demonstrated by Specker (1971), giving a recursive relation R without recursive infinite

[32] We do not have a concrete model construction to demonstrate this in mind, however.

R-homogenous sets. Much of the study since then has been focused on finding constructively or intuitionistically true 'Ramsey-like' principles, see e.g. Veldman and Bezem (1993), Berardi and Steila (2014), and Marzion (2016). The only work of constructive reverse mathematics in this area which we are aware of is Berardi and Steila (2017) who prove the standard Ramsey principle for at least two colours and recursive relations equivalent to Σ^0_3-LLPO over HA. As far as we know, the additive Ramsey principle, specifically, has not been considered in the constructive setting except by Lichter and Smolka (2018). The characterisation of the additive Ramsey principle, both in terms of LPO and in terms of ILPO + TWKL, seems to be novel. However, the direction of LPO $\rightarrow$ AR might already be present, in some form, in Berardi and Steila (2017).[33]

The work of Lichter and Smolka (2018) is a direct predecessor of our results. They carry out a constructive analysis of infinite word automata and S1S (monadic second-order logic with single successor) using Rocq. In our presentation, this corresponds to the formulation over CIC with purely logical connectives (i.e. $\exists$ and $\vee$). They prove various results over unrestricted infinite words equivalent to the additive Ramsey principle, most notably S1S satisfaction and satisfiability, correctness of automata complementation and decidability of automata acceptance. Restricting to periodic words, all of these results are provable constructively. The core of our findings (equivalence with additive Ramsey and constructivisation via restriction to periodic branches) directly parallel theirs. Indeed, using the usual argument of the decidability of the global trace condition via automata constructions (see e.g. theorem 4.4 of Wehr (2021)), these results can be presented as a sort of 'corollary' to (Lichter and Smolka 2018). The 'detour' through reset proofs we take in this chapter is only needed[34] to obtain the equivalences with LPO and the conjunction of ILPO and TWKL. Specifically the equivalence with LPO occupies a 'blind spot' of Lichter and Smolka (2018) because they work in a setting without unique choice, in which the 'collapse' with LPO does not occur.

Another similar work is Provenzano (2025). They study the classical reverse mathematics of an ill-founded proof system over an ill-founded, countably branching language. They prove consistency, cut-elimination and soundness of the system to each be equivalent to Π^1_3-RFN(Π^1_1-CA$_0$), a rather strong reflection

[33] Because the authors study the principle in the setting of Heyting arithmetic, it is not fully clear to us to what extend their results apply to the 'higher order' constructive foundations we consider here.

[34] In the sense that we have not found a more direct proof strategy. Of course, such a strategy might still exist.

principle. They also study the setting in which the language is restricted to binary branching: the strength of consistency and soundness remains the same but cut-elimination becomes equivalent to the somewhat weak Π^1_2-RFN(ACA_0).

8.7 Technical lemmas

In this section, we give proofs of lemmas which were employed in earlier sections whose proofs we deem too technical and uninformative to include in the main sections.

We begin with results needed in section 8.2. The following were used to prove that every sound reset annotation certifies the proofhood of the annotated preproof.

Proposition 8.7.1. Let (ρ, θ) be reset cycle along an $\mathcal{R}$-preproof Π. Let there be $a \in \bigcap_{i<|\rho|} \Theta_i$ and let at least two a-resets take place along the reset cycle. Then for every $x \in \iota(\lambda(\rho_0))$ with $a \in \sigma_0(x)$ there exists $y \in \iota(\lambda(\rho_0))$ with $a \in \sigma_0(y)$ such that $(x, 1, y) \in \iota(\rho)$.

Proof. We first note two observations: Let $(\Theta, \sigma) \overset{\tau}{\rightsquigarrow}_g (\Theta', \sigma')$ and let $a \in \Theta \cap \Theta'$. For every $\sigma'(y) \ni a$ there is $\sigma(x) \ni a$ such that $(x, b, y) \in \tau$. Furthermore,

(a) if a is not reset along $\overset{\tau}{\rightsquigarrow}_g$ then $\sigma(x) \leq \sigma'(y)$ and $b = 1$ if $\sigma(x) < \sigma'(y)$.

(b) if a is reset along $\overset{\tau}{\rightsquigarrow}_g$ then $\sigma(x) \geq \sigma'(y)$ and $b = 1$ if $\sigma(x) = \sigma'(y)$.

Observe that (a) generalises to any sequence of $\vec{\tau}$-steps without an a reset.

Write $\tau_i := \iota(\rho_i, \rho_{i+1})$. To obtain the result, let a be reset at $\theta_i \overset{\tau_i}{\rightsquigarrow}_g \theta_{i+1}$ and $\theta_j \overset{\tau_j}{\rightsquigarrow}_g \theta_{j+1}$ with $i < j$ and at no $\theta_k \overset{\tau_k}{\rightsquigarrow}_g \theta_{k+1}$ with $i < k < j$. Then we may divide the reset cycle as follows:

$$\theta_0 \overset{\tau_0\ldots\tau_{i-1}}{\rightsquigarrow_g} \theta_i \overset{\tau_i}{\rightsquigarrow}_g \theta_{i+1} \overset{\tau_{i+1}\ldots\tau_{j-1}}{\rightsquigarrow_g} \theta_j \overset{\tau_j}{\rightsquigarrow}_g \theta_{j+1} \overset{\tau_{j+1}\ldots\tau_{|\rho|-2}}{\rightsquigarrow_g} \theta_{|\rho|-1} = \theta_0$$

Towards the result, let $\sigma_0(y) \ni a$. Then by the above observations, there are $\sigma_{j+1}(y') \ni a$, $\sigma_j(z) \ni a$, $\sigma_{i+1}(x') \ni a$ and $\sigma_0(x) \ni a$ such that

$$(x, b_1, x') \in \tau_i \circ \ldots \circ \tau_0 \qquad (x', b_2, z) \in \tau_{j-1} \circ \ldots \circ \tau_{i+1} \qquad (z, b_3, y') \in \tau_j$$

$$(y', b_4, y) \in \tau_{|\rho|-2} \circ \ldots \tau_{j+1}.$$

Thus, we already know that $(x, b_1 \vee b_2 \vee b_3 \vee b_4, y) \in \iota(\rho)$. It remains to show that $b_1 \vee b_2 \vee b_3 \vee b_4 = 1$. Towards this, suppose $b_3 = 0$, meaning by (b) that $\sigma_j(z) > \sigma_{j+1}(y')$. Because $\overset{\tau_j}{\rightsquigarrow}_g$ incurred an a-reset, that means that a must be already covered in $\sigma_j(z)$ i.e., $\max \sigma_j(z) \neq a$. On the other hand, because $\overset{\tau_i}{\rightsquigarrow}_g$

also incurs an a-reset, a must be uncovered in $\sigma_{i+1}(x')$ i.e., $\max \sigma_{i+1}(x') = a$. By (a), this can only mean that $\sigma_{i+1}(x') < \sigma_j(z)$, as no a-resets take place between them, and thus that $b_2 = 1$ as desired. □

Proposition 8.7.2. Let $R \subseteq n \times n$ with $n > 0$ be surjective and decidable. Then there exists a cycle $x \in n^{>1}$ through R i.e., $x_0 = x_{|x|-1}$ and $(x_i, x_{i+1}) \in R$.

Proof. We call a sequence $u \in n^*$ a path through R if $(u_i, u_{i+1}) \in R$ for all $i < |u| - 1$. We prove per induction on $n - |X|$ the following claim: Let $X \subseteq n$ and $u \in n^*$ a path through R such that every $m \in X$ occurs in u, then there is a cycle through R.

Base: Then $X = n$. As R is surjective, there is $(x, u_0) \in R$. As $x \in X$ we know $u'x < u$ for some $u' \in n^*$. Then $xu'x$ is a cycle through R.

Step: If $X = \emptyset$, we may apply the IH with $u := 0$ and $X := \{0\}$. Suppose $X \neq \emptyset$ and thus $|u| > 0$. As R is surjective, there is $(x, u_0) \in R$. If $x \notin X$, we may apply the IH with $X := X \cup \{x\}$ and $u := xu$. If $x \in X$ then $u'x < u$ for some $u' \in n^*$ and $xu'x$ is a cycle through R.

□

We continue by proving that failed cycles in reset annotations give rise to branches in the original preproof which serve as a counterexample to proofhood. This technical result actually requires a technical lemma of its own.

Lemma 8.7.3.

1. Let $(\Theta, \sigma) \overset{\tau}{\rightsquigarrow}_g (\Theta', \sigma')$ and $(x, b, y) \in \tau$. Then $\min \sigma(y) \leq_\Theta \min \sigma(x)$.
2. Let $(\Theta, \sigma) \overset{\tau}{\rightsquigarrow}_g (\Theta', \sigma')$. If $\gamma \in \Theta \cap \Theta'$ then $\{\gamma' \leq_{\Theta'} \gamma\}$ is a prefix of Θ.
3. Let $(\Theta_i, \sigma_i)_{i \leq n}$, $(\tau_i)_{i<n}$ and $(x_i)_{i \leq n}$, $(b_i)_{i<n}$ be given such that $(\Theta_i, \sigma_i) \overset{\tau_i}{\rightsquigarrow}_g (\Theta_{i+1}, \sigma_{i+1})$ and $(x_i, b_i, x_{i+1}) \in \tau_i$ for every $i < n$. Then $\min \sigma_n(x_n) \leq_{\Theta_0} \min \sigma_i(x_i) \leq_{\Theta_0} \min \sigma_0(x_0)$ for any $i \leq n$.
4. Let $(\Theta, \sigma) \overset{\tau}{\rightsquigarrow}_g (\Theta', \sigma')$ and $(x, b, y) \in \tau$. If σ_p is a prefix of $\sigma(x), \sigma'(y)$ and $\sigma(x) \setminus \sigma_p \neq \emptyset$ then on of the following must hold:
   a. $\sigma'(y) = \sigma_p$ and $\max \sigma_p$ was reset along $(\Theta, \sigma) \overset{\tau}{\rightsquigarrow}_g (\Theta', \sigma')$
   b. $\min(\sigma'(y) \setminus \sigma_p) \leq_\Theta \min(\sigma(x) \setminus \sigma_p)$
5. Let $(\Theta_i, \sigma_i)_{i \leq n}$, $(\tau_i)_{i<n}$ and $(x_i)_{i \leq n}$, $(b_i)_{i<n}$ be given such that $(\Theta_i, \sigma_i) \overset{\tau_i}{\rightsquigarrow}_g (\Theta_{i+1}, \sigma_{i+1})$ and $(x_i, b_i, x_{i+1}) \in \tau_i$ for every $i < n$. Further, let σ_p be a prefix of $\sigma_i(x_i)$ for every $i \leq n$ and let $\sigma_0(x_0) \setminus \sigma_p \neq \emptyset$. Then one of the following must hold:

(i) there is $i < n$ such that $\max \sigma_p$ is reset at $(\Theta_i, \sigma_i) \overset{\tau_i}{\rightsquigarrow}_g (\Theta_{i+1}, \sigma_{i+1})$

(ii) $\sigma_p \cup \{\min(\sigma_0(x_0) \setminus \sigma_p)\}$ is a prefix of $\sigma_i(x_i)$ for every $i \leq n$

(iii) $\sigma_n(x_n) \setminus \sigma_p \neq \emptyset$ and $\min(\sigma_n(x_n) \setminus \sigma_p) <_{\Theta_0} \min(\sigma_0(x_0) \setminus \sigma_p)$

Then $\min \sigma_n(x_n) \leq_{\Theta_0} \min \sigma_i(x_i) \leq_{\Theta_0} \min \sigma_0(x_0)$ for any $i \leq n$.

Proof.

1. We know that $\sigma(x)$ is moved into $\overline{\sigma}(y, -)$ by the MOVE and potentially extended into $\sigma_e \in \sigma_c(y, -)$ by COVER. Observe that $\min_{\Theta \oplus \Theta_c} \sigma(x) = \min_{\Theta \oplus \Theta_c} \sigma_e$. Ultimately, $\sigma'(y)$ is chosen from among $\sigma_c(y, -)$ via THIN, meaning $\min \sigma'(y) \leq \sigma_e$ (otherwise, σ_e should be chosen over $\sigma'(y) \sqsubset \sigma_T(y)$). As $\sigma_T(y) \neq \emptyset$, the POPULATE step does not affect y and RESET does not affect $\min \sigma(y)$.
2. This is because any such $\gamma' \in \Theta'$ must originate from Θ and cannot have been 'added' during the COVER and POPULATE steps, as these only introduce $\gamma' >_{\Theta'} \gamma$.
3. Proof by induction on n. $n = 0$ is trivial. Thus, suppose for the inductive step that $\min \sigma_{n+1}(x_{n+1}) \leq_{\Theta_1} \min \sigma_{i+1}(x_{i+1}) \leq_{\Theta_1} \min \sigma_1(x_1)$ per IH. By 1. we know that $\min \sigma_1(x_1) <_{\Theta_0} \min \sigma_0(x_0)$. Now simply observe that if $\gamma \leq_{\Theta_1} \sigma_1(x_1)$ then $\gamma <_{\Theta_0} \sigma_1(x_1)$ by 2.
4. Perform a case-distinction:

   $\sigma'(y) = \sigma_p$: The chip $\min(\sigma(x) \setminus \sigma_p)$ cannot have been removed from the stack by thinning, hence it must have been removed by a reset. As $\max \sigma_p \in \sigma'(y)$, it must have been a $\max \sigma_p$ reset.

   $\sigma'(y) \neq \sigma_p$: Write $\gamma_x := \min(\sigma(x) \setminus \sigma_p)$, $\gamma_y := \min(\sigma'(y) \setminus \sigma_p)$. Then the only way $\gamma_x \neq \gamma_y$ is possible if a thinning has 'replaced' the stack. But this means that $\gamma_y <_{\Theta \oplus \Theta_c} \gamma_x$. As $\gamma_x \in \Theta$ this must mean $\gamma_y <_\Theta \gamma_x$ as claimed.
5. Proof by induction on n. For $n = 0$, (ii) trivially holds. For the inductive step, suppose claim held for a suffix of length n. We first perform a case-distinction on $(\Theta_0, \sigma_0) \overset{\tau_0}{\rightsquigarrow}_g (\Theta_1, \sigma_1)$ according to 4.:

   a): That means $\max \sigma_p$ was reset along $(\Theta_0, \sigma_0) \overset{\tau_0}{\rightsquigarrow}_g (\Theta_1, \sigma_1)$ and case (i) of our claim holds.

   b): That means $\sigma_1(x_1) \setminus \sigma_p \neq \emptyset$ and, writing $\gamma_i := \min(\sigma_i(x_i) \setminus \sigma_p)$, $\gamma_1 \leq_{\Theta_0} \gamma_0$. Furthermore, the IH applies for the sequence from 1 to $n + 1$. Case distinction on the outcome of the IH:

   (i): In that case (i) also applies from 0 to $n + 1$.

(ii): We perform a case distinction on $\gamma_1 \leq_{\Theta_0} \gamma_0$:

$\gamma_1 = \gamma_0$: Then $\sigma_p \cup \{\gamma_0\} = \sigma_p \cup \{\gamma_1\}$ is a prefix of the sequence from 0 to $n+1$ and (ii) applies.

$\gamma_1 <_{\Theta_0} \gamma_0$: Then (iii) applies: $\gamma_{n+1} = \gamma_1 <_{\Theta_0} \gamma_0$.

(iii): By 2. this means that $\gamma_{n+1} <_{\Theta_0} \gamma_1$ and thus $\gamma_{n+1} <_{\Theta_0} \gamma_0$, meaning (iii) still applies.

□

Proposition 8.7.4. If (ρ, θ) is a failed reset cycle of an $\mathcal{R}$-preproof Π then there exists a periodic branch of Π which is not progressing.

Proof. The branch in question is $[\varepsilon, \rho_0]\rho^\omega$. Suppose it was progressing. W.l.o.g. that means there exists a trace $(x_i)_{i\leq n}(b_i)_{i<n}$ along ρ^k for some k such that $x_0 = x_n$ and $b_s = 1$ for some $s < n$. We make some observations.

(i) $\min \sigma_i(x_i) = \min \sigma_j(x_j)$ *for every* $i, j \leq n$: As $x_0 = x_n$ we know that $\sigma_0(x_0) = \sigma_n(x_n)$. By lemma 8.7.3.3. it thus follows that $\min \sigma_0(x_0) \leq_{\Theta_0} \min \sigma_i(x_i) \leq_{\Theta_0} \sigma_0(x_0)$, meaning $\min \sigma_0(x_0) \leq_{\Theta_0}= \sigma_i(x_i)$, for any $i \leq n$.

(ii) *There exists a maximal, non-empty* σ^* *which is a prefix to every* $\sigma_i(x_i)$*:* We know that $\{\min \sigma_0(x_0)\}$ is such a prefix. The prefixes are linearly ordered by the prefix ordering, so there must exist a maximal one.

Now we perform a case distinction:

$\sigma_0(x_0) = \sigma^*$: Because $(x_s, b_s, x_{s+1}) \in \tau_s$ we know that $\sigma_{s+1}(x_{s+1}) \setminus \sigma^* \neq \emptyset$ as the cover rule must add a new chip to $\sigma_{s+1}(x_{s+1})$. Now apply lemma 8.7.3.5 and perform a case distinction on the outcome:

(i) It follows from lemma 8.7.3.2 that $\{\gamma <_{\Theta_0} \max \sigma^*\} \subseteq \Theta_0$ is an invariant along the cycle ρ^k, meaning it is also an invariant along ρ^k. If $\max \sigma^*$ is reset along the cycle ρ, this means ρ was no failed reset circle in the first place, a contradiction.

(ii), (iii): Either case implies that $|\sigma_n(x_n)| > |\sigma^*|$ but per assumption $\sigma_n(x_n) = \sigma_0(x_0) = \sigma^*$, a contradiction.

$\sigma_0(x_0) \neq \sigma^*$: Then apply lemma 8.7.3.5 to the whole cycle ρ^k and perform a case distinction on the outcome to derive contradictions:

(i) As in the previous case (i), this would make ρ a successful reset cycle.

(ii) This contradicts σ^* being the maximal invariant.

(iii) Because of $\sigma_n(x_n) = \sigma_0(x_0)$, this would imply $\min(\sigma_0(x_0) \setminus \sigma^*) <_{\Theta_0} \min(\sigma_0(x_0) \setminus \sigma^*)$, contradicting irreflexivity of $<_{\Theta_0}$.

□

We close by proving the correctness of the construction $\gamma(\sigma)$ employed in the reduction theorem 8.4.5. Again, this correctness proof actually requires a small lemma of its own.

Proposition 8.7.5. Let Π be an $\mathcal{R}$-preproof, π a branch along it and $(x_i \in \iota(\lambda(\pi_{m+i})))_{i\in\omega}$ an infinitely progressing trace along it. Then there exists a strictly increasing $\rho : \omega \to \omega$ such that for every $i \in \omega$ if there is $\rho(n) = i$ then $(x_i, 1, x_{i+1}) \in \iota(\pi_{m+i}, \pi_{m+i+1})$ and if $i \notin \operatorname{im}(\rho)$ then $(x_i, 0, x_{i+1}) \in \iota(\pi_{m+1}, \pi_{m+i+1})$.

Proof. Recall that $(x_i \in \iota(\lambda(\pi_{m+i})))_{i\in\omega}$ being progressing means that for any $i \in \omega$ there exists $j > i$ such that j *progresses* i.e., $(x_j, 1, x_{j+1}) \in \iota(\pi_{m+j}, \pi_{m+j+1})$. This means that there is a function $\gamma : \omega \to \omega$ such that $\gamma(i) > i$ and $\gamma(i)$ progresses for every $i \in \omega$. Then we can define

$$\rho(0) := \min\{j \leq \gamma(0) \mid j \text{ progresses}\}$$
$$\rho(i+1) := \min\{\rho(i) < j \leq \gamma(\rho(i)) \mid j \text{ progresses}\}$$

which is well defined by the properties of γ. □

Claim 8.7.6. Every $\sigma \in \Gamma^\omega$ has a factorisation iff $\gamma(\sigma)$ has a progressing trace.

Proof.

$\to$: Suppose there was a strictly increasing $\tau : \omega \to \omega$ and a $g \in \Gamma$ such that $g = \sum_{i=\tau(n)}^{i<\tau(n+1)} \sigma_i$. Consider the sequence $x \in \iota(\star)^\omega$ given below:

$$x_j := \left(g, \sum_{i=\tau(n)}^{i<\tau(n)+m} \sigma_i\right) \text{ where } \tau(0) + \left\lceil \frac{j}{2} \right\rceil = \tau(n) + m \leq \tau(n+1)$$

and we set $\sum_{i=k}^{i<k} \sigma_i := \bullet$. We begin by proving that it is a trace along $\gamma(\sigma)$ starting at offset $2\tau(0)$ i.e., we prove that

$$(x_j, b, x_{j+1}) \in \iota(\gamma(\sigma)_{2\tau(0)+j}, \gamma(\sigma)_{2\tau(0)+j+im})$$

for every $j \in \omega$:

j ODD: Then the step from $\gamma(\sigma)_{2\tau(0)+j}$ to $\gamma(\sigma)_{2\tau(0)+j+1}$ is a step from bud to companion and $\iota(\gamma(\sigma)_{2\tau(0)+j}, \gamma(\sigma)_{2\tau(0)+j+1}) = 1_{\iota(\star)}$. Per definition $x_j = x_{j+1}$ and thus $(x_j, 0, x_{j+1}) \in 1_{\iota(\star)}$, as desired.

j EVEN: We know there must be unique m, n such that $\tau(0) + \lceil \frac{j+1}{2} \rceil = \tau(n)+m \leq \tau(n+1)$. Then the step from $\gamma(\sigma)_{2\tau(0)+j}$ to $\gamma(\sigma)_{2\tau(0)+j+1}$ is from ε to $(k-1) \in \omega^*$ with $\sigma_{\tau(n)+m-1} = g_k$ and

$$\iota(\gamma(\sigma)_{2\tau(0)+j}, \gamma(\sigma)_{2\tau(0)+j+1}) = \dagger_k.$$

We again distinguish two cases:

$m = 0$: Then $x_j = (g, \sum_{i=\tau(n-1)}^{i<\tau(n)-1} \sigma_i)$ and $\sigma_{\tau(n)-1} = g_k$ and $x_{j+1} = (g, \bullet)$. The second case of $\dagger_k$ applies i.e., $(x_j, 1, x_{j+1}) \in \dagger_k$ because $(\sum_{i=\tau(n-1)}^{i<\tau(n)-1} \sigma_i) + \sigma_{\tau(n)-1} = g$ per the factorisation.

$m > 0$: Then $x_j = (g, \sum_{i=\tau(n)}^{i<\tau(n)+m-1} \sigma_i)$ and furthermore $x_{j+1} = (g, \sum_{i=\tau(n)}^{i<\tau(n)+m} \sigma_i)$. Hence the first case of $\dagger_k$ i.e., $(x_j, 0, x_{j+1}) \in \dagger_k$.

Lastly, observe that $(x_i \in \iota(\star))_{i\in\omega}$ is progressing: the case of j even and $m = 0$ causes the trace to progress, which happens infinitely often because the $\tau(n)$ are finite.

$\longleftarrow$: Suppose $(x_i \in \iota(\star))_{i\in\omega}$ was a progressing trace along $\gamma(\sigma)$ with offset m. By proposition 8.7.5, we may assume a strictly increasing $\rho : \omega \to \omega$ which marks all progress points along $(x_i)_{i\in\omega}$. That is,

$$(x_i, b, x_{i+1}) \in \iota(\gamma(\sigma)_{m+i}, \gamma(\sigma)_{m+i+1}) \text{ where } b = \begin{cases} 1 & \exists n.\ i = \rho(n) \\ 0 & \text{otherwise} \end{cases}.$$

Per definition of the $\dagger_k$, we know there is some $g \in \Gamma$ such that that $x_i = (g, a_i)$ for all $i \in \omega$. Let w.l.o.g. $a_0 = \bullet$ and m be even and thus $\gamma(\sigma)_m = \varepsilon$. That means $a_{2(i+1)} = a_{2i+1}$ for all $i \in \omega$ because the step from $\gamma(\sigma)_{m+2i+1}$ to $\gamma(\sigma)_{m+2(i+1)}$ leads from a bud to the root via a cycle. On the other hand, $\gamma(\sigma)_{m+2i} = k \in \omega^*$ such that $\sigma_{\frac{m}{2}+i} = g_{k+1}$. This also means that all $\rho(n)$ must be even. With this, we know for $2i > \rho(0)$ that we have

$$a_{2i} = \sum_{j=\frac{\rho(n)}{2}+1}^{j<i} \sigma_{\frac{m}{2}+j} \quad \text{where } \rho(n) < 2i \leq \rho(n+1)$$

which we prove per induction on $2i - \rho(n)$ where n is as above:

$2i = \rho(n) + 2$: The equation above requires that $a_{2i} = \bullet$. Because the trace progresses from $x_{\rho(n)}$ to $x_{\rho(n)+1} = x_{2i-1}$, we know that $a_{2i-1} = \bullet$ from the definition of the $\dagger_k$s. As observed above, we know that $a_{2i} = a_{2i-1} = \bullet$.

$\rho(n) + 2 < 2i \leq \rho(n+1)$: We know that $a_{2i-2} = \sum_{j=\frac{\rho(n)}{2}+1}^{j<i-1} \sigma_{\frac{m}{2}+j}$ per IH. Because $2i \leq \rho(n+1)$, we know that the trace does not progress from x_{2i-2} to x_{2i-1}, in other words we know that $a_{2i-1} = a_{2i-2} + \sigma_{\frac{m}{2}+i-1}$. As $a_{2i} = a_{2i-1}$, and observing that $i - 1 \geq \frac{\rho(n)}{2} + 1$, the equation is thus satisfied.

Using the observations above and because every step from $x_{\rho(n)}$ to $x_{\rho(n)+1}$ progresses, we thus know that

$$g = a_{\rho(n)} + \sigma_{\frac{m+\rho(n)}{2}} = \sum_{j=\frac{\rho(n-1)}{2}+1}^{j\leq\frac{\rho(n)}{2}} \sigma_{\frac{m}{2}+j}$$

for every $n > 1$. Thus $\tau(n) := \frac{\rho(n)}{2} + 1$ is a factorisation of σ into g as desired.

□

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Sammanfattning

Cykliska bevis representeras av grafer snarare än bevisträd och kräver ytterligare sundhetsvillkor för att urskilja bevis från andra härledningar. Genom att utnyttja en abstrakt beskrivning av cykliska bevissystem presenterar denna avhandling på ett enhetligt vis resultat rörande olika metoder att representera samma bevissystem.

Vi bevisar att alla traditionella cykliska bevissystem kan representeras som bevissystem av två andra typer: som återställningsbevis (reset proofs), som gör sundhetsvillkoret lokalt genom att utnyttja en metod för annoteringar av sekventer, och som stackkontrollerade bevis (stack-controlled proofs), ett begrepp som presenteras i denna avhandlingen, vari den cykliska och induktiva strukturen av bevis sammanfaller.

Genom att utnyttja Stackkontrollerade bevis presenterar vi ytterligare resultat: en ny översättning mellan cyklisk och 'vanlig' Heytingaritmetik; en korrespondens mellan fragment av cyklisk Peanoaritmetik och de induktiva fragmenten $I\Pi_n$; ett bevisteoretiskt konsistensargument samt en ordinalannotering för cyklisk Peanoaritmetik; ett generellt sundhetsargument för cykliska bevissystem som undviker att använda sig av klassiska principer.

I avhandlingen utnyttjas även återställningsbevis för att presentera en konstruktiv analys av den vanligaste typen av sundhetskriterium för cykliska bevissystem. Detta kriterium är ekvivalent med den additiva versionen av Ramseys sats, samt med en kombination av två nya principer, och är i många fall också ekvivalent med den begränsade allvetandeprincipen (LPO). Sundhetskriteriet kan dessutom göras helt konstruktivt genom att begränsa villkoret till periodiska vägar genom grafen.

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